TABLE 3.9.N.3-2 (SHEET 7 OF 8)

Valve Number	System	Actuated By	Size (in.)	Type/ANS Safety Class	Normal Position	Basis
8923A/B[(b)]	SIS	Motor	6	Gate/2	Open	(3)
8920	SIS	Motor	1 1/2	Globe/2	Open	(3)
8924	SIS	Motor	6	Gate/2	Open	(3)
1204U6090	SIS	ΔP	8	Check/2	NA	(3)
1204U6083	SIS	ΔP	10	Check/1	NA	(1) (3)
1204U6084	SIS	ΔP	10	Check/1	NA	(1) (3)
1204U6085	SIS	ΔP	10	Check/1	NA	(1) (3)
1204U6086	SIS	ΔP	10	Check/1	NA	(1) (3)
1204U6124	SIS	ΔP	6	Check/1	NA	(1) (3)
1204U6125	SIS	ΔP	6	Check/1	NA	(1) (3)
1204U6126	SIS	ΔP	6	Check/1	NA	(1) (3)
1204U6127	SIS	ΔP	6	Check/1	NA	(1) (3)
1204U6079	SIS	ΔP	10	Check/1	NA	(1) (3)
1204U6080	SIS	ΔP	10	Check/1	NA	(1) (3)
1204U6081	SIS	ΔP	10	Check/1	NA	(1) (3)
1204U6082	SIS	ΔP	10	Check/1	NA	(1) (3)
1205U6001	SIS	ΔP	12	Check/2	NA	(3) (4)
1205U6002	SIS	ΔP	12	Check/2	NA	(3) (4)
8964	SIS	Air	3/4	Globe/2	Closed	(2)
2402U4017	SIS	ΔP	1	Check/2	NA	(2)
1208U6436	SIS	ΔP	8	Check/2	NA	(3)
1204U6163	SIS	ΔP	8	Check/2	NA	(3)

TABLE 3.9.N.3-2 (SHEET 8 OF 8)

Valve Number	System	Actuated By	Size (in.)	Type/ANS Safety Class	Normal Position	Basis
HV-943A/B	SIS	Solenoid	1	Globe/2	Closed	(4)
7126	WPS	Air	3/4	Diaphragm/2	Open	(2)
7136	WPS	Air	3	Diaphragm/2	Open	(2)
7150	WPS	Air	3/4	Diaphragm/2	Open	(2)
7699	WPS	Air	3	Diaphragm/2	Open	(2)
9001A/B	CSS	Motor	8	Gate/2	Closed	(2) (5)
9002A/B	CSS	Motor	10	Gate/2	Closed	(2) (5)
9003A/B	CSS	Motor	10	Gate/2	Closed	(5)
1206U6015	CSS	ΔP	8	Check/2	NA	(2) (5)
1206U6016	CSS	ΔP	8	Check/2	NA	(2) (5)
9017A/B	CSS	Motor	10	Gate/2	Open	(5)
1206U6001	CSS	ΔP	10	Check/2	NA	(5)
1206U6008	CSS	ΔP	10	Check/2	NA	(5)
7603A/B/C/D	SGBD	Air	3	Globe/2	Open	(2) (3)

a. Basis are defined as follows:
   1. Reactor coolant system pressure boundary isolation.
   2. Containment isolation.
   3. ECCS safeguards operations.
   4. Safety grade cold shutdown operation.
   5. CSS safeguards operations.

b. For these valves, the capability to perform their safety related function will be demonstrated through inclusion in plant maintenance programs, plant procedures, and/or Technical Specifications.

c. Receives a nonsafety-related control grade signal, failure to close has no significant impact on plant safety.

d. Unit 1 only.

NOTE: The valve table as shown reflects the inservice test program as modified by the second 10-year interval update. The updated IST program will be fully implemented by May 31, 1998.

TABLE 3.9.N.3-3

LOAD COMBINATIONS AND ACCEPTANCE CRITERIA FOR
PRESSURIZER SAFETY AND RELIEF VALVE PIPING - UPSTREAM OF VALVES

Combination	Plant/System Operating Condition	Load Combination	Piping Allowable Stress Intensity
1	Normal	N	$1.5\ S_m$
2	Upset	$N + OBE + SOT_U$	$1.8\ S_m/1.5\ S_y$
3	Emergency	$N + SOT_E$	$2.25\ S_m/1.8\ S_y$
4	Faulted	$N + SSE + SOT_F$	$3.0\ S_m$

1. Table 3.9.N.3-5 contains SOT definitions and other load abbreviations.

2. SRSS is to be used for combining dynamic load responses.

TABLE 3.9.N.3-4

LOAD COMBINATIONS AND ACCEPTANCE CRITERIA FOR
PRESSURIZER SAFETY AND RELIEF VALVE PIPING
SEISMICALLY DESIGNED DOWNSTREAM PORTION

Combination	Plant/System Operating Condition	Load Combination	Piping Allowable Stress Intensity
1	Normal	N	1.0 S_h
2	Upset	N + SOT_U	1.2 S_h
3	Upset	N + OBE + SOT_U	1.8 S_h
4	Emergency	N + SOT_E	1.8 S_h
5	Faulted	N + SSE + SOT_F	2.4 S_h

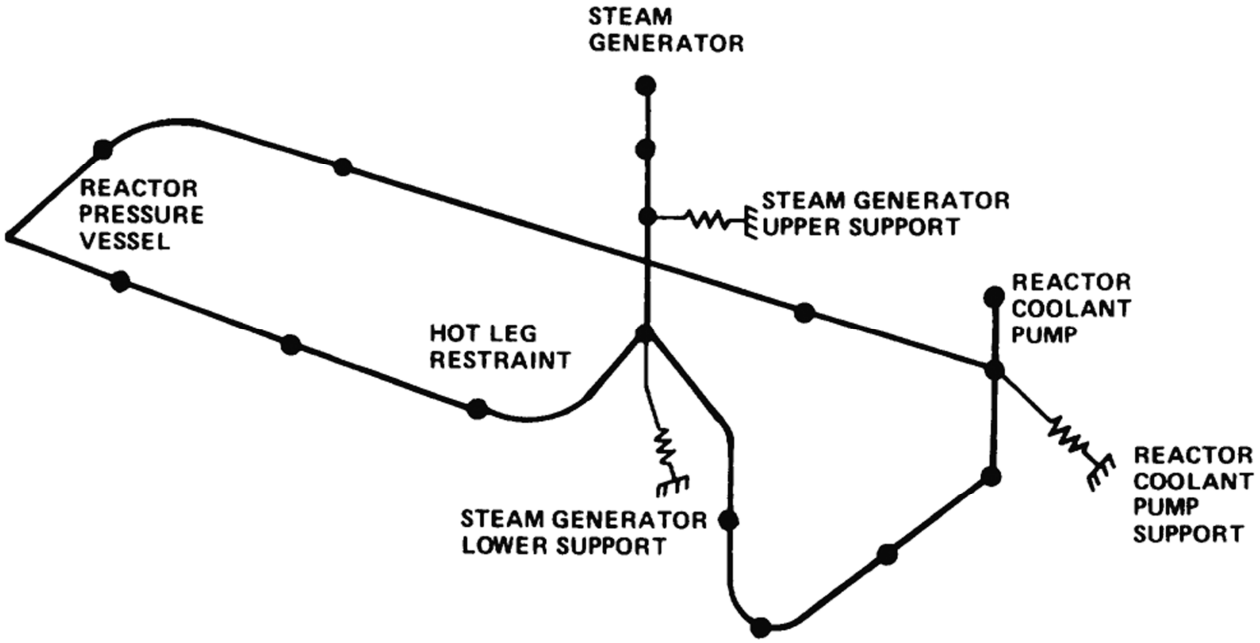
1. Table 3.9.N.3-5 contains SOT definitions and other load abbreviations.

2. SRSS is to be used for combining dynamic load responses.

TABLE 3.9.N.3-5

DEFINITIONS OF LOAD ABBREVIATIONS

N	=	Sustained loads during normal plant operation
SOT	=	System operating transient
SOT_U	=	Relief valve discharge transient
SOT_E	=	Safety valve discharge transient
SOT_F	=	Max (SOT_U, SOT_E), or transition flow
OBE	=	Operating basis earthquake
SSE	=	Safe shutdown earthquake
S_h	=	Basic material allowable stress at maximum (hot) temperature
S_m	=	Allowable design stress intensity
S_y	=	Yield strength value



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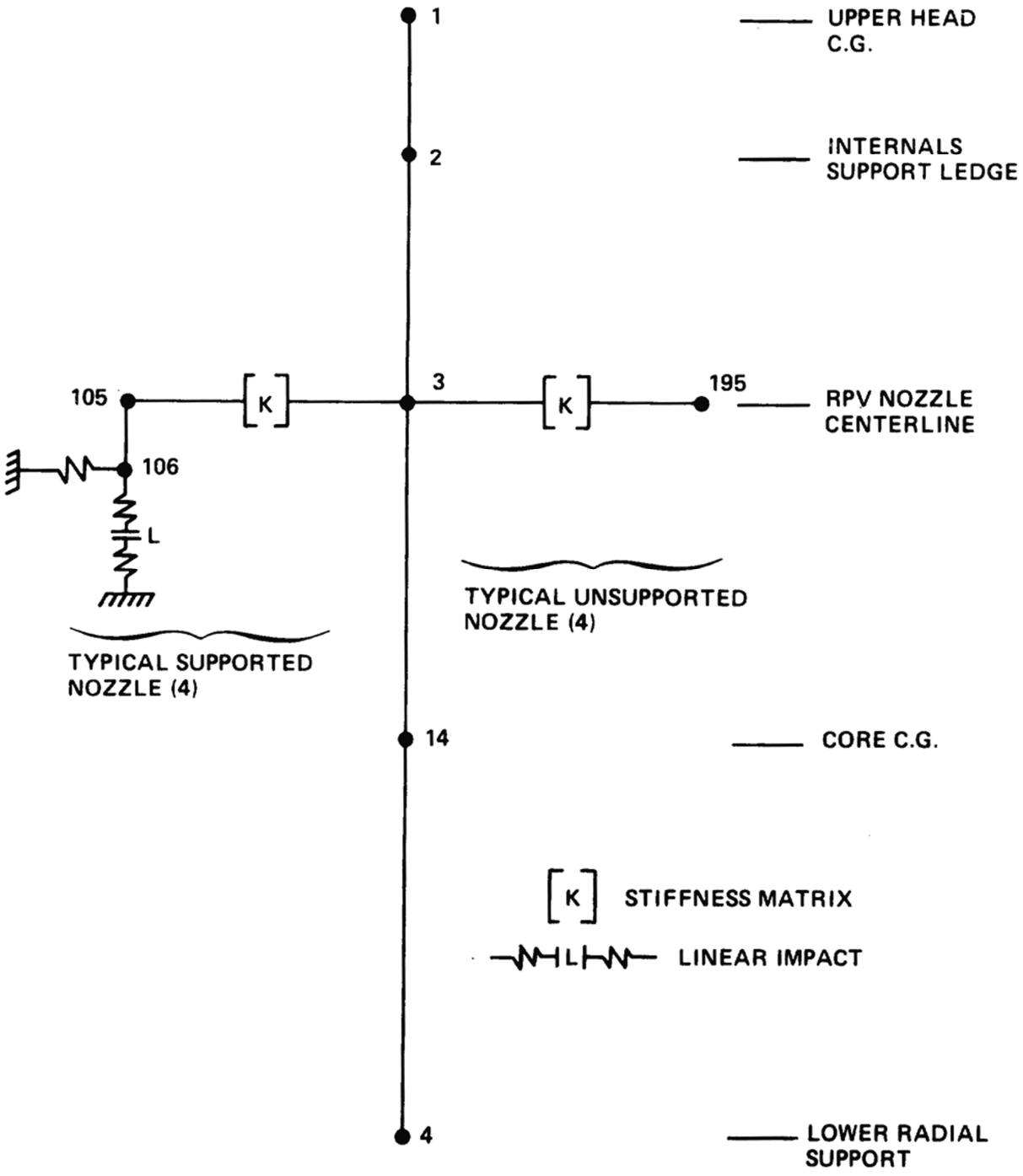
SOUTHERN COMPANY *Energy to Serve Your World®* VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	REACTOR COOLANT LOOP SUPPORTS SYSTEM DYNAMIC STRUCTURAL MODEL
	FIGURE 3.9.N.1–1



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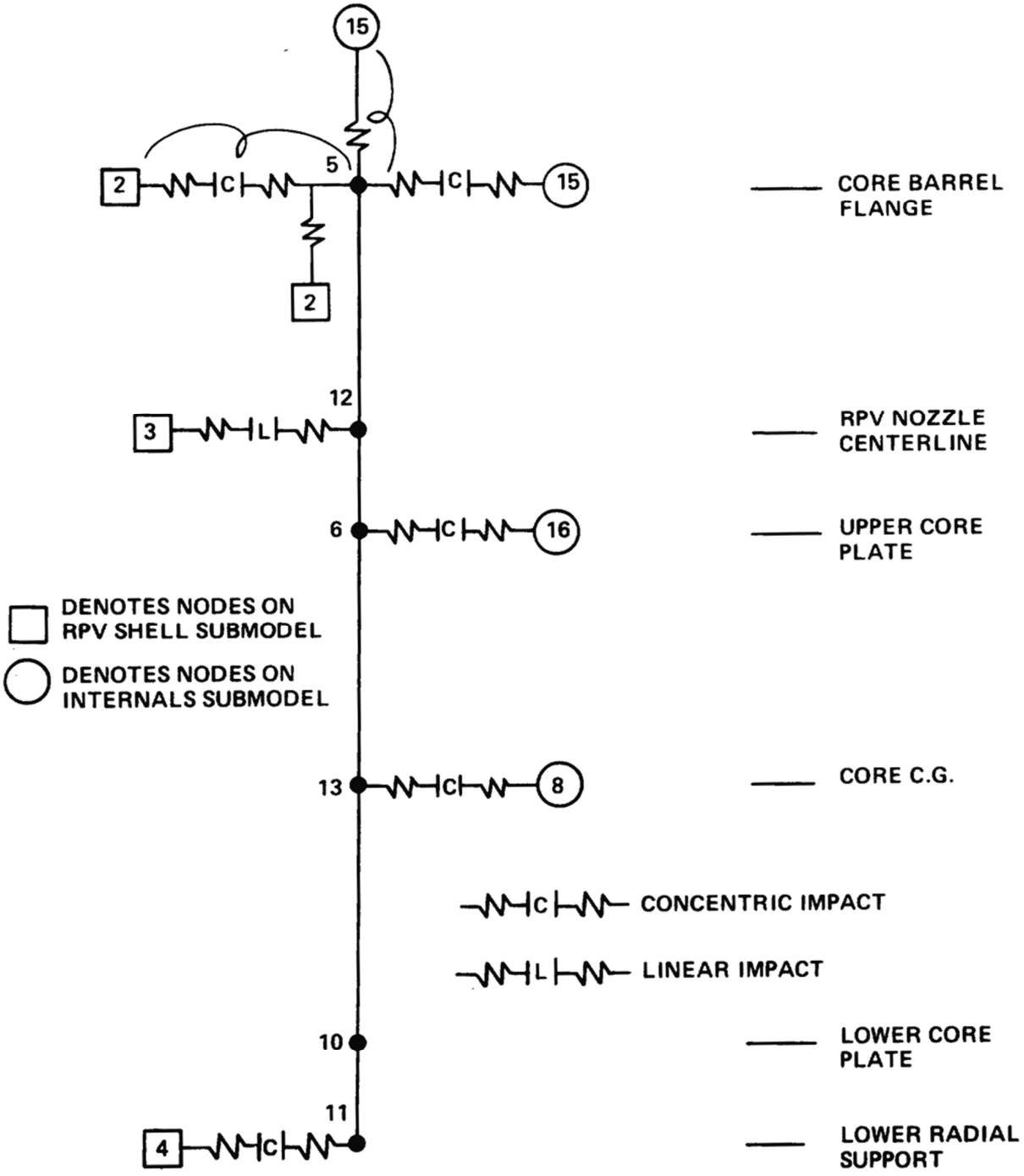
SOUTHERN COMPANY *Energy to Serve Your World®* VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	RPV SHELL SUBMODEL
	FIGURE 3.9.N.1–2



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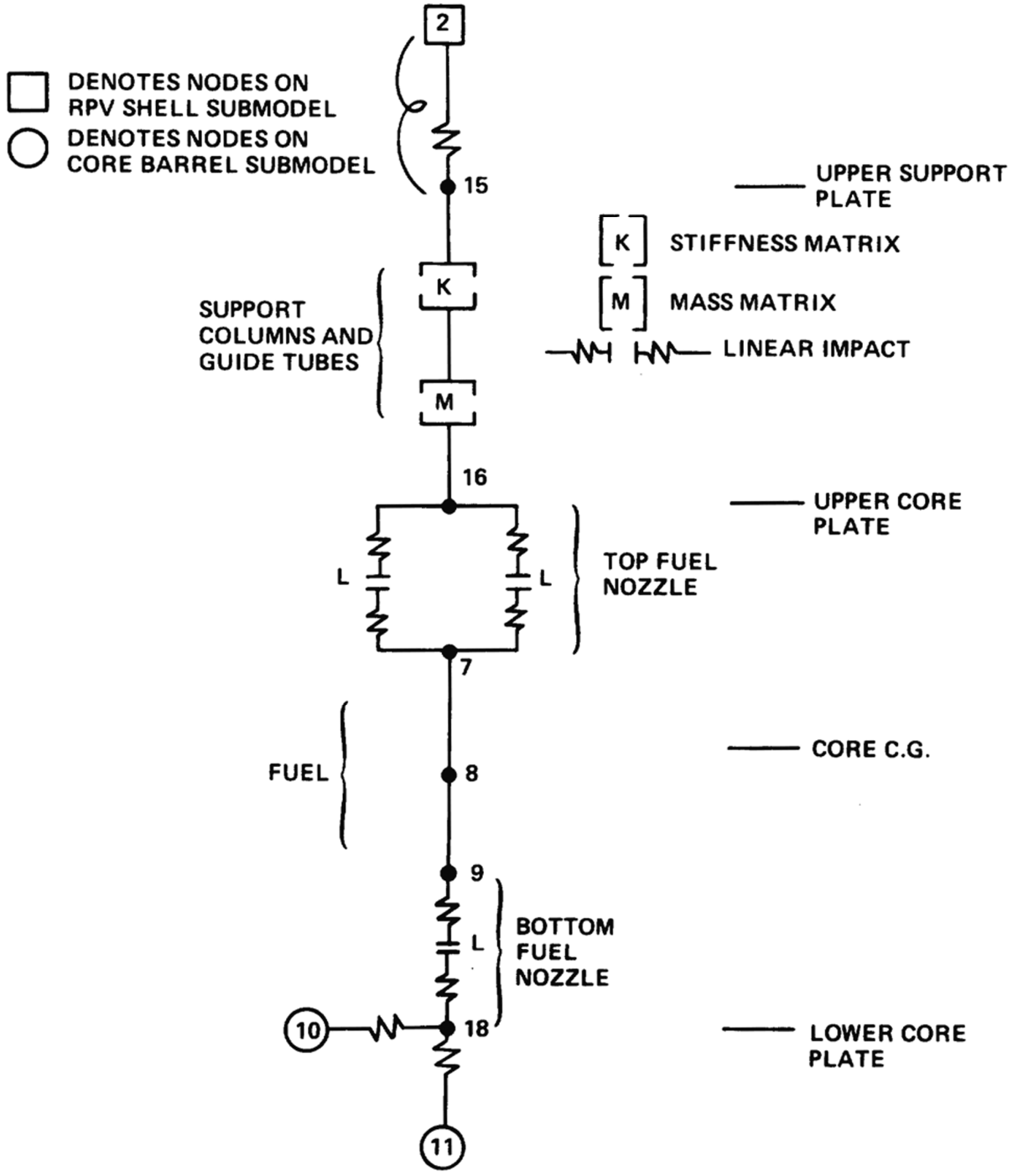
SOUTHERN COMPANY *Energy to Serve Your World®* VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	CORE BARREL SUBMODEL
	FIGURE 3.9.N.1–3



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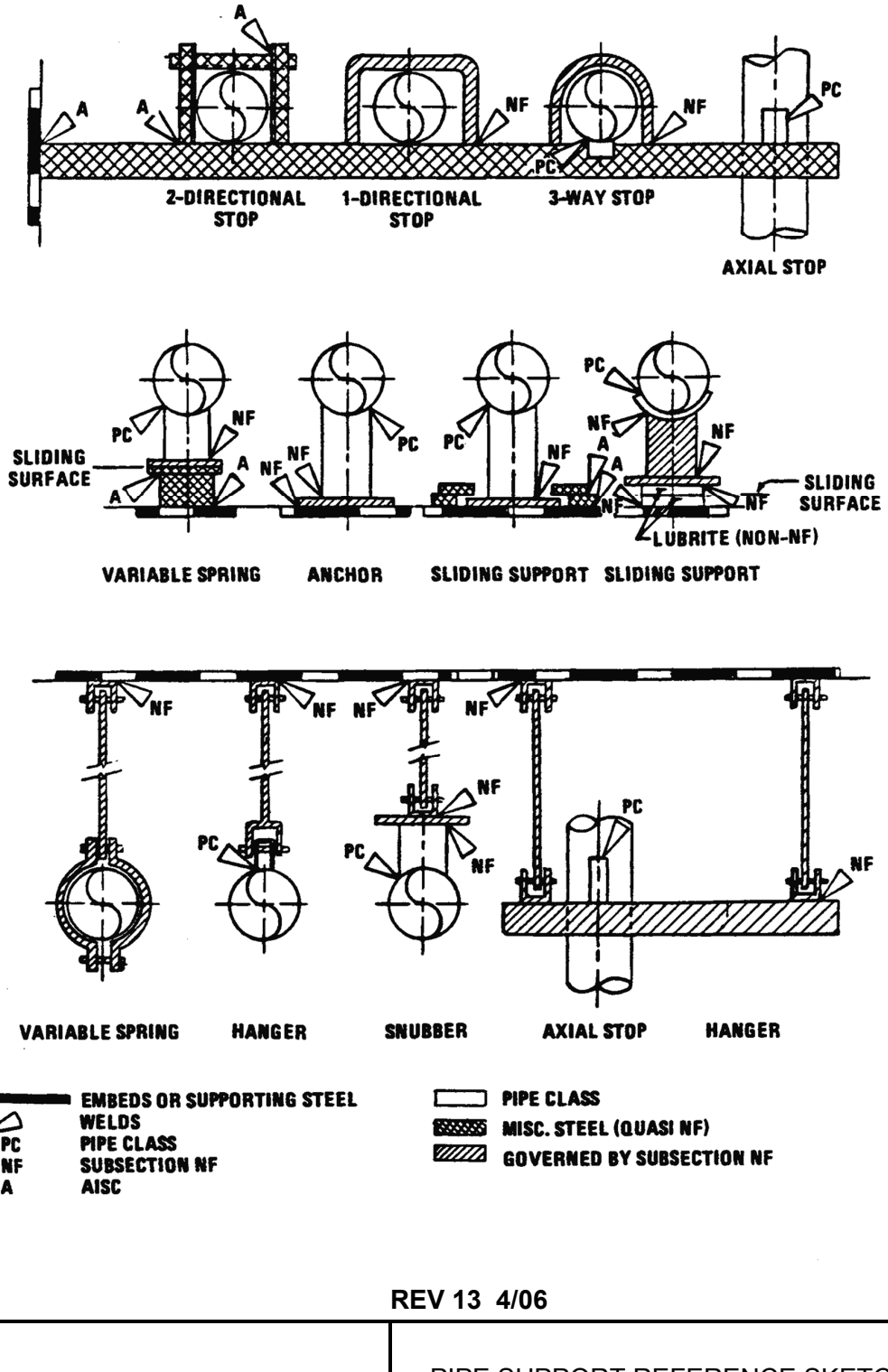
SOUTHERN COMPANY *Energy to Serve Your World®* VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	INTERNALS SUBMODEL
	FIGURE 3.9.N.1–4



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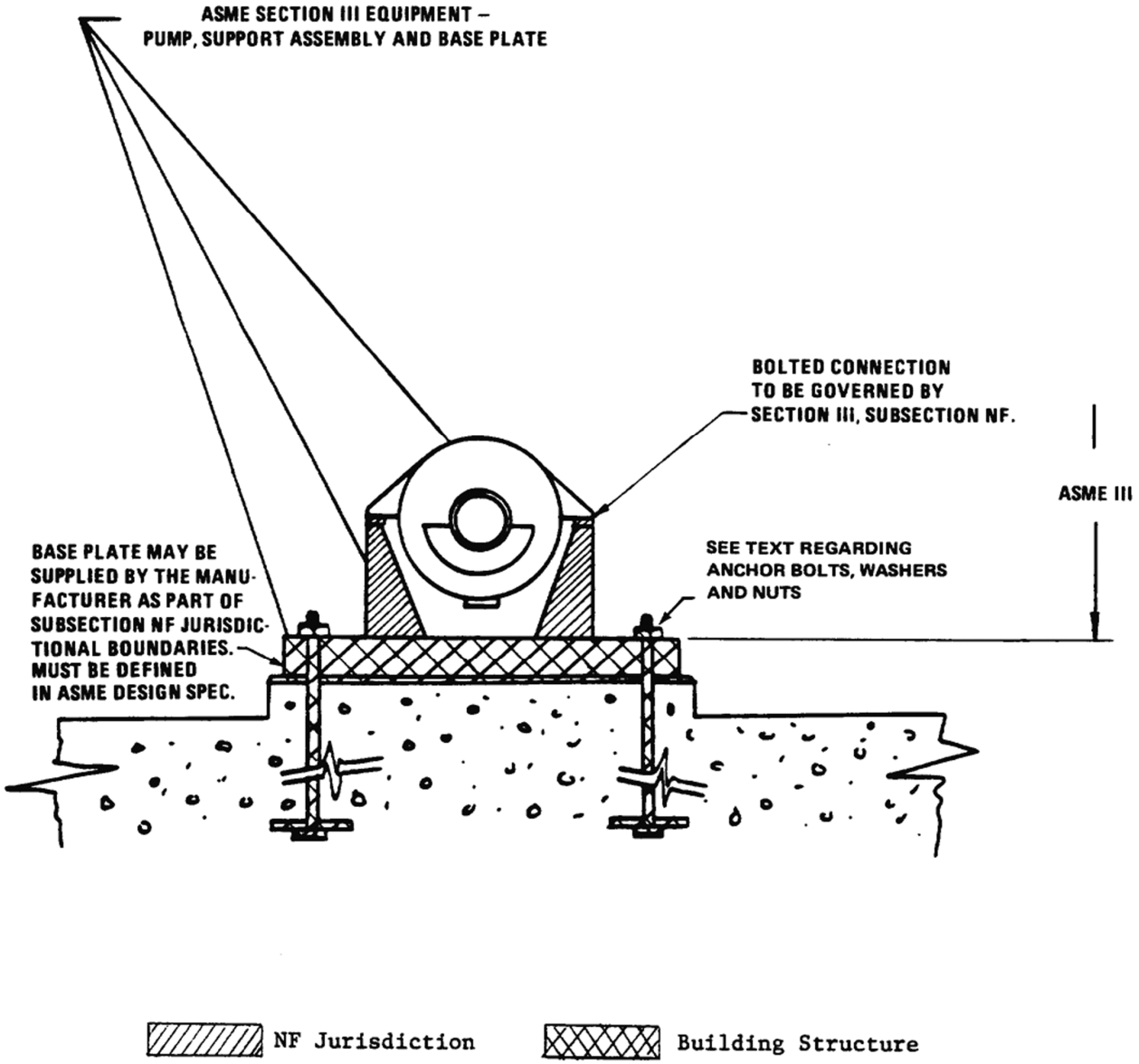
SOUTHERN COMPANY *Energy to Serve Your World®* VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	PIPE SUPPORT REFERENCE SKETCHES FIGURE 3.9.B.3–1



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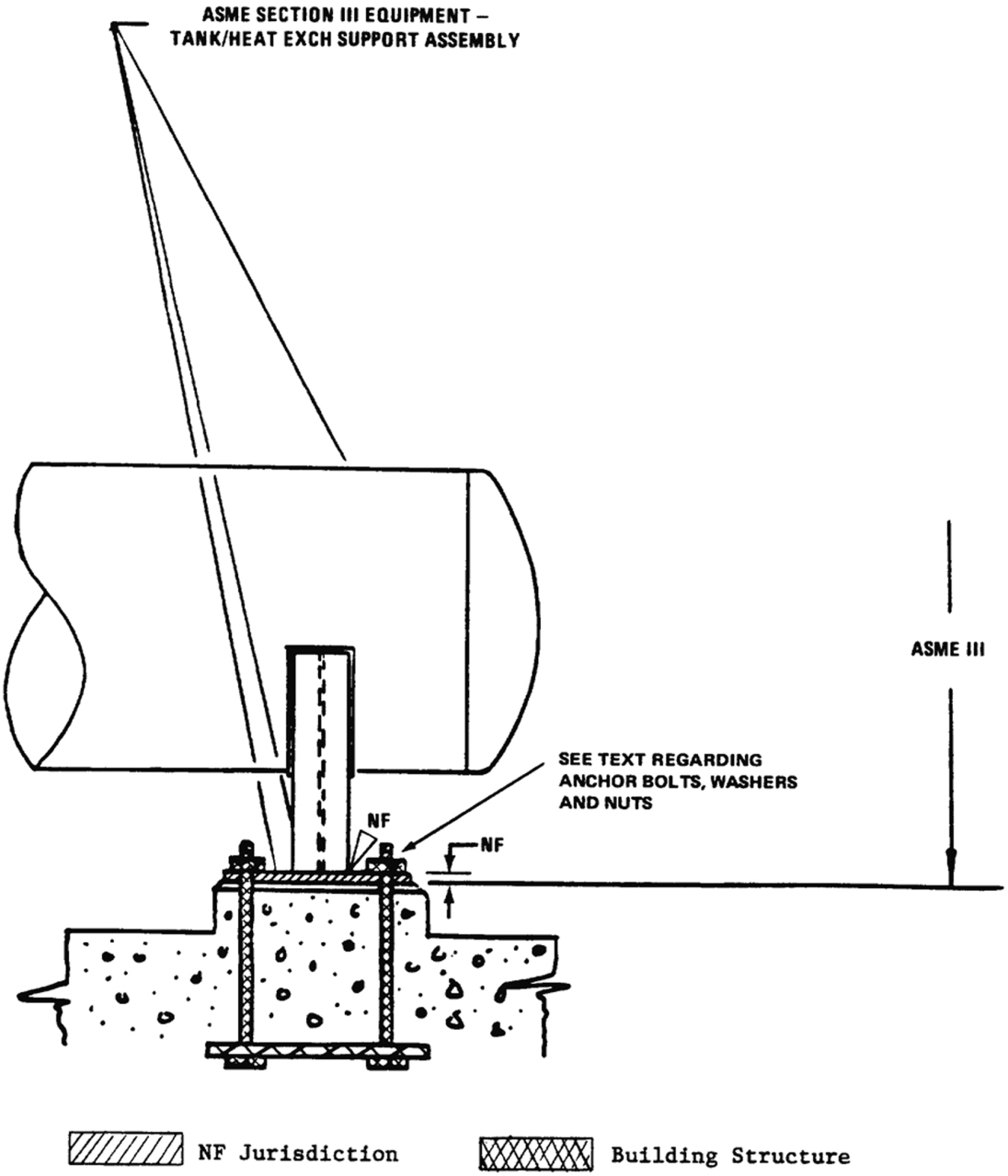
UNIT 1 AND UNIT 2

PUMP COMPONENTS SHOWING ASME SECTION III JURISDICTIONAL BOUNDARY (TYPICAL SKETCH)

FIGURE 3.9.B.3–2



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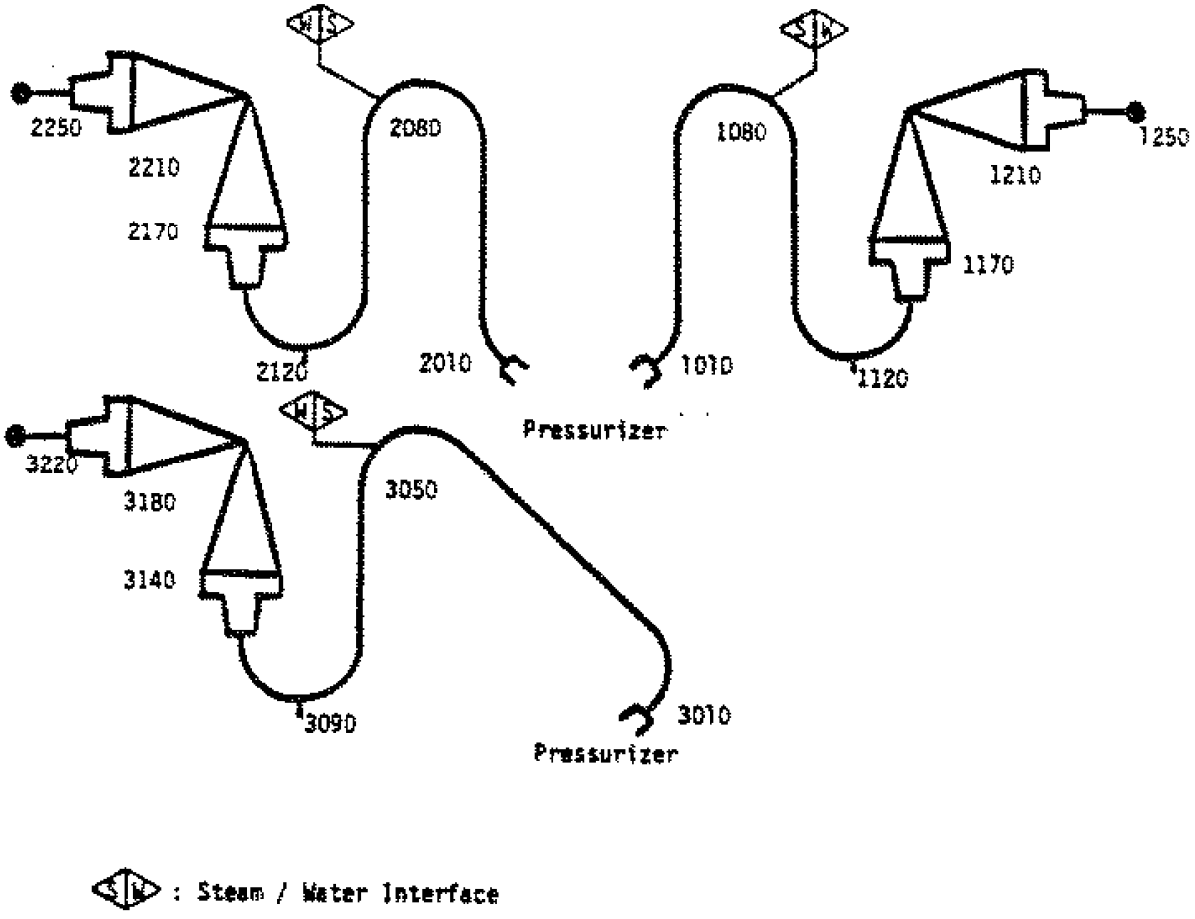
UNIT 1 AND UNIT 2

COMPONENTS SUCH AS HEAT EXCHANGERS & TANKS SHOWING ASME SECTION III NF JURISDICTIONAL BOUNDARY (TYPICAL SKETCH)

FIGURE 3.9.B.3–3



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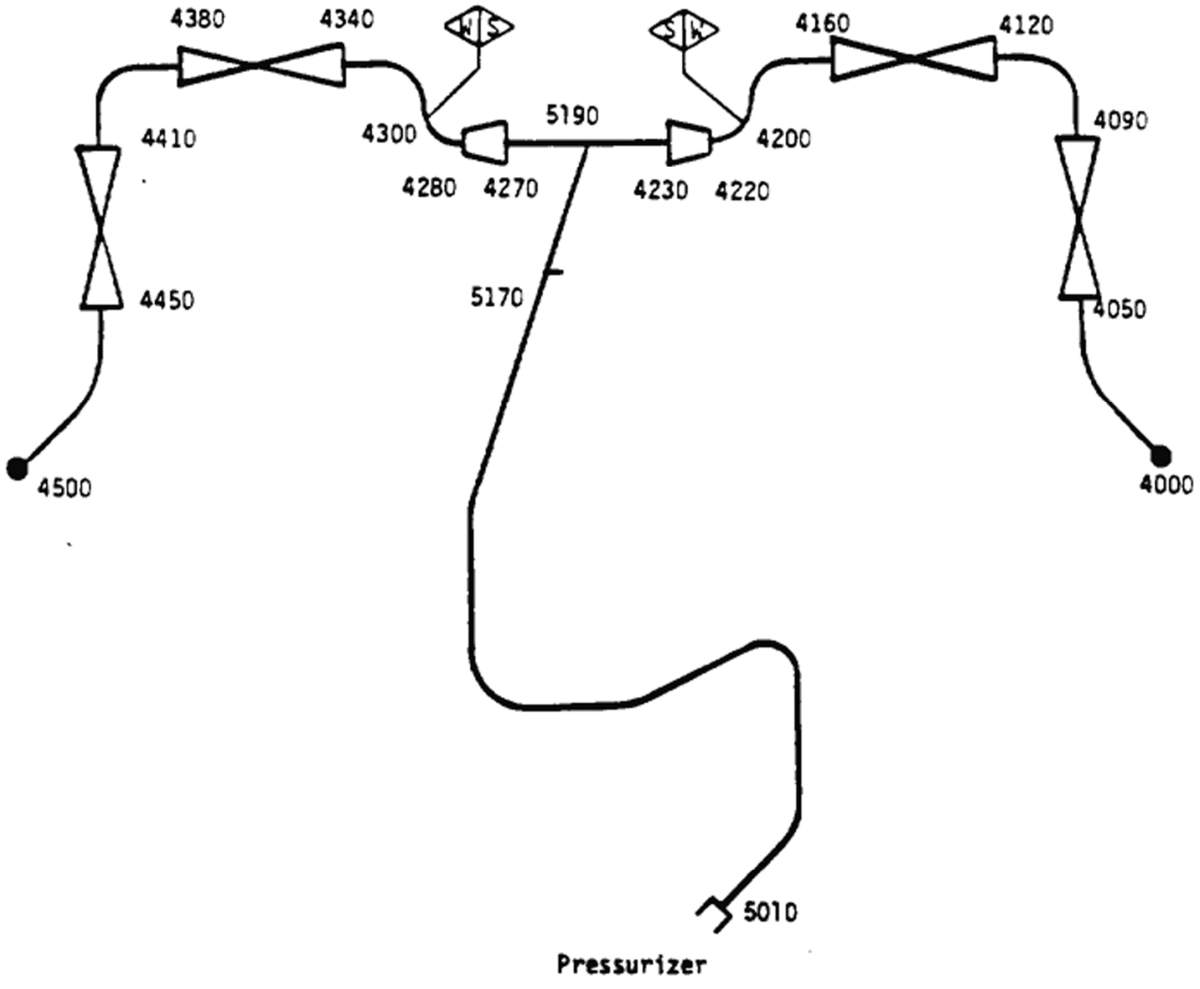
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	CLASS 1 PIPING LAYOUT– SAFETY LINES
	FIGURE 3.9.N.3–1



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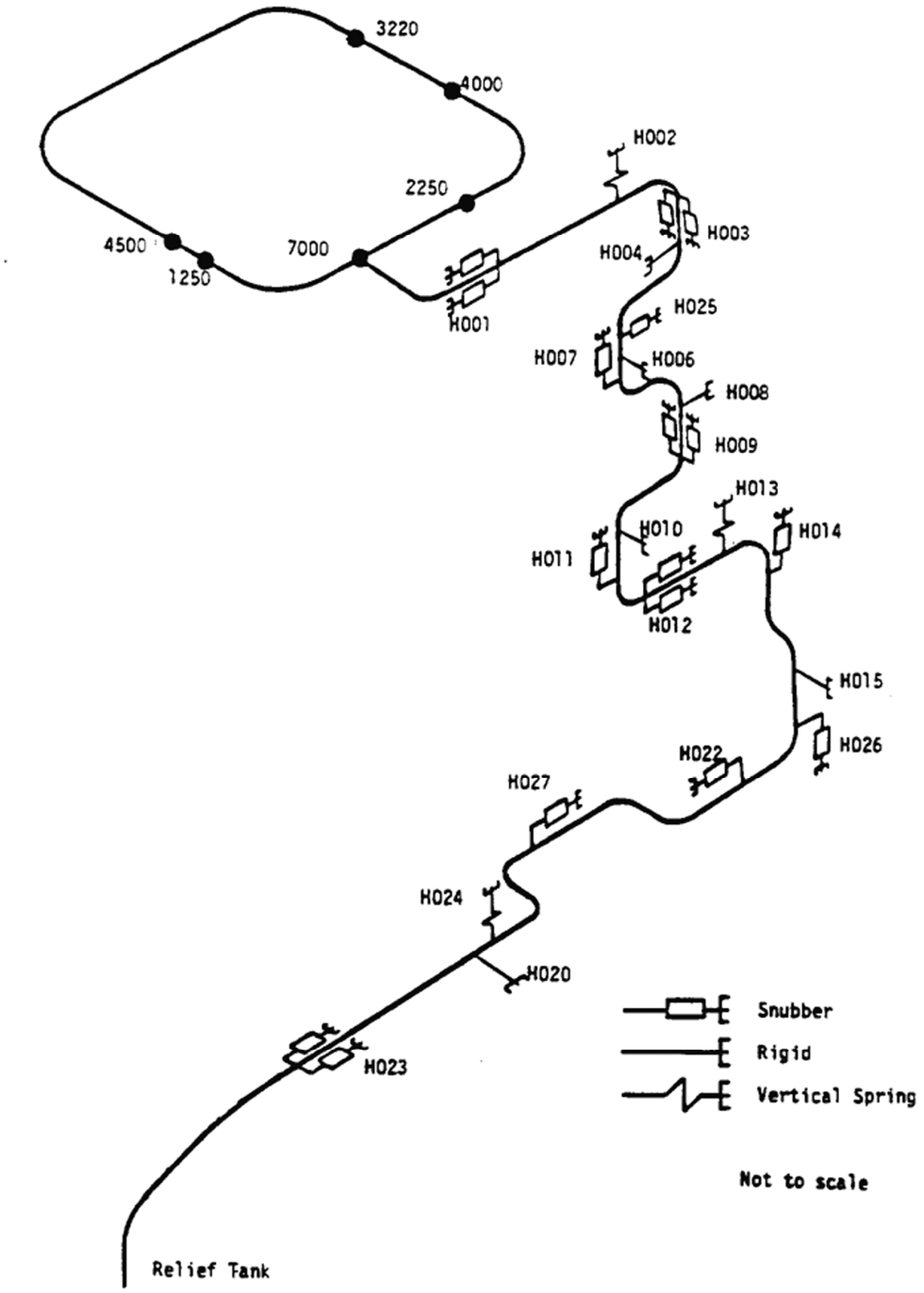
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	CLASS 1 PIPING LAYOUT– RELIEF LINES
	FIGURE 3.9.N.3–2



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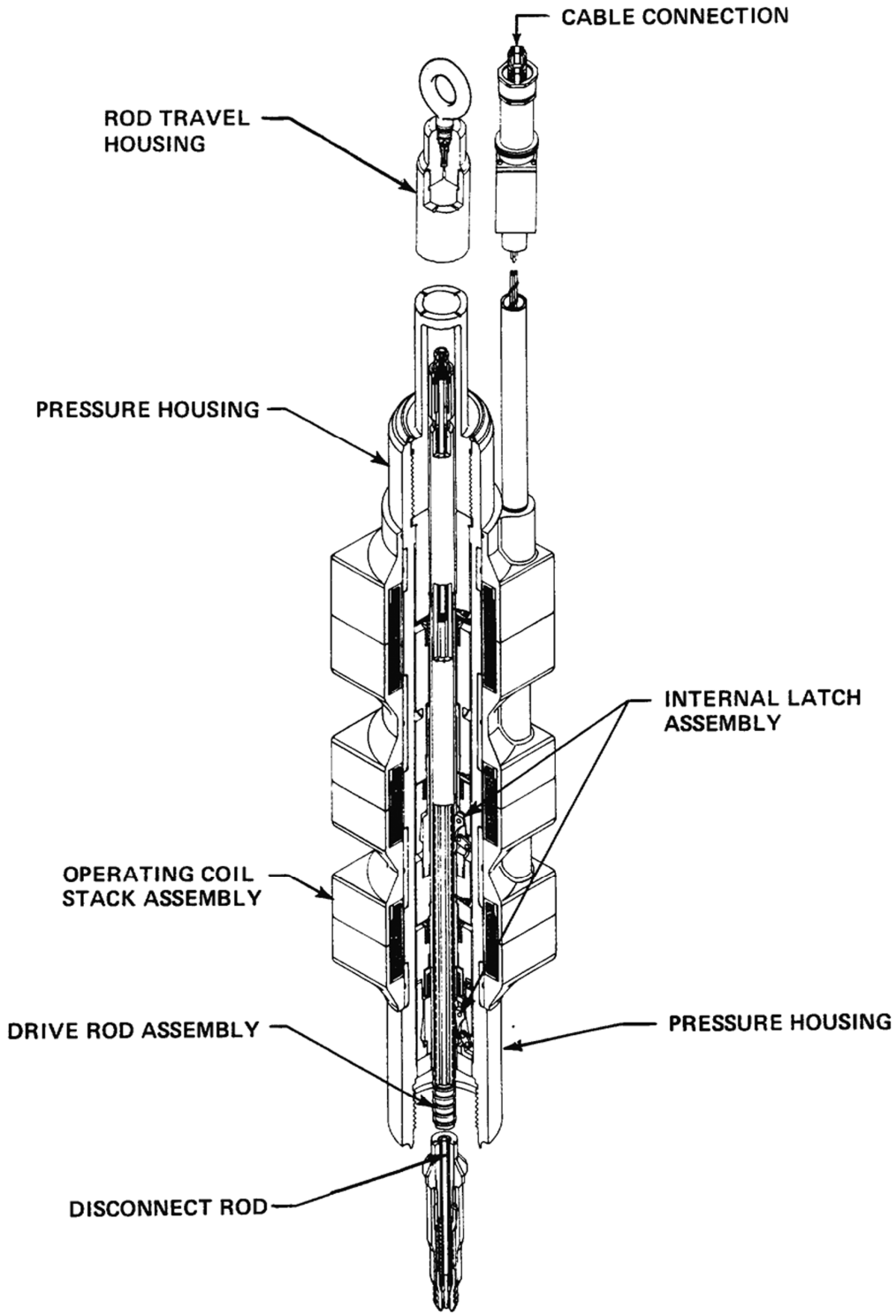
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	DOWNSTREAM PIPING LAYOUT– SAFETY AND RELIEF LINES
	FIGURE 3.9.N.3–3



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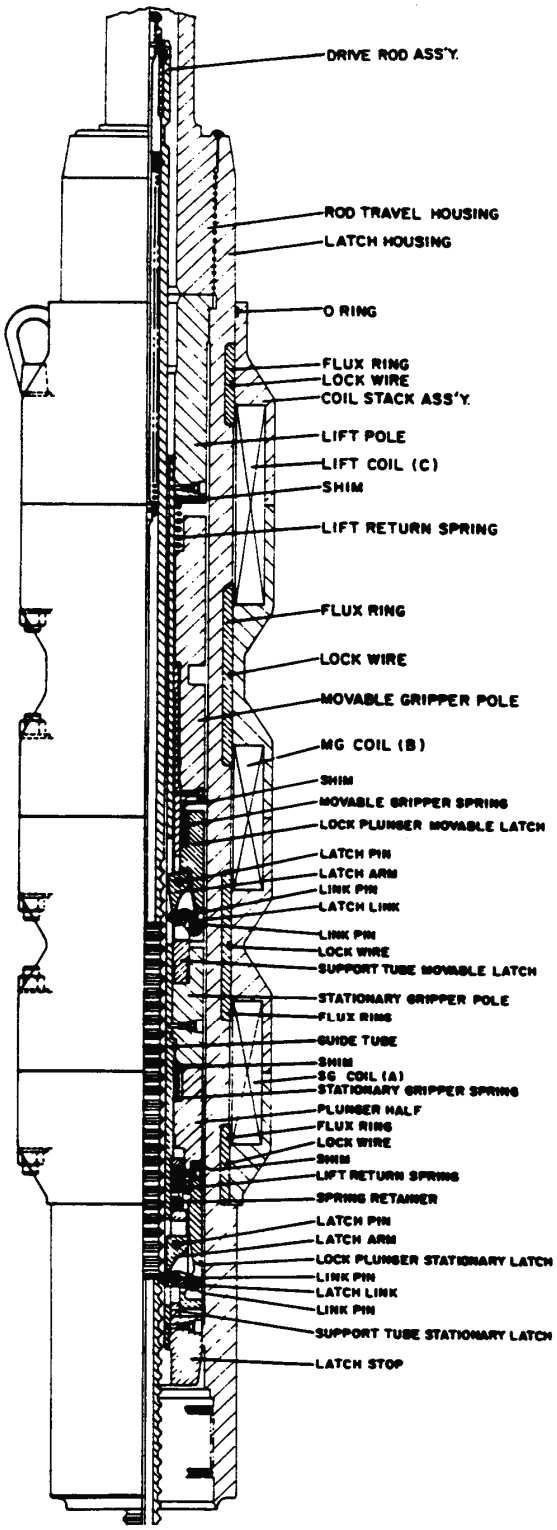
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	FULL–LENGTH CONTROL ROD DRIVE MECHANISM
	FIGURE 3.9.4-1



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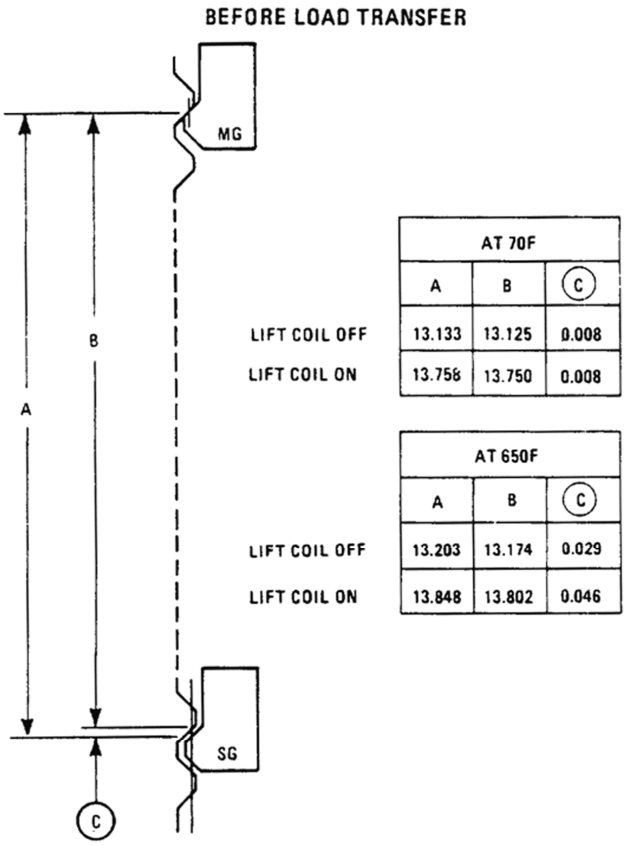
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

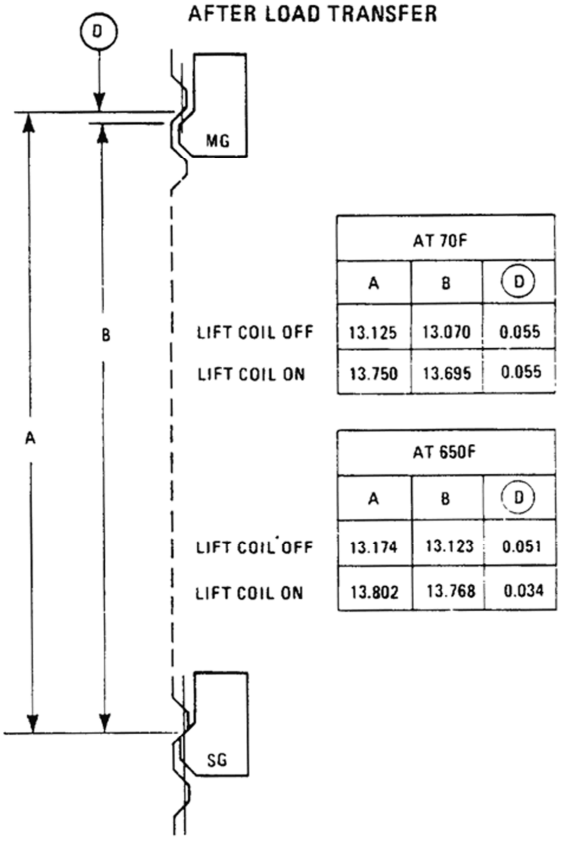
FULL–LENGTH CONTROL ROD
DRIVE MECHANISM SCHEMATIC

FIGURE 3.9.4–2



	AT 70F		
	A	B	C
LIFT COIL OFF	13.133	13.125	0.008
LIFT COIL ON	13.758	13.750	0.008

	AT 650F		
	A	B	C
LIFT COIL OFF	13.203	13.174	0.029
LIFT COIL ON	13.848	13.802	0.046



	AT 70F		
	A	B	D
LIFT COIL OFF	13.125	13.070	0.055
LIFT COIL ON	13.750	13.695	0.055

	AT 650F		
	A	B	D
LIFT COIL OFF	13.174	13.123	0.051
LIFT COIL ON	13.802	13.768	0.034

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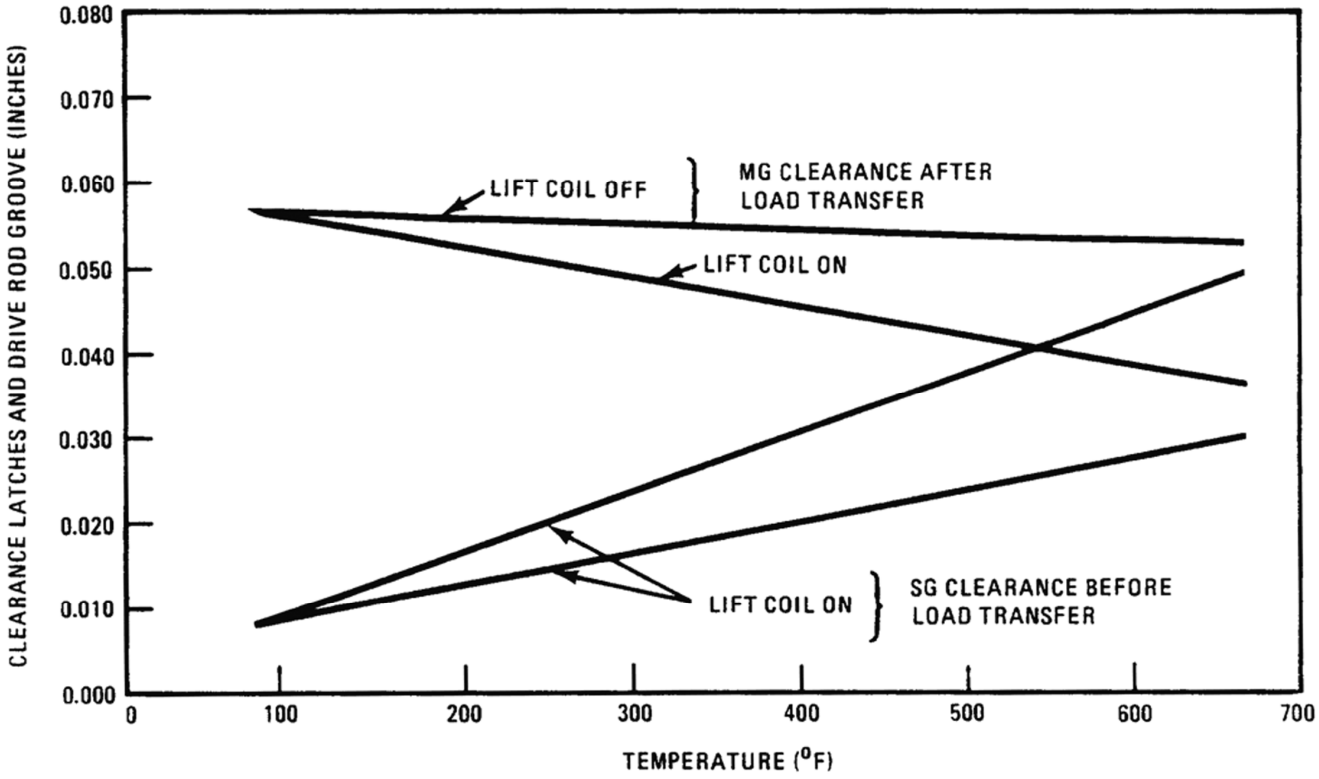
UNIT 1 AND UNIT 2

NOMINAL LATCH CLEARANCE AT MINIMUM AND MAXIMUM TEMPERATURES

FIGURE 3.9.4–3



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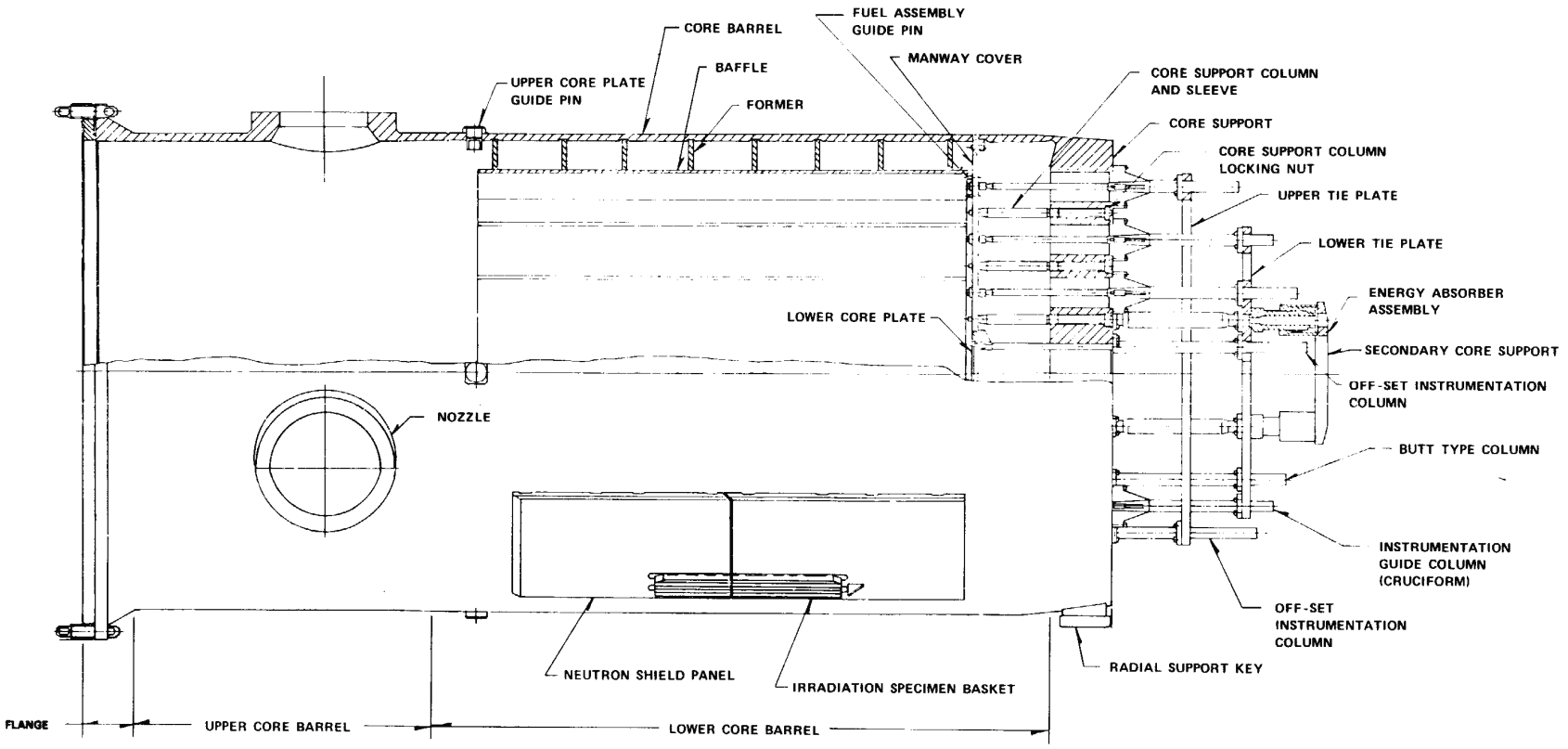
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

CONTROL ROD DRIVE MECHANISM LATCH CLEARANCE THERMAL EFFECT

FIGURE 3.9.4–4



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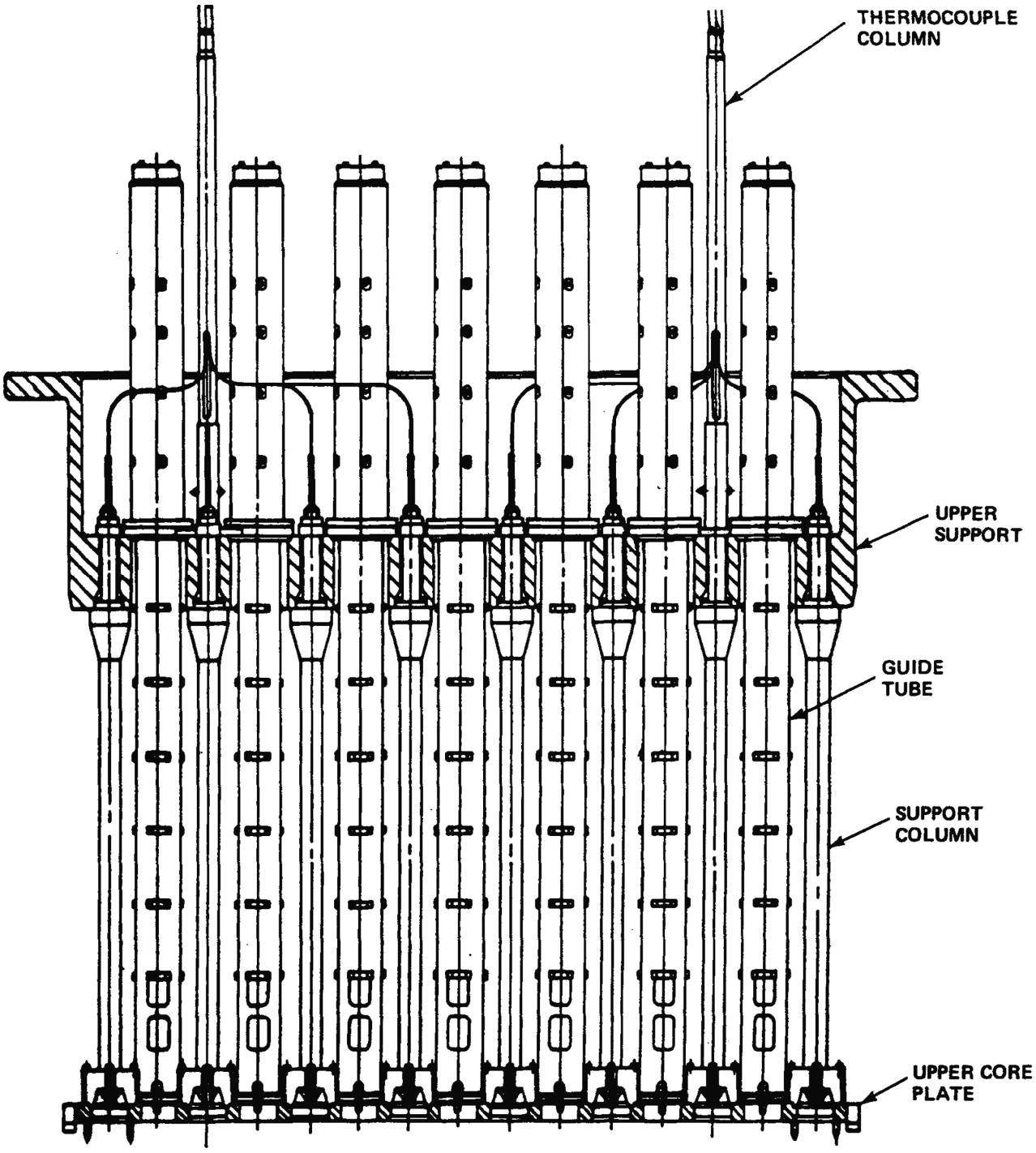
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LOWER INTERNALS ASSEMBLY

FIGURE 3.9.5-1



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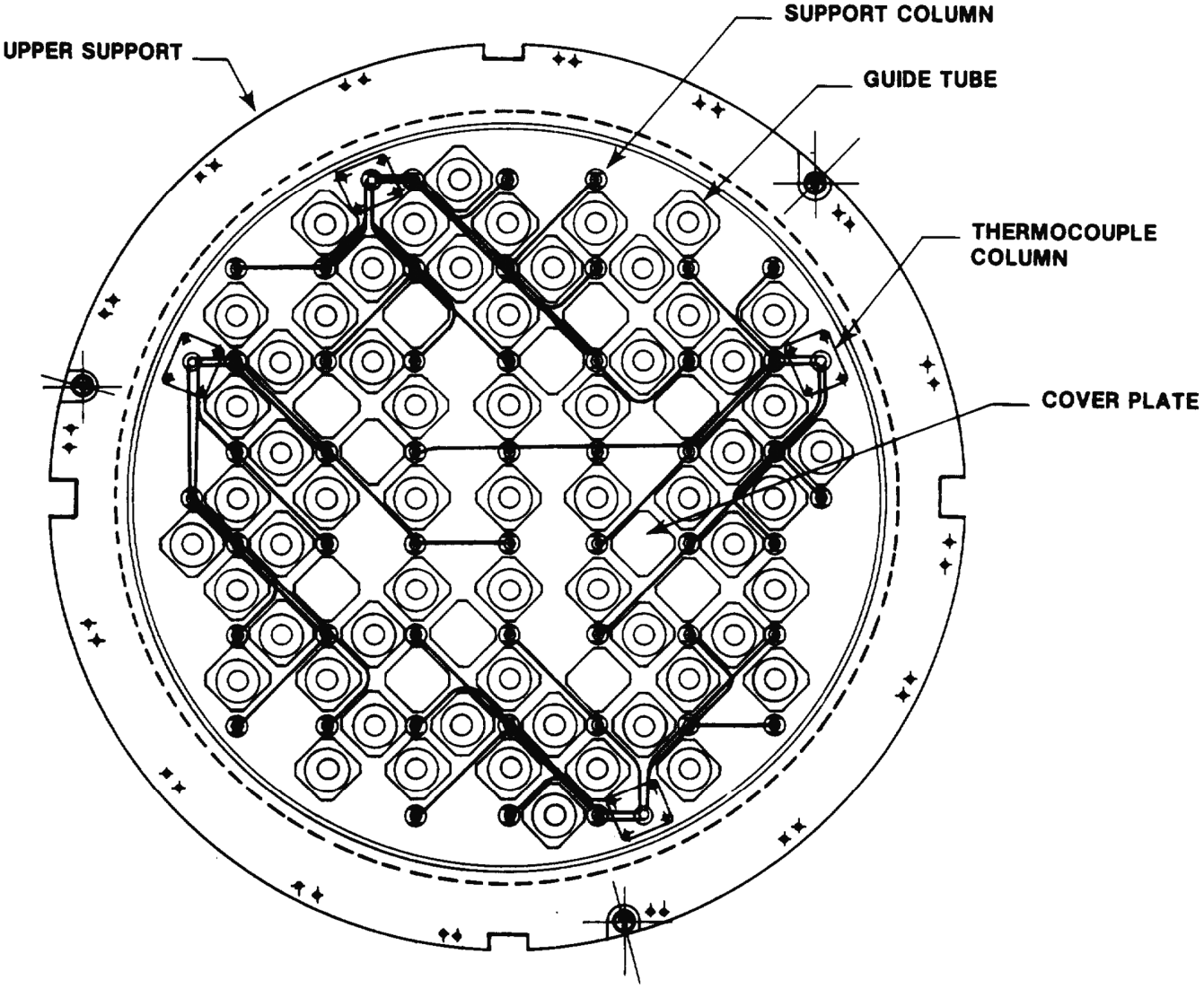
UNIT 1 AND UNIT 2

UPPER CORE SUPPORT ASSEMBLY

FIGURE 3.9.5–2



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ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PLAN VIEW OF UPPER CORE SUPPORT ASSEMBLY

FIGURE 3.9.5–3

3.10.B SEISMIC QUALIFICATION OF SEISMIC CATEGORY 1 MECHANICAL AND ELECTRICAL EQUIPMENT

This section presents information to demonstrate that safety-related mechanical and electrical equipment, instrumentation, and, where applicable, their supports classified as Seismic Category 1 are capable of performing their designated safety-related functions, considering the full range of normal and accident loadings in the event of an earthquake. The information presented includes:

- Identification of the Seismic Category 1 equipment.
- The criteria and methods of seismic qualification.
- Definition of the applicable seismic environment.
- The designated safety-related functional requirements.
- The definition of other normal and accident loadings.
- Documentation of the qualification process which demonstrates the required seismic capability.

Seismic Category 1 structures, components, and systems are identified in table 3.2.2-1. A master list and summary of the seismic qualification of the balance of plant (BOP) safety-related, Seismic Category 1 mechanical and electrical equipment will be maintained as part of the equipment qualification file. Refer to section 3.10.N for information concerning the seismic qualification of Seismic Category 1 mechanical and electrical equipment within the Westinghouse nuclear steam supply system (NSSS).

3.10.B.1 SEISMIC QUALIFICATION CRITERIA

The extent to which the VEGP design criteria meet the general requirements for the seismic qualification of safety-related mechanical and electrical equipment, as discussed in General Design Criteria (GDC) 1, 2, 4, 14, and 30 of Appendix A to 10 CFR 50, is discussed in section 3.1.

The general methods of implementing the requirements of Appendix A to 10 CFR 100 as it relates to qualifying equipment to withstand the effects of natural phenomena, such as earthquakes, and the requirements of Appendix B to 10 CFR 50 are discussed in section 2.5 and chapter 17, respectively.

The seismic qualification and documentation procedures used for safety-related equipment and their supports are in conformance with Institute of Electrical and Electronics Engineers (IEEE) Standard 344-1975, Recommended Practices for Seismic Qualification for Class 1E Equipment for Nuclear Power Generating Stations, and Nuclear Regulatory Commission (NRC) Regulatory Guide 1.100. The VEGP position regarding conformance to NRC Regulatory Guide 1.100 is given in section1.9.

Seismic Category 1 safety-related mechanical and electrical equipment is qualified to withstand the effects of seismic loads resulting from the operating basis earthquake (OBE) and the safe shutdown earthquake (SSE), considering the full range of normal and accident loadings. The parameters used to develop seismic loadings and the seismic criteria for qualification of Seismic Category 1 structures, systems, and components are described in section 3.7.

The acceptance criteria for qualification of Seismic Category 1 safety-related mechanical and electrical equipment are specified to the supplier in the design purchase specification.

Testing is the preferred method to qualify equipment. Both dynamic as well as static test approaches are used to assure structural integrity and operability of mechanical and electrical equipment in the event of an SSE preceded by a number of occurrences of the OBE. Test fixtures are designed to simulate the actual service mounting. Test samples are selected according to type, load level, and size, as well as other pertinent factors on a prototype basis.

Analysis using mathematical modeling techniques correlated to tests performed on similar equipment or structures and/or verified analytical approaches are also used to qualify equipment. Combined analysis and testing may also be used to qualify equipment.

The analytical approach to seismic qualification without testing is used:

- If only maintaining structural integrity is required for the safety function.
- If the equipment is too large or heavy to obtain a representative test input at existing test facilities. (The essential control devices and electrical parts of large equipment are tested separately if required.)
- If the interfaces (e.g., interconnecting cables to the cabinet or other complex inputs) cannot be conservatively considered during testing.
- If the response of the equipment is essentially linear or has a simple nonlinear behavior which can be predicted by conservative analytical methods.

A combination of testing and analysis is used when complete testing is not practical and is incorporated into a test and analysis operability program.

Equipment that has been previously qualified by means of test and analysis equivalent to those described herein are acceptable provided that proper documentation is submitted.

Analysis and/or tests are performed for all Seismic Category 1 safety-related mechanical and electrical equipment to assure their structural and functional capability as required for the seismic criteria described in section 3.7.

The guidance provided in the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code, Section III, is followed in the design of safety-related Seismic Category 1 mechanical equipment to assure the structural integrity of pressure boundary components.

3.10.N SEISMIC AND DYNAMIC QUALIFICATION OF SEISMIC CATEGORY 1 MECHANICAL AND ELECTRICAL EQUIPMENT

The classification of Seismic Category 1 includes three basic types of equipment:

A. Seismic Category 1 instrumentation and electrical equipment.

B. Seismic Category 1 mechanical equipment that must perform a mechanical motion during the course of accomplishing a system safety function. These devices are defined as active and include the control rod drive mechanisms (CRDMs), snubbers, and certain pumps and valves within the nuclear steam supply system (NSSS) scope of supply.

C. Other Seismic Category 1 mechanical equipment whose only safety function is to maintain structural integrity.

This section presents information to demonstrate that mechanical equipment, electrical equipment, instrumentation, and, where applicable, their supports classified as Seismic Category 1 are capable of performing their designated safety-related functions under the full range of normal and accident (including seismic) loadings. This equipment includes devices associated with systems that are essential to the safe shutdown of the reactor containment isolation, reactor core cooling, and containment and reactor heat removal, or otherwise are essential in preventing significant release of radioactive material to the environment or mitigating the consequences of accidents. The information presented includes:

- Identification of the Seismic Category 1 instrumentation, electrical equipment, and appropriate mechanical equipment that are within the scope of the Westinghouse NSSS.
- The qualification criteria employed for each type of equipment.
- The designated safety-related functional requirements.
- Definition of the applicable seismic environment.
- Definition of other normal and accident loadings.
- Documentation of the qualification process employed to demonstrate the required structural integrity and operability of mechanical and electrical equipment and instrumentation in the event of a safe shutdown earthquake (SSE) after a number of postulated occurrences of the operating basis earthquake (OBE) in combination with other relevant dynamic and static loads.

3.10.N.1 SEISMIC AND DYNAMIC QUALIFICATION CRITERIA

3.10.N.1.1 Qualification Standards

The methods of meeting the general requirements for the seismic and dynamic qualification of Seismic Category 1 mechanical and electrical equipment and instrumentation as described by General Design Criteria (GDC) 1, 2, 4, 14, 23, and 30 are described in section 3.1. The general methods of implementing the requirements of Appendix B to 10 CFR 50 are described in chapter 17. The general methods of implementing the requirements of Appendix A to 10 CFR 100 as they relate to qualifying equipment to withstand the effects of natural phenomena such as earthquakes are discussed in section 2.5.

The Nuclear Regulatory Commission (NRC) recommendations concerning the methods to be employed for seismic qualification of instrumentation and electrical equipment are contained in Regulatory Guide 1.100, Seismic Qualification of Electric Equipment for Nuclear Power Plants, which endorses the Institute of Electrical and Electronics Engineers (IEEE) Standard 344-1975, Recommended Practices for Seismic Qualification of Class 1E Equipment for Nuclear Power Generating Stations.

Westinghouse Electric Corporation meets this standard, as modified by Regulatory Guide 1.100, by either type testing, analysis, or an appropriate combination of these methods. Westinghouse meets this commitment by employing the methodology described in WCAPs 8587 and 9714.[(1)(2)]

The guidance provided in the ASME Boiler and Pressure Vessel Code Section III is followed in the design of Seismic Category 1 mechanical equipment to assure the structural integrity of pressure boundary components. In addition, Westinghouse has implemented an operability program for active pumps and valves as originally required by Regulatory Guide 1.48, Design

Limits and Loading Combinations for Seismic Category 1 Fluid System Components. Regulatory Guide 1.148 is addressed in section 1.9.

3.10.N.1.2 Performance Requirements for Seismic Qualification

WCAP-8587 Supplement 1[(3)] contains an equipment qualification data package (EQDP) for every item of instrumentation and electrical equipment classified as Seismic Category 1 within the Westinghouse NSSS scope of supply. Table 3.2.2-1 identifies the Seismic Category 1 electrical equipment and instrumentation supplied by Westinghouse and the equipment qualification central file identifies the applicable EQDP in reference 3 or the appropriate qualification test report. Each EQDP contains a section entitled "Performance Specification." This specification establishes the safety-related functional requirements of the equipment to be demonstrated during and after a seismic event. The test response spectrum employed by Westinghouse for generic seismic qualification is also identified in the specification, as applicable. The spectra employed have been selected to envelop the plant specific required response spectra defined in section 3.7.

For active Seismic Category 1 mechanical components, the performance requirements are defined in the appropriate design and equipment specifications. System functional requirements are described in other sections of the FSAR that define other performance requirements for these components. For active pumps and valves, additional requirements are discussed in paragraph 3.10.N.2.2, and the equipment qualification reports are referenced in the equipment qualification central file. For other Seismic Category 1 mechanical components, the only performance requirement is to maintain structural integrity under all appropriate loading conditions.

A master list and summary of seismic qualification of safety-related Category I electrical and mechanical equipment will be maintained as part of the equipment qualification file.

3.10.N.1.3 Acceptance Criteria

Seismic and dynamic loading qualification must demonstrate that Category 1 instrumentation and electrical equipment as well as active pumps and valves are capable of performing their designated safety-related functions under all plant loading conditions including the SSE. The qualification must also demonstrate the structural integrity of other Seismic Category 1 pumps, valves, mechanical supports, and structures at the OBE level. Some permanent deformation of supports and structures is acceptable at the SSE level, provided that the ability to perform the designated safety-related functions is not impaired.

3.10.N.1.4 References

1. Butterworth, G., and Miller, R. B., "Methodology for Qualifying Westinghouse WRD Supplied NSSS Safety-Related Electrical Equipment," WCAP-8587 Revision 6, April 1983.

2. Kelly, R. E., and McInerney, J. J., "Methodology for the Seismic Qualification of Westinghouse WRD Supplied Equipment," WCAP-9714, May 1980.

3. "Equipment Qualification Data Package," Supplement 1 to WCAP-8587, April 1983.

3.10.B.2 METHODS AND PROCEDURES FOR QUALIFYING MECHANICAL AND ELECTRICAL EQUIPMENT

3.10.B.2.1 Means of Qualification

Institute of Electrical and Electronics Engineers (IEEE) Standard 344-1975, Recommended Practices for Seismic Qualification for Class 1E Equipment for Nuclear Power Generating Stations, and Nuclear Regulatory Commission (NRC) Regulatory Guide 1.100 are used for seismic qualifications. See section 3.10.N for information regarding the Westinghouse nuclear steam supply system (NSSS) equipment.

The horizontal and vertical operating basis earthquake (OBE) and safe shutdown earthquake (SSE) required response spectra (RRS) curves, as discussed in section 3.7, form the basis for the seismic qualification of equipment, systems, and components. The RRS curves are identified with the building elevation. These curves are integrated into the purchase specification, along with the equipment location or locations, and the acceptance criteria for the safety-related functions for each item of equipment.

Seismic qualification of equipment is demonstrated by one of the following:

A. Qualifying a complete assembly identical to that being installed.

B. Qualifying a complete assembly similar to that being installed with adequate justification for differences.

C. Qualifying the individual devices and/or components separately and demonstrating the adequacy of the complete equipment assembly.

All interfaces and the effects of the amplification within the equipment due to the interfaces and supporting structure are considered in the seismic qualification.

Seismic qualification plans/procedures are prepared by the equipment suppliers and submitted for review before testing and/or analysis are performed. The seismic qualification reports, when prepared by the supplier and submitted for review, demonstrate (in accordance with subsection 3.10.B.1) that the equipment performs its required safety-related function before, during, and after (as required) multiple occurrences of the OBE followed by one SSE. The OBEs shall contain a minimum of 50 c of maximum stress.

For components that have been previously tested to generic criteria, multifrequency inputs developed by the test laboratory are reviewed to assure conservatism in amplitude and frequency content and to assure that the test response spectra (TRS) envelops the RRS over the critical frequency range. Test reports are reviewed to confirm the operability of essential equipment.

For active mechanical equipment (i.e., pumps and valves) a combination of test and analysis is used to demonstrate operability and structural integrity of components. Other Seismic Category 1 safety-related mechanical equipment is qualified by analysis to demonstrate structural integrity. The criteria used to decide which method is appropriate are discussed in paragraph 3.9.B.2.2.

Qualification of BOP safety-related active pumps is based on an evaluation of the entire pump/motor (including coupling system) assembly considering its interface with the mounting structure. Simultaneous dynamic interactions between pump, motor, and pedestal/mounting structure are addressed through seismic analysis. Frequency search tests of the pump/motor

assembly are used to verify the mathematical modeling technique used for equipment qualification.

Load combinations, combining of dynamic responses for mechanical equipment, and the pump and valve operability assurance program are discussed in subsection 3.9.B.3.

The pump and valve operability assurance program is discussed in paragraph 3.9.B.3.2.

3.10.B.2.2 Method of Qualification

The methods for seismic qualification are listed below:

- Analysis.
- Test.
- Combination of analysis and test.

3.10.B.2.2.1 Analysis

Mathematical analyses without testing are acceptable if the structural integrity alone ensures the intended design function of the equipment. (See subsection 3.10.B.1.) The procedures used are in accordance with Section 5 of IEEE 344-1975.

When an equivalent static coefficient analysis is performed, justification for its use is provided by the supplier. See paragraph 3.7.B.3.5 for additional information on use of equivalent static load method of analysis. All equipment qualification performed by mathematical analysis is based on mathematical methods correlated with tests of similar equipment or structures or verified analytical techniques.

Analytical results are evaluated for mechanical strength, fatigue, alignment, and noninterruption of function as related to the functional requirements of the equipment during an SSE event. Maximum stresses under all loading are computed and compared with the allowables. Interference effects as well as interaction effects are considered in the analysis when significant.

3.10.B.2.2.2 Testing

Seismic qualification by testing is performed using either multifrequency or single frequency inputs. These test inputs and methods are in accordance with IEEE 344-75, Section 6.

The multifrequency test method is used for floor- and wall-mounted equipment. In addition, in special cases it is used for equipment mounted on structural steel, piping, ducts, or other types of supports or equipment where an analysis or test has been performed to determine the RRS at the equipment mounting location. The test or analysis considers the dynamic amplification characteristics of the support system.

For equipment qualified by multifrequency testing, the measured TRS envelops the RRS in the critical frequency range as shown in the appropriate BOP equipment qualification data package (EQDP).

Single-frequency tests are used for line-mounted equipment, which includes equipment mounted in piping systems and in ducts. The equipment is tested to a required input motion

(RIM). The RIM is the peak acceleration of the input motion (sine wave or sine beats) as a fraction of input frequency.

The piping and duct systems are designed and supported to limit the peak acceleration experienced by the equipment to a value less than the specified RIM acceleration.

Single-frequency tests may also be used for other types of equipment as permitted by IEEE 344-75 and Regulatory Guide 1.100.

3.10.B.2.2.3 Combined Test and Analysis

When the equipment could not be qualified practically by analysis or testing because or its size or complexity, combined analysis and testing were utilized. This method of qualification is applied to equipment such as cabinets that may contain several different configurations of internally mounted devices.

The combined analysis and test method is in accordance with Section 7 of IEEE 344-75, and the equipment qualification method of paragraphs 3.10.B.2.2.1 and 3.10.B.2.2.2 apply.

Equipment that has been previously qualified by means of tests and analyses equivalent to those described herein is acceptable if proper documentation is provided.

3.10.B.2.2.4 Test Sequence Verification

As defined in Part B of Regulatory Guide 1.100, IEEE 344-1975 is an ancillary standard of IEEE 323-1974. In accordance with this standard, seismic testing as part of the overall qualification is performed in its proper sequence as indicated in Section 6 of IEEE 323-1974.

3.10.N.2 METHODS AND PROCEDURES FOR QUALIFYING ELECTRICAL EQUIPMENT, INSTRUMENTATION, AND MECHANICAL COMPONENTS

Seismic qualification of Seismic Category 1 instrumentation and electrical equipment is demonstrated by either type testing or a combination of test and analysis methods. The choice of qualification method employed by Westinghouse for a particular item of equipment is based upon many factors including practicability, complexity of equipment, economics, and availability of previous seismic qualification to earlier standards. The qualification method employed for a particular item of instrumentation or electrical equipment is identified in the individual equipment qualification data package (EQDP) of reference 1.

For active pumps and valves Westinghouse utilizes a combination of test and analysis to demonstrate the structural integrity and operability of such components. Other Seismic Category 1 mechanical equipment is qualified by analysis to demonstrate structural integrity.

Westinghouse methods of load combination and methods of combining dynamic responses for mechanical equipment are discussed in subsection 3.9.N.3. For instrumentation and electrical equipment, the only dynamic loads considered in testing are seismic loads and vibratory loads where applicable. Other dynamic loads which instrumentation and electrical equipment may be subjected to are enveloped by this testing or are addressed by analysis.

Provided below is a description of the methods used to qualify Seismic Category 1 equipment.

3.10.N.2.1 Seismic Qualification of Instrumentation and Electrical Equipment

3.10.N.2.1.1 Type Testing

Prior to the implementation of Institute of Electrical and Electronics Engineers (IEEE) 344-1975, Westinghouse utilized single-axis sine-beat inputs as specified by IEEE 344-1971 to qualify equipment. Westinghouse demonstrated the conservatism of this test method with respect to the modified methods of testing for complex equipment recommended by IEEE 344-1975 through a supplemental test program. As a result, no additional qualification testing was required on a limited number of equipment items defined in Table 7.1 of reference 2.

For other Seismic Category 1 instrumentation and electrical equipment, seismic qualification by test is performed in accordance with IEEE 344-1975. Where testing is utilized, multifrequency, multiaxis inputs are developed by the general procedures outlined in reference 3. The test results contained in the individual EQDPs of reference 1 demonstrate that the measured test response spectrum envelops the generic required response spectrum defined in Section 1 of the EQDP. Qualification for VEGP use is established by verification that the generic required response spectrum specified by Westinghouse envelops the applicable VEGP response spectrum.

Alternative test methods, such as single-frequency, single-axis inputs for line-mounted equipment, are used in selected cases as permitted by IEEE 344-1975 and Regulatory Guide 1.100. These methods are further described in reference 3.

3.10.N.2.1.2 Test and Analysis

Westinghouse also utilizes a combination of test and analysis to qualify Seismic Category 1 instrumentation and electrical equipment. The test methods utilized are similar to those described above for type testing. Available test results are employed in combination with the analysis methods described in IEEE 344-1975 to demonstrate seismic qualification. The analytical methods used include both static and dynamic techniques which are described in detail in reference 3 and the qualification reports identified in the equipment qualification central file.

3.10.N.2.2 Seismic and Operability Qualification of Active Mechanical Equipment

Active mechanical equipment is qualified for both structural integrity and operability under all its intended service conditions by a combination of test and analysis. These methods address such loading conditions as thermal transients, flow loads where significant, and degraded flow conditions if applicable. The test and analysis methods utilized in qualification of these components provide adequate assurance of operability under all required plant conditions.

Qualification methods utilized for active pumps and valves are described below. The qualification methods utilized for control rod drive mechanisms are described in section 3.9.N, and the qualification methods utilized for snubbers are described in section 3.9.N and section 5.2.

3.10.N.2.2.1 Nuclear Steam Supply System Pumps

All active pumps listed in table 3.9.N.3-1 are qualified for operability by first being subjected to rigid tests both prior to installation in the plant and after installation in the plant. The in-shop tests include:

A. Hydrostatic tests of pressure-retaining parts to 150 percent of the design pressure times the ratio of material allowable stress at room temperature to the allowable stress value at the design temperature.

B. Seal leakage tests.

C. Performance tests to determine total developed head, minimum and maximum head, net positive suction head (NPSH) requirements, and other pump parameters.

Also monitored during these operating tests are bearing temperatures and vibration levels. Bearing temperature limits are determined by the manufacturer based on the bearing material, clearances, oil type, and rotational speed. These limits are approved by Westinghouse. After the pump is installed in the plant, it undergoes the cold hydro tests, hot functional tests, and, where applicable, periodic inservice inspection and operation. These tests demonstrate that the pump will function as required during all normal operating conditions for the design life of the plant. Except as noted, those pumps listed in table 3.9.N.3-1 will be included in the Inservice Testing Program (ASME Code Section XI). For those pumps which are not included in the Inservice Testing Program and are listed in table 3.9.N.3-1, the capability to perform their safety related function will be demonstrated through inclusion in plant maintenance programs and/or plant procedures.

In addition to these tests, the safety-related active pumps are qualified for operability by assuring that the pump will start, continue operating, and not be damaged during the faulted condition.

The pump manufacturer is required to show by analysis correlated by tests, prototype tests, or existing documented data that the pump will perform its safety function when subjected to loads imposed by the maximum seismic accelerations and the maximum faulted nozzle loads. It is required that test or dynamic analysis be used to show that the lowest natural frequency of the pump is greater than 33 Hz. The pump, when having a natural frequency above 33 Hz, is considered essentially rigid. This frequency is sufficiently high to avoid problems with amplification between the component and structure for all seismic areas. A static shaft deflection analysis of the rotor is performed with the conservative safe shutdown earthquake (SSE) accelerations of 2.1 g in two orthogonal horizontal directions and 2.1 g vertical acting simultaneously. The deflections determined from the static shaft analysis are compared to the allowable rotor clearances. The nature of seismic disturbances dictates that the maximum contact (if it occurs) will be of short duration. If rubbing or impact is predicted, it is required that it be shown by prototype tests or existing documented data that the pump will not be damaged or cease to perform its design function. The effect of impact on the operation of the pump is evaluated by comparison of the impacting surfaces of the pump to similar surfaces of pumps which have been or will be tested.

In order to avoid damage during the faulted plant condition, the stresses caused by the combination of normal operating loads, SSE, and dynamic system loads are limited as specified in tables 3.9.B.3-2 through 3.9.B.3-7. In addition, the pump casing stresses caused by the maximum faulted nozzle loads are limited to the stresses outlined in table 3.9.B.3-5. The changes in operating rotor clearances caused by casing distortions due to these nozzle loads are considered. The maximum seismic nozzle loads combined with the loads imposed by the

seismic accelerations are also considered in an analysis of the pump supports. Furthermore, the calculated misalignment is shown to be less than that misalignment which could cause pump malfunction. The stresses in the supports are below those in tables 3.9.B.3-6 and 3.9.B.3-7; thus the support distortion is elastic and of short duration (equal to the duration of the seismic event).

Performing these analyses with the conservative loads stated and with the restrictive stress limits of table 3.9.B.3-5 as allowables assures that critical parts of the pump will not be damaged during the short duration of the faulted condition and that, therefore, the reliability of the pump for post-faulted condition operation will not be impaired by the seismic event. To complete the seismic qualification procedures, the pump motor is qualified for operation during the maximum seismic event.

In many instances, pumps and motors are seismically qualified independently of each other. Where applicable, loads that may be transmitted between pump and motor as a result of seismic interaction are included in the qualification. Most pump assemblies employ a coupling between the driver and the pump. Westinghouse specifies flexible, limited end float couplings for use in nuclear applications. This design compensates for end or axial movement of the coupled shafts preventing either shaft from exerting excessive thrust on the other. Under severe loading conditions such as a strong seismic event, a small amount of thrust is transmitted through the coupled shafts. Westinghouse pump designs include thrust collars to absorb forces in the shaft axial direction. This prohibits the pump from tansmitting thrust loads to the motor. Most motor designs use ball or journal bearing, either of which are very effective in limiting shaft axial movement. Should loads be transmitted from the motor to the pump, again the thrust collar would provide the ability to absorb them.

Westinghouse has performed dynamic seismic testing on a multistage centrifugal pump/gear/motor assembly. The couplings performed superbly throughout the test. Disassembly and visual examination with tolerance measuring showed no excessive wear or damage resulting from the seismic events.

The program above gives the required assurance that the safety-related pump/motor assemblies will not be damaged and will continue operating under SSE loadings and, therefore, will perform their intended functions. The proposed requirements take into account the complex characteristics of the pump and are sufficient to demonstrate and assure the seismic operability of the active pumps.

Since the pump is not damaged during the faulted condition, the functional ability of active pumps after the faulted condition is assured, since only normal operating loads and steady-state nozzle loads exist. Since it is demonstrated that the pumps would not be damaged during the faulted condition, the postfaulted-condition operating loads will be identical to the normal plant operating loads. This is assured by requiring that the imposed nozzle loads (steady-state loads) for normal conditions and postfaulted conditions are limited by the magnitudes of the normal condition nozzle loads. The postfaulted-condition ability of the pumps to function under these applied loads is proven during the normal operating plant conditions for active pumps.

3.10.N.2.2.2 Nuclear Steam Supply System Valves

Safety-related active valves, listed in table 3.9.N.3-2 must perform their mechanical motion in times of an accident. Assurance is supplied that these valves will operate during a seismic event. Tests and analyses are conducted to qualify active valves.

The safety-related valves are subjected to a series of stringent tests prior to service and during the plant life. Prior to installation, the following tests are performed: shell hydrostatic test to

American Society of Mechanical Engineers (ASME) Section III requirements, back-seat and main seat leakage tests, disc hydrostatic tests, and operational tests to verify that the valve will open and close. For the qualification of motor operators for environmental conditions refer to section 3.11.N. After installation, the valves undergo hydro tests, construction acceptance tests, and preoperational tests. Where applicable, periodic inservice inspections, and periodic inservice operations are performed in situ to verify and assure the functional ability of the valve. These tests guarantee reliability of the valve for the design life of the plant. Except as noted, those valves listed in Table 3.9.N.3-2 will be included in the Inservice Testing Program (ASME Code, Section XI). For those valves which are not included in the Inservice Testing Program and are listed in table 3.9.N.3-2, the capability to perform their safety related function will be demonstrated through inclusion in plant maintenance programs, plant procedures, and/or Technical Specifications. The valves are constructed in accordance with the ASME Boiler and Pressure Vessel Code, Section III. On active valves, an analysis of the extended structure is performed for static equivalent seismic SSE loads applied at the center of gravity of the extended structure. The maximum stress limits used for active Class 1, 2, and 3 valves are shown in subsection 3.9.B.3.

In addition to these tests and analyses, representative valves of each design type are tested for verification of operability during a simulated plant faulted-condition event by demonstrating operational capabilities within the specified limits. A representative valve of a specific design type is identified for this testing by the specification (e.g., globe valve, motor-operated valve, etc.) for that particular type of valve. A stratification of design is further made based upon the valve size, pressure rating, type of operator, and previous operability testing to evaluate the need for additional testing of a particular design type. The testing procedures are described below.

The valve is mounted in a manner which conservatively represents typical valve installations. The valve includes the operator pilot, solenoid valves, and limit switches when such devices are normally attached to the valve in service. The faulted-condition nozzle loads are considered in the test in either of two ways: loads equivalent to the faulted-condition nozzle loads are simultaneously applied to the valve through its mounting during the below described test, or by analysis, the nozzle loads are shown not to affect the operability of the valve. Interface requirements are specified to limit nozzle loads such that deflection or deformation of the valve materials will not affect the operability of the valve. The operability of a rigid valve (natural frequency equal to or greater than 33 Hz) is demonstrated by satisfying the following criteria:

A. The actuator and yoke of the valve system are statically deflected by applying a load or loads equivalent to the resultant faulted seismic loads at the extended structure center of gravity in the direction of the lowest determined natural frequency. The design pressure of the valve will be simultaneously applied to the valve during the static deflection tests.

B. The valve is cycled while in the deflected position. The time required to open or close the valve in the deflected position will be compared to similar data taken in the undeflected condition to evaluate the significance of any change.

C. Motor operators, external limit switches, and pilot solenoid valves necessary for operation are qualified by IEEE 344-1975 as described in paragraph 3.10.N.2.1.1.

The accelerations which are used for the static valve qualification shall be equivalent, as justified by analysis, to 2.1 g in two orthogonal horizontal directions and 2.1 g vertical. The piping design must maintain the operator accelerations to these levels.

If the natural frequency of the valve is less than 33 Hz, a dynamic analysis of the valve will be performed to determine the equivalent acceleration which will be applied during the static test. The analysis will provide the amplification of the input acceleration considering the natural frequency of the valve and the frequency content of the applicable plant piping response spectra. The adjusted accelerations will be determined using the same conservatisms contained in the 2.1-g-horizontal and 2.1-g-vertical accelerations used for rigid valves. The adjusted acceleration will then be used in the static deflection test and the valve operability will be assured by the methods outlined above, using the modified acceleration input.

The above testing program applies to valves with extended structures. The testing is conducted on a representative number of valves. Valves from each of the primary safety-related design types are tested. Valve sizes, which cover the range of sizes in service, are qualified by the tests, and the results are used to qualify all valves within the intermediate range of sizes.

Valves that are safety-related but can be classified as not having an extended structure, such as check valves and safety valves, are considered separately.

Check valves are characteristically simple in design, and their operation will not be affected by seismic accelerations or the maximum applied nozzle loads. The check valve design is compact, and there are no extended structures or masses whose motion could cause distortion which could restrict operation of the valve. The nozzle loads due to maximum seismic excitation will not affect the functional ability of the valve, since the valve disc is typically designed to be isolated from the body wall. The clearance supplied by the design around the disc will prevent the disc from becoming bound or restricted due to any body distortions caused by nozzle loads. Therefore, the design of these valves is such that once the structural integrity of the valve is assured using standard methods, the ability of the valve to operate is assured by the design features. The valve will also undergo in-shop hydrostatic test, in-shop seat leakage test, and periodic in situ valve exercising and inspection to assure the functional ability of the valve.

For NSSS check valves, generic testing has been performed to determine the performance characteristics for various sizes of check valves. The performance characteristics include flow required to open, pressure drop, etc. These tests demonstrate the valves will be fully open during the design conditions, therefore precluding cycling of the valve which results in wear.

In addition, the ability of the valve to open is assured by its inherent design characteristics. The swing check design and the clearance between the disc hanger assembly and body preclude the possibility of binding.

The methodology used for system layout is per Westinghouse document 1.12, "Systems Standard Design Criteria NSSS Layout Guidelines." In addition, valve sizing is determined by line size and flowrates at which the valve is required to operate.

The pressurizer safety valves are qualified by the following procedures (these valves are also subjected to tests and analysis similar to check valves): stress and deformation analyses of critical items which may affect operability for faulted condition loads, in-shop hydrostatic and seat leakage tests, and periodic in situ valve inspection. In addition to these tests, a static load equivalent to that applied by the faulted condition is applied at the top of the bonnet, and the pressure is increased until the valve mechanism actuates. Successful actuation within the design requirements of the valve assures its overpressurization safety capabilities during a seismic event.

Using these methods, all the safety-related valves in the system are qualified for operability during a faulted event. These methods outlined above conservatively simulate the seismic event and assure that the active valves will perform their safety-related function when necessary.

Degraded conditions as discussed in the Standard Review Plan Section 3.10, paragraph II.1.a(2) are minimized during system design to preclude the presence of debris, impurities, and containments in the fluid system. For example, containment sump design considers the presence of debris as described in Final Safety Analysis Report subsection 6.2.2. Other degraded conditions, such as motive power fluctuations, air pressure, etc., are addressed by testing or analysis showing that sufficient margin was included in the design of the equipment to perform its function.

For the types of valves supplied for nuclear steam supply system active service, supplemental tests and analytical data confirm that flow-induced loadings and thermal loads, when considered in conjunction with seismic and operating loads, have no significant effect on valve operability. The same is true for valve end loads.

Conformance to Regulatory Guide 1.148, concerning active valve assemblies, is addressed in table 3.9.B.3-10. In addition, procurement specifications for replacement components (active safety related) will be consistent with the original purchased equipment to the extent of conformance discussed.

3.10.N.2.2.3 Pump Motor and Valve Operator Qualification

Active pump motors, vital pump appurtenances, active valve motor operators, limit switches, and solenoid valves are seismically qualified in accordance with IEEE 344-1975 as discussed in the appropriate EQDPs of reference 1.

3.10.N.2.3 Seismic Qualification of Other Seismic Category 1 Mechanical Equipment

For Seismic Category 1 mechanical equipment not defined as active, Westinghouse utilizes analysis to demonstrate structural integrity. The analysis methods used by Westinghouse are described in sections 3.7.N and 3.9.N and reference 3.

3.10.N.2.4 References

1. "Equipment Qualification Data Package," Supplement 1 to WCAP-8587, April 1983.
2. Butterworth, G., and Miller, R. B., "Methodology for Qualifying Westinghouse WRD Supplied NSSS Safety-Related Electrical Equipment," WCAP-8587 Revision 6, April 1983.
3. Kelly, R. E., and McInerney, J. J., "Methodology for the Seismic Qualification of Westinghouse WRD Supplied Equipment," WCAP-9714, May 1980.

3.10.B.3 METHODS AND PROCEDURES OF ANALYSIS OR TESTING OF SUPPORTS OF MECHANICAL AND ELECTRICAL EQUIPMENT AND INSTRUMENTATION

Analyses or tests are performed for all safety-related, Seismic Category 1 electrical and mechanical equipment supports to ensure their structural capability to withstand seismic excitation.

Information concerning the structural integrity of pressure- retaining components, their supports, and core supports is presented in subsection 3.9.B.3. The following bases are used in the design and analysis of cable tray supports and instrument tubing supports.

A. The methods used in the seismic analysis of cable tray supports are described in subsection 3.7.B.3. The amplification of seismic loads due to the flexibility of the supporting system, if any, is accounted for in the design of the cable trays.

B. The Seismic Category 1 instrument tubing systems are supported so that the allowable stresses permitted by Section III of American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code are not exceeded when the tubing is subjected to the loads specified in section 3.9 and Regulatory Guide 1.48 for Class 2 and 3 piping.

For field-mounted instruments the supports are tested or analyzed to meet the following:

- The field mounting supports for Seismic Category 1 instruments excluding line-mounted instruments have a fundamental frequency of 33 Hz or greater, with the weight of the instrument included. If, however, the mounting should be flexible (i.e., frequency <33 Hz), the dynamics of the support are considered in the qualification of the supported instrument.
- The stress level in the mounting support or piping does not exceed the material allowable stress when subjected to the maximum acceleration level of the mounting location. The weight of the instrument is included.

In some cases, panels and racks supporting Seismic Category 1 devices are tested and/or analyzed with equipment installed. If the equipment is in an inoperative mode during the support test, the response at the equipment mounting location is monitored. In such a case, devices are qualified separately, and the actual input to the equipment is more conservative in amplitude and frequency content than the response monitored at the equipment location. The required response spectra (RRS) for devices (i.e., in-cabinet response spectra) are generated and as shown in the individual EQDPs applicable to the device and the test response spectra to which the device is qualified envelops the RRS measured at the device mounting location.

3.10.N.3 METHOD AND PROCEDURES FOR QUALIFYING SUPPORTS OF ELECTRICAL EQUIPMENT, INSTRUMENTATION, AND MECHANICAL COMPONENTS

The equipment qualification data packages (EQDPs) contained in reference 1 identify the equipment mounting employed for qualification purposes and establish interface requirements for the equipment to ensure subsequent in-plant installation does not prejudice the qualification established by Westinghouse.

Westinghouse provides interface requirements to the installer for the mounting of mechanical equipment. Where Westinghouse supplies component supports, the support configuration is addressed in the seismic and dynamic qualification. The criteria used to ensure the structural integrity of component supports are defined in section 3.9.N.

3.10.N.3.1 Test Configurations

Westinghouse seismic qualification testing configurations are designed to represent typical plant installation for the tested component. Interface requirements are defined based on the test

configuration and other design requirements. Installation is then completed in accordance with the component interface and installation requirements. Any dynamic coupling effects that result from mounting the component in accordance with these interface criteria are adequately simulated during the test program.

3.10.N.3.2 Reference

1. "Equipment Qualification Data Package," Supplement 1 to WCAP-8587, April 1983.

3.10.B.4 OPERATING LICENSE REVIEW

3.10.B.4.1 Qualification and Documentation Procedures

Qualification and documentation procedures for safety-related Seismic Category 1 mechanical and electrical equipment are in accordance with the recommendations contained in the Institute of Electrical and Electronics Engineers (IEEE) Standard 344-1975 and Regulatory Guide 1.100.

EQDPs for balance of plant equipment are prepared for all safety-related Seismic Category 1 electrical and mechanical equipment. Balance of plant EQDPs include qualification plan/procedures and reports documenting that criteria established in subsections 3.10.B.1, 3.10.B.2, and 3.10.B.3 have been satisfied.

3.10.B.4.2 Standard Review Plan Evaluation

The following summary describes the Standard Review Plan differences in regard to seismic and dynamic qualification of mechanical and electrical equipment:

A. The Standard Review Plan requires that equipment be tested in the operational condition and that loadings simulating normal plant conditions should be superimposed on seismic and dynamic loads. This includes flow-induced loads and degraded flow conditions. VEGP tests for nuclear steam supply system (NSSS) equipment are made in the operational conditions where practical, simulated as appropriate, or addressed by analysis. Flow loads are not superimposed on seismic loads for valve operability tests.

   Full operational conditions are simulated because of the impracticality of using actual conditions during testing. Flow loads for valve static operability tests are calculated, and, if significant, the effects are simulated in the test.

B. Evaluation is performed to ensure that test configurations conservatively simulate actual field mountings.

   Seismic qualification testing configurations are designed to represent typical plant installation. Interface requirements are simulated during the test in accordance with field installation. For generic testing, evaluation is performed to ensure test configurations conservatively simulate actual field mounting.

C. The VEGP does not apply end loadings to active valves during static deflection tests. End loads are evaluated in the active valve analysis.

D. The VEGP does not analyze valve discs for Δp or impact energy resultant from a loss-of-coolant accident (LOCA), except for certain cases where a significant impact from the LOCA is expected.

Westinghouse performs design verification on vendor-supplied valves to ensure that the valves are designed properly and meet the stress acceptance criteria in the equipment specification and in the American Society of Mechanical Engineers Code standards. For balance of plant vendors, loadings simulating Δp conditions are calculated and included in the static operability test, if significant.

E. The VEGP does not utilize Regulatory Guide 1.92 guidance for combination of multimodal or multidirectional responses.

For balance of plant equipment, VEGP uses a Bechtel topical report (BC-TOP-4A) as guidance for combination of multimodal response. This report meets the intent of Regulatory Guide 1.92. For NSSS equipment, Westinghouse uses methods previously justified and accepted by the Nuclear Regulatory Commission.

F. The program for environmental qualification of active mechanical equipment is based on a combination of design, testing, and analysis of critical components as discussed in the Final Safety Analysis Report paragraph 3.11.B.2.

G. The VEGP documentation file conforms to the Standard Review Plan guidelines with the exception of certain NSSS documentation which remains in an auditable file at Westinghouse.

Information that Westinghouse considers proprietary will be summarized in EQDPs located in a central file. Detailed documentation will be available for auditing at Westinghouse.

3.10.N.4 OPERATING LICENSE REVIEW

The results of tests and analyses that ensure that the criteria established in subsection 3.10.N.1 have been satisfied, employing the qualification methods described in subsections 3.10.N.2 and 3.10.N.3, are included in the individual equipment qualification data packages and test reports referenced in the equipment qualification central file.

3.10.N.4.1 Documentation

Seismic qualification of equipment is documented in test reports, analysis reports, calculation notes, etc., contained in the Westinghouse files. Westinghouse satisfies existing regulatory requirements for documentation as further described in chapter 17.

3.10.N.4.2 Standard Review Plan Evaluation

The following summary describes the Standard Review Plan differences in regard to seismic and dynamic qualification of mechanical and electrical equipment:

A. The Standard Review Plan requires that equipment be tested in the operational condition and that loadings simulating normal plant conditions should be superimposed on seismic and dynamic loads. This includes flow-induced loads and degraded flow conditions. VEGP tests for nuclear steam supply system (NSSS) equipment are made in the operational conditions where practical, simulated as appropriate, or addressed by analysis. Flow loads are not superimposed on seismic loads for valve operability tests.

   Full operational conditions are simulated because of the impracticality of using actual conditions during testing. Flow loads for valve static operability tests are calculated, and, if significant, the effects are simulated in the test.

B. Evaluation is performed to ensure that test configurations conservatively simulate actual field mountings.

   Seismic qualification testing configurations are designed to represent typical plant installation. Interface requirements are simulated during the test in accordance with field installation. For generic testing, evaluation is performed to ensure test configurations conservatively simulate actual field mounting.

C. The VEGP does not apply end loadings to active valves during static deflection tests. End loads are evaluated in the active valve analysis.

   End loads are evaluated in the valve analysis.

D. The VEGP does not analyze valve discs for Δp or impact energy resultant from a loss-of-coolant accident (LOCA), except for certain cases where a significant impact from the LOCA is expected.

   Westinghouse performs design verification on vendor-supplied valves to ensure that the valves are designed properly and meet the stress acceptance criteria in the equipment specification and in the American Society of Mechanical Engineers Code standards. For balance of plant vendors, loadings simulating Δp conditions are calculated and included in the static operability test, if significant.

E. The VEGP does not utilize Regulatory Guide 1.92 guidance for combination of multimodal or multidirectional responses.

   For balance of plant equipment, VEGP uses a Bechtel topical report (BC-TOP-4A) as guidance for combination of multimodal response. This report meets the intent of Regulatory Guide 1.92. For NSSS equipment, Westinghouse uses methods previously justified and accepted by the Nuclear Regulatory Commission.

F. The program for environmental qualification of active mechanical equipment is based on a combination of design, testing, trending, and analysis of critical components as discussed in Final Safety Analysis Report paragraph 3.11.B.2. This program will be supported by periodic plant testing and maintenance/surveillance programs. This program meets the intent of the Standard Review Plan.

G. The VEGP documentation file conforms to the Standard Review Plan guidelines with the exception of certain NSSS documentation which remains in an auditable file at Westinghouse.

   Information that Westinghouse considers proprietary will be summarized in equipment qualification documentation packages located in a central file. Detailed documentation will be available for auditing at Westinghouse.

3.11.B ENVIRONMENTAL DESIGN OF MECHANICAL AND ELECTRICAL EQUIPMENT

This section provides environmental qualification[a] information which verifies the capability of safety-related mechanical and electrical equipment to perform its design functions under all normal, abnormal, and design basis accident (DBA) conditions. This section provides the information for the balance of plant (BOP) equipment. Section 3.11.N provides the information for the nuclear steam supply system (NSSS). Mechanical and electrical equipment covered by this section includes equipment located in a harsh environment and associated with systems that are essential to emergency reactor shutdown, containment isolation, reactor core cooling, and containment and reactor heat removal.

With the exception of diaphragm valves, which are addressed in table 1.9-1, the replacement/ refurbishment of safety-related mechanical and electrical equipment in a mild environment and safety-related mechanical equipment in a harsh environment is based on a combination of design life, trending, and periodic maintenance and surveillance.

The information in this section includes a definition of applicable environmental conditions, requirement for documentation of the qualification tests and analysis performed on equipment, and demonstration of adequacy of the equipment qualification program.

The seismic qualification of mechanical and electrical equipment is presented in section 3.10.B.

3.11.B.1 EQUIPMENT IDENTIFICATION AND ENVIRONMENTAL CONDITIONS

Table 3.11.B.1-1 and figure 3.11.B.1-1 identify applicable normal, abnormal, and DBA environmental conditions conforming to 10 CFR 50, Appendix A, General Design Criterion 4. These environmental conditions are associated with various plant areas by building, by an environmental designator. The environmental designator is composed of the following elements:

- IA and IB - Containment (upper and lower).
- II - Main steam isolation valve (MSIV)/main feedwater isolation valve (MFIV) area.
- III - Outside areas.
- IV - Diesel generator structures.
- V - Equipment building.
- VI - Auxiliary feedwater pump area.
- VII - Fuel handling building.
- VIII - Auxiliary building.
- IX - Control building.

[a] The renewed operating licenses authorize an additional 20-year period of extended operation for both VEGP units resulting in a plant operating life of 60 years. The EQ program is credited to continue to manage aging effects associated with the EQ equipment for the period of extended operation (see subsections 19.3.1 and 19.4.3). Applicable EQ evaluations based on a 40-year design life were evaluated as time-limited aging analyses (TLAAs) for license renewal and will be revised as necessary to reflect the 60-year plant operating life before the units enter the period of extended operation.

X - Turbine building.

XI - Not used.

XII - Nuclear service cooling water structure.

XIII - Radwaste structures (abandoned in-place).

For mild environments, the area conditions do not change as the result of an accident; as a result, there are no degrading environmental effects that could lead to common mode failure of safety-related equipment.

The environmental conditions identified in table 3.11.B.1-1 are defined as follows:

A. Normal operating environmental conditions are defined as those conditions that exist during routine plant operations for which the equipment is expected to perform its safety functions, as required, on a continuous basis.

B. Abnormal/test environmental conditions are those plant conditions for which the equipment is designed to operate for a period of time without accelerating normal periodic tests, inspections, and maintenance schedules for that equipment. These conditions are reviewed on a case-by-case basis for severity and duration when required. The maximum and minimum conditions identified as the abnormal condition are based on the design temperature limits for the affected areas.

C. DBA and post-DBA conditions are those plant conditions resulting from various postulated equipment and piping failures during which safety-related equipment must operate without impairment of the safety-related function. The DBA and post-DBA conditions as shown on figure 3.11.B.1-1 are enveloping values. Qualification may be based on component specific calculations when necessary.

Compatibility of equipment with the specified environmental conditions is assured by the following:

A. Systems and components required to mitigate the consequences of a DBA or to perform safe shutdown operation are qualified to remain functional after exposure to the environmental conditions in table 3.11.B.1-1.

B. Safety-related systems and components have a qualified life goal of 41 years[a] (including 1 year of post-DBA operating time). Demonstration of qualified life by test and/or analysis with adequate justification is provided by equipment suppliers, including the effects of aging when applicable. For critical components susceptible to aging, a qualified life was established which includes the effects of the total integrated radiation dose experienced at their respective locations within the plant. When a 41-year[a] qualified life was not possible, a shorter qualified life was established, and a replacement program was implemented.

[a] The renewed operating licenses authorize an additional 20-year period of extended operation for both VEGP units resulting in a plant operating life of 60 years. The EQ program is credited to continue to manage aging effects associated with the EQ equipment for the period of extended operation (see subsections 19.3.1 and 19.4.3). Applicable EQ evaluations based on a 40-year design life were evaluated as TLAAs for license renewal and will be revised as necessary to reflect the 60-year plant operating life before the units enter the period of extended operation.

C. Equipment qualification has taken into account the most severe environmental conditions resulting from the design basis high-energy line break. Included in these conditions are the short-term peak transient temperature following a main steam line break (MSLB), radiation exposure, temperature, and chemical spray due to a loss-of-coolant accident (LOCA) within the reactor containment. For equipment located inside containment, the surface temperature does not exceed the saturation temperature at containment peak steam partial pressure.

For equipment located outside containment, the effects of post-LOCA recirculating fluids outside containment are included in the total integrated doses to the equipment in the affected areas. Postulated high energy line failures (as defined in paragraph 3.6.2.1.2) are assumed in all areas where high energy lines (greater than 1 in.) are routed. The main steam and main feedwater isolation valve areas have been evaluated using VEGP specific blowdown data which includes superheat.(1) The main steam piping in the MSIV compartments is designed to the break exclusion (superpipe) criteria of Branch Technical Position MEB 3-1 item B.1.b for the portions of piping passing through the primary containment and extending to the first five-way restraint past the MSIVs as discussed in paragraph 3.6.2.1.1.D.

The following cases were reanalyzed using a VEGP specific model and input.

- Two power levels were assumed: 70% and 102% of 3579 MWt.
- Ten break sizes were assumed: from a larger break of 1.0 ft^2 to a smaller break of 0.1 ft^2 at increments of 0.1 ft^2.

These break sizes include the following:

  — 1.0 ft^2 is the largest postulated break in superpipe.

  — 0.5 ft^2 is the largest branchline break.

- Two auxiliary feedwater (AFW) conditions were assumed: three AFW pumps (nominal and maximum flow) and two AFW pumps (nominal flow)

Cases were run with and without the turbine-driven pump available to model superpipe breaks where a single failure is not considered and branchline breaks where a single failure is considered. Additional cases were run reflecting minimum and maximum initial steam generator inventory.

The analyses of the environmental response of each MSIV compartment to MSLBs with superheated steam blowdown is consistent with the requirements of NUREG-0588. These analyses were completed using the EPRI computer code "GOTHIC." The GOTHIC code calculates heat transfer to the surrounding structure surfaces.

A facility response evaluation was performed to determine if the equipment was essential for an MSLB in the area, and the environmental qualification test reports for the essential equipment were reviewed to ensure that the equipment was qualified for the MSLB event. For four components (MSIVs, MSIV bypass valves, steam generator atmospheric relief valves, and auxiliary feedwater discharge valves), the maximum MSLB environmental temperatures achieved during the qualification tests did not envelope the maximum MSLB environmental temperature profiles considering superheat developed for the control building and auxiliary building MSIV compartments. A thermal lag analysis was performed on these components to demonstrate that the actual safety-related component

temperature achieved under the VEGP superheated MSLB conditions is less than the component temperature reached in the qualification testing program.

The essential equipment for an MSLB in the auxiliary and control building MSIV compartments has successfully completed environmental qualification test programs which, in conjunction with thermal lag analysis, demonstrate that the equipment is qualified for the maximum MSLB environmental temperature postulated in these compartments. It is concluded that no required safety components are precluded from performing their safety function in the event of an MSLB in either of the MSIV compartments. Therefore, no safety implications exist to prevent safe shutdown of the VEGP.

Where post-DBA functional requirements after performing the safety-related function are different from the DBA requirements, the following are considered:

1. Such components inside the containment consider the effects of temperature, pressure, relative humidity, radiation, and chemical parameters during the post-DBA period.

2. Such components outside the containment are designed for the required temperature pressure and other environmental conditions.

Specific supporting qualification information for each mechanical and electrical equipment component is contained in the applicable BOP equipment qualification data package (EQDP).

D. Rooms with safety-related equipment necessary for safe shutdown were evaluated to assure that the temperatures would be within acceptable values in order to determine the necessity for temperature surveillance. Rooms that met any of the following criteria do not require surveillance:

- Rooms with safety-related HVAC systems.
- Rooms with a high temperature alarm that annunciates in the main control room when the normal design temperature is exceeded. High temperature alarms require the same level of action for exceeding the design temperature limits as those described in section 16.3, Area Temperature Monitoring.
- Rooms classified as mild environment where the calculated room temperature with a loss of normal HVAC for 7 days is less than 150 F.
- Rooms with natural ventilation and designed for ambient conditions (e.g., NSCW pumphouse).

Rooms that contain safety-related equipment necessary for safe shutdown that do not meet any of the above criteria were also evaluated based on historical records of temperatures. These records indicated that the temperatures remain within their environmental qualification limits with the normally provided HVAC in operation, but that upon failure of the HVAC it is possible for the room temperature to exceed the values assumed for equipment environmental qualification. HVAC systems were operated in a degraded condition during high ambient conditions during the summer to determine what minimum ventilation is required to maintain temperatures below the limits listed in FSAR table 16.3-6. Therefore, surveillance of these rooms is conducted in accordance with section 16.3, Requirement 7 - Area Temperature Monitoring, in the event of a loss of the normal HVAC.

The qualified life of environmentally qualified equipment is based on the environmental conditions indicated in table 3.11.B.1-1 and/or the results of temperature monitoring. Area temperature monitoring in conjunction with the evaluation described above provides reasonable

assurance that the normal temperature used for environmental qualification of safety-related equipment necessary for safe shutdown is not exceeded without an evaluation of equipment operability.

Nonactive mechanical equipment whose only safety function is structural integrity is designed in accordance with the American Society of Mechanical Engineers Code. The accident and post-accident environmental effects are considered in the design of such structural components as pump casings and valve bodies. The environmental qualification program, therefore, is restricted to evaluating the design of critical nonmetallic subcomponents of active devices in a harsh environment where failure could result in loss of the active component.

Active mechanical equipment (pumps and valves) is qualified for operability as discussed in subsection 3.9.B.3 and section 3.10.B. This operability program, combined with the qualification of the electrical appurtenances (motors, valve operators, solenoids, limit switches etc.), demonstrates qualification under all required environmental conditions. Active mechanical equipment is defined as that equipment that must perform a mechanical motion as part of its safety function.

3.11.B.1.1 Standard Review Plan Evaluation

The VEGP addresses the guidelines of NUREG-0588 by providing environmental qualification of mechanical equipment using a combination of test and analysis rather than by testing alone.

The environmental qualifications of mechanical equipment is addressed through the stringent selection of materials for use under adverse environmental conditions. This selection is supported by partial type testing and materials analysis and evaluation to confirm the adequacy of the materials used. If sufficient documentation is not found on materials, then equipment is qualified by testing using the guidance of NUREG-0588.

3.11.B.1.2 Reference

1. Georgia Power Company letter from D. O. Foster to NRC, dated June 25, 1986.

3.11.N ENVIRONMENTAL DESIGN OF MECHANICAL AND ELECTRICAL EQUIPMENT

This section presents information to demonstrate that the mechanical and electrical portions of the engineered safety features (ESF) and the reactor protection systems are capable of performing their designated safety-related functions while exposed to applicable normal, abnormal, test, accident, and post-accident environmental conditions. The information presented includes identification of the safety-related equipment that is within the scope of the Westinghouse nuclear steam supply system (NSSS) and, for each item of equipment, the designated safety-related functional requirements, definition of the applicable environmental parameters, and documentation of the qualification process employed to demonstrate the required environmental capability. The seismic qualification of NSSS-supplied safety-related mechanical and electrical equipment is presented in section 3.10.N.

3.11.N.1 EQUIPMENT IDENTIFICATION AND ENVIRONMENTAL CONDITIONS

3.11.N.1.1 Equipment Identification

A complete list of safety-related electrical and active mechanical equipment within the Westinghouse NSSS scope of supply that is essential to emergency reactor shutdown, containment isolation, reactor core cooling, or containment and reactor heat removal or that is otherwise essential in preventing significant release of radioactive material to the environment is provided in table 3.2.2-1. A master list of safety-related electrical and mechanical equipment along with a summary of electrical equipment qualification results will be maintained as part of the equipment qualification central file.

3.11.N.1.2 Definition of Environmental Conditions

The plant-specific normal, abnormal, accident and post-accident conditions are defined in subsection 3.11.B.1. The parameters considered for the Westinghouse qualification program are included in the equipment qualification data packages (EQDPs) found in WCAP-8587, Supplement 1.

3.11.N.1.3 Equipment Operability Times

For Westinghouse NSSS supplied Class 1E electrical and active mechanical equipment, post-accident operability times are generally defined as follows:

Equipment	Required Post-Accident Operability
Equipment necessary to perform trip functions	5 min
Equipment that is located outside containment, is accessible, and can be repaired, replaced, or recalibrated	2 weeks
Equipment located inside containment that is inaccessible and is required for post-accident monitoring	4 months (This number is based on an acceptable amount of time to allow the instrument to be repaired, replaced, or recalibrated, or for an equivalent indication to be obtained.)
Equipment that is located inside containment, is inaccessible, or cannot be repaired, replaced, or recalibrated	1 year
Equipment located in a mild environment following an accident	Continuous

Specific information for each device qualified as part of Westinghouse's Institute of Electrical and Electronic Engineers (IEEE) 323-1974 qualification program is contained in the appropriate EQDP.

In general, the active mechanical component is qualified for operability as discussed in section 3.10.N utilizing test, analysis, or a combination of tests and analyses. This operability program, combined with the qualification of the electrical appurtenances (motors and valve operators, etc.) discussed in the appropriate EQDPs, demonstrates qualification for the required post-accident times.

3.11.N.1.4 Standard Review Plan Evaluation

The VEGP addresses the guidelines of NUREG-0588 by providing environmental qualification of mechanical equipment using a combination of test and analysis rather than by testing alone.

The environmental qualification of mechanical equipment is addressed through the stringent selection of materials for use under adverse environmental conditions. The selection is supported by partial type testing and materials analysis and evaluation to confirm the adequacy of the materials used. If sufficient documentation is not found on materials, then equipment is qualified by testing using the guidance of NUREG-0588.

3.11.B.2 QUALIFICATION TESTS AND ANALYSES

Qualification of safety-related equipment located in a harsh environment is based on type testing of actual or similar equipment or by analysis. Testing is used to demonstrate the ability of the equipment to perform its required safety-related function for a definable period. The type test includes, as a minimum, thermal and mechanical aging, radiation, and exposure to extremes of environmental, seismic, and vibration effects. Type testing is done with representative samples of the production line equipment and simulated service conditions generally in accordance with the sequence indicated in the Institute of Electrical and Electronics Engineers (IEEE) Standard 323-1974 to the specified service conditions, including margin, and takes into account normal and abnormal plant operation and DBA and post-DBA operations.

A master list of safety-related electrical and mechanical equipment along with a summary of electrical equipment qualification results will be maintained as part of the equipment qualification file for harsh environment only.

IEEE Standard 323-1974, IEEE Standard for Qualifying Class 1E Equipment for Nuclear Power Generating Stations, and NRC Regulatory Guide 1.89, Qualification of Class 1E Equipment for Nuclear Power Plants, are used to establish acceptance criteria for environmental qualification for all safety-related equipment located in a harsh environment. Other regulatory guides providing guidance for meeting the requirements of 10 CFR 50 Appendix A, General Design Criteria 1, 4, 23, and 50 and Appendix B, Criterion III to 10 CFR 50 include Regulatory Guide 1.30, Quality Assurance Requirements for the Installation, Inspection, and Testing of Instrumentation and Electric Equipment; Regulatory Guide 1.40, Qualification Tests of Continuous-Duty Motors Installed Inside the Containment of Water-Cooled Nuclear Power Plants; Regulatory Guide 1.63, Electrical Penetration Assemblies in Containment Structures for Light-Water-Cooled Nuclear Power Plants; Regulatory Guide 1.73, Qualification Tests of Electric Valve Operators Installed Inside the Containment of Nuclear Power Plants; and Regulatory Guide 1.100, Seismic Qualification of Electric Equipment for Nuclear Power Plants; and Regulatory Guide 1.131, Qualification Tests of Electric Cables, Field Splices, and Connections for Light-Water-Cooled Nuclear Power Plants.

Additional information regarding conformance with each of these Regulatory Guides is given in section 1.9.

When sufficiently reliable data and proven analytical methods are available, environmental qualification may be based on analysis.

Qualification by analysis includes justification of the methods, theories, and assumptions used (i.e., mathematical or logical proof based on actual test data) that the equipment meets or exceeds its specified performance when subjected to normal, abnormal, and design basis event (DBE) environmental conditions.

For mechanical equipment in the harsh environmental, post-DBA environmental effects on structural components such as valve and pump casings and bodies is considered negligible. Critical subcomponents are evaluated and qualified as required. Organic materials, such as gaskets, seals, and O-rings, are evaluated for their function and evaluated separately. Combined test and analysis therefore is the qualification approach used for this equipment if located in harsh environments.

With the exception of diaphragm valves, which are addressed in table 1.9-1, replacement/refurbishment intervals for safety-related equipment located in a mild environment and safety-related mechanical equipment in a harsh environment are based on a combination of design life, trending, and periodic maintenance and surveillance.

Prior to fuel loading, VEGP will implement administrative controls of component qualification involving an equipment qualification file, handling of documentation, internal acceptance review procedures, and a maintenance surveillance program.

Planned maintenance and surveillance is a VEGP program that schedules equipment maintenance, calibration, and surveillance activities. Its purpose is to maintain equipment in a condition safe for operations, minimize unplanned outages due to breakdown, and provide a mechanism by which greater than anticipated degradation of safety-related equipment can be detected and remedied. The program is being developed using personnel experience in the area of maintenance and surveillance of electrical, mechanical, and instrumentation and controls equipment.

Under the program, a planned maintenance and surveillance checklist is prepared for each piece of safety-related equipment that identifies the maintenance and surveillance tasks to be performed. If the task requires removing components for internal inspection, an equipment-specific procedure is referenced describing how the removal and inspection is to be performed. If the task only requires visual inspection of the outer areas of the equipment, it is described on the checklist. The content of the program is derived from the following sources:

- Manufacturer/vendor recommendations.
- Lubrication requirements.
- Calibration requirements.
- Field verification of equipment descriptions.
- Industry experience.
- Qualification test/analysis results.

The information is then used in preparing equipment qualification data packages (EQDPs) which include maintenance and surveillance requirements specified by vendors in their test reports. Safety-related equipment located in a mild environment does not require environmental qualification documentation for inclusion in the EQDPs.

All EQDPs were originally transmitted to the equipment qualification task force (EQTF), which reviewed the vendor's information and the architect-engineer's evaluation for completeness and validity. Upon approval, the EQTF transmitted a copy of the EQDP to the equipment

qualification (EQ) group. The EQ group identified all equipment tag numbers included in the EQDP and transmitted this information along with the maintenance and surveillance requirements from the EQDP to the maintenance performance teams. Maintenance performance teams then prepared replacement schedules, planned maintenance and surveillance checklists, and procedures (if necessary).

When required, corrective maintenance will be performed to assure that equipment will operate satisfactorily. Such corrective maintenance will become part of an equipment history file. Proper documentation of corrective maintenance actions will highlight recurring situations in similar equipment and will provide data to identify component past-performance trends.

Furthermore, equipment or component failures detected in other nuclear power plants will be available to VEGP through industry event reports, Nuclear Regulatory Commission inspection and enforcement bulletins, information notices, letters, and directives, and manufacturer's information notices. VEGP will review these reports to determine their applicability and will modify its maintenance and surveillance program accordingly.

In addition to that required by the planned maintenance and surveillance checklists, surveillance and operability testings are performed by the VEGP inservice testing and the Technical Specifications surveillance programs to address mechanical aging. The inservice testing is conducted in accordance with American Society of Mechanical Engineers (ASME) Section XI. All safety-related pumps with a Class 1E power source and selected safety-related active valves required by ASME Section XI are tested on a regular basis. The pump testing includes measurement of differential pressure at a predetermined flow rate, vibration amplitude, bearing temperatures and inlet pressure. The valve testing includes determination of leak rates for certain isolation valves, stroke times for certain power operated valves, fail safe verification, position indication verification, safety valve set point verification, and opening force or torque for check valves with external lever arms. The Technical Specification surveillance program covers all equipment required by the VEGP Technical Specifications. The VEGP Technical Specifications specify requirements for the test frequency, acceptability of testing, and measured parameters.

As described in FSAR paragraph 3.11.B.1.1 and in conformance with NUREG-0588 guidelines, VEGP has prepared and issued specifications identifying qualification methods and requirements for safety-related mechanical and electrical equipment. This specification requires suppliers to meet IEEE 323-1974 section 6 paragraph 6.3.2 sequence for safety-related equipment located in harsh environments. Subsections of paragraph 6.3.2 regarding "Aging" or "Qualified-Life" do not apply to safety-related equipment located in mild environments. The specification also required suppliers to simulate mechanical aging by operating equipment to simulate the expected mechanical wear.

For qualification tests of electric valve operators, the specification requires vendors to follow the direction given in Section 4 of IEEE Standard 382-1972 as committed to in VEGP FSAR. The vendors are also required to provide complete justification that the sequence used is the most severe for the item being tested.

For NSSS equipment, the impact of mechanical aging effects on equipment qualification has been addressed in the design process and/or as part of specific qualification test program. Significant mechanical aging effects can result from system piping vibration and cycling of equipments. Equipment where such effects may be significant are active pumps and line-mounted equipment (active valves, resistance temperature detectors, etc.).

Relative to system/piping vibration, FSAR paragraph 3.9.B.2 describes the preoperational piping vibration program which is being implemented on VEGP. This program assures that piping system vibration limits are within ASME code limits. This testing program includes transient

testing such as the opening and closing of valves and the stopping and starting of pumps. This program assures that piping vibration effects are not significant.

Relative to active valves, the testing program described above assures that piping vibration effects on the mechanical portions of the valve are insignificant. For valve appurtenances, vibration aging is included in the qualification test program.

Requirements for the cycling of valves are included in valve specifications. This requirement has been factored into valve designs by selecting proper materials, assuring proper clearances, and internal part interfaces. Also, several valve vendors have performed valve cycle tests to demonstrate that the Class 1 valves have performed and design transients for Class 2 and 3 valves are evaluated. Finally, the valves supplied to VEGP by Westinghouse are similar to these and other Westinghouse plants which have been operating for extended periods of time.

3.11.N.2 QUALIFICATION TESTS AND ANALYSIS

3.11.N.2.1 Environmental Qualification Criteria

The methods of meeting the general requirements for environmental design and qualification of safety-related equipment as described by General Design Criteria (GDC) 1, 2, 4, 23, and 50 are described in section 3.1. Additional specific information concerning the implementation of GDC 23 and 50 is provided in paragraph 7.2.2.2 and section 6.2, respectively. The general methods of implementing the requirements of Appendix B to 10 CFR 50 are described in chapter 17. Regulatory Guides 1.40, 1.73, and 1.89 concerning environmental qualification are addressed in section 1.9. Administrative control of component qualification is addressed in paragraph 3.11.B.2.

3.11.N.2.2 Environmental Design of Mechanical Equipment

Westinghouse-supplied safety-related mechanical components have been designed to perform their required safety functions under the appropriate environmental effects of normal, abnormal, accident, and post-accident conditions as required by GDC 4. For mild environments, the area conditions do not change as the result of an accident; as a result, there are no degrading environmental effects that could lead to common mode failure of safety-related equipment. However, mechanical equipment located in harsh environmental zones must be designed to perform under the appropriate environmental conditions.

For safety-related mechanical equipment, there are two basic categories of components as follows:

A. Active equipment - equipment that must perform a mechanical motion as part of its safety function.

B. Nonactive equipment - equipment whose only safety function is structural integrity. Nonactive components are designed for structural integrity in accordance with Section III of the American Society of Mechanical Engineers (ASME) Code as discussed in section 3.9.N.

Based on the above information, the VEGP program for environmental qualification of mechanical equipment discussed in this section pertains strictly to active components that must perform their safety functions in a harsh environment. The post-accident environmental effects are considered to be negligible on structural components such as pump casings and valve

bodies. Therefore, the program is restricted to evaluating the design of critical subcomponents of active devices whose failure could result in loss of the active components.

The design of these components is based in part on the stringent selection of materials utilized in safety-grade mechanical components. Organic materials, such as gaskets, seals, and O-rings, are selected for use based on their capability to perform in a nuclear environment. In some cases, partial type tests have been performed to confirm the adequacy of selected materials or subcomponents for use under adverse environmental conditions. Mechanical aging of safety-related pumps and valves is addressed in paragraph 3.11.B.2.

The program for environmental qualification of active mechanical components is based on a combination of design, test, trending, and analysis of critical subcomponents, which is supported by periodic plant test and maintenance/surveillance programs. This program addresses the requirements of GDC 4 and provides adequate assurance that active components will perform their required functions under all normal, abnormal, accident, and post-accident environmental conditions.

3.11.N.2.3 Environmental Design of Electrical Equipment

The Westinghouse approach for environmental qualification of nuclear steam supply system (NSSS)-supplied Class 1E equipment is outlined in reference 2. This methodology was developed based on the guidelines provided in the Institute of Electrical and Electronics Engineers (IEEE) Standards 323-1974, IEEE Standard for Qualifying Class 1E Equipment for Nuclear Power Generating Stations, and 344-1975, IEEE Recommended Practices for Seismic Qualifications of Class 1E Equipment for Nuclear Power Generating Stations.

Westinghouse meets the IEEE Standard 323-1974 (augmented by Regulatory Guide 1.89), including IEEE Standard 323a-1975, the Nuclear Power Engineering Committee Position Statement of July 24, 1975, by either type test, operating experience, analysis, or an appropriate combination of these methods. Westinghouse meets this commitment by employing the methodology described in WCAP-8587. The Westinghouse program outlined in WCAP-8587 has been approved as an acceptable program for demonstrating qualification of Class 1E equipment to the requirements of Regulatory Guides 1.89 and 1.100 and IEEE Standards 323-1974 and 344-1975(3).

3.11.N.2.4 References

1. Buttersworth, G., and Miller, R. B., "Methodology for Qualifying Westinghouse WRD Supplied NSSS Safety-Related Electrical Equipment," WCAP-8587, Revision 3, May 5, 1980.
2. Methodology for Qualifying Westinghouse WRD Supplied NSSS Safety-Related Electrical Equipment, WCAP-8587.
3. NRC letter from C. O. Thomas to E. P. Rahe, Jr., Westinghouse Electric Corporation, dated November 10, 1983.

3.11.B.3 QUALIFICATION TEST RESULTS

The results of the qualification program for the balance of plant (BOP) equipment are provided in the appropriate BOP equipment qualification data packages (EQDP).

Each BOP EQDP contains qualification procedures and reports demonstrating qualification and provides adequate assurance that safety-related equipment can perform its required safety functions under all required plant conditions.

In addition, table 3.11.B.3-1 provides information relating VEGP conformance to the requirements in NUREG-0588, Interim Staff Position on Environmental Qualification of Safety-Related Electrical Equipment.

3.11.N.3 QUALIFICATION PROGRAM RESULTS

Qualification program test results for the nuclear steam supply system supplied Class 1E equipment are provided in the equipment qualification central file.

The program for the environmental qualification of safety-related mechanical equipment is outlined in paragraph 3.11.N.2.2. This program was developed to meet the intent of General Design Criterion 4 and provides adequate assurance that safety-related mechanical equipment can perform its required safety functions under all required plant conditions. Qualification results are provided in the equipment qualification central file.

A comparison of NUREG-0588 against WCAP-8587, Supplement 1, is contained in table 3.11.N.3-1.

3.11.4 LOSS OF VENTILATION

The maximum temperatures considered in the sizing of air-conditioning systems serving safety-related systems are determined by analysis of the following factors:

A. Outdoor design temperatures for the VEGP site (both wet bulb and dry bulb readings).

B. Conservative piping thermal loads for the room, using maximum operating temperatures for the pipe contents and maximum footage of active pipe for each mode of operation.

C. Conservative electrical heat loads, assuming full lighting for the room and using the maximum control and equipment resistance losses for each mode of operation.

D. Conservative heat transfer from miscellaneous equipment surfaces, if applicable, e.g., outer surface of the diesel generator.

E. Conservative heat transfer from the surfaces of open pools and tanks, if applicable, using the maximum operating temperature of the contents.

F. Conservative heat transfer from the surfaces of the room including walls, floor, and ceiling or roof.

Safety-related air-conditioning, air handling, or ventilation systems described in section 9.4 are powered from Class 1E electrical power supplies and are provided for the following areas:

- Control room envelope.
- Control building auxiliary relay rooms.
- Control building safety features electrical equipment rooms (including Class 1E battery rooms).

- Class 1E motor control center and switchgear rooms.
- Piping penetration areas.
- Class 1E electrical tunnels.
- Residual heat removal pump, safety injection pump, and containment spray pump rooms.
- Control building control room ESF chiller rooms.
- Centrifugal charging pump rooms.
- Remote shutdown panel rooms.
- Diesel generator buildings.
- Component cooling water pump rooms.
- Auxiliary feedwater pump rooms.
- Spent fuel pool heat exchanger and pump rooms.
- Fuel handling building.
- Containment.

Safety-related air-conditioning systems are designed such that the single failure of an active component, after a design basis accident (DBA), cannot impair the ability of the systems located within the area served by the redundant train of air-conditioning equipment to fulfill their safety functions. Should one train of the safety-related air-conditioning system become inoperative during normal operation, sufficient equipment capacity is still available to mitigate the consequences of a DBA.

Four redundant 100-percent-capacity essential air filtration units are provided in the control building for the common control room complex. It is not considered a credible event to lose all control room air-conditioning simultaneously.

The worst-case environments are presented in table 3.11.B.1-1.

3.11.5 ESTIMATED CHEMICAL AND RADIATION ENVIRONMENT

The plant-specific estimates of the radiation dose incurred by equipment during normal operation and the estimated doses and chemical conditions following a loss-of-coolant accident are defined in table 3.11.B.1-1.

3.11.5.1 Chemical Environment

Engineered safety feature (ESF) systems and components are qualified to perform their safety-related functions in the temperature and pressure conditions described in table 3.11.B.1-1. In addition, safety-related components inside the containment are designed to perform their safety-related functions in long-term contact with a combined boric acid- trisodium phosphate solution recirculated through the emergency core cooling system and the containment spray system.

3.11.5.2 Radiation Environment

Safety-related equipment is qualified to perform safety-related functions in the radiation environments present during normal and design basis accident conditions. The normal operational exposure is based upon design source terms presented in chapter 11 and subsection 12.2.1, and the equipment and shielding configurations presented in section 12.3. Post-accident ESF system and component radiation exposures are dependent on the location of the equipment in the plant and include the effect of recirculatory fluid for equipment outside the containment. Source terms and other accident parameters are presented in subsection 12.2.1 and chapter 15 and are consistent with the recommendations of Regulatory Guides 1.89, 1.4, and 1.7. The maximum combined integrated radiation dose for inside containment is based on the integrated effects (beta and gamma) of the normally expected radiation environment over the equipment's installed life plus that associated with the most severe design basis event during or following which the equipment is required to remain functional. Source term and methodology are consistent with that discussed in NUREG-0588. (See table 3.11.B.2-1.) Normal and accident radiation exposures based on the above assumptions are presented in table 3.11.B.1-1.

TABLE 3.11.B.1-1 (SHEET 1 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test[b]		DBA/Post-DBA[c]			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[h]	Pressure[h]	Int Dose (rads)[d]	Normal (%)	DBA (%)
SPRAYED VAPOR REGION[e]											
IB-R-301H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-302H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-101H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-102H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-103H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-104H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-105H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-106H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-109H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-111H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-A01H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
UNSPRAYED VAPOR REGION[e]											
IB-R-107H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-108H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-110H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-115H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-A02H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-A03H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-A04H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-A05H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-A06H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-A07H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-A08H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 6	Sheet 11	1.8×10^8	50	100
IB-R-A09H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100

CONTAINMENT BUILDING
(Sheet 1 of 4)

TABLE 3.11.B.1-1 (SHEET 2 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test[b]		DBA/Post-DBA[c]			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[h]	Pressure[h]	Int Dose (rads)[d]	Normal (%)	DBA (%)
UNSPRAYED VAPOR REGION[e]											
IB-R-B03H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-B10H	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 9	Sheet 11	1.8×10^8	50	100
IB-R-B12H[f]	1/2	amb	14.7 psia	-	-	11.7 psia	-	-	-	-	-
IB-R-B13H[f]	1/2	amb	14.7 psia	-	-	11.7 psia	-	-	-	-	-
IB-R-B14H[f]	1/2	amb	14.7 psia	-	-	11.7 psia	-		-	-	-
IB-R-B15H[f]	1/2	amb	14.7 psia	-	-	11.7 psia	-		-	-	-
SUMP REGION[e]											
IA-R-B01H[g]	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 10	Sheet 11	1.8×10^8	50	100
IA-R-B02H[g]	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 10	Sheet 11	1.8×10^8	50	100
IA-R-B04H[g]	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 10	Sheet 11	1.8×10^8	50	100
IA-R-B05H[g]	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 10	Sheet 11	1.8×10^8	50	100
IA-R-B06H[g]	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 10	Sheet 11	1.8×10^8	50	100
IA-R-B07H[g]	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 10	Sheet 11	1.8×10^8	50	100
IA-R-B08H[g]	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 10	Sheet 11	1.8×10^8	50	100
IA-R-B09H[g]	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 10	Sheet 11	1.8×10^8	50	100
IA-R-B11H[g]	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 10	Sheet 11	1.8×10^8	50	100
IA-R-B16H[g]	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 10	Sheet 11	1.8×10^8	50	100
IA-R-C03H[g]	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 10	Sheet 11	1.8×10^8	50	100
IA-R-C10H[g]	1/2	120	17.7-13.2 psia	2×10^6	120/60	60 psig	Sheet 10	Sheet 11	1.8×10^8	50	100

TABLE 3.11.B.1-1 (SHEET 3 OF 92)
ENVIRONMENTAL CONDITIONS

NOTES

a. H = Harsh environment due to existence of either of the following conditions:

- Temperature increases due to the pipe break.
- Total integrated dose (TID) > 1×10^4 rad.

- = No high-energy line or safety-related equipment located in the room.

b. The containment test pressure is 60 psig.

c. Includes normal doses.

d. Integrated dose = 3×10^7 rads (γ), 1.5×10^8 rads (β). (Calculated for center of containment building but envelope all areas.)

e. Spray Exposure

The containment building consists of three general regions, defined as follows:

- Sprayed Vapor Region - That area above the operating deck that would be exposed to the direct effects of the spray system.

- Unsprayed Vapor Region - That area of containment that is below the operating deck and above the containment flood level that may be exposed to the indirect effects of the spray system.

TABLE 3.11.B.1-1 (SHEET 4 OF 92)
ENVIRONMENTAL CONDITIONS

NOTES (continued)

- Sump Region - That area of containment that would be flooded post-accident.

  Sprayed and unsprayed vapor regions pH at injection phase (0-100 min) = 4.5.

  Sprayed and unsprayed vapor regions pH at recirculation phase (100 min - 24 h) = 7.5 - 10.5.

  Sump region pH at 0 min - 24 h = 7.5 - 10.5.

  Chemicals - Spray solution is boric acid (2400 - 2600 ppm boron) plus sufficient trisodium phosphate (TSP) to achieve the desired pH.

f. Tendon access shafts which are outside the containment pressure boundary.

g. If equipment is located above flood level, sheet 9 shall be used for temperature. (Flood level is el 181 ft 2 in.)

h. Sheet numbers refer to figure 3.11.B.1-1.

TABLE 3.11.B.1-1 (SHEET 5 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[c]	Pressure[c]	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING											
II-R-A08H	2	115	atm	1×10^3	126/17[b]	11.7 psia	Sheet 7	Sheet 8	1×10^3	90	100
II-R-A09H	2	115	atm	5×10^5	126/17[b]	11.7 psia	Sheet 7	Sheet 8	5×10^6	90	100
II-R-A56H	1/com	115	atm	5×10^5	126/17[b]	11.7 psia	Sheet 7	Sheet 8	5×10^6	90	100
II-R-A57H	1/com	115	atm	1×10^3	126/17[b]	11.7 psia	Sheet 7	Sheet 8	1×10^3	90	100
II-R-121H	1/com	115	atm	1×10^3	126/17[b]	11.7 psia	Sheet 7	Sheet 8	1×10^3	90	100
II-R-122H	1/com	115	atm	5×10^5	126/17[b]	11.7 psia	Sheet 7	Sheet 8	5×10^6	90	100
II-R-123H	1/com	115	atm	5×10^5	126/17[b]	11.7 psia	Sheet 7	Sheet 8	5×10^6	90	100
II-R-121H	2	115	atm	1×10^3	126/17[b]	11.7 psia	Sheet 7	Sheet 8	1×10^3	90	100
II-R-122H	2	115	atm	5×10^5	126/17[b]	11.7 psia	Sheet 7	Sheet 8	5×10^6	90	100
II-R-123H	2	115	atm	5×10^5	126/17[b]	11.7 psia	Sheet 7	Sheet 8	5×10^6	90	100
AUXILIARY BUILDING											
II-R-A06H	1/com	115	atm	1×10^3	126/17[b]	11.7 psia	Sheet 7	Sheet 8	1×10^3	90	100
II-R-A11H	1/com	115	atm	5×10^5	126/17[b]	11.7 psia	Sheet 7	Sheet 8	5×10^6	90	100
II-R-A12H	1/com	115	atm	5×10^5	126/17[b]	11.7 psia	Sheet 7	Sheet 8	5×10^6	90	100
II-R-A20H	1/com	115	atm	1×10^3	126/17[b]	11.7 psia	Sheet 7	Sheet 8	1×10^3	90	100
II-R-A99H	2	115	atm	1×10^3	126/17[b]	11.7 psia	Sheet 7	Sheet 8	1×10^3	90	100
II-R-A104H	2	115	atm	5×10^5	126/17[b]	11.7 psia	Sheet 7	Sheet 8	5×10^6	90	100
II-R-A105H	2	115	atm	5×10^5	126/17[b]	11.7 psia	Sheet 7	Sheet 8	5×10^6	90	100
II-R-A107H	2	115	atm	1×10^3	126/17[b]	11.7 psia	Sheet 7	Sheet 8	1×10^3	90	100
II-R-108H	1/com	115	atm	5×10^5	126/17[b]	11.7 psia	Sheet 7	Sheet 8	5×10^6	90	100
II-R-159H	2	115	atm	5×10^5	126/17[b]	11.7 psia	Sheet 7	Sheet 8	5×10^6	90	100

MSIV AREA
(Sheet 1 of 2)

TABLE 3.11.B.1-1 (SHEET 6 OF 92)
ENVIRONMENTAL CONDITIONS

NOTES

a. H = Harsh environment due to existence of either of the following conditions:

- Temperature increases due to the pipe break.
- TID > 1 x 10^4 rad.

\- = No high-energy line or safety-related equipment located in the room.

b. During plant operation, the minimum room temperature is 47°F.

c. Sheet numbers refer to figure 3.11.B.1-1.

MSIV AREA
(Sheet 2 of 2)

TABLE 3.11.B.1-1 (SHEET 7 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
EQUIPMENT BUILDING											
V-R-114	1/com	120	atm	1×10^3	135/40	11.7 psia	-	-	1×10^3	90	-
V-R-116	1/com	120	atm	1×10^3	135/40	11.7 psia	-	-	1×10^3	90	-
V-R-117H[b]	1/com	120	atm	5×10^5	125/40	11.7 psia	-	-	1×10^7	90	-
V-R-118	1/com	120	atm	1×10^3	120/40	11.7 psia	-	-	5×10^3	90	-
V-R-120	1/com	120	atm	1×10^3	120/40	11.7 psia	-	-	1×10^3	90	-
V-R-124	1/com	120	atm	1×10^3	120/40	11.7 psia	-	-	1×10^3	90	-
V-R-125H[b]	1/com	120	atm	1×10^3	122/40	11.7 psia	-	-	1×10^7	90	-
V-R-114	2	120	atm	1×10^3	135/40	11.7 psia	-	-	1×10^3	90	-
V-R-116	2	120	atm	1×10^3	135/40	11.7 psia	-	-	1×10^3	90	-
V-R-117H[b]	2	120	atm	5×10^5	125/40	11.7 psia	-	-	1×10^7	90	-
V-R-118	2	120	atm	1×10^3	120/40	11.7 psia	-	-	5×10^3	90	-
V-R-120	2	120	atm	1×10^3	120/40	11.7 psia	-	-	1×10^3	90	-
V-R-124	2	120	atm	1×10^3	120/40	11.7 psia	-	-	1×10^3	90	-
V-R-125H[b]	2	120	atm	1×10^3	122/40	11.7 psia	-	-	1×10^7	90	-

EQUIPMENT BUILDING
(Sheet 1 of 2)

TABLE 3.11.B.1-1 (SHEET 8 OF 92)
ENVIRONMENTAL CONDITIONS

NOTES

a. H = Harsh environment due to existence of either of the following conditions:

- Temperature increases due to the pipe break.
- TID > 1 x 10^4 rad.

\- = No high-energy line or safety-related equipment located in the room.

b. Room is included in the area temperature monitoring program (FSAR 16.3).

EQUIPMENT BUILDING
(Sheet 2 of 2)

TABLE 3.11.B.1-1 (SHEET 9 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL D											
VIII-R-D01H[b]	2	100	atm[c]	5×10^5	120/40	atm	120	-	5×10^5	60	-
VIII-R-D02H[b]	2	100	atm[c]	5×10^5	120/40[d]	atm	130	-	5×10^6	60	-
VIII-R-D03H[b]	2	120	atm[c]	5×10^5	130/40[e]	atm	-	-	5×10^7	60	-
VIII-R-D04H[b]	2	100	atm[c]	5×10^5	120/40[d]	atm	120	-	2×10^6	60	-
VIII-R-D05H[b]	2	100	atm[c]	5×10^5	120/40[d]	atm	120	-	2×10^6	60	-
VIII-R-D06	2	100	atm[c]	1×10^3	106/40	atm	-	-	5×10^3	60	-
VIII-R-D07	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-D08	2	100	atm[c]	1×10^3	120/65	atm	-	-	1×10^3	60	-
VIII-R-D09H	2	100	atm[c]	5×10^5	120/65	atm	-	-	1×10^8	60	-
VIII-R-D10H	2	100	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/A	1×10^3	60	100
VIII-R-D11	2	100	atm[c]	1×10^3	105/40	atm	Sheet 5	Sheet 6/C	1×10^3	60	100
VIII-R-D12H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	1×10^8	60	100
VIII-R-D13H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D14H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	1×10^8	60	100
VIII-R-D15H	1/com	100	atm[c]	1×10^3	120/40	atm	Sheet 3	Sheet 6/C	1×10^3	60	100
VIII-R-D16H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	5×10^5	60	100
VIII-R-D17H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	1×10^8	60	100
VIII-R-D18H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	1×10^8	60	100
VIII-R-D19	2	100	atm[c]	1×10^3	120/40	atm	Sheet 3	Sheet 6/C	1×10^3	60	100

TABLE 3.11.B.1-1 (SHEET 10 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL D (Continued)											
VIII-R-D20H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	5×10^5	60	100
VIII-R-D21H[b]	2	100	atm[c]	1×10^7	120/40[d]	atm	-	-	5×10^7	60	-
VIII-R-D22H[b]	2	100	atm[c]	1×10^7	120/40[d]	atm	-	-	5×10^7	60	-
VIII-R-D23	void										
VIII-R-D24H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D25H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D26H	1/com	100	atm[c]	5×10^5	112/40	atm	-	-	1×10^8	60	-
VIII-R-D27H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 3	Sheet 6/D	1×10^8	60	100
VIII-R-D28H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D29H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D30H	2	100	atm[c]	5×10^4	120/40	atm	Sheet 3	Sheet 6/C	5×10^5	60	100
VIII-R-D31	void										
VIII-R-D32	2	100	atm[c]	1×10^3	110/40	atm	Sheet 5	Sheet 6/C	1×10^3	60	100
VIII-R-D33H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D34H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	1×10^8	60	100
VIII-R-D35H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	1×10^8	60	100
VIII-R-D36H	1/com	100	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-D37H	2	100	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-D38H	1/com	100	atm[c]	5×10^4	120/40	atm	Sheet 5	Sheet 6/C	5×10^4	60	100
VIII-R-D39H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	1×10^8	60	100

TABLE 3.11.B.1-1 (SHEET 11 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL D (Continued)											
VIII-R-D40H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	1×10^8	60	100
VIII-R-D41	2	100	atm[c]	1×10^3	110/40	atm	113	-	5×10^3	60	100
VIII-R-D42H	1/com	100	atm[c]	5×10^4	120/40	atm	Sheet 5	Sheet 6/C	5×10^4	60	100
VIII-R-D43H	1/com	100	atm[c]	1×10^3	120/40	atm	Sheet 1	Sheet 6/C	1×10^3	60	100
VIII-R-D44H	1/com	100[f]	atm[c]	1×10^3	120/40	atm	Sheet 3	Sheet 6/C	1×10^3	60	100
VIII-R-D45H	1/com	100	atm[c]	5×10^5	102/40	atm	Sheet 2	Sheet 6/C	1×10^8	60	100
VIII-R-D46H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D47H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 2	Sheet 6/C	1×10^8	60	100
VIII-R-D48H[b]	1/com	100	atm[c]	1×10^7	120/40[d]	atm	-	-	5×10^7	60	-
VIII-R-D49H[b]	1/com	100	atm[c]	1×10^7	120/40[d]	atm	-	-	5×10^7	60	-
VIII-R-D50H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D51H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D52H	1/com	100	atm[c]	1×10^3	116/40	atm	Sheet 2	Sheet 6/C	1×10^3	60	100
VIII-R-D53H[h]	1/com	100	atm[c]	1×10^3	110/40	atm	Sheet 5	Sheet 6/E	5×10^3	60	100
VIII-R-D54H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	1×10^8	60	100
VIII-R-D55H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	5×10^5	60	100
VIII-R-D56H	1/com	100	atm[c]	1×10^3	110/40	atm	Sheet 5	Sheet 6/C	1×10^3	60	100
VIII-R-D57H	1/com	100	atm[c]	5×10^5	120/40	atm	-	Sheet 6/C	1×10^8	60	-
VIII-R-D58H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 5	-	1×10^8	60	100

TABLE 3.11.B.1-1 (SHEET 12 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL D (Continued)											
VIII-R-D59H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	1×10^8	60	100
VIII-R-D60H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	5×10^5	60	100
VIII-R-D61H	1/com	100[f]	atm[c]	1×10^3	120/40	atm	Sheet 3	Sheet 6/C	1×10^3	60	100
VIII-R-D62H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	1×10^8	60	100
VIII-R-D63H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D64H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	1×10^8	60	100
VIII-R-D65H	1/com	100[f]	atm[c]	5×10^4	120/40	atm	Sheet 3	Sheet 6/D	5×10^5	60	100
VIII-R-D66H	1/com	100[f]	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/D	1×10^3	60	100
VIII-R-D67H[h]	1/com	100	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/C	1×10^3	60	100
VIII-R-D68	1/com	100	atm[c]	1×10^3	106/40	atm	-	-	1×10^3	60	-
VIII-R-D69H	1/com	100	atm[c]	5×10^5	120/65	atm	-	-	1×10^8	60	-
VIII-R-D70	1/com	100	atm[c]	1×10^3	120/65	atm	-	-	1×10^3	60	-
VIII-R-D71	1/com	100[f]	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-D72H[h]	1/com	100	atm[c]	1×10^5	125/65	atm	-	-	1×10^5	60	-
VIII-R-D73H[h]	1/com	100	atm[c]	5×10^5	121/40	atm	-	-	5×10^5	60	-
VIII-R-D74H[b]	1/com	100	atm[c]	5×10^5	120/40	atm	120	-	5×10^5	60	-
VIII-R-D75H	1/com	100	atm[c]	1×10^3	106/40	atm	-	-	5×10^3	60	-
VIII-R-D76H[b]	1/com	100	atm[c]	5×10^5	120/40[d]	atm	120	-	2×10^6	60	-
VIII-R-D77H[b]	1/com	100	atm[c]	5×10^5	120/40[d]	atm	120	-	2×10^6	60	-

TABLE 3.11.B.1-1 (SHEET 13 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL D (Continued)											
VIII-R-D78H[b]	1/com	120	atm[c]	5×10^5	130/40[e]	atm	-	-	5×10^7	60	-
VIII-R-D79H[b]	1/com	100	atm[c]	5×10^5	120/40[d]	atm	130	-	5×10^6	60	-
VIII-R-D80H	1/com	100	atm[c]	5×10^5	112/40	atm	-	-	5×10^5	60	-
VIII-R-D81H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D82H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D83H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D84H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D85H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D86H	1/com	100[i]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D87H	1/com	100[i]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D88H	1/com	100[i]	atm[c]	5×10^5	110/40	atm	-	-	1×10^8	60	-
VIII-R-D89H	1/com	100[i]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D90H	1/com	100[i]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D91H	1/com	100[i]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D92H	1/com	100[i]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D93H	1/com	100[i]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D94H	1/com	100[i]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D95H	1/com	100[i]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-D96H	1/com	100[i]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-

TABLE 3.11.B.1-1 (SHEET 14 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL D (Continued)											
VIII-R-D97H	1/com	100	atm[c]	1×10^3	120/40	atm	Sheet 2	Sheet 6/E	1×10^3	60	100
VIII-R-D98H[b]	2	100	atm[c]	5×10^5	120/40[e]	atm	120	-	5×10^7	60	-
VIII-R-D99H[b]	2	100	atm[c]	5×10^5	112/40[e]	atm	Sheet 12/A Sheet 13	Sheet 6/C	5×10^7	60	100
VIII-R-D100H[b]	1/com	100	atm[c]	5×10^5	120/40[e]	atm	120	-	5×10^7	60	-
VIII-R-D101H[b]	1/com	100	atm[c]	5×10^5	120/40[e]	atm	Sheet 12/A Sheet 13	Sheet 6/C	5×10^7	60	100
VIII-R-D102H	1/com	100	atm[c]	5×10^5	110/40	atm	Sheet 5	Sheet 6/C	1×10^8	60	100
VIII-R-D103	void										
VIII-R-D104[b]	2	100	atm[c]	1×10^3	120/40[d]	atm	-	-	5×10^3	60	-
VIII-R-D105[b]	1/com	100	atm[c]	1×10^3	120/40[d]	atm	-	-	5×10^3	60	-
VIII-R-D106H[h]	2	100	atm[c]	1×10^5	125/65	atm	-	-	1×10^5	60	-
VIII-R-D107H[h]	2	100	atm[c]	5×10^5	105/40	atm	-	-	5×10^5	60	-
VIII-R-D108	2	100	atm[c]	1×10^3	106/40	atm	-	-	1×10^3	60	-
VIII-R-D109	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-D110	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-D111H	2	100	atm[c]	5×10^5	110/40	atm	Sheet 5	Sheet 6/C	1×10^8	60	100
VIII-R-D112H	2	100	atm[c]	5×10^5	106/40	atm	-	-	5×10^6	60	-
VIII-R-D113H[h]	2	100	atm[c]	5×10^5	103/40	atm	-	-	1×10^6	60	-
VIII-R-D114H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	5×10^5	60	100
VIII-R-D115H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	1×10^8	60	100

TABLE 3.11.B.1-1 (SHEET 15 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL D (Continued)											
VIII-R-D116H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	1×10^8	60	100
VIII-R-D117H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/C	1×10^8	60	100
VIII-R-D118	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-D119H[h]	1/com	100	atm[c]	1×10^5	121/65	atm	Sheet 5	Sheet 6/C	1×10^5	60	100
VIII-R-D120H	2	100	atm[c]	5×10^5	120/40	atm	-	-	5×10^5	60	-
VIII-R-D121H[h]	1/com	100	atm[c]	5×10^5	103/40	atm	-	-	1×10^6	60	-
VIII-R-D122H	1/com	100[f]	atm[c]	5×10^5	106/40	atm	-	-	5×10^6	60	-
VIII-R-D123H[h]	2	100	atm[c]	1×10^5	121/65	atm	-	-	1×10^5	60	-
VIII-R-D124	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-D125H	1/com	100[f]	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/C	1×10^3	60	100
VIII-R-D126H	1/com	100[f]	atm[c]	5×10^4	120/40	atm	Sheet 3	Sheet 6/D	5×10^6	60	100
VIII-R-D127H	2	100	atm[c]	5×10^4	120/40	atm	Sheet 3	Sheet 6/C	5×10^6	60	100
VIII-R-D128H[b]	1/com	100	atm[c]	5×10^5	111/40	atm	106	-	5×10^6	60	-
VIII-R-D129H[b]	2	100	atm[c]	5×10^5	111/40	atm	106	-	5×10^6	60	-
VIII-R-D130H[b]	1/com	100	atm[c]	5×10^5	110/40	atm	-	-	1×10^6	60	-
VIII-R-D131H[b]	2	100	atm[c]	5×10^5	110/40	atm	-	-	1×10^6	60	-
VIII-R-D132H	1/com	100[f]	atm[c]	5×10^5	104/40	atm	-	-	1×10^6	60	-
VIII-R-D133H	2	100	atm[c]	5×10^5	104/40	atm	-	-	1×10^6	60	-
VIII-UCD001H	2	120	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100

TABLE 3.11.B.1-1 (SHEET 16 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL D (Continued)											
VIII-UCD002H	2	120	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCD003H	1/com	120[f]	atm[c]	5×10^5	120/40	atm	Sheet 4	Sheet 6/A	1×10^8	60	100
VIII-UCD004H	1/com	120[f]	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCD005H	1/com	120[f]	atm[c]	5×10^5	120/40	atm	Sheet 4	Sheet 6/A	1×10^8	60	100
VIII-UCD006H	1	120	atm[c]	5×10^5	120/40	atm	Sheet 4	Sheet 6/A	1×10^8	60	100
VIII-UCD006H	2	120[f]	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCD007H	1/com	120[f]	atm[c]	5×10^5	120/40	atm	Sheet 4	Sheet 6/A	1×10^8	60	100
VIII-UCD008H	1/com	120[f]	atm[c]	5×10^5	120/40	atm	Sheet 4	Sheet 6/A	1×10^8	60	100
VIII-UCD009H	1/com	120[f]	atm[c]	5×10^5	120/40	atm	Sheet 4	Sheet 6/A	1×10^8	60	100
VIII-UCD010H	1/com	120[f]	atm[c]	5×10^5	120/40	atm	Sheet 12/B	Sheet 6/C	1×10^8	60	100
VIII-UCD011H	2	120	atm[c]	5×10^5	120/40	atm	Sheet 12/B	Sheet 6/C	1×10^8	60	100
AUXILIARY BUILDING - LEVEL C											
VIII-R-C01H[b]	2	100	atm[c]	5×10^5	120/40[d]	atm	Sheet 2	Sheet 6/A	5×10^5	60	100
VIII-R-C02H[b]	2	115	atm[c]	1×10^7	120/40[d]	atm	Sheet 2	Sheet 6/A	1×10^7	60	100
VIII-R-C03H[b]	2	114	atm[c]	1×10^3	120/40[d]	atm	Sheet 2	Sheet 6/A	1×10^4	60	100
VIII-R-C04H[b]	2	100	atm[c]	5×10^5	105/40	atm	102, Sheet 2	Sheet 6/A	5×10^5	60	100
VIII-R-C05H[b]	2	106	atm[c]	5×10^5	120/40[e]	atm	Sheet 2 Sheet 6	Sheet 6/A	5×10^7	60	100
VIII-R-C06H	2	128	atm[c]	1×10^3	120/40	atm	154	-	5×10^4	60	-
VIII-R-C07H[b]	2	100	atm[c]	1×10^3	120/40[d]	atm	-	-	5×10^4	60	-

TABLE 3.11.B.1-1 (SHEET 17 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL C (Continued)											
VIII-R-C08H	2	100	atm[c]	1×10^3	120/40	atm	Sheet 2	Sheet 6/A	1×10^3	60	100
VIII-R-C09H[b]	2	115	atm[c]	5×10^5	120/40[d]	atm	-	-	1×10^7	60	-
VIII-R-C10H[h]	2	100	atm[c]	1×10^3	109/40	atm	-	-	1×10^7	60	-
VIII-R-C11H[b]	2	104	atm[c]	5×10^5	120/65[d]	atm	120	-	1×10^6	60	-
VIII-R-C12H[b]	2	116	atm[c]	5×10^5	120/40[d]	atm	117	-	5×10^6	60	-
VIII-R-C13	2	100	atm[c]	1×10^3	120/40	atm	-	-	5×10^3	60	-
VIII-R-C14	2	100	atm[c]	1×10^3	120/40	atm	-	-	5×10^3	60	-
VIII-R-C15	2	100	atm[c]	1×10^3	120/40	atm	-	-	5×10^3	60	-
VIII-R-C16H[b]	2	100	atm[c]	5×10^5	120/65[d]	atm	-	-	1×10^7	60	-
VIII-R-C17H[b]	2	100	atm[c]	5×10^5	120/65[d]	atm	-	-	1×10^7	60	-
VIII-R-C18H[b]	2	115	atm[c]	5×10^5	120/65[d]	atm	Sheet 17	-	1×10^6	60	-
VIII-R-C19	2	100	atm[c]	1×10^3	106/40	atm	-	-	1×10^4	60	-
VIII-R-C20H	2	100	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/A	1×10^3	60	100
VIII-R-C21H	2	100	atm[c]	1×10^3	120/40	atm	Sheet 4	Sheet 6/D	1×10^3	60	100
VIII-R-C22H	2	100	atm[c]	5×10^4	120/40	atm	Sheet 3	Sheet 6/D	5×10^4	60	100
VIII-R-C23H[b]	2	102	atm[c]	5×10^5	120/40[e]	atm	Sheet 12/A Sheet 13	Sheet 6/C	1×10^7	60	100
VIII-R-C24H	2	100	atm[c]	1×10^3	122/40	atm	Sheet 4	Sheet 6/D	1×10^4	60	100
VIII-R-C25H[b]	2	100	atm[c]	1×10^7	120/40[d]	atm	112	-	5×10^7	60	-
VIII-R-C26H[b]	2	100	atm[c]	1×10^7	120/40[d]	atm	112	-	5×10^7	60	-

TABLE 3.11.B.1-1 (SHEET 18 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL C (Continued)											
VIII-R-C27H	2	100	atm[c]	5×10^4	104/40	atm	-	-	5×10^7	60	-
VIII-R-C28H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 3	Sheet 6/D	1×10^8	60	100
VIII-R-C29H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 3	Sheet 6/D	1×10^8	60	100
VIII-R-C30H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 4	Sheet 6/A	5×10^6	60	100
VIII-R-C31H	2	100	atm[c]	1×10^3	120/40	atm	Sheet 4	Sheet 6/D	1×10^3	60	100
VIII-R-C32H	2	100	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/A	1×10^3	60	100
VIII-R-C33H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-C34H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-C35H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-C36H	2	100	atm[c]	5×10^5	120/40	atm	-	-	5×10^6	60	-
VIII-R-C37H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-R-C38H[b]	2	100	atm[c]	5×10^4	104/40	atm	-	-	5×10^6	60	-
VIII-R-C39	2	104	atm[c]	1×10^3	120/40	atm	-	-	5×10^3	60	-
VIII-R-C40H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 1	Sheet 6/A	1×10^8	60	100
VIII-R-C41H	2	100	atm[c]	5×10^5	118/40	atm	Sheet 1	Sheet 6/A	1×10^8	60	100
VIII-R-C42H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 2	Sheet 6/A	5×10^5	60	100
VIII-R-C43H	2	100	atm[c]	5×10^5	121/40	atm	Sheet 2	Sheet 6/D	1×10^8	60	100
VIII-R-C44H	2	100	atm[c]	1×10^3	120/40	atm	Sheet 4	Sheet 6/D	5×10^3	60	100
VIII-R-C45H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 3	Sheet 6/A	1×10^8	60	100
VIII-R-C46H	2	100	atm[c]	5×10^5	110/40	atm	Sheet 3	Sheet 6/A	1×10^8	60	100

TABLE 3.11.B.1-1 (SHEET 19 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL C (Continued)											
VIII-R-C47H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 3	Sheet 6/A	1×10^8	60	100
VIII-R-C48H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 3	Sheet 6/A	1×10^8	60	100
VIII-R-C49H	2	100	atm[c]	1×10^3	113/40	atm	Sheet 3	Sheet 6/A	1×10^3	60	100
VIII-R-C50H	2	100	atm[c]	5×10^5	107/40	atm	Sheet 3	Sheet 6/D	1×10^8	60	100
VIII-R-C51H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/D	5×10^5	60	100
VIII-R-C52H	2	100	atm[c]	5×10^5	100/40	atm	-	-	1×10^8	60	-
VIII-R-C53H	1/com	100[f]	atm[c]	5×10^5	120/40	atm	Sheet 4	Sheet 6/A	1×10^8	60	100
VIII-R-C54H	2	100	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-C55H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 3	Sheet 6/D	1×10^8	60	100
VIII-R-C56H	2	100	atm[c]	5×10^5	105/40	atm	Sheet 3	Sheet 6/D	1×10^8	60	100
VIII-R-C57H	2	100	atm[c]	5×10^4	120/40	atm	Sheet 3	Sheet 6/D	5×10^4	60	100
VIII-R-C58H	1/com	100	atm[c]	1×10^3	120/40	atm	Sheet 4	Sheet 6/D	1×10^3	60	100
VIII-R-C59H	1/com	100[f]	atm[c]	1×10^3	120/40	atm	Sheet 3	Sheet 6/C	1×10^3	60	100
VIII-R-C60H	1/com	100[f]	atm[c]	5×10^5	120/40	atm	Sheet 4	Sheet 6/D	5×10^6	60	100
VIII-R-C61H	1/com	100	atm[c]	5×10^5	120/65	atm	Sheet 1	Sheet 6/D	1×10^8	60	100
VIII-R-C62H	1/com	100	atm[c]	5×10^4	120/40	atm	Sheet 1	Sheet 6/D	1×10^5	60	100
VIII-R-C63H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 1	Sheet 6/D	1×10^8	60	100
VIII-R-C64H	1/com	100	atm[c]	5×10^5	129/65	atm	Sheet 2	Sheet 6/D	1×10^8	60	100

TABLE 3.11.B.1-1 (SHEET 20 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL C (Continued)											
VIII-R-C65H	1/com	100	atm[c]	5×10^5	118/65	atm	Sheet 2	Sheet 6/D	1×10^8	60	100
VIII-R-C66H	1/com	100	atm[c]	5×10^5	124/40	atm	Sheet 2	Sheet 6/D	1×10^8	60	100
VIII-R-C67H	1/com	100	atm[c]	5×10^5	121/40	atm	Sheet 2	Sheet 6/D	1×10^8	60	100
VIII-R-C68H	1/com	100	atm[c]	5×10^5	110/40	atm	Sheet 3	Sheet 6/D	1×10^8	60	100
VIII-R-C69H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 3	Sheet 6/D	1×10^8	60	100
VIII-R-C70H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 3	Sheet 6/D	1×10^8	60	100
VIII-R-C71H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 3	Sheet 6/D	1×10^8	60	100
VIII-R-C72H	1/com	100	atm[c]	5×10^5	105/40	atm	Sheet 3	Sheet 6/D	1×10^8	60	100
VIII-R-C73H	1/com	100	atm[c]	5×10^4	120/40	atm	Sheet 3	Sheet 6/D	5×10^4	60	100
VIII-R-C74H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 3	Sheet 6/D	1×10^8	60	100
VIII-R-C75H	1/com	100	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-C76H	1/com	100[f]	atm[c]	5×10^5	120/40	atm	Sheet 4	Sheet 6/A	1×10^8	60	100
VIII-R-C77H[b]	1/com	100	atm[c]	5×10^5	100/40	atm	-	-	1×10^8	60	-
VIII-R-C78H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/D	5×10^5	60	100
VIII-R-C79H	1/com	100	atm[c]	5×10^5	107/40	atm	Sheet 3	Sheet 6/D	1×10^8	60	100
VIII-R-C80H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-C81H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-C82H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-C83H[h]	1/com	100	atm[c]	1×10^3	113/40	atm	Sheet 4	Sheet 6/D	1×10^3	60	100

TABLE 3.11.B.1-1 (SHEET 21 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL C (Continued)											
VIII-R-C84H[b]	1/com	100	atm[c]	1×10^7	104/40	atm	Sheet 16	-	5×10^7	60	-
VIII-R-C85H	1/com	100	atm[c]	1×10^3	120/40	atm	Sheet 4	Sheet 6/D	5×10^3	60	100
VIII-R-C86H	1/com	100[f]	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-R-C87H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 4	Sheet 6/D	5×10^5	60	100
VIII-R-C88	1/com	104	atm[c]	1×10^3	120/40	atm	-	-	5×10^3	60	-
VIII-R-C89H[b]	1/com	100	atm[c]	5×10^4	104/40	atm	-	-	5×10^6	60	-
VIII-R-C90H[b]	1/com	100	atm[c]	1×10^7	120/40[d]	atm	112	-	5×10^7	60	-
VIII-R-C91H[b]	1/com	100	atm[c]	1×10^7	120/40[d]	atm	112	-	5×10^7	60	-
VIII-R-C92H[b]	1/com	100	atm[c]	5×10^4	104/40	atm	-	-	5×10^7	60	-
VIII-R-C93H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 3	Sheet 6/D	1×10^8	60	100
VIII-R-C94H	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 3	Sheet 6/D	1×10^8	60	100
VIII-R-C95H[b]	1/com	100	atm[c]	5×10^5	120/40[e]	atm	Sheet 12/A Sheet 13	Sheet 6/C	1×10^7	60	100
VIII-R-C96H[b]	1/com	100	atm[c]	1×10^7	100/40	atm	Sheet 16	-	5×10^7	60	-
VIII-R-C97H	1/com	100	atm[c]	1×10^3	122/40	atm	Sheet 4	Sheet 6/D	1×10^4	60	100
VIII-R-C98H	1/com	100	atm[c]	1×10^3	120/40	atm	Sheet 4	Sheet 6/D	1×10^3	60	100
VIII-R-C99H	1/com	100[e]	atm[c]	5×10^4	120/40	atm	Sheet 3	Sheet 6/D	5×10^4	60	100
VIII-R-C100H	1/com	100	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/D	1×10^3	60	100
VIII-R-C101H	1/com	100	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/D	1×10^3	60	100
VIII-R-C102H	1/com	100[e]	atm[c]	1×10^3	120/40	atm	Sheet 4	Sheet 6/D	1×10^3	60	100

TABLE 3.11.B.1-1 (SHEET 22 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL C (Continued)											
VIII-R-C103H[b]	1/com	106	atm[c]	5×10^5	120/40[e]	atm	Sheet 2 Sheet 16	Sheet 6/A	5×10^7	60	100
VIII-R-C104H[b]	1/com	128	atm[c]	1×10^3	128/40	atm	154	-	5×10^4	60	-
VIII-R-C105H[b]	1/com	100	atm[c]	5×10^5	120/40	atm	102/Sheet 2	Sheet 6/A	5×10^5	60	100
VIII-R-C106H[b]	1/com	100	atm[c]	1×10^3	105/40	atm	Sheet 2	Sheet 6/A	1×10^4	60	100
VIII-R-C107H[b]	1/com	100	atm[c]	5×10^5	120/40[d]	atm	Sheet 2	Sheet 6/A	5×10^5	60	100
VIII-R-C108H[b]	1/com	115	atm[c]	1×10^7	120/40[d]	atm	Sheet 2	Sheet 6/A	1×10^7	60	100
VIII-R-C109H[b]	1/com	100	atm[c]	1×10^3	120/40[d]	atm	-	-	5×10^4	60	-
VIII-R-C110H[b]	1/com	100[f]	atm[c]	1×10^3	120/40[d]	atm	Sheet 2	Sheet 6/A	1×10^3	60	100
VIII-R-C111H[b]	1/com	116	atm[c]	5×10^5	120/40	atm	117	-	5×10^6	60	-
VIII-R-C112H[b]	1/com	115	atm[c]	5×10^5	120/40[d]	atm	-	-	1×10^7	60	-
VIII-R-C113H[h]	1/com	100	atm[c]	1×10^3	109/40	atm	-	-	1×10^7	60	-
VIII-R-C114H[b]	1/com	104	atm[c]	5×10^5	120/65	atm	120	-	1×10^6	60	-
VIII-R-C115H[b]	1/com	100	atm[c]	5×10^5	120/65[d]	atm	-	-	1×10^7	60	-
VIII-R-C116	1/com	100	atm[c]	1×10^3	120/40[d]	atm	-	-	5×10^3	60	-
VIII-R-C117	1/com	100[i]	atm[c]	1×10^3	120/40	atm	-	-	5×10^3	60	-
VIII-R-C118H[b]	1/com	100	atm[c]	5×10^5	120/65[d]	atm	-	-	1×10^7	60	-
VIII-R-C119H[b]	1/com	115	atm[c]	5×10^5	120/65[d]	atm	Sheet 17	-	1×10^6	60	-
VIII-R-C120	1/com	100	atm[c]	1×10^3	106/40	atm	-	-	1×10^4	60	-
VIII-R-C121	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	5×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 23 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL C (Continued)											
VIII-R-C122H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	5×10^6	60	-
VIII-R-C123H	2	100	atm[c]	1×10^7	104/40	atm	Sheet 16	-	5×10^7	60	-
VIII-R-C124H[b]	2	100	atm[c]	5×10^5	120/40[e]	atm	120/Sheet 2	Sheet 6/A	1×10^7	60	100
VIII-R-C125H	1/com	115	atm[c]	1×10^7	120/40	atm	Sheet 2	Sheet 6/A	1×10^7	60	100
VIII-R-C126H[b]	2	100	atm[c]	5×10^5	120/40[e]	atm	120/Sheet 2	Sheet 6/A	5×10^7	60	100
VIII-R-C127H[b]	2	100	atm[c]	1×10^7	104/40	atm	Sheet 16	-	5×10^7	60	-
VIII-R-C128H[b]	2	100	atm[c]	5×10^5	100/40	atm	120/Sheet 2	Sheet 6/A	5×10^7	60	100
VIII-R-C129H	2	100	atm[c]	1×10^3	120/40	atm	Sheet 2	Sheet 6/A	5×10^7	60	100
VIII-R-C130H	1/com	100[f]	atm[c]	1×10^3	120/40	atm	Sheet 2	Sheet 6/A	5×10^7	60	100
VIII-R-C131H[b]	1/com	100	atm[c]	5×10^5	100/40	atm	120/Sheet 2	Sheet 6/A	5×10^7	60	100
VIII-R-C132	void										
VIII-R-C133H[b]	1/com	100	atm[c]	5×10^5	120/40[e]	atm	120/Sheet 2	Sheet 6/A	5×10^7	60	100
VIII-R-C134H[b]	1/com	100	atm[c]	5×10^5	120/40[e]	atm	120/Sheet 2	Sheet 6/A	1×10^7	60	100
VIII-R-C135H	2	115	atm[c]	1×10^7	120/40	atm	Sheet 2	Sheet 6/A	1×10^7	60	100
VIII-R-C136	void										
VIII-R-C137	void										
VIII-R-C138H	2	100	atm[c]	1×10^9	120/40	atm		-	1×10^9	60	-
VIII-R-C139H	2	100	atm[c]	1×10^9	120/40	atm		-	1×10^9	60	-
VIII-R-C140H	2	100	atm[c]	1×10^9	120/40	atm		-	1×10^9	60	-

TABLE 3.11.B.1-1 (SHEET 24 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL C (Continued)											
VIII-R-C141H	1/com	100[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-C142H	1/com	100[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-C143H	1/com	100[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-C144H	1/com	100[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-C145H	1/com	100[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-C146H	1/com	100[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-C147H	2	100[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-C148	spare										
VIII-R-C149H	2	100	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-UCC01H	1	130	atm[c]	5×10^5	130/40	atm	Sheet 4	Sheet 6/A	1×10^8	60	100
VIII-UCC01H	2	130	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCC02H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCC03H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 4	Sheet 6/A	1×10^8	60	100
VIII-UCC04H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 4	Sheet 6/A	1×10^8	60	100
VIII-UCC05H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 4	Sheet 6/A	1×10^8	60	100
VIII-UCC06H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 4	Sheet 6/A	1×10^8	60	100
VIII-UCC07H	1/com	130[f] 130	atm[c]	5×10^5	130/40	atm	Sheet 4	Sheet 6/A	1×10^8	60	100
VIII-UCC08H	2	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCC09H	1/com	130	atm[c]	5×10^5	130/40	atm	Sheet 4	Sheet 6/A	1×10^8	60	100
VIII-UCC10H	2	130[i]	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCC11H	1/com		atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/D	1×10^8	60	100

TABLE 3.11.B.1-1 (SHEET 25 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL C (Continued)											
VIII-UCC12H	1/com	140[f]	atm[c]	5×10^5	140/40	atm	Sheet 5	Sheet 6/D	1×10^8	60	100
VIII-UCC13H	1/com	140[f]	atm[c]	1×10^9	140/40	atm	-	-	1×10^9	60	-
VIII-UCC14H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 12/B	Sheet 6/A	1×10^8	60	100
VIII-UCC15H	2	130	atm[c]	5×10^5	130/40	atm	Sheet 12/B	Sheet 6/A	1×10^8	60	100
VIII-UCC16H	2	140 140	atm[c]	5×10^5	140/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCC17H	2		atm[c]	1×10^9	140/40	atm	-	-	1×10^9	60	-
AUXILIARY BUILDING - LEVEL B											
VIII-R-B01	1/com	100[i]	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-B02H	1/com	100	atm[c]	1×10^4	120/40	atm	-	-	5×10^5	60	-
VIII-R-B03H[b]	1/com	100	atm[c]	1×10^7	104/40	atm	Sheet 16	-	5×10^7	60	-
VIII-R-B04H[b]	1/com	100	atm[c]	5×10^4	120/40[d]	atm	110	-	1×10^5	60	-
VIII-R-B05H[b]	1/com	100	atm[c]	5×10^5	120/40[d]	atm	-	-	1×10^7	60	-
VIII-R-B06H[b]	1/com	100	atm[c]	5×10^5	120/40[d]	atm	-	-	1×10^7	60	-
VIII-R-B07H[b]	1/com	100	atm[c]	5×10^5	120/40[d]	atm	Sheet 2 Sheet 15	Sheet 6/C	5×10^5	60	100
VIII-R-B08H[b]	1/com	100	atm[c]	5×10^5	120/40[d] 104/40	atm	Sheet 2 Sheet 15	Sheet 6/C	5×10^5	60	100
VIII-R-B09H[b]	1/com	100	atm[c]	5×10^4	128/40	atm	115	-	1×10^7	60	-
VIII-R-B10H[b]	1/com	128	atm[c]	5×10^4	120/40[d]	atm	154	-	1×10^8	60	-
VIII-R-B11H[b]	1/com	100	atm[c]	5×10^5		atm	115	-	1×10^7	60	-

TABLE 3.11.B.1-1 (SHEET 26 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL B (Continued)											
VIII-R-B12H[b]	1/com	100	atm[c]	5×10^4	104/40	atm	115	-	1×10^7	60	-
VIII-R-B13H[b]	1/com	100	atm[c]	5×10^4	120/40[d]	atm	115	-	5×10^6	60	-
VIII-R-B14H[b]	1/com	100	atm[c]	5×10^5	104/40	atm	-	-	1×10^8	60	-
VIII-R-B15H[b]	1/com	100	atm[c]	5×10^5	120/40[d]	atm	-	-	4×10^6	60	-
VIII-R-B16H[b]	1/com	100	atm[c]	1×10^3	120/40[d]	atm	-	-	1×10^5	60	-
VIII-R-B17H[b]	1/com	100	atm[c]	1×10^3	114/40	atm	-	-	5×10^6	60	-
VIII-R-B18H[b]	1/com	100	atm[c]	5×10^5	100/40	atm	-	-	1×10^8	60	-
VIII-R-B19H[b]	1/com	100	atm[c]	5×10^5	120/40[d]	atm	-	-	4×10^6	60	-
VIII-R-B20H[b]	1/com	112	atm[c]	1×10^7	120/40[d]	atm	-	-	5×10^7	60	-
VIII-R-B21H	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	5×10^6	60	-
VIII-R-B22H	1/com	100	atm[c]	1×10^3	114/40	atm	-	-	5×10^6	60	-
VIII-R-B23H[h]	1/com	100	atm[c]	5×10^5	117/40	atm	-	-	1×10^7	60	-
VIII-R-B24H	1/com	100	atm[c]	1×10^3	115/40	atm	-	-	5×10^5	60	-
VIII-R-B25H[b]	1/com	100	atm[c]	5×10^5	120/40[d]	atm	-	-	1×10^8	60	-
VIII-R-B26H[b]	1/com	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/D	5×10^6	60	100
VIII-R-B27H	1/com	100[f]	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/D	1×10^3	60	100
VIII-R-B28	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-B29H	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^6	60	-
VIII-R-B30H	1/com	100[f]	atm[c]	5×10^4	120/40	atm	Sheet 5	Sheet 6/D	5×10^6	60	100

TABLE 3.11.B.1-1 (SHEET 27 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL B (Continued)											
VIII-R-B311H	1/com	100	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/D	1×10^3	60	100
VIII-R-B32H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B33H	1/com	100[f]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B34H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B35H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B36H	1/com	100[f]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B37H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B38H	1/com	100[f]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B39H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B40H	1/com	100[f]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B41H	1/com	100[f]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B42H	1/com	100[f]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B43H	1/com	100[f]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B44H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B45H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B46H	1/com	100[f]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B47H	1/com	100[f]	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B48H	1/com	100	atm[c]	5×10^4	120/40	atm	-	-	5×10^4	60	-
VIII-R-B49H	1/com	100	atm[c]	5×10^4	120/40	atm	-	-	5×10^4	60	-
VIII-R-B50H[b]	1/com	120	atm[c]	5×10^5	130/40	atm	Sheet 12/A Sheet 13	Sheet 6/C	5×10^7	60	100

TABLE 3.11.B.1-1 (SHEET 28 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL B (Continued)											
VIII-R-B51H[b]	1/com	100	atm[c]	1×10^7	120/40[e]	atm	Sheet 16	-	1×10^8	60	-
VIII-R-B52	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	5×10^3	60	-
VIII-R-B53H[b]	1/com	100	atm[c]	1×10^7	120/40[e]	atm	Sheet 16	-	1×10^8	60	-
VIII-R-B54	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-B55	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	5×10^3	60	-
VIII-R-B56	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-B57	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-B58H	1/com	100	atm[c]	5×10^5	109/40	atm	113	-	1×10^8	60	100
VIII-R-B59H	1/com	100	atm[c]	5×10^5	110/40	atm	113	-	1×10^8	60	100
VIII-R-B60H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B61H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B62	1/com	104	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-B63H	1/com	100	atm[c]	5×10^4	120/40	atm	-	-	5×10^4	60	-
VIII-R-B64H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B65H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B66H	1/com	100	atm[c]	5×10^4	120/40	atm	-	-	1×10^8	60	-
VIII-R-B67H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B68H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B69H	1/com	100[f]	atm[c]	1×10^3	120/40	atm	Sheet 3	Sheet 6/C	1×10^3	60	100

TABLE 3.11.B.1-1 (SHEET 29 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL B (Continued)											
VIII-R-B70	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-B71	1/com	100[f]	atm[c]	1×10^3	120/40	atm	-	-	1×10^4	60	-
VIII-R-B72H	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	5×10^5	60	-
VIII-R-B73H[b]	2	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/A	5×10^6	60	100
VIII-R-B74H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B75H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B76H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B77	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^4	60	-
VIII-R-B78H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B79H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B80H	2	100	atm[c]	5×10^4	120/40	atm	-	-	5×10^4	60	-
VIII-R-B81H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B82H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B83H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B84H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B85H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B86H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B87H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B88H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B89H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-

TABLE 3.11.B.1-1 (SHEET 30 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL B (Continued)											
VIII-R-B90H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B91H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B92	2	100	atm[c]	1×10^3	103/40	atm	-	-	1×10^3	60	-
VIII-R-B93H	2	100	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/A	1×10^3	60	100
VIII-R-B94H	2	100	atm[c]	5×10^4	120/40	atm	Sheet 5	Sheet 6/D	5×10^6	60	100
VIII-R-B95H	2	100	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/A	1×10^3	60	100
VIII-R-B96H	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^6	60	-
VIII-R-B97H	2	100	atm[c]	5×10^4	120/40	atm	-	-	1×10^8	60	-
VIII-R-B98H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B99H	2	100	atm[c]	5×10^5	120/40	atm		-	1×10^8	60	-
VIII-R-B100H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B101H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-B102H	2	100	atm[c]	5×10^4	120/40	atm	-	-	5×10^4	60	-
VIII-R-B103H	2	100	atm[c]	5×10^4	120/40	atm	-	-	5×10^4	60	-
VIII-R-B104H	2	100	atm[c]	5×10^4	120/40	atm	-	-	1×10^5	60	-
VIII-R-B105	2	100	atm[c]	1×10^3	120/40	atm	-	-	5×10^3	60	-
VIII-R-B106H[b]	2	100	atm[c]	1×10^7	120/40[e]	atm	Sheet 16	-	1×10^8	60	-
VIII-R-B107	2	100	atm[c]	1×10^3	120/40	atm	-	-	5×10^3	60	-
VIII-R-B108H[b]	2	100	atm[c]	1×10^7	120/40[e]	atm	Sheet 16	-	1×10^8	60	-
VIII-R-B109	2	104	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 31 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp $(°F)_{(i)}$	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL B (Continued)											
VIII-R-B110H	2	120	atm[c]	5×10^5	130/40[e]	atm	Sheet 12/A	Sheet 6/C	5×10^7	60	100
VIII-R-B111H	2	100	atm[c]	1×10^3	120/40	atm	Sheet 13	-	5×10^6	60	-
VIII-R-B112H	2	100	atm[c]	1×10^3	115/40	atm	-	-	5×10^5	60	-
VIII-R-B113H[h]	2	100	atm[c]	5×10^5	117/40	atm	-	-	1×10^7	60	-
VIII-R-B114H	2	100	atm[c]	1×10^3	114/40	atm	-	-	5×10^6	60	-
VIII-R-B115H[b]	2	100	atm[c]	5×10^5	120/40[d]	atm	-	-	1×10^8	60	-
VIII-R-B116H[b]	2	112	atm[c]	1×10^7	120/40[d]	atm	-	-	5×10^7	60	-
VIII-R-B117H[b]	2	100	atm[c]	5×10^5	120/40[d]	atm	-	-	4×10^6	60	-
VIII-R-B118H[b]	2	100	atm[c]	5×10^5	104/40	atm	-	-	1×10^8	60	-
VIII-R-B119H[b]	2	100	atm[c]	5×10^5	120/40[d]	atm	-	-	4×10^6	60	-
VIII-R-B120H[b]	2	100	atm[c]	5×10^5	104/40	atm	-	-	1×10^8	60	-
VIII-R-B121H[b]	2	100	atm[c]	1×10^3	114/40	atm	-	-	5×10^6	60	-
VIII-R-B122H[b]	2	100	atm[c]	1×10^3	120/40[d]	atm	-	-	1×10^5	60	-
VIII-R-B123H[b]	2	100	atm[c]	5×10^4	120/40[d]	atm	115	-	5×10^6	60	-
VIII-R-B124H[b]	2	100	atm[c]	1×10^7	104/40	atm	Sheet 16	-	5×10^7	60	-
VIII-R-B125H[b]	2	100	atm[c]	5×10^4	120/40[d]	atm	110	-	1×10^5	60	-
VIII-R-B126H[b]	2	100	atm[c]	5×10^5	120/40[d]	atm	-	-	1×10^7	60	-
VIII-R-B127H[b]	2	100	atm[c]	5×10^5	120/40[d]	atm	115	-	1×10^7	60	-
VIII-R-B128H[b]	2	128	atm[c]	5×10^4	128/40	atm	154	-	1×10^8	60	-
VIII-R-B129H[b]	2	100	atm[c]	5×10^4	104/40	atm	115	-	1×10^7	60	-

TABLE 3.11.B.1-1 (SHEET 32 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL B (Continued)											
VIII-R-B130H[b]	2	100	atm[c]	5×10^5	120/40[d]	atm	Sheet 2 Sheet 15	Sheet 6/C	5×10^5	60	100
VIII-R-B131H[b]	2	100	atm[c]	5×10^5	120/40[d]	atm	Sheet 2 Sheet 15	Sheet 6/C	5×10^5	60	100
VIII-R-B132H[b]	2	100	atm[c]	5×10^4	104/40	atm	115	-	1×10^7	60	-
VIII-R-B133H	2	100	atm[c]	5×10^5	120/40[d]	atm	-	-	1×10^7	60	-
VIII-R-B134	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-B135H	2	100	atm[c]	1×10^4	120/40	atm	-	-	5×10^5	60	-
VIII-R-B136H	2	100	atm[c]	1×10^3	120/40	atm	-	-	5×10^5	60	-
VIII-R-B137H[b]	1/com	100	atm[c]	5×10^5	120/65	atm	Sheet 3 Sheet 14	Sheet 6/C	1×10^8	60	100
VIII-R-B138H[b]	2	100	atm[c]	5×10^5	120/65	atm	Sheet 3 Sheet 14	Sheet 6/C	1×10^8	60	100
VIII-R-B139H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	-	-	1×10^7	60	-
VIII-R-B140H	1/com	120[f]	atm[c]	5×10^5	120/40	atm	-	-	1×10^7	60	-
VIII-R-B141H	1/com	120[f]	atm[c]	5×10^5	120/40	atm	-	-	1×10^7	60	-
VIII-R-B142H	1/com	120[f]	atm[c]	5×10^5	120/40	atm	-	-	1×10^7	60	-
VIII-R-B143H	1/com	120[f]	atm[c]	5×10^5	120/40	atm	-	-	1×10^7	60	-
VIII-R-B144H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	-	-	1×10^7	60	-
VIII-R-B145H	1/com	130[f]	atm[c]	1×10^9	130/40	atm	-	-	1×10^9	60	-
VIII-R-B146H	1/com	120[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-B147H	1/com	120[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-B148H	1/com	120[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-

TABLE 3.11.B.1-1 (SHEET 33 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL B											
VIII-R-B149H	1/com	120[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-B150H	1/com	130[f]	atm[c]	1×10^9	130/40	atm	-	-	1×10^9	60	-
VIII-R-B151H[b]	1/com	115	atm[c]	1×10^9	115/40	atm	-	-	1×10^9	60	-
VIII-R-B152H[b]	1/com	120	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-B153H[b]	1/com	120[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-B154H	1/com	120[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-B155H[b]	1/com	137	atm[c]	1×10^9	137/40	atm	-	-	1×10^9	60	-
VIII-R-B156H	1/com	140[f]	atm[c]	1×10^9	140/40	atm	-	-	1×10^9	60	-
VIII-R-B157H	1/com	155[f]	atm[c]	1×10^9	155/40	atm	-	-	1×10^9	60	-
VIII-R-B158H[b]	1/com	115	atm[c]	5×10^5	120/40[d]	atm	Sheet 16	-	1×10^7	60	-
VIII-R-B159H[b]	1/com	113	atm[c]	5×10^5	120/40[d]	atm	Sheet 16	-	1×10^7	60	-
VIII-R-B160H	1/com	120[f]	atm[c]	5×10^5	120/40	atm	-	-	1×10^7	60	-
VIII-R-B161H	1/com	120[f]	atm[c]	5×10^5	120/40	atm	-	-	1×10^7	60	-
VIII-R-B162H[b]	1/com	135	atm[c]	5×10^5	135/40	atm	-	-	1×10^7	60	-
VIII-R-B163H	1/com	140[f]	atm[c]	5×10^5	140/40	atm	-	-	1×10^7	60	-
VIII-R-B164H	1/com	155[f]	atm[c]	5×10^5	155/40	atm	-	-	1×10^7	60	-
VIII-R-B165H	2	130	atm[c]	5×10^5	130/40	atm	-	-	1×10^7	60	-
VIII-R-B166	spare										
VIII-R-B167	spare										

TABLE 3.11.B.1-1 (SHEET 34 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL B											
VIII-R-B168H	2	120	atm[c]	5×10^5	120/40	atm	-	-	1×10^7	60	-
VIII-R-B169H	2	120	atm[c]	5×10^5	120/40	atm	-	-	1×10^7	60	-
VIII-R-B170H	2	130	atm[c]	5×10^5	130/40	atm	-	-	1×10^7	60	-
VIII-R-B171H	2	130	atm[c]	1×10^9	130/40	atm	-	-	1×10^9	60	-
VIII-R-B172	spare										
VIII-R-B173	spare										
VIII-R-B174H	2	120	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-B175H	2	120	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-B176H	2	130	atm[c]	1×10^9	130/40	atm	-	-	1×10^9	60	-
VIII-R-B177H	2	140	atm[c]	1×10^9	140/40	atm	-	-	1×10^9	60	-
VIII-R-B178H[b]	2	137	atm[c]	1×10^9	137/40	atm	-	-	1×10^9	60	-
VIII-R-B179H	2	120	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-B180H	2	120	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-B181H[b]	2	120	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-B182H[b]	2	115	atm[c]	1×10^9	115/40	atm	-	-	1×10^9	60	-
VIII-R-B183H	2	140	atm[c]	5×10^5	140/40	atm	-	-	1×10^7	60	-
VIII-R-B184H[b]	2	135	atm[c]	5×10^5	135/40	atm	-	-	1×10^7	60	-
VIII-R-B185H	2	120	atm[c]	5×10^5	120/40	atm	-	-	1×10^7	60	-
VIII-R-B186H	2	120	atm[c]	5×10^5	120/40	atm	-	-	1×10^7	60	-
VIII-R-B187H[b]	2	113	atm[c]	5×10^5	120/40[d]	atm	Sheet 16	-	1×10^7	60	-

TABLE 3.11.B.1-1 (SHEET 35 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL B (Continued)											
VIII-R-B188H[b]	2	115	atm[c]	5×10^5	120/40[d]	atm	Sheet 16	-	1×10^7	60	-
VIII-UCB01H	1/2	130	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCB02H	2	130	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCB03H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCB04H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	-	-	1×10^8	60	-
VIII-UCB05H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCB06H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCB07H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	-	-	1×10^8	60	-
VIII-UCB08H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCB09H	2	130[f]	atm[c]	5×10^5	130/40	atm	-	-	1×10^8	60	-
VIII-UCB10H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCB11H[b]	1/com	130[f]	atm[c]	5×10^5	130/40	atm	-	-	1×10^8	60	-
VIII-UCB12H	2	130	atm[c]	5×10^5	130/40	atm	-	-	1×10^8	60	-
VIII-UCB13H[b]	1/com	130	atm[c]	5×10^5	130/40	atm	Sheet 2	Sheet 6/C	1×10^8	60	100
VIII-UCB14H[b]	2	130	atm[c]	5×10^5	130/40	atm	Sheet 2	Sheet 6/C	1×10^8	60	100
AUXILIARY BUILDING - LEVEL A											
VIII-R-A01	1/com	100[f]	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-A02	1/com	100[f]	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-A03[b]	1/com	100	atm[c]	1×10^3	120/40[d]	atm	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 36 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL A (Continued)											
VIII-R-A04	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-A05[b]	1/com	100	atm[c]	1×10^3	120/40[d]	atm	-	-	1×10^3	60	-
VIII-R-A07H[b]	1/com	116	atm[c]	1×10^7	120/40[d]	atm	Sheet 12/A	24 psia[g]	5×10^7	60	100
VIII-R-A08H[b]	1/com	119	atm[c]	5×10^5	120/40[d]	atm	Sheet 12/A	20 psia	1×10^8	60	100
VIII-R-A09H[b]	1/com	103	atm[c]	5×10^5	120/40[d]	atm	120	Sheet 6C[g]	1×10^8	60	100
VIII-R-A10H[b]	1/com	120	atm[c]	1×10^4	120/40[d]	atm	Sheet 12/A 126	Sheet 6/C	1×10^4	60	100
VIII-R-A13H[b]	1/com	100	atm[c]	5×10^5	120/40[d]	atm	Sheet 12/A Sheet 3 Sheet 15	Sheet 6/C	5×10^7	60	100
VIII-R-A14H	1/com	100	atm[c]	5×10^4	112/40	atm	-	-	5×10^5	60	-
VIII-R-A15[b]	1/com	128	atm[c]	1×10^3	128/40	atm	154	-	1×10^4	60	-
VIII-R-A16H[b]	1/com	100[f]	atm[c]	5×10^5	120/40	atm	-	-	1×10^7	60	-
VIII-R-A17	1/com	100	atm[c]	1×10^3	112/40	atm	-	-	1×10^4	60	-
VIII-R-A18H[b]	2	100	atm[c]	5×10^5	120/40[d]	atm	Sheet 3 Sheet 15	Sheet 6/C	5×10^7	60	100
VIII-R-A19H[b]	1/com	115	atm[c]	5×10^4	115/40	atm	140	-	1×10^7	60	-
VIII-R-A21H	1/com	100[f]	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/D	1×10^3	60	-
VIII-R-A22	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-A23H	1/com	100	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/D	1×10^3	60	-
VIII-R-A24H	1/com	100[f]	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/D	5×10^3	60	100
VIII-R-A25H	1/com	100[f]	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/D	5×10^6	60	100
VIII-R-A26H[b]	1/com	115	atm[c]	5×10^5	120/40[d]	atm	128	-	1×10^8	60	-

TABLE 3.11.B.1-1 (SHEET 37 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL A (Continued)											
VIII-R-A27H	1/com	100	atm[c]	5×10^5	109/40	atm	-	-	1×10^8	60	-
VIII-R-A28H	1/com	100	atm[c]	5×10^5	109/40	atm	113	-	1×10^8	60	100
VIII-R-A29H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A30	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	5×10^3	60	-
VIII-R-A31H	1/com	100	atm[c]	5×10^5	109/40	atm	-	-	5×10^5	60	-
VIII-R-A32H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A33H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A34H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A35H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A36H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A37H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A38H	1/com	100	atm[c]	5×10^5	112/40	atm	-	-	5×10^5	60	-
VIII-R-A39H	2	100	atm[c]	5×10^4	120/40	atm	-	-	5×10^5	60	-
VIII-R-A40	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	5×10^3	60	-
VIII-R-A40A	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	5×10^3	60	-
VIII-R-A40B	2	100	atm[c]	1×10^3	120/40	atm	-	-	5×10^3	60	-
VIII-R-A41H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A42H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A43H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A44H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A45H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-

TABLE 3.11.B.1-1 (SHEET 38 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL A (Continued)											
VIII-R-A46H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A47H[h]	1/com	100	atm[c]	5×10^5	120/40	atm	113	-	1×10^8	60	100
VIII-R-A48H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A49H[b]	1/com	128	atm[c]	5×10^5	120/40[d]	atm	Sheet 12/A Sheet 13	Sheet 6/C	1×10^8	60	100
VIII-R-A50	1/com	100	atm[c]	1×10^3	107/40	atm	-	-	5×10^3	60	-
VIII-R-A51	1/com	104	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-A52	1/com	100	atm[c]	1×10^3	112/40	atm	-	-	1×10^3	60	-
VIII-R-A53H[b]	1/com	100	atm[c]	1×10^4	120/40[d]	atm	-	-	5×10^5	60	-
VIII-R-A54	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-A55H	1/com	100[f]	atm[c]	1×10^3	120/40	atm	Sheet 3	Sheet 6/C	1×10^5	60	100
VIII-R-A56	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-A57	1/com	100[f]	atm[c]	6×10^3	120/40	atm	-	-	1×10^4	60	-
VIII-R-A58H	1/com	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A59[b]	2	128	atm[c]	1×10^3	120/40	atm	154	-	1×10^4	60	-
VIII-R-A60	void										
VIII-R-A61	2	100	atm[c]	1×10^3	112/40	atm	-	-	1×10^4	60	-
VIII-R-A62H	2	100	atm[c]	5×10^5	120/40	atm	-	-	5×10^5	60	-
VIII-R-A63H[b]	2	115	atm[c]	5×10^5	115/40	atm	140	-	1×10^7	60	-
VIII-R-A64H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-

TABLE 3.11.B.1-1 (SHEET 39 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL A (Continued)											
VIII-R-A65H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A66H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A67H	2	100	atm[c]	5×10^5	109/40	atm	-	-	5×10^5	60	-
VIII-R-A68	2	100	atm[c]	6×10^3	120/40	atm	-	-	1×10^4	60	-
VIII-R-A69	2	100	atm[c]	1×10^3	120/40	atm	-	-	5×10^3	60	-
VIII-R-A70H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A71H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A72H	2	100	atm[c]	5×10^5	109/40	atm	113	-	1×10^8	60	100
VIII-R-A73H	2	100	atm[c]	5×10^5	109/40	atm	-	-	1×10^8	60	-
VIII-R-A74H	2	100	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/A	1×10^3	60	100
VIII-R-A75	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-A76H	2	100	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/A	1×10^3	60	100
VIII-R-A77H	2	100	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/A	5×10^3	60	100
VIII-R-A78H[b]	2	115	atm[c]	5×10^5	120/40[d]	atm	128	-	1×10^8	60	-
VIII-R-A79H	2	100	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/A	5×10^6	60	100
VIII-R-A80H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A81H[b]	2	128	atm[c]	5×10^5	140/40[e]	atm	Sheet 12/A Sheet 13	Sheet 6/C	1×10^8	60	100
VIII-R-A82H[h]	2	100	atm[c]	5×10^5	107/40	atm	-	-	1×10^8	60	-
VIII-R-A83H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-

TABLE 3.11.B.1-1 (SHEET 40 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL A											
VIII-R-A84H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A85H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A86H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A87H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A88H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^8	60	-
VIII-R-A89	2	100	atm[c]	1×10^3	120/40	atm	-	-	5×10^3	60	-
VIII-R-A90	2	100	atm[c]	1×10^3	112/40	atm	-	-	1×10^3	60	-
VIII-R-A91H[b]	2	100	atm[c]	1×10^4	120/40[d]	atm	-	-	5×10^5	60	-
VIII-R-A92	2	104	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-A93	2	100	atm[c]	1×10^3	107/40	atm	-	-	5×10^3	60	-
VIII-R-A94	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-A95	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-A96[b]	2	100	atm[c]	1×10^3	120/40[d]	atm	-	-	1×10^3	60	-
VIII-R-A97	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-A98[b]	2	100	atm[c]	1×10^3	120/40[d]	atm	-	-	1×10^3	60	-
VIII-R-A100H[b]	2	116	atm[c]	1×10^7	120/40[d]	atm	Sheet 12/A	24 psia[f]	5×10^7	60	100
VIII-R-A101H[b]	2	119	atm[c]	5×10^5	120/40[d]	atm	Sheet 12/A	20 psia[f]	1×10^8	60	100
VIII-R-A102H[b]	2	100	atm[c]	1×10^4	120/40	atm	126	Sheet 6/C	1×10^4	60	100
VIII-R-A103H[b]	2	103	atm[c]	5×10^5	120/40[d]	atm	Sheet 12/A 120 Sheet 12/A	Sheet 6/C	1×10^8	60	100

TABLE 3.11.B.1-1 (SHEET 41 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL A (Continued)											
VIII-R-A106H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^7	60	-
VIII-R-A108	void										
VIII-R-A109H	1/com	100[f]	atm[c]	5×10^5	120/40	atm	-	-	1×10^7	60	-
VIII-R-A110H	2	100	atm[c]	5×10^5	120/40	atm	-	-	1×10^7	60	-
VIII-R-A111H	1/com	115[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A112H	1/com	115[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A113H	1/com	115[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A114H	1/com	115[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A115H	1/com	115[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^3	60	-
VIII-R-A116H	1/com	115[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A117H	1/com	115[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A118H	1/com	115[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A119H	1/com	115[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A120H	1/com	115[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A121H	1/com	115[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A122H	1/com	115[f]	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A123H	2	115	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A124H	2	115	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A125H	2	115	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A126H	2	115	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-

TABLE 3.11.B.1-1 (SHEET 42 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL A (Continued)											
VIII-R-A127H	2	115	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A128H	2	115	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A129H	2	115	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A130H	2	115	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A131H	2	115	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A132H	2	115	atm[c]	1×10^9	120/40	atm	-	-	1×10^9	60	-
VIII-R-A133H	2	130	atm[c]	1×10^9	130/40	atm	-	-	1×10^9	60	-
VIII-R-A134H	2	130	atm[c]	1×10^9	130/40	atm	-	-	1×10^9	60	-
VIII-UCA01H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCA02H	2	130	atm[c]	5×10^5	130/40	atm	-	-	1×10^8	60	-
VIII-UCA03H	2	130	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCA04H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCA05H	1/2	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCA06H	1/2	130	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCA07H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCA08H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCA09H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCA10H	1/com	130[f]	atm[c]	5×10^5	130/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-UCA11H	1/com	120[f]	atm[c]	5×10^5	120/40	atm	Sheet 12/A	Sheet 6/A	1×10^8	60	100
VIII-UCA12H	1/com	120[f]	atm[c]	1×10^7	120/40	atm	Sheet 12/A	20 psia[f]	5×10^7	60	100

TABLE 3.11.B.1-1 (SHEET 43 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL A (Continued)											
VIII-UCA13H	2	120	atm[c]	5×10^5	120/40	atm	Sheet 12/A	Sheet 6/B	1×10^8	60	100
VIII-UCA14H	2	120	atm[c]	1×10^7	120/40	atm	Sheet 12/A	20 psia[f]	5×10^7	60	100
AUXILIARY BUILDING - LEVEL 1											
VIII-R-101	1/com	100[f]	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-102	1/com	100[f]	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-103	1/com	100[f]	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-104	1/com	100	atm[c]	1×10^3	108/40	atm	-	-	1×10^3	60	-
VIII-R-105	1/com	100	atm[c]	1×10^3	107/40	atm	-	-	1×10^3	60	-
VIII-R-106H[b]	1/com	115	atm[c]	1×10^7	120/40[d]	atm	-	-	5×10^7	60	-
VIII-R-107H[b]	1/com	131	atm[c]	5×10^5	131/40	atm	-	-	1×10^8	60	-
VIII-R-109	void										
VIII-R-110	1/com	100	atm[c]	1×10^4	104/40	atm	-	-	1×10^4	60	-
VIII-R-111[b]	1/com	128	atm[c]	1×10^3	128/40	atm	154	-	1×10^3	60	-
VIII-R-112	1/com	100[f]	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-113	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-114	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-115	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-116[b]	1/com	85	atm[c]	1×10^3	120/40[d]	atm	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 44 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL 1 (Continued)											
VIII-R-117	1/com	100[f]	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-118[b]	1/com	85	atm[c]	1×10^3	120/40[d]	atm	-	-	1×10^3	60	-
VIII-R-120	1/com	100	atm[c]	1×10^3	112/40	atm	-	-	1×10^3	60	-
VIII-R-121	1/com	104	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-122	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-123	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-124	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-125	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-126	1/com	100[f]	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-127	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-128	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-129	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-130	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-131	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-132	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-133	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-134H[b]	1/com	117	atm[c]	5×10^5	120/40	atm	119	-	1×10^8	60	-
VIII-R-135H[b]	1/com	106	atm[c]	1×10^7	120/40[d]	atm	-	-	5×10^7	60	-
VIII-R-136H[b]	1/com	100	atm[c]	1×10^3	120/40	atm	Sheet 12/B Sheet 13	Sheet 6/C	1×10^3	60	100

TABLE 3.11.B.1-1 (SHEET 45 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL 1 (Continued)											
VIII-R-137	2	100	atm[c]	1×10^3	114/40	atm	-	-	1×10^3	60	-
VIII-R-138	2	104	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-139	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-140	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-141	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-142	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-143H	2	100	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/A	1×10^3	60	100
VIII-R-144	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-145H	2	100	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/A	1×10^3	60	100
VIII-R-146	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-147[b]	2	85	atm[c]	1×10^3	120/40[d]	atm	-	-	1×10^3	60	-
VIII-R-148	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-149[b]	2	85	atm[c]	1×10^3	120/40[d]	atm	-	-	1×10^3	60	-
VIII-R-150	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-151	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-152	2	100	atm[c]	1×10^3	108/40	atm	-	-	1×10^3	60	-
VIII-R-153	2	100	atm[c]	1×10^3	107/40	atm	-	-	1×10^3	60	-
VIII-R-154H[b]	2	115	atm[c]	1×10^7	120/40[d]	atm	-	-	5×10^7	60	-
VIII-R-155H[b]	2	131	atm[c]	5×10^5	131/40	atm	-	-	1×10^8	60	-

TABLE 3.11.B.1-1 (SHEET 46 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL 1 (Continued)											
VIII-R-156[b]	2	128	atm[c]	1×10^3	128/40	atm	154	-	1×10^3	60	-
VIII-R-157	2	100	atm[c]	1×10^4	104/40	atm	-	-	1×10^4	60	-
VIII-R-158	void										
VIII-R-160H[b]	2	106	atm[c]	1×10^7	120/40[d]	atm	-	-	5×10^7	60	-
VIII-R-161H[b]	2	117	atm[c]	1×10^5	117/40	atm	119	-	1×10^8	60	-
VIII-R-162H[b]	2	100	atm[c]	1×10^3	120/40	atm	Sheet 12/B Sheet 13	Sheet 6/C	1×10^3	60	100
VIII-R-UC101H[b]	1/com	104[f]	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VIII-R-UC102H[b]	2	104	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
AUXILIARY BUILDING - LEVEL 2											
VIII-R-201	1/com	100[f]	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-202	1/com	100	atm[c]	1×10^3	107/40	atm	-	-	1×10^3	60	-
VIII-R-203	1/com	100	atm[c]	1×10^3	107/40	atm	-	-	1×10^3	60	-
VIII-R-204	1/com	100	atm[c]	1×10^3	116/40	atm	-	-	1×10^3	60	-
VIII-R-205[b]	1/com	128	atm[c]	1×10^3	128/40	atm	154	-	1×10^3	60	-
VIII-R-206	1/com	100[f]	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-207[b]	1/com	100	atm[c]	1×10^3	120/40[d]	atm	-	-	1×10^3	60	-
VIII-R-208H[b]	1/com	100	atm[c]	1×10^3	120/40	atm	Sheet 12/B Sheet 13	Sheet 6/C	1×10^3	60	100
VIII-R-209H[b]	1/com	100	atm[c]	1×10^3	105/40[g]	atm	-	-	1×10^6	60	-
VIII-R-210H[b]	1/com	100	atm[c]	1×10^3	104/40	atm	-	-	1×10^6	60	-
VIII-R-211	1/com	104	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 47 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[i]	Pressure[i]	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL 2 (Continued)											
VIII-R-212H[h]	1/com	100	atm[c]	1×10^3	127/40	atm	-	-	5×10^5	60	-
VIII-R-213	void										
VIII-R-214	spare										
VIII-R-215	1/com	100[f]	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-216	1/com	100[f]	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-217	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-218	2	104	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-219H[b]	2	100	atm[c]	1×10^3	104/40	atm	-	-	1×10^6	60	-
VIII-R-220H[b]	2	104	atm[c]	1×10^3	105/40	atm	-	-	1×10^6	60	-
VIII-R-221H[h]	2	100	atm[c]	1×10^3	127/40	atm	-	-	5×10^5	60	-
VIII-R-222	2	100	atm[c]	1×10^3	120/40	atm	Sheet 12/B Sheet 13	Sheet 6/C	1×10^3	60	100
VIII-R-223[b]	2	100	atm[c]	1×10^3	120/40[d]	atm	-	-	1×10^3	60	-
VIII-R-224H	2	100	atm[c]	1×10^3	120/40	atm	Sheet 5	Sheet 6/A	1×10^3	60	100
VIII-R-225	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-226	2	100	atm[c]	1×10^3	107/40	atm	-	-	1×10^3	60	-
VIII-R-227	2	100	atm[c]	1×10^3	107/40	atm	-	-	1×10^3	60	-
VIII-R-228[b]	2	100	atm[c]	1×10^3	128/40	atm	154	-	1×10^3	60	-
VIII-R-229	2	100	atm[c]	1×10^3	116/40	atm	-	-	1×10^3	60	-
VIII-R-230	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 48 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY BUILDING - LEVEL 3											
VIII-R-301	1/com	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-302	1/com	100[f]	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-303	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-304	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-305	1/com	100[f]	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-306	2	100	atm[c]	1×10^3	120/40	atm	-	-	1×10^3	60	-
VIII-R-307	1	104	atm[c]	1×10^3	-	-	-	-	1×10^3	-	-
VIII-R-308	2	104	atm[c]	1×10^3	-	-	-	-	1×10^3	-	-

TABLE 3.11.B.1-1 (SHEET 49 OF 92)
ENVIRONMENTAL CONDITIONS

NOTES

a. H = Harsh environment due to existence of either of the following conditions:

- Temperature increases due to the pipe break.
- TID > 1 x 10^4 rad.

\- = No high-energy line or safety-related equipment located in the room.

b. Indicates rooms that are served by Class 1E environmental support systems.

c. Within negative pressure boundary.

d. Maximum abnormal room temperature value rounded up to 120°F per reference 6.2.2.3. Consult reference 6.2.2.3 for actual temperature.

e. Maximum room temperature is due to one train RHR cooldown.

f. Normal room temperature is an assumed value which is based on bulk building or surrounding area temperature. This room does not contain safety-related equipment/instrumentation.

g. Short spike <1 s. For remainder of transient, see figure 6/C.

h. Room is included in the area temperature monitoring program (reference FSAR section 16.3).

i. Sheet numbers refer to figure 3.11.B.1-1. Alpha designators refer to specific curves on these sheets.

TABLE 3.11.B.1-1 (SHEET 50 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[n]	Pressure[n]	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL C											
IX-R-C01	1/com	104	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-C02	1/com	104	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-C03	1/com	104	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-C04[b]	2	104	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-C05[b]	2	104	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-C006[b]	2	104	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-C07H	1/com	104	atm	5×10^5	120/55	-	Sheet 5	Sheet 6/A	1×10^8	60	100
IX-R-C08[b]	2	104	atm	1×10^3	120/55	-	-	-	1×10^3	60	100
IX-R-C09	1/com	104	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
CONTROL BUILDING - LEVEL B											
IX-R-B01[b]	2	85	atm	1×10^3	100/65	-	-	-	5×10^3	60	-
IX-R-B02H[m]	2	100	atm	1×10^3	104/50	-	-	-	5×10^4	60	-
IX-R-B03	2	100	atm	1×10^3	103/55	-	-	-	1×10^4	60	-
IX-R-B04[b]	2	85	atm	1×10^3	100/65[c]	-	-	-	1×10^3	60	-
IX-R-B05[b]	2	100	atm	1×10^3	101/65	-	-	-	5×10^3	60	-
IX-R-B06[b]	2	100	atm	1×10^3	106/55	-	-	-	1×10^3	60	-
IX-R-B07	2	100	atm	1×10^3	124/55	-	-	-	1×10^3	60	-
IX-R-B08	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B09	2	100	atm	1×10^3	105/55	-	-	-	1×10^4	60	-

TABLE 3.11.B.1-1 (SHEET 51 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL B (Continued)											
IX-R-B10H[m]	2	100	atm	1×10^3	107/50	atm	-	-	5×10^4	60	-
IX-R-B11H	2	100	atm	1×10^3	103/50	atm	-	-	5×10^4	60	-
IX-R-B12	2	100	atm	1×10^3	120/55	-	-	-	1×10^4	60	-
IX-R-B13	2	100	atm	1×10^3	120/55	-	-	-	5×10^3	60	-
IX-R-B14	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B15	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B16[b]	2	90	atm	1×10^3	100/65	-	-	-	1×10^3	60	-
IX-R-B17[b]	2	90	atm	1×10^3	100/65	-	-	-	1×10^3	60	-
IX-R-B18[b]	2	85	atm	1×10^3	100/65[d]	-	-	-	5×10^4	60	-
IX-R-B19H[m]	2	100	atm	1×10^3	106/50	atm					
IX-R-B20	spare		atm	1×10^3			-	-	1×10^3	60	-
IX-R-B21	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B22	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B23	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B24	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B25[b]	2	75	atm	1×10^3	80/70	-	-	-	1×10^3	50	-
IX-R-B26[b]	2	100	atm	1×10^3	100/65	-	-	-	1×10^3	50	-
IX-R-B27[b]	2	75	atm	1×10^3	80/70	-	-	-	1×10^3	50	-
IX-R-B28[b]	2	80	atm	1×10^3	80/70	atm	-	-	1×10^3	60	-
IX-R-B29[b]	2	100	atm	1×10^3	100/65[e]	-	-	-	1×10^3	50	-

TABLE 3.11.B.1-1 (SHEET 52 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL B (Continued)											
IX-R-B30	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B31[b]	2	100	atm	1×10^3	100/55	-	-	-	1×10^3	50	-
IX-R-B32[b]	2	75	atm	1×10^3	80/70[f]	-	-	-	1×10^3	50	-
IX-R-B33	2	100	atm	1×10^3	123/65	-	-	-	1×10^3	60	-
IX-R-B34	2	100	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-B35	2	100	atm	1×10^3	120/65	-	-	-	1×10^3	60	-
IX-R-B36[b]	2	100	atm	1×10^3	100/55[g]	-	-	-	1×10^3	50	-
IX-R-B37[b]	2	75	atm	1×10^3	80/70	-	-	-	1×10^3	50	-
IX-R-B38	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B39	1/com	100	atm	1×10^3	104/65	-	-	-	1×10^3	60	-
IX-R-B40	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B41	1/com	104	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B42	1/com	100	atm	1×10^3	104/65	-	-	-	1×10^3	60	-
IX-R-B43	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B44[b]	1/com	75	atm	1×10^3	80/70	-	-	-	1×10^3	50	-
IX-R-B45	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B46	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B47[b]	1/com	85	atm	1×10^3	92/55[g]	-	-	-	1×10^3	50	-
IX-R-B48[b]	1/com	100	atm	1×10^3	100/55	-	-	-	1×10^3	50	-
IX-R-B49[b]	1/com	75	atm	1×10^3	80/70[f]	-	-	-	1×10^3	50	-
IX-R-B50	1/com	100	atm	1×10^3	124/65	-	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 53 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL B (Continued)											
IX-R-B51	1/com	104	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B52[b]	1/com	85	atm	1×10^3	100/65[e]	-	-	-	1×10^3	50	-
IX-R-B53[b]	1/com	80	atm	1×10^3	80/70	atm	-	-	1×10^3	60	-
IX-R-B54[b]	1/com	75	atm	1×10^3	80/70	-	-	-	1×10^3	50	-
IX-R-B55[b]	1/com	100	atm	1×10^3	100/65	-	-	-	1×10^3	50	-
IX-R-B56[b]	1/com	75	atm	1×10^3	76/70	-	-	-	1×10^3	50	-
IX-R-B57	spare						-	-			
IX-R-B58	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B59	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B60[b]	1/com	90	atm	1×10^3	100/65	-	-	-	1×10^3	60	-
IX-R-B61[b]	1/com	85	atm	1×10^3	100/65[d]				1×10^3	60	-
IX-R-B62[b]	1/com	90	atm	1×10^3	100/65	-	-	-	1×10^3	60	-
IX-R-B63	1/com	100	atm	1×10^3	120/55	-	-	-	5×10^3	60	-
IX-R-B64	spare						-	-			
IX-R-B65H[m]	1/com	100	atm	1×10^3	106/50	atm	-	-	5×10^4	60	-
IX-R-B66	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^4	60	-
IX-R-B67H[b]	1/com	100	atm	1×10^3	103/50	atm	-	-	5×10^4	60	-
IX-R-B68	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B69	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B70	1/com	100	atm	1×10^3	106/55	-	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 54 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[n]	Pressure[n]	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL B (Continued)											
IX-R-B71	1/com	100	atm	1×10^3	124/55	-	-	-	1×10^3	60	-
IX-R-B72	1/com	100	atm	1×10^3	120/65	-	-	-	1×10^3	60	-
IX-R-B73	1/com	100	atm	1×10^3	105/55	-	-	-	1×10^4	60	-
IX-R-B74H[m]	1/com	100	atm	1×10^3	107/50	atm	-	-	5×10^4	60	-
IX-R-B75	1/com	100	atm	1×10^3	101/65	-	-	-	5×10^3	60	-
IX-R-B76[b]	1/com	85	atm	1×10^3	86/65[c]	-	-	-	1×10^3	60	-
IX-R-B77	1/com	100	atm	1×10^3	103/55	atm	-	-	1×10^4	60	
IX-R-B78H[m]	1/com	100	atm	1×10^3	104/50	-	-	-	5×10^4	60	-
IX-R-B79[b]	1/com	85	atm	1×10^3	100/65	-	-	-	5×10^3	60	-
IX-R-B80	2	100	atm	1×10^3	120/55	-	-	-	5×10^3	60	-
IX-R-B81	1/com	100	atm	1×10^3	120/55	-	-	-	5×10^3	60	-
IX-R-B82	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B83	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B84[b]	1/com	85	atm	1×10^3	100/65	atm	-	-	1×10^3	60	-
IX-R-B85[b]	2	85	atm	1×10^3	100/65	atm	-	-	1×10^3	60	-
IX-R-B86	2	104	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B87	2	104	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B88	1/com	100	atm	1×10^3	106/50	-	-	-	1×10^3	60	-
IX-R-B89	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B90	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-B91	1	100	atm	1×10^3	104/50	-	-	-	5×10^3	60	-
IX-R-B92	2	100	atm	1×10^3	104/50	-	-	-	5×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 55 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL A											
IX-R-A01	2	100	atm	1×10^3	115/65	-	-	-	1×10^4	60	-
IX-R-A02H[m]	2	100	atm	1×10^3	107/50	atm	-	-	1×10^5	60	-
IX-R-A03	2	100	atm	1×10^3	101/65	-	-	-	5×10^3	60	-
IX-R-A04	2	100	atm	1×10^3	103/65	-	-	-	5×10^3	60	-
IX-R-A05	spare										
IX-R-A06	2	100	atm	1×10^3	101/65	-	-	-	1×10^3	60	-
IX-R-A07	2	100	atm	1×10^3	107/55	-	-	-	1×10^3	60	-
IX-R-A10H[m]	2	100	atm	1×10^3	106/50	atm	-	-	1×10^5	60	-
IX-R-A11	2	100	atm	1×10^3	114/65	-	-	-	1×10^4	60	-
IX-R-A12H	2	100	atm	1×10^3	106/50	atm	-	-	1×10^5	60	-
IX-R-A13	2	100	atm	1×10^3	120/55	-	-	-	1×10^4	60	-
IX-R-A14	2	100	atm	1×10^3	103/55	-	-	-	1×10^4	60	-
IX-R-A15[b]	2	100	atm	1×10^3	100/65[h]	-	-	-	5×10^3	60	-
IX-R-A16[b]	2	100	atm	1×10^3	100/65[i]	-	-	-	1×10^3	60	-
IX-R-A17	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A18	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A19	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A20	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A21	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A22[b]	2	100	atm	1×10^3	100/65[i]	-	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 56 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL A (Continued)											
IX-R-A23	2	100	atm	1×10^3	102/55	-	-	-	1×10^3	60	-
IX-R-A24[b]	2	75	atm	1×10^3	80/70	atm	-	-	1×10^3	60	-
IX-R-A25	spare										
IX-R-A26	spare										
IX-R-A27	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A28	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A29	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A30	2	75	atm	1×10^3	104/60	atm	-	-	1×10^3	50	-
IX-R-A31	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A32	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A33	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A34	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A35	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A36	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A37	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A38	1/com	75	atm	1×10^3	104/60	atm	-	-	1×10^3	50	-
IX-R-A39	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A40	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A41	spare										
IX-R-A42	spare										
IX-R-A43[b]	1/com	75	atm	1×10^3	80/70	atm	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 57 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL A (Continued)											
IX-R-A44	1/com	100	atm	1×10^3	102/55	-	-	-	1×10^3	60	-
IX-R-A45[b]	1/com	100	atm	1×10^3	100/65[i]	-	-	-	1×10^3	60	-
IX-R-A46	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A47	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A48[b]	1/com	100	atm	1×10^3	100/65[l]	-	-	-	1×10^3	60	-
IX-R-A49	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A50[b]	1/com	100	atm	1×10^3	100/65[h]	-	-	-	5×10^3	60	-
IX-R-A51	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^4	60	-
IX-R-A52	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^4	60	-
IX-R-A53H[b]	1/com	100	atm	1×10^3	106/60	atm	-	-	1×10^5	60	-
IX-R-A54	1/com	100	atm	1×10^3	114/65	-	-	-	1×10^4	60	-
IX-R-A55H[m]	1/com	100	atm	1×10^3	106/50	atm	-	-	1×10^5	60	-
IX-R-A58	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A59	1/com	100	atm	1×10^3	101/55	-	-	-	1×10^3	60	-
IX-R-A60	1/com	100	atm	1×10^3	101/65	-	-	-	1×10^3	60	-
IX-R-A61	1/com	100	atm	1×10^3	101/65	-	-	-	5×10^3	60	-
IX-R-A62	1/com	100	atm	1×10^3	103/55	-	-	-	5×10^3	60	-
IX-R-A63	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A64	1/com	100	atm	1×10^3	115/55	-	-	-	1×10^4	60	-
IX-R-65H[b]	1/com	100	atm	1×10^3	107/50	atm	-	-	1×10^5	60	-

TABLE 3.11.B.1-1 (SHEET 58 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[n]	Pressure[n]	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL A (Continued)											
IX-R-A66	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A67	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A68	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A69	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A70	1/com	104	atm	1×10^3	113/55	-	-	-	1×10^3	60	-
IX-R-A71	1/com	104	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A72	2	104	atm	1×10^3	113/55	-	-	-	1×10^3	60	-
IX-R-A73	2	104	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A74	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-A75[b]	1/com	75	atm	1×10^3	76/70	atm	-	-	1×10^3	60	-
IX-R-A76[b]	2	75	atm	1×10^3	80/70	atm	-	-	1×10^3	60	-
IX-R-A77[b]	1/com	85	atm	1×10^3	100/65	-	-	-	5×10^3	60	-
IX-R-A78	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^4	60	-
IX-R-A79[b]	2	85	atm	1×10^3	100/65	-	-	-	5×10^3	60	-
IX-R-A80	2	100	atm	1×10^3	120/55	-	-	-	1×10^4	60	-
IX-R-A81[b]	2	100	atm	1×10^3	100/55	-	-	-	1×10^3	60	-
IX-R-A82[b]	1/com	100	atm	1×10^3	100/55	-	-	-	1×10^3	60	-
CONTROL BUILDING - LEVEL 1											
IX-R-101	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-102	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 59 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[(a)]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL 1 (Continued)											
IX-R-103	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-104	1/com	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-105	1/com	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-106	1/com	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-107	1/com	75	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-108	1/com	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-109	1/com	75	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-110	1/com	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-111	1/com	75	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-112	1/com	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-113	1/com	75	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-114	1/com	75	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-115	1/com	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-116	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-117	2	80	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-118	2	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-119	1/com	75	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-120	1/com	80	atm	1×10^3	87/55	-	-	-	1×10^3	60	-
IX-R-121	1/com	80	atm	1×10^3	86/65	-	-	-	1×10^3	60	-
IX-R-122	1/com	80	atm	1×10^3	86/65	-	-	-	1×10^3	60	-
IX-R-123	1/com	80	atm	1×10^3	86/65	-	-	-	1×10^3	60	-
IX-R-124	1/com	80	atm	1×10^3	86/55	-	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 60 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[(a)]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL 1 (Continued)											
IX-R-125	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-126	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-127	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-128	1/com	80	atm	1×10^3	87/55	-	-	-	1×10^3	60	-
IX-R-129	1/com	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-130	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-131	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-132	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-133	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-134	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-135	1/com	75	atm	1×10^3	82/70	-	-	-	1×10^3	60	-
IX-R-136	1/com	75	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-137	1/com	75	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-138	1/com	80	atm	1×10^3	87/65	-	-	-	1×10^3	60	-
IX-R-139	1/com	75	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-140	1/com	75	atm	1×10^3	83/70	atm	-	-	1×10^3	60	-
IX-R-141	1/com	80	atm	1×10^3	87/60	-	-	-	1×10^3	60	-
IX-R-142	1/com	80	atm	1×10^3	104/55	atm	-	-	1×10^3	60	-
IX-R-143	1/com	75	atm	1×10^3	83/65	atm	-	-	1×10^3	60	-
IX-R-144	1/com	75	atm	1×10^3	83/65	atm	-	-	1×10^3	60	-
IX-R-145	1/com	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-146	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 61 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA[c]			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[n]	Pressure[n]	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL 1 (Continued)											
IX-R-147	1/com	80	atm	1×10^3	87/55	-	-	-	1×10^3	60	-
IX-R-148	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-149	1/com	80	atm	1×10^3	87/55	-	-	-	1×10^3	60	-
IX-R-150	1/com	75	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-151	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-152	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-153	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-154	1/com	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-155	1/com	75	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-156[b]	1/com	85	+1/8" WG	1×10^3	85/60	atm[k]	85	(k)	1×10^3	60	-
IX-R-157[b]	1/com	85	+1/8" WG	1×10^3	85/70	atm[k]	85	-	1×10^3	60	-
IX-R-158[b]	1/com	85	+1/8" WG	1×10^3	85/70	atm[k]	85	-	1×10^3	60	-
IX-R-159	1/com	80	atm	1×10^3	104/55	atm	-	-	1×10^3	60	-
IX-R-160[b]	1/com	85	+1/8" WG	1×10^3	93/70	atm[k]	93	(k)	1×10^3	60	-
IX-R-161[b]	1/com	85	+1/8" WG	1×10^3	85/70	atm[k]	85	(k)	1×10^3	60	-
IX-R-162[b]	1/com	85	+1/8" WG	1×10^3	85/70	atm[k]	85	(k)	1×10^3	60	-
IX-R-163[b]	1/com	85	+1/8" WG	1×10^3	85/70[j]	atm[k]	85	(k)	1×10^3	50	-
IX-R-164[b]	2	85	+1/8" WG	1×10^3	85/70[j]	atm[k]	85	(k)	1×10^3	50	-
IX-R-165	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-166	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-167	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 62 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[(a)]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL 1 (Continued)											
IX-R-168	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-169	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-170	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-171	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-172	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-173	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-174	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-175	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-176	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-177	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-178	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-179	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-180	1/com	100	+1/8" WG	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-181	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-182	1/com	100	+1/8" WG	1×10^3	120/55	atm	-	-	1×10^3	60	-
IX-R-183	1/com	75	+1/8" WG	1×10^3	104/60	atm	-	-	1×10^3	60	-
IX-R-184	1/com	75	+1/8" WG	1×10^3	104/55	atm	-	-	1×10^3	60	-
IX-R-185	1/com	75	+1/8" WG	1×10^3	104/60	atm	-	-	1×10^3	60	-
IX-R-186	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-187	1/com	75	+1/8" WG	1×10^3	104/55	atm	-	-	1×10^3	60	-
IX-R-188	1/com	75	+1/8" WG	1×10^3	104/55	atm	-	-	1×10^3	60	-
IX-R-189	1/com	75	+1/8" WG	1×10^3	104/55	atm	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 63 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[n]	Pressure[n]	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL 1 (Continued)											
IX-R-190	1/com	75	+1/8" WG	1×10^3	120/55	atm	-	-	1×10^3	60	-
IX-R-191	1/com	80	+1/8" WG	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-192	1/com	100	+1/8" WG	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-193	1/com	75	+1/8" WG	1×10^3	104/55	atm	-	-	1×10^3	60	-
IX-R-194	1/com	100	+1/8" WG	1×10^3	120/55	atm	-	-	1×10^3	60	-
IX-R-195	1/com	100	+1/8" WG	1×10^3	120/55	atm	-	-	1×10^3	60	-
IX-R-196	1/com	100	+1/8" WG	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-197	1/com	100	+1/8" WG	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-198	1/com	100	+1/8" WG	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-199	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-199A	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-199B	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
Sample chase H	1/com	130	atm	5×10^5	130/40	-	Sheet 5	Sheet 6/A	1×10^8	60	100
CONTROL BUILDING - LEVEL 2											
IX-R-201	2	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-202	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-203	2	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-204	2	80	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-205	2	80	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-206	2	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-207	2	80	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-208	2	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 64 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL 2 (Continued)											
IX-R-209	2	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-210	2	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-211	2	80	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-212	2	80	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-213	2	80	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-214	2	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-215	2	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-216	2	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-217	2	80	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-218	2	80	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-219	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-220	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-221	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-222	2	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-223[b]	2	100	atm	1×10^3	100/65	-	-	-	1×10^3	60	-
IX-R-224	2	100	atm	1×10^3	102/55	-	-	-	1×10^3	60	-
IX-R-225	1/com	100	atm	1×10^3	102/55	-	-	-	1×10^3	60	-
IX-R-226[b]	1/com	100	atm	1×10^3	100/65	-	-	-	1×10^3	60	-
IX-R-227	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-228	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-229	1/com	75	atm	1×10^3	104/60	-	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 65 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[(a)]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL 2 (Continued)											
IX-R-230	1/com	80	atm	1×10^3	90/65	-	-	-	1×10^3	60	-
IX-R-231	1/com	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-232	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-233	1/com	100	atm	1×10^3	101/65	-	-	-	1×10^3	60	-
IX-R-234	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-235	spare										
IX-R-236	spare										
IX-R-237	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-238	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-239	1/com	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-240	spare										
IX-R-241	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-242	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-243	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-244	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-245	1/com	80	atm	1×10^3	90/65	-	-	-	1×10^3	60	-
IX-R-246	1/com	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-247	1/com	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-248	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-249	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-250	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-251	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 66 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[n]	Pressure[n]	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL 2 (Continued)											
IX-R-252	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-253	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-254	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-255	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-256	spare										
IX-R-257	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-258	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-259	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-260	spare										
IX-R-261	spare										
IX-R-262	spare										
IX-R-263	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-264	2	80	atm	1×10^3	90/65	-	-	-	1×10^3	60	-
IX-R-265	spare										
IX-R-266	1/com	75	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-267	1/com	75	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-268	1/com	75	atm	1×10^3	104/60	-	-	-	1×10^3	60	-
IX-R-269	2	80	atm	1×10^3	104/55	-	-	-	1×10^3	60	-
IX-R-270	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-271	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-272	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-273	1/com	100	atm	1×10^3	120/55	-	-		1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 67 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[n]	Pressure[n]	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL 2 (Continued)											
IX-R-274	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-275	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-276	2	130	atm	1×10^3	130/40	-	-	-	1×10^3	60	-
IX-R-277	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-UC201H	1/com	130	atm	5×10^5	130/40	-	Sheet 5	Sheet 6/A	1×10^8	60	100
IX-UC202H	2	130	atm	5×10^5	130/40	-	Sheet 5	Sheet 6/A	1×10^8	60	100
CONTROL BUILDING - LEVEL 3											
IX-R-301	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-302	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-303	1/com	100	atm	1×10^3	102/55	-	-	-	1×10^3	60	-
IX-R-304	1/com	100	atm	1×10^3	107/65	-	-	-	1×10^8	60	-
IX-R-305[b]	1/com	104	atm	1×10^3	110/65	-	-	-	5×10^3	60	-
IX-R-306	spare										
IX-R-307	2	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-308[b]	1/com	104	atm	1×10^3	110/65	-	-	-	1×10^3	60	-
IX-R-309	1/com	100	atm	1×10^3	110/55	-	-	-	1×10^3	60	-
IX-R-310[b]	1/com	104	atm	1×10^3	110/65	-	-	-	1×10^3	60	-
IX-R-311[b]	1/com	104	atm	1×10^3	110/65	-	-	-	5×10^3	60	-
IX-R-312[b]	1/com	104	atm	1×10^3	110/65	-	-	-	5×10^3	60	-
IX-R-313[b]	1/com	104	atm	1×10^3	110/65	-	-	-	1×10^3	60	-
IX-R-314	void										
IX-R-315	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-316	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-317	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 68 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[n]	Pressure[n]	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL 3 (Continued)											
IX-R-318	spare										
IX-R-319	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-320[b]	1/com	104	atm	1×10^3	110/65	-	-	-	1×10^3	60	-
IX-R-321[b]	1/com	104	atm	1×10^3	110/65	-	-	-	1×10^3	60	-
IX-R-322[b]	1/com	100	atm	1×10^3	100/55	-	-	-	1×10^3	60	-
IX-R-323	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-324	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-325[b]	1/com	100	atm	1×10^3	118/55	-	-	-	1×10^3	60	-
IX-R-326	spare										
IX-R-327	spare										
IX-R-328	1/com	100	atm	1×10^3	120/40	-	-	-	1×10^3	100	-
IX-R-329	1/com	100	atm	1×10^3	120/40	-	-	-	1×10^3	60	-
IX-R-330	1/com	100	atm	1×10^3	120/40	-	-	-	1×10^3	60	-
CONTROL BUILDING - LEVEL 4											
IX-R-401	2	100	atm	1×10^3	120/40	11.7 psia	-	-	1×10^3	60	-
IX-R-402	2	100	atm	1×10^3	120/40	11.7 psia	-	-	1×10^3	60	-
IX-R-403	1/com	100	atm	1×10^3	120/55	11.7 psia	-	-	1×10^3	60	-
IX-R-404	1/com	100	atm	1×10^3	120/40	11.7 psia	-	-	1×10^3	60	-
IX-R-405	1/com	100	atm	1×10^3	120/40	11.7 psia	-	-	1×10^3	60	-
IX-R-406	1/com	80	atm	1×10^3	104/60	11.7 psia	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 69 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[(a)]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
CONTROL BUILDING - LEVEL 4 (Continued)											
IX-R-407	1/com	100	atm	1×10^3	120/55	11.7 psia	-	-	1×10^3	60	-
IX-R-408	1/com	100	atm	1×10^3	120/55	11.7 psia	-	-	1×10^3	60	-
IX-R-409	1/com	80	atm	1×10^3	104/60	11.7 psia	-	-	1×10^3	60	-
IX-R-410	1/com	100	atm	1×10^3	120/55	11.7 psia	-	-	1×10^3	60	-
CONTROL BUILDING - LEVEL 5											
IX-R-501	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-
IX-R-502	1/com	100	atm	1×10^3	120/55	-	-	-	1×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 70 OF 92)
ENVIRONMENTAL CONDITIONS

NOTES

a. H = Harsh environment due to existence of either of the following conditions:

- Temperature increases due to the pipe break.
- TID > 1 x 10^4 rad.

- = No high-energy line or safety-related equipment located in the room.

b. Indicates rooms that are served by Class 1E environmental support systems.

c. Maximum room temperature 30 min after loss of all cooling is 112.7°F.

d. Maximum room temperature 30 min after loss of all cooling is 121.0°F.

e. Maximum room temperature 30 min after loss of all cooling is 123.7°F.

f. Maximum room temperature 30 min after loss of all cooling is 82.3°F.

g. Maximum room temperature 30 min after loss of all cooling is 122.0°F.

h. Maximum room temperature 30 min after loss of all cooling is 106.7°F.

i. Maximum room temperature 30 min after loss of all cooling is 106.0°F.

j. Maximum room temperature 30 min after loss of all cooling is 92.8°F.

TABLE 3.11.B.1-1 (SHEET 71 OF 92)
ENVIRONMENTAL CONDITIONS

NOTES (Continued)

k. Indicates rooms which have an abnormal and DBA/post-DBA pressure of at least +1/8 in. WG with respect to adjacent areas when Class 1E environment support system is operated in emergency mode and atmospheric pressure when operated in isolation mode.

l. Maximum room temperature 30 min after loss of all cooling is 109.0°F.

m. Room is included in the area temperature monitoring program (FSAR 16.3).

n. Sheet numbers refer to figure 3.11.B.1-1. Alpha designators refer to specific curves on these sheets.

CONTROL BUILDING
(Sheet 22 of 22)

TABLE 3.11.B.1-1 (SHEET 72 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[g]	Pressure[g]	Int Dose (rads)	Normal (%)	DBA (%)
FUEL HANDLING BUILDING - LEVEL C											
VII-R-C01H[b]	2	104	atm[c]	5×10^5	110/40	atm	Sheet 16	-	1×10^7	60	-
VII-R-C02H[b]	2	104	atm[c]	5×10^5	117/40	atm	Sheet 16	-	5×10^7	60	-
VII-R-C03H[b]	2	104	atm[c]	5×10^5	120/40[d]	atm	Sheet 16	-	5×10^6	60	-
VII-R-C04H	2	104	atm[c]	1×10^4	120/40	atm	-	-	5×10^5	60	-
VII-R-C05H	1/com	104	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VII-R-C06H	1/com	104	atm[c]	1×10^4	120/40	atm	-	-	5×10^5	60	-
VII-R-C07H[b]	1/com	104	atm[c]	5×10^5	110/40	atm	Sheet 16	-	1×10^7	60	-
VII-R-C08H[b]	1/com	104	atm[c]	5×10^5	117/40	atm	Sheet 16	-	5×10^7	60	-
VII-R-C09H[b]	1/com	104	atm[c]	5×10^5	120/40[d]	atm	Sheet 16	-	5×10^6	60	-
FUEL HANDLING BUILDING - LEVEL B											
VII-R-B01H[b]	2	104	atm[c]	5×10^5	120/40[e]	atm	Sheet 14	-	1×10^6	60	-
VII-R-B02H	2	104	atm[c]	1×10^4	120/40	atm	-	-	5×10^5	60	-
VII-R-B03H[b]	2	104	atm[c]	5×10^5	119/40	atm	Sheet 14	-	5×10^7	60	-
VII-R-B04H	1/com	104	atm[c]	5×10^5	120/40	atm	-	-	1×10^6	60	-
VII-R-B05H	1/com	104	atm[c]	5×10^5	120/40	atm	-	-	5×10^7	60	-
VII-R-B06H[f]	2	104	atm[c]	1×10^4	107/40	atm	-	-	5×10^5	60	-
VII-R-B07H	1/com	104	atm[c]	5×10^5	120/40	atm	-	-	1×10^6	60	-
VII-R-B08H[f]	1/com	104	atm[c]	1×10^4	107/40	atm	-	-	5×10^5	60	-

TABLE 3.11.B.1-1 (SHEET 73 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[g]	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
FUEL HANDLING BUILDING - LEVEL B (Continued)											
VII-R-B09H	1/com	104	atm[c]	5×10^5	120/40	atm	-	-	5×10^7	60	-
VII-R-B10H	1/com	104	atm[c]	5×10^5	120/40	atm	-	-	1×10^6	60	-
VII-R-B11H[b]	1/com	100	atm[c]	5×10^5	120/40[e]	atm	Sheet 14	-	1×10^6	60	-
VII-R-B12H[b]	1/com	104	atm[c]	1×10^4	120/40	atm	-	-	5×10^5	60	-
VII-R-B13H[b]	1/com	104	atm[c]	5×10^5	119/40	atm	Sheet 14	-	5×10^7	60	-
FUEL HANDLING BUILDING - LEVEL A											
VII-R-A01H[b]	2	104	atm[c]	5×10^5	120/40[e]	atm	Sheet 16	-	4×10^6	60	-
VII-R-A02H	2	104	atm[c]	1×10^4	120/40	atm	-	-	1×10^6	60	-
VII-R-A03H	2	104	atm[c]	1×10^4	120/40	atm	-	-	1×10^6	60	-
VII-R-A04H[b]	2	104	atm[c]	1×10^4	108/40	atm	-	-	1×10^6	60	-
VII-R-A05H	2	104	atm[c]	1×10^4	105/40	atm	-	-	5×10^5	60	-
VII-R-A06H	1/com	104	atm[c]	1×10^4	105/40	atm	-	-	5×10^5	60	-
VII-R-A07H[b]	1/com	104	atm[c]	1×10^4	108/40	atm	-	-	1×10^6	60	-
VII-R-A08H	1/com	104	atm[c]	1×10^4	120/40	atm	-	-	1×10^6	60	-
VII-R-A09H	1/com	104	atm[c]	1×10^4	120/40	atm	-	-	1×10^6	60	-
VII-R-A10H[b]	1/com	104	atm[c]	5×10^5	120/40[e]	atm	Sheet 16	-	4×10^6	60	-
VII-R-A11H	1/com	104	atm[c]	1×10^4	120/40	atm	-	-	5×10^5	60	-
VII-R-A12H	1/com	104	atm[c]	1×10^4	120/40	atm	-	-	5×10^5	60	-

TABLE 3.11.B.1-1 (SHEET 74 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[g]	Pressure[g]	Int Dose (rads)	Normal (%)	DBA (%)
FUEL HANDLING BUILDING - LEVEL A (Continued)											
VII-R-A13H	2	104	atm[c]	1×10^4	120/40	atm	-	-	1×10^8	60	-
VII-R-A14H	1/com	104	atm[c]	1×10^4	120/40	atm	-	-	1×10^8	60	-
FUEL HANDLING BUILDING - LEVEL 1											
VII-R-101H	2	104	atm[c]	1×10^4	120/40	atm	-	-	5×10^5	60	-
VII-R-102H	1/com	104	atm[c]	1×10^4	120/40	atm	-	-	5×10^5	60	-
VII-R-103H	1/com	104	atm[c]	1×10^4	120/40	atm	-	-	5×10^5	60	-
VII-R-104H	1/com	104	atm[c]	1×10^4	120/40	atm	-	-	5×10^5	60	-
VII-R-105[b]	1/com	104	atm[c]	1×10^4	120/40	atm	-	-	1×10^4	60	-
VII-R-106H	1/com	104	atm[c]	1×10^4	120/40	atm	-	-	5×10^5	60	-
VII-R-107H	1/com	104	atm[c]	1×10^4	120/40	atm	-	-	5×10^5	60	-
VII-R-108[b]	1/com	104	atm[c]	1×10^4	120/40	atm	-	-	1×10^4	60	-
VII-R-109H	1/com	104	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
VII-R-110H	2	104	atm[c]	5×10^5	120/40	atm	Sheet 5	Sheet 6/A	1×10^8	60	100
FUEL HANDLING BUILDING - LEVEL 3											
VII-R-301[b]	1/com	104	atm[c]	1×10^4	114/40	atm	-	-	1×10^4	60	-
VII-R-302H	1/com	104	atm[c]	1×10^4	131/40	atm	-	-	5×10^5	60	-
VII-R-303[b]	1/com	104	atm[c]	2×10^3	111/40	atm	-	-	3×10^3	60	-
VII-R-304[b]	1/com	104	atm[c]	2×10^3	112/40	atm	-	-	3×10^3	60	-
VII-R-305H	1/com	104	atm[c]	2×10^3	107/40	atm	-	-	3×10^3	60	-

TABLE 3.11.B.1-1 (SHEET 75 OF 92)
ENVIRONMENTAL CONDITIONS

NOTES

a. H = Harsh environment due to existence of either of the following conditions:

- Temperature increases due to the pipe break.
- TID > 1 x 10^4 rad.

\- = No high-energy line or safety-related equipment located in the room.

b. Indicates rooms that are served by Class 1E environmental support systems.

c. Within negative pressure boundary.

d. Maximum abnormal room temperature value rounded up to 120°F per reference 6.2.2.3. Consult reference 6.2.2.3 for actual temperature.

e. Maximum temperature is due to one train RHR cooldown.

f. Room is included in the area temperature monitoring program (FSAR 16.3).

g. Sheet numbers refer to figure 3.11.B.1-1. Alpha designators refer to specific curves on these sheets.

FUEL HANDLING BUILDING
(Sheet 4 of 4)

TABLE 3.11.B.1-1 (SHEET 76 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
AUXILIARY FEEDWATER PUMPHOUSE											
VI-R-101[b]	1	100	atm	1×10^3	120/40	11.7 psia	-	-	1×10^3	90	-
VI-R-102[b]	1	100	atm	1×10^3	120/40	11.7 psia	-	-	1×10^3	90	-
VI-R-103	1	(c)	atm	1×10^3	120/40	11.7 psia	-	-	1×10^3	90	-
VI-R-104H	1	104	atm	1×10^3	120/40	11.7 psia	(e)	(e)	1×10^3	90	-
VI-R-105H	1	104	atm	1×10^3	120/40	11.7 psia	(e)	(e)	1×10^3	90	-
VI-R-106H[b]	1	104[d]	atm	1×10^3	122/40	11.7 psia	(e)	(e)	1×10^3	90	-
VI-R-101[b]	2	100	atm	1×10^3	120/40	11.7 psia	-	-	1×10^3	90	-
VI-R-102[b]	2	100	atm	1×10^3	120/40	11.7 psia	-	-	1×10^3	90	-
VI-R-103	2	(c)	atm	1×10^3	120/40	11.7 psia	-	-	1×10^3	90	-
VI-R-104H	2	104	atm	1×10^3	120/40	11.7 psia	(e)	(e)	1×10^3	90	-
VI-R-105H	2	104[c]	atm	1×10^3	120/40	11.7 psia	(e)	(e)	1×10^3	90	-
VI-R-106H[b]	2	104[d]	atm	1×10^3	120/40	11.7 psia	(e)	(e)	1×10^3	90	-
VI-R-201	1	(d)	atm	1×10^3	120/40	11.7 psia	-	-	1×10^3	90	-
VI-R-201	2	(d)	atm	1×10^3	120/40	11.7 psia	-	-	1×10^3	90	-

AUXILIARY FEEDWATER PUMPHOUSE
(Sheet 1 of 2)

TABLE 3.11.B.1-1 (SHEET 77 OF 92)
ENVIRONMENTAL CONDITIONS

NOTES

a. H = Harsh environment due to existence of either of the following conditions:

- Temperature increases due to the pipe break.
- TID > 1 x 10^4 rad.

\- = No high-energy line or safety-related equipment located in the room.

b. Indicates rooms that are served by Class 1E environmental support systems.

c. No ventilation provided.

d. Temperature in these rooms reaches 122°F when the turbine-driven AFW pump is in operation.

e. Postulated pipe breaks in the steam supply line to the turbine-driven pump would impact only safety-related equipment associated with the pump. These postulated breaks would disable the pump; therefore, this equipment would not be required to function and qualification of this equipment to the pipe break environment is not required.

AUXILIARY FEEDWATER PUMPHOUSE
(Sheet 2 of 2)

TABLE 3.11.B.1-1 (SHEET 78 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
DIESEL GENERATOR BUILDING - LEVEL 1											
IV-R-101[b]	1/com	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-102	1/com	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-103[b]	1/com	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-104	1/com	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
DIESEL GENERATOR BUILDING - LEVEL 2											
IV-R-201	1/com	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-202	1/com	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-203[b]	1/com	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-204[b]	1/com	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-205	1/com	120	atm	1×10^3	120/17	11.7 psia	-	-	1×10^3	90	-
IV-R-206	1/com	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-207	1/com	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-208[b]	1/com	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-209[b]	1/com	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-210	1/com	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
DIESEL GENERATOR BUILDING - LEVEL 1											
IV-R-101[b]	2	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-102	2	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-103[b]	2	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-104	2	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-

TABLE 3.11.B.1-1 (SHEET 79 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
DIESEL GENERATOR BUILDING - LEVEL 2											
IV-R-201	2	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-202	2	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-203[b]	2	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-204[b]	2	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-205	2	120	atm	1×10^3	120/17	11.7 psia	-	-	1×10^3	90	-
IV-R-206	2	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-207	2	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-208[b]	2	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-209[b]	2	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-210	2	120	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-

TABLE 3.11.B.1-1 (SHEET 80 OF 92)
ENVIRONMENTAL CONDITIONS

NOTES

a. H = Harsh environment due to existence of either of the following conditions:

- Temperature increases due to the pipe break.
- TID > 1 x 10^4 rad.

\- = No high-energy line or safety-related equipment located in the room.

b. Indicates rooms that are served by Class 1E environmental support systems.

DIESEL GENERATOR BUILDING
(Sheet 3 of 3)

TABLE 3.11.B.1-1 (SHEET 81 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[(a)]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
DIESEL FUEL OIL STORAGE TANK VALVE BOX - LEVEL A											
IV-R-A01	1/com	amb	atm	1×10^3	94/30	11.7 psia	-	-	1×10^3	90	-
IV-R-A02	1/com	amb	atm	1×10^3	94/30	11.7 psia	-	-	1×10^3	90	-
IV-R-A03	1/com	amb	atm	1×10^3	94/30	11.7 psia	-	-	1×10^3	90	-
IV-R-A01	2	amb	atm	1×10^3	94/30	11.7 psia	-	-	1×10^3	90	-
IV-R-A02	2	amb	atm	1×10^3	94/30	11.7 psia	-	-	1×10^3	90	-
IV-R-A03	2	amb	atm	1×10^3	94/30	11.7 psia	-	-	1×10^3	90	-
DIESEL FUEL OIL STORAGE TANK VALVE BOX - LEVEL 1											
IV-R-101	1/com	amb	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-
IV-R-101	2	amb	atm	1×10^3	120/50	11.7 psia	-	-	1×10^3	90	-

DIESEL FUEL OIL STORAGE TANK VALVE BOX
(Sheet 1 of 2)

TABLE 3.11.B.1-1 (SHEET 82 OF 92)
ENVIRONMENTAL CONDITIONS

NOTES

a. H = Harsh environment due to existence of either of the following conditions:

- Temperature increases due to the pipe break.
- TID > 1 x 10^4 rad.

\- = No high-energy line or safety-related equipment located in the room.

DIESEL FUEL OIL STORAGE TANK VALVE BOX
(Sheet 2 of 2)

TABLE 3.11.B.1-1 (SHEET 83 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
NSCW PUMPHOUSE - TRAIN A											
XII-R-101	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-102	1/com	amb[c]	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-103	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-104	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-105	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-106	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-107	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-108	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-109	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-110	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-111	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
NSCW PUMPHOUSE - TRAIN B											
XII-R-201	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-202	1/com	amb[c]	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-203	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-204	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-205	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-206	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-207	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-208	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-209	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-

TABLE 3.11.B.1-1 (SHEET 84 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
NSCW PUMPHOUSE - TRAIN B (Cont.)											
XII-R-210	1/com	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-211	1/com	amb amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-301	2	amb[b][c]	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-302	2	amb amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-303	2	amb amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-304	2	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-305	2	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-306	2	amb[b]	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-307	2	amb[c]	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-308	2		atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-309	2		atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-310	2		atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-311	2		atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
NSCW PUMPHOUSE - TRAIN A											
XII-R-401	2	amb[b]	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-402	2	amb[c]	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-403	2	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-404	2	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-405	2	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-406	2	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-407	2	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-

TABLE 3.11.B.1-1 (SHEET 85 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
NSCW PUMPHOUSE - TRAIN A (Cont.)											
XII-R-408	2	amb amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-409	2	amb	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-410	2	amb[b][c]	atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-
XII-R-411	2		atm	1×10^3	104/5	11.7 psia	-	-	1×10^3	90	-

NSCW PUMPHOUSE
(Sheet 3 of 4)

TABLE 3.11.B.1-1 (SHEET 86 OF 92)
ENVIRONMENTAL CONDITIONS

NOTES

a. H = Harsh environment due to existence of either of the following conditions:

- Temperature increases due to the pipe break.
- TID > 1 x 10^4 rad.

\- = No high-energy line or safety-related equipment located in the room.

b. No ventilation provided.

c. Minimum room temperature is 40°F.

NSCW PUMPHOUSE
(Sheet 4 of 4)

TABLE 3.11.B.1-1 (SHEET 87 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[f]	Pressure[f]	Int Dose (rads)	Normal (%)	DBA (%)
OUTSIDE AREAS - MAIN STEAM TUNNELS											
III-1T1H	1/com	amb	atm	1×10^3	-	11.7 psia	370[d]	56 psia[e]	1×10^3	-	100
III-2T1H	2	amb	atm	1×10^3	-	11.7 psia	370[d]	56 psia[e]	1×10^3	-	100
OUTSIDE AREAS - NSCW PIPING TUNNELS											
III-1T2A	1/com	amb[c]	atm	1×10^3	-	11.7 psia	-	-	1×10^3	-	-
III-1T2B	1/com	amb[c]	tm	1×10^3	-	11.7 psia	-	-	1×10^3	-	-
III-2T2A	2	amb[c]	atm	1×10^3	-	11.7 psia	-	-	1×10^3	-	-
III-2T2B	2	amb[c]	atm	1×10^3	-	11.7 psia	-	-	1×10^3	-	-
OUTSIDE AREAS - DIESEL GENERATOR COOLING TUNNEL											
III-1T3A	1/com	amb[c]	atm	1×10^3	-	11.7 psia	-	-	1×10^3	-	-
III-1T3B	1/com	amb	atm	1×10^3	-	11.7 psia	-	-	1×10^3	-	-
III-2T3A	2	amb[c]	atm	1×10^3	-	11.7 psia	-	-	1×10^3	-	-
III-2T3B	2	amb[c]	atm	1×10^3	-	11.7 psia	-	-	1×10^3	-	-
OUTSIDE AREAS - DIESEL GENERATOR ELECTRICAL TUNNELS											
III-1T4A[b]	1/com	amb	atm	1×10^3	104/40	11.7 psia	-	-	1×10^3	-	-
III-1T4B[b]	1/com	amb	atm	1×10^3	104/40	11.7 psia	-	-	1×10^3	-	-
III-2T4A[b]	2	amb[c]	atm	1×10^3	104/40	11.7 psia	-	-	1×10^3	-	-
III-2T4B[b]	2	amb[c]	atm	1×10^3	104/40	11.7 psia	-	-	1×10^3	-	-
III-1T8A	1/com	amb	atm	1×10^3	-	11.7 psia	-	-	1×10^3	-	-
III-1T8BH	1/com	amb	atm	1×10^3	-	11.7 psia	Sheet 2	Sheet 6/A	1×10^3	-	100

OUTSIDE AREAS
(Sheet 1 of 4)

TABLE 3.11.B.1-1 (SHEET 88 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
OUTSIDE AREAS - NSCW TOWER CABLE TUNNELS											
III-1T5A[b]	1/com	amb	atm	1×10^3	104/40	11.7 psia	-	-	1×10^3	-	-
III-1T5B[b]	1/com	amb	atm	1×10^3	104/40	11.7 psia	-	-	1×10^3	-	-
III-2T5A[b]	2	amb[c]	atm	1×10^3	-	11.7 psia	-	-	1×10^3	-	-
III-2T5B[b]	2	amb[c]	atm	1×10^3	-	11.7 psia	-	-	1×10^3	-	-
OUTSIDE AREAS - AUXILIARY FEEDWATER TUNNELS											
III-1T6AH	1/com	amb	atm	1×10^3	-	11.7 psia	304[d]	31.0 psia[e]	1×10^3	-	-
III-1T6BH	1/com	amb	atm	1×10^3	-	11.7 psia	326[d]	37.0 psia[e]	1×10^3	-	-
III-2T6AH	2	amb	atm	1×10^3	-	11.7 psia	304[d]	31.0 psia[e]	1×10^3	-	-
III-2T6BH	2	amb	atm	1×10^3	-	11.7 psia	326[d]	37.0 psia[e]	1×10^3	-	-
OUTSIDE AREAS - RWST AND RMWST TUNNELS											
III-RST	1/com	amb	atm	1×10^3	104/40	11.7 psia	-	-	1×10^3	-	-
III-RST	2	amb	atm	1×10^3	104/40	11.7 psia	-	-	1×10^3	-	-
OUTSIDE AREAS - ELECTRIC STEAM BOILER TUNNEL											
III-ESBT	1/com	amb	atm	1×10^3	-	11.7 psia	-	-	1×10^3	-	-
OUTSIDE AREAS - REACTOR MAKEUP WATER STORAGE TANK											
III-R-101	1/com	amb[c]	atm	1×10^3	atm	11.7 psia	-	-	1×10^3	-	-
III-R-101	2	amb[c]	atm	1×10^3	atm	11.7 psia	-	-	1×10^3	-	-

TABLE 3.11.B.1-1 (SHEET 89 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
OUTSIDE AREAS - REFUELING WATER STORAGE TANK											
III-R-101	1/com	amb[c]	atm	1×10^3	atm	11.7 psia	-	-	1×10^3	-	-
III-R-101	2	amb[c]	atm	1×10^3	atm	11.7 psia	-	-	1×10^3	-	-

TABLE 3.11.B.1-1 (SHEET 90 OF 92)
ENVIRONMENTAL CONDITIONS

NOTES

a. H = Harsh environment due to existence of either of the following conditions:

- Temperature increases due to the pipe break.
- TID > 1 x 10^4 rad.

\- = No high-energy line or safety-related equipment located in the room.

b. Indicates rooms that are served by Class 1E environmental support systems.

c. Minimum room temperature is 40°F.

d. Maximum room temperature.

e. Maximum room pressure.

f. Sheet numbers refer to figure 3.11.B.1-1. Alpha designators refer to specific curves on these sheets.

OUTSIDE AREAS
(Sheet 4 of 4)

TABLE 3.11.B.1-1 (SHEET 91 OF 92)
ENVIRONMENTAL CONDITIONS

Environmental Designator[a]	Unit	Normal			Abnormal/Test		DBA/Post-DBA			Relative Humidity (max)	
		Temp (°F)	Pressure	Int Dose (rads)	Temp (°F) (max/min)	Pressure	Temp (°F)[b]	Pressure	Int Dose (rads)	Normal (%)	DBA (%)
TURBINE BUILDING											
X-R-Level 1H	1/2	104	atm	1×10^3	104	11.7 psia	Sheet 18	17.7 psia	1×10^3	60	100
X-R-Level 2H	1/2	104	atm	1×10^3	104	11.7 psia	Sheet 18	17.7 psia	1×10^3	60	100

TURBINE BUILDING
(Sheet 1 of 2)

TABLE 3.11.B.1-1 (SHEET 92 OF 92)
ENVIRONMENTAL CONDITIONS

NOTES

a. H = Harsh environment due to existence of either of the following conditions:

- Temperature increases due to the pipe break.
- TID > 1×10^4 rad.

\- = No high-energy line or safety-related equipment located in the room.

b. Sheet numbers refer to figure 3.11.B.1-1. Alpha designators refer to specific curves on these sheets.

TURBINE BUILDING
(Sheet 2 of 2)

TABLE 3.11.B.3-1 (SHEET 1 OF 14)

CONFORMANCE TO THE REQUIREMENTS OF NUREG-0588

Requirements per NUREG-0588	VEGP Conformance
1. Establishment of the Qualification Parameters for Design Basis Events (DBEs)	
1.1 Temperature and Pressure Conditions Inside Containment - Loss-of-Coolant Accident (LOCA)	
(1) The time-dependent temperature and pressure established for the design of the containment structure and found acceptable by the staff may be used for environmental qualification of equipment.	(1) The time-dependent temperature and pressure profile used for qualifying VEGP equipment located inside containment was developed specifically for equipment qualification purposes. The methodology incorporated in the development of this profile is more conservative than those used in the structural design calculations.
(2) Acceptable methods for calculating and establishing the containment pressure and temperature envelopes to which equipment should be qualified are summarized below. Acceptable methods for calculating mass and energy release rates are summarized in Appendix A. Pressurized Water Reactors (PWRs) Dry containment - Calculate LOCA containment environment using CONTEMPT-LT or equivalent industry codes. Additional guidance is provided in Standard Review Plan (SRP), Section 6.2.1.1.A, NUREG-75/087.	(2) The containment pressure and temperature values used to qualify the equipment were derived from the Bechtel standard computer program COPATTA. This program is used for VEGP and has been reviewed and found to be acceptable by the NRC for other Bechtel projects.
(3) Not applicable.	
(4) The test profiles included in Appendix A to Institute of Electrical and Electronics Engineers (IEEE) Std. 323-1974 should not be considered an acceptable alternative in lieu of using plant-specific containment temperature and pressure design profiles unless plant-specific analysis is provided to verify the adequacy of those profiles.	(4) IEEE Std. 323-1974 profiles, when used, are verified to envelop VEGP specific requirements.

TABLE 3.11.B.3-1 (SHEET 2 OF 14)

Requirements per NUREG-0588	VEGP Conformance
1.2 Temperature and Pressure Conditions Inside Containment - Main Steam Line Break (MSLB)	
(1) The environmental parameters used for equipment qualification should be calculated with a plant-specific model reviewed and approved by the staff.	(1) Conform. Refer to 1.1(2) response
(2) Models that are acceptable for calculating containment parameters are listed in section 1.1(2).	(2) Conform. Refer to 1.1(2) response.
(3) Not applicable.	
(4) The test profiles included in Appendix A to IEEE Std. 323-1974 should not be considered an acceptable alternative in lieu of using plant-specific containment temperature and pressure design profiles unless plant-specific analysis is provided to verify the adequacy of those profiles.	(4) VEGP is comparing the plant requirement with the test results to verify the adequacy.
(5) Where qualification has been completed but only LOCA conditions were considered, it must be demonstrated that the LOCA qualification conditions exceed or are equivalent to the maximum calculated MSLB conditions. The following technique is acceptable:	(5) VEGP containment profile includes both LOCA/MSLB requirements for the qualification effort.
(a) Calculate the peak temperature envelope from an MSLB using a model based on the staff's approved assumptions defined in section 1.1(2).	Conforms when employed.
(b) Show that the peak surface temperature of the component to be qualified does not exceed the LOCA qualification temperature by the method discussed in item 2 of Appendix B.	Conforms when employed.
(c) If the calculated surface temperature exceeds the qualification temperature, the staff requires requalification testing be performed with appropriate margins, or qualified physical protection be provided to assure that the surface temperature will not exceed the actual qualification temperature.	

TABLE 3.11.B.3-1 (SHEET 3 OF 14)

Requirements per NUREG-0588	VEGP Conformance
1.3 Effects of Chemical Spray	
The effects of caustic spray should be addressed for the equipment qualification. The concentration of caustics used for qualification should be equivalent to or more severe than those used in the plant containment spray system. If the chemical composition of the caustic spray can be affected by the equipment malfunctions, the most severe caustic spray environment that results from a single failure in the spray system should be assumed. See SRP Section 6.5.2 (NUREG-75/087), paragraph II, item (e) for caustic spray solution guidelines.	Chemical spray is addressed by requiring the most severe concentration that will be present. Vendor qualification is required to meet or exceed the required chemical spray concentration.
1.4 Radiation Conditions Inside and Outside Containment	
The radiation environment for qualification of equipment should be based on the normally expected radiation environment over the equipment qualified life, plus that associated with the most severe design basis accident (DBA) during or following that which equipment must remain functional. It should be assumed that the DBA-related environmental conditions occur at the end of the equipment qualified life.	The normal total integrated dose and DBA dose are considered in the qualification of safety-related equipment. Operability after irradiation to the calculated dose plus margin is verified.
The sample calculations in Appendix D and the following positions provide an acceptable approach for establishing radiation limits for qualifications. Additional radiation margins identified in Section 6.3.1.5 of IEEE Std. 323-1974 for qualification type testing are not required if these methods are used.	
(1) The source term to be used in determining the radiation environment associated with the design basis LOCA should be taken as an instantaneous release from the fuel to the atmosphere of 100 percent of the noble gases, 50 percent of the iodines, and 1 percent of the remaining fission products.	(1) Conform.

TABLE 3.11.B.3-1 (SHEET 4 OF 14)

Requirements per NUREG-0588	VEGP Conformance
For all other non-LOCA DBA conditions, a source term involving an instantaneous release from the fuel to the atmosphere of 10 percent of the noble gases (except Kr-85 for which a release of 30 percent should be assumed) and 10 percent of the iodine is acceptable.	
(2) The calculation of the radiation environment associated with DBAs should take into account the time-dependent transport of released fission products within various regions of containment and auxiliary structures.	(2) Conform. Calculations are on file.
(3) The initial distribution of activity within the containment should be based on a mechanistically rational assumption. The assumption of uniform distribution of activity throughout the containment at time zero is not appropriate.	(3) Conform.
(4) Effects of engineered safety features systems, such as containment ventilation and filtration systems, which act to remove airborne activity and redistribute activity within containment, should be calculated using the same assumptions used in the calculation of offsite dose. See SRP Section 15.6.5 (NUREG-75/087) and the related sections referenced in the appendices to that section.	(4) Conform.
(5) Natural deposition (i.e., plateout) of airborne activity should be determined using a mechanistic model and best estimates for the model parameters. The assumption of 50-percent instantaneous plateout of the iodine released from the core should not be made. Removal of iodine from surfaces by steam condensate flow or washoff by the containment spray may be assumed if such effects can be justified and quantified by analysis or experiment.	(5) Conform.

TABLE 3.11.B.3-1 (SHEET 5 OF 14)

	Requirements per NUREG-0588	VEGP Conformance
(6)	For unshielded equipment located in the containment, the gamma dose and dose rate should be equal to the dose and dose rate at the centerpoint of the containment plus the contribution from location dependent sources such as the sump water and plateout, unless it can be shown by analyses that location and shielding of the equipment reduces the dose and dose rate.	(6) Conform.
(7)	For unshielded equipment, the beta doses at the surface of the equipment should be the sum of the airborne and plateout sources. The airborne beta dose should be taken as the beta dose circulated for a point at the containment center.	(7) Airborne beta doses are calculated for a hemispherical geometry.
(8)	Shielded components need be qualified only to the gamma radiation levels required, provided an analysis or test shows that the sensitive portions of the component or equipment are not exposed to beta radiation or that the effects of beta radiation heating and ionization have no deleterious effects on component performance.	(8) Conform.
(9)	Cables arranged in cable trays in the containment should be assumed to be exposed to half the beta radiation dose calculated for a point at the center of the containment plus the gamma ray dose calculated in accordance with Section 1.4(6). This reduction in beta dose is allowed because of the localized shielding by other cables plus the cable tray itself.	(9) Beta is accounted for by testing with a gamma dose to irradiate cables inside containment to the calculated total integrated dose plus margin.
(10)	Paints and coatings should be assumed to be exposed to both beta and gamma rays in assessing their resistance to radiation. Plateout activity should be assumed to remain on the equipment surface unless the effects of the removal mechanisms, such as spray washoff or stream condensate flow, can be justified and quantified by analysis or experiment.	(10) Plateout with no washoff is included in the conservative qualification doses used for VEGP. Paints and coatings used for inside containment and decontaminable areas meet American National Standards Institute (ANSI) 101.2 and 101.4 requirements.

TABLE 3.11.B.3-1 (SHEET 6 OF 14)

	Requirements per NUREG-0588	VEGP Conformance
(11)	Components of the emergency core cooling system (ECCS) located outside containment (e.g., pumps, valves, seals, and electrical equipment) should be qualified to withstand the radiation equivalent to that penetrating the containment plus the exposure from the sump fluid using assumptions consistent with the requirements stated in Appendix K to 10 CFR 50.	(11) Conform. BOP equipment is qualified based on this requirement. NSSS equipment supplied by Westinghouse, including its qualification, is reviewed and verified to meet this requirement.
(12)	Equipment that may be exposed to radiation doses below 10^4 rads should not be considered to be exempt from radiation qualification unless analysis supported by test data is provided to verify that these levels will not degrade the operability of the equipment below acceptable values.	(12) For low levels of radiation exposure, the literature search conducted by EPRI-NP-2129, November 1981, for organic material indicates the presence of a radiation threshold level of 10^4 rads for electrical equipment and 10^3 rads for electronic equipment. Equipment is evaluated for all levels of radiation exposure.
(13)	The staff will accept a given component to be qualified provided it can be shown that the component has been qualified to integrated beta and gamma doses which are equal to or higher than those levels resulting from an analysis similar in nature and scope to that included in Appendix D (which uses the source term given in item (1) above) and that the component incorporates appropriate factors pertinent to the plant design and operating characteristics, as given in these general guidelines.	(13) Conform. Test reports are reviewed to assure that the radiation levels are met and no degradation will affect the performance of the equipment.
(14)	When a conservative analysis has not been provided by the applicant for staff review, the staff will use the radiation environment guidelines contained in Appendix D, suitably corrected for differences in reactor power level, type, containment, size, and other appropriate factors.	(14) Conform. Analysis used was conservative.
1.5	Environmental Conditions for Outside Containment	
(1)	Equipment located outside containment that could be subjected to high-energy pipe breaks should be qualified to the conditions resulting from the accident for the duration required.	(1) Conform. Equipment is qualified to the effects of high-energy line breaks. Postulated temperature and radiation levels have been calculated. The applicable environmental designator for each piece of safety-related equipment is used to qualify test equipment.

TABLE 3.11.B.3-1 (SHEET 7 OF 14)

Requirements per NUREG-0588	VEGP Conformance
The techniques to calculate the environmental parameters described in section 1.1 through 1.4 above should be applied.	
(2) Equipment located in general plant areas outside containment where equipment is not subjected to a DBA environment should be qualified to the normal and abnormal range of environmental conditions postulated to occur at the equipment location.	(2) Normal and abnormal conditions for all plant areas were established. These conditions were used to qualify the equipment located in each area. However, per the VEGP revised mild EQ position, environmental qualification shall not apply to safety-related equipment located in mild environments.
(3) Equipment not served by Class 1E environmental support systems, or served by Class 1E support systems that may be secured during plant operation or shutdown, should be qualified to the limiting environmental conditions that are postulated for the location, assuming a loss of the environmental support system.	(3) Equipment essential for safe shutdown or accident mitigation is qualified to the limiting environmental conditions that are postulated assuming loss of the environmental support systems.
2. Qualification Methods	
2.1 Selection of Methods	
(1) Qualification methods should conform to the requirements defined in IEEE Std. 323-1974.	(1) The VEGP program for safety-related equipment located in harsh environments conforms to IEEE Std. 323-1974. Subsections of IEEE Std. 323-1974 Section 6 regarding "Aging" or "Qualified Life" do not apply to safety-related equipment located in mild environments.
(2) The choice of the methods selected is largely a matter of technical judgment and availability of information that supports the conclusions reached. Experience has shown that qualification of equipment subjected to an accident environment without test data is not adequate to demonstrate functional operability. In general, the staff will not accept analysis in lieu of test data unless (a) testing of the component is impractical due to size limitations, and (b) partial type test data is provided to support the analytical assumptions and conclusions reached.	(2) Type testing in accordance to IEEE Std. 323-1974 is the preferred method of qualification. Vendors perform partial testing as a basis for other methodology used, such as analysis.

TABLE 3.11.B.3-1 (SHEET 8 OF 14)

Requirements per NUREG-0588	VEGP Conformance
(3) The environmental qualification of equipment exposed to DBA environments should conform to the following positions. The bases should be provided for the time interval required for operability of this equipment. The operability and failure criteria should be specified and the safety margins defined.	(3) The VEGP program addresses operability of the equipment, including margins necessary to meet specified criteria.
(a) Equipment that must function in order to mitigate any accident should be qualified by test to demonstrate its operability for the time required in the environmental conditions resulting from that accident.	(a) The VEGP program allows several qualification methods for safety-related equipment. However, testing is the preferred method. Mechanical equipment is qualified by analyses or a combination of test and analyses.
(b) Any equipment (safety-related or nonsafety-related) that need not function in order to mitigate any accident but that must not fail in a manner detrimental to plant safety should be qualified by test to demonstrate its capability to withstand any accident environment for the time during which it must not fail.	(b) See (a) above. Nonsafety-related equipment which must maintain its structural integrity in order not to fail in a manner detrimental to plant safety is designed to withstand the accident environment.
(c) Equipment that need not function in order to mitigate any accident and whose failure in any mode in any accident environment is not detrimental to plant safety need only be qualified for its nonaccident service environment. Although actual type testing is preferred, other methods when justified may be found acceptable. The bases should be provided for concluding that such equipment is not required to function in order to mitigate any accident and that its failure in any mode in any accident environment is not detrimental to plant safety.	(c) Conform. Testing is the preferred method. Other methods are acceptable on a case-by-case basis with justification.
(4) For environmental qualification of equipment subject to events other than a DBA which result in abnormal environmental conditions, actual type testing is preferred. However, analysis or operating history or any applicable combination thereof coupled with partial type test data may be found acceptable, subject to the applicability and detail of information provided.	(4) Conform.

TABLE 3.11.B.3-1 (SHEET 9 OF 14)

	Requirements per NUREG-0588	VEGP Conformance
2.2	Qualification by Test	
	(1) The failure criteria should be established prior to testing.	(1) Qualification test plans and procedures are reviewed to assure that acceptance criteria are established prior to that test.
	(2) Test results should demonstrate that the equipment can perform its required function for all service conditions postulated (with margin) during its installed life.	(2) Conform.
	(3) The items described in Section 6.3 of IEEE Std. 323-1974 supplemented by items (4) through (12) below constitute acceptable guidelines for establishing test procedures.	(3) Conform.
	(4) When establishing the simulated environmental profile for qualifying equipment located inside containment, it is preferred that a single profile be used that envelops the environmental conditions resulting from any DBE during any mode of plant operation (e.g., a profile that envelops the conditions produced by the MSLB and LOCA).	(4) VEGP uses a single profile for the accident conditions for equipment inside containment. On a case-by-case basis, the equipment may be tested to another profile if and only if the effects on the equipment envelop the effects resulting from the VEGP profile.
	(5) Equipment should be located above flood level or protected against submergence by locating the equipment in qualified watertight enclosures. Where equipment is located in watertight enclosures, qualification by test or analysis should be used to demonstrate the adequacy of such protection. Where equipment could be submerged, it should be identified and demonstrated to be qualified by test for the duration required.	(5) The location of safety-related components was compared to the calculated flood level in the surrounding area. Those items identified as being below the flood level were reviewed for acceptability. In the majority of cases, it was confirmed that the submerged component was not required to be functional for the event that could cause the flooding and that failure of the component would be acceptable. In the remaining cases, it was shown that the component would perform its required function (e.g., input to an alarm) prior to the time it would become submerged. In such cases, it was also confirmed that subsequent submergence would not result in unacceptable consequences. In those cases in which failure of the component would be unacceptable, corrective actions were made to preclude submergence.

TABLE 3.11.B.3-1 (SHEET 10 OF 14)

Requirements per NUREG-0588		VEGP Conformance
		The circuitry for submerged components was reviewed for electrical interfaces with other components. It was judged that the circuit breakers have been properly coordinated, so that submergence of the component would not result in the failure of other connected devices.
(6)	The temperature to which equipment is qualified, when exposed to the simulated accident environment, should be defined by thermocouple readings on or as close as practical to the surface of the component being qualified.	(6) The test chamber temperature simulating the DBA conditions is measured and monitored on or as close as practical to the surface of the component being qualified.
(7)	Performance characteristics of equipment should be verified before, after, and periodically during testing throughout its range of required operability.	(7) Baseline functional tests and performance tests after a test sequence and periodic functional checks are performed during the test sequence to demonstrate operability.
(8)	Caustic spray should be incorporated during simulated event testing at the maximum pressure and at the temperature conditions that would occur when the onsite spray systems actuate.	(8) The DBE is simulated by test, which includes the chemical spray concentration and pH requirements. Caustic spray effects during the DBE are also addressed by analysis with justification.
(9)	The operability status of equipment should be monitored continuously during testing. For long-term testing, however, monitoring at discrete intervals should be justified if used.	(9) The VEGP program requires monitoring of the test equipment based on its function. Intermittent monitoring is acceptable depending upon the function being simulated.
(10)	Expected extremes in power supply voltage range and frequency should be applied during simulated event environmental testing.	(10) The VEGP program addresses operability at environmental extremes, including voltage.
(11)	Dust environments should be addressed when establishing qualification service conditions.	(11) Conform.
(12)	Cobalt-60 is an acceptable gamma radiation source for environmental qualification.	(12) Cobalt 60 is the source used for qualification
2.3	Test Sequence	
(1)	The test sequence should conform fully to the guidelines established in Section 6.3.2 of IEEE Std. 323-1974. The test procedures should ensure that the same piece of equipment is used throughout the test sequence to and that the test simulates as closely as practicable the postulated accident environment	(1) Qualification reports identify the test sequence used in accordance to IEEE-325-1974 or daughter standards. Vendors are required to provide justification if other test sequences are used. If separate specimens are used to eliminate parts of the sequence, justification is required.

TABLE 3.11.B.3-1 (SHEET 11 OF 14)

Requirements per NUREG-0588	VEGP Conformance
2.4 Other Qualification Methods	
Qualification by analysis or operating experience implemented, as described in IEEE Std. 323-1974 and other ancillary standards, may be found acceptable. The adequacy of these methods will be evaluated on the basis of the quality and detail of the information submitted in support of the assumptions made and the specific function and location of the equipment. These methods are most suitable for equipment where testing is precluded by physical size of the equipment being qualified. It is required that, when these methods are employed, some partial type tests on vital components of the equipment be provided in support of these methods.	Conform.
3. Margins	
(1) Quantified margins should be applied to the design parameters discussed in Section 1 to assure that the postulated accident conditions have been enveloped during testing. These margins should be applied in addition to any margins (conservatism) applied during the derivation of the specified plant parameters.	(1) Margins are in accordance to IEEE Std. 323-1974. Qualification reports are reviewed, and margins are quantified to assure that adequate conservatism exists.
(2) In lieu of other proposed margins that may be found acceptable, the suggested values indicated in IEEE Std. 823-1974, Section 6.3.1.5, should be used as a guide. (Note exceptions stated in Section 1.4.)	(2) Conform.
(3) When the qualification envelope in Appendix C is used, the only required margins are those accounting for the inaccuracies in the test equipment. Sufficient conservatism has already been included to account for un-certainties such as production errors and errors associated with defining satisfactory performance (e.g., when only a small number of units are tested).	(3) This is not applicable to VEGP. Appendix C is for boiling water reactors and condenser containments.

TABLE 3.11.B.3-1 (SHEET 12 OF 14)

	Requirements per NUREG-0588	VEGP Conformance
(4)	Some equipment may be required by the design to only perform its safety function within a short time period into the event (i.e., within seconds or minutes); and, once its function is complete, subsequent failures are shown not to be detrimental to plant safety. Other equipment may not be required to perform a safety function but must not fail within a short time period into the event and subsequent failures are also shown not to be detrimental to plant safety. Equipment in these categories is required to remain functional in the accident environment for a period of at least 1 h in excess of the time assumed in the accident analysis. For all other equipment (e.g., post-accident monitoring, recombiners, etc.), the 10-percent time margin identified in Section 6.3.1.5 of IEEE Std. 323-1974 may be used.	(4) Equipment is qualified for the period before, during, and after the simulated DBA. For fast-acting equipment, functional operability is monitored at various times during the test. Acceptability of failures after the initial function is complete is reviewed on the case-by-case basis depending on the specific safety-related function of the equipment and the level of margin attained.
4. Aging		
(1)	Aging effects on all equipment, regardless of its location in the plant, should be considered and included in the qualification program.	(1) The effect of aging on all environmentally degradable parts of safety-related equipment located in harsh environments is considered. Subsections of paragraph 6.3.2 of IEEE Std. 323-1974 regarding "Aging" or "Qualified Life" do not apply to safety-related equipment located in mild environments.
(2)	The degrading influences discussed in Section 6.3.3, 6.3.4, and 6.3.5 of IEEE Std. 323-1974 and the electrical and mechanical stresses associated with cyclic operation of equipment should be considered and included as part of the aging progress.	(2) Mechanical and/or electrical cycling simulating the expected number of operational cycles for a reasonable period of component life is included in the program.
(3)	Synergistic effects should be considered in the accelerated aging programs. Investigation should be performed to assure that no known synergistic effects have been identified on materials that are included in the materials that are included in the equipment being qualified. Where synergistic effects have been identified, they should be accounted for in the qualification programs. Refer to NUREG/CR-0276 (SAND 78-0799) and NUREG/CR-0401 (SAND 78-1452), Qualification Testing Evaluation Quarterly Reports, for additional information.	(3) The only known synergistic effect that has been identified is for the cable.

TABLE 3.11.B.3-1 (SHEET 13 OF 14)

	Requirements per NUREG-0588		VEGP Conformance
(4)	The Arrhenius methodology is considered an acceptable method of addressing accelerated aging. Other aging methods that can be supported by type tests will be evaluated on a case-by-case basis.	(4)	Arrhenius methodology is an acceptable method. Other methods are accepted on a case-by-case basis.
(5)	Known material phase changes and reactions should be defined to ensure that no known changes occur within the extrapolation limits.	(5)	Extrapolations are within known material physical limits.
(6)	The aging acceleration rate used during qualification testing and the basis upon which the rate was established should be described and justified.	(6)	Conform.
(7)	Periodic surveillance testing under normal service conditions is not considered an acceptable method for ongoing qualification, unless the plant design includes provisions for subjecting the equipment to the limiting service environment conditions (specified in Section 3(7) of IEEE Std. 279-1971) during such testing.	(7)	Ongoing qualification is not used for equipment in a harsh environment.
(8)	Effects of relative humidity need not be considered in the aging of electrical cable insulation.	(8)	Conform.
(9)	The qualified life of the equipment (and/or component as applicable) and the basis for its selection should be defined.	(9)	Conform.
(10)	Qualified life should be established on the basis of the severity of the testing performed, the conservatisms employed in the extrapolation of data, the operating history, and in other methods that may be reasonably assumed, coupled with good engineering judgment.	(10)	Conform.

TABLE 3.11.B.3-1 (SHEET 14 OF 14)

Requirements per NUREG-0588	VEGP Conformance
5. Qualification Documentation	
(1) The staff endorses the requirements stated in IEEE Std. 323-1974 that, "The qualification documentation shall verify that each type of electrical equipment is qualified for its application and meets its specified performance requirements. The basis of qualification shall be explained to show the relationship of all facets of proof needed to support adequacy of the complete equipment. Data used to demonstrate the qualification of the equipment shall be pertinent to the application and organized in an auditable form."	(1) VEGP documentation of safety-related equipment will be in an auditable file.
(2) The guidelines for documentation in IEEE Std. 323-1974 when fully implemented are acceptable. The documentation should include sufficient information to address the required information identified in Appendix E. A certificate of conformance by itself is not acceptable unless it is accompanied by test data and information on the qualification program.	(2) Conform.

TABLE 3.11.N.3-1 (SHEET 1 OF 16)

COMPARISON OF NSSS ENVIRONMENTAL QUALIFICATION PROGRAM TO NUREG-0588

	NUREG-0588 CATEGORY 1 POSITION	NSSS PROGRAM
1.0	Establishment of the Qualification Parameters for Design Basis Events	
1.1	Temperature and Pressure Conditions Inside Containment - Loss of Coolant	
1.1.1	The time-dependent temperature and pressure established for the design of the containment structure and found acceptable by the staff may be used for environmental qualification of equipment.	The containment structural design has been based on the results of an analysis performed by Westinghouse employing methodology described below. The results of this analysis are reported in section 6.2.
1.1.2	Acceptable methods for calculating and establishing the containment pressure and temperature envelopes to which equipment should be qualified are summarized below. Acceptable methods for calculating mass and energy release rates are summarized in Appendix A.	Westinghouse employs the methodology described in WCAP-8312A for calculating the LOCA mass and energy release. Appendix A to NUREG-0588 indicates that this methodology is acceptable to the staff.
	Pressurized Water Reactors (PWRs)	
	Dry containment - Calculate LOCA containment environment using CONTEMPT-LT or equivalent industry codes. Additional guidance is provided in Standard Review Plan (SRP), Section 6.2.1.1.A, NUREG-75/087.	Westinghouse employs the COCO model described in WCAP-8327 to establish the containment pressure and temperature time-dependent variations following LOCA.
	Ice condenser containment – Calculate LOCA containment environment using LOTIC or equivalent industry codes. Additional guidance is provided in SRP Section 6.2.1.1.B, NUREG-75/087.	NA
1.1.3	In lieu of using the plant-specific containment temperature and pressure design profiles for boiling water reactor (BWR) and ice condenser type plants, the generic envelope shown in Appendix C may be used for qualification testing.	NA

TABLE 3.11.N.3-1 (SHEET 2 OF 16)

	NUREG-0588 CATEGORY 1 POSITION	NSSS PROGRAM
1.1.4	The test profiles include in Appendix A to IEEE Std. 323-1974 should not be considered an acceptable specific containment temperature and pressure design profiles unless plant-specific analysis is provided to verify the adequacy of those profiles.	Westinghouse does not employ the test profiles included in Appendix A to IEEE Std. 323-1974.
1.2	Temperature and Pressure Conditions Inside Containment - Main Steam Line Break (MSLB)	
1.2.1	The environmental parameters used for equipment qualification should be calculated with a plant-specific model reviewed and approved by the staff.	Westinghouse employs the methodology described in WCAP-8822 for calculating the mass and energy release following an MSLB. Westinghouse has completed the mass and energy release calculations for VEGP assuming entrainment.
1.2.2	Models that are acceptable for calculating containment parameters are listed in Section 1.1(2).	Westinghouse employs the COCO model described in WCAP-8327 and WCAP-8936 to establish the containment pressure and temperature time-dependent variations following an MSLB. Westinghouse submitted these reports for generic staff review. The staff has accepted these methods on specific applications. Westinghouse will continue to use these models for containment analysis.
1.2.3	In lieu of using the plant-specific containment temperature and pressure design profiles for BWR and ice condenser plants, the generic envelope shown in Appendix C may be used.	NA

TABLE 3.11.N.3-1 (SHEET 3 OF 16)

	NUREG-0588 CATEGORY 1 POSITION	NSSS PROGRAM
1.2.4	The test profiles included in Appendix A to IEEE Std. 323-1974 should not be considered an acceptable alternative in lieu of using plant-specific containment temperature and pressure design profiles, unless plant-specific analysis is provided to verify the adequacy of those profiles.	Westinghouse does not employ the test profiles included in Appendix A to IEEE Std. 323-1974.
1.2.5	Where qualification has been completed but only LOCA conditions were considered, then it must be demonstrated that the LOCA qualification conditions exceed or are equivalent to the maximum calculated MSLB conditions. The following technique is acceptable:	
1.2.5.a	Calculate the peak temperature from an MSLB using a model based on the staff's approved assumptions defined in Section 1.1.(2).	
1.2.5.b	Show that the peak surface temperature of the component to be qualified does not exceed the LOCA qualification temperature by the method discussed in item 2 of Appendix B.	Westinghouse has established a single target qualification envelope for equipment that is required to perform safety functions in a hostile environment resulting from a primary or secondary side break. This envelope has been selected to envelop, with margin, the anticipated range of LOCA and MSLB transients. As a consequence, there is no intent in the design of the program described in WCAP-8587 to employ this type of analysis to justify qualification for equipment required to perform a safety function in a hostile environment.
1.2.5.c	If the calculated surface temperature exceeds the qualification temperature, the staff requires that (a) requalification testing be performed with appropriate margins, or (b) qualified physical protection be provided to assure that the surface temperature will not exceed the actual qualification temperature. For the plants that are currently being reviewed or will be submitted for an operating license review within 6 months from issue date of this report, compliance with items (a) or (b) above may represent a substantial impact. For those plants, the staff will consider additional information submitted by the applicant to demonstrate that the equipment can maintain its functional operability if its surface temperature rises to the value calculated.	

TABLE 3.11.N.3-1 (SHEET 4 OF 16)

	NUREG-0588 CATEGORY 1 POSITION	NSSS PROGRAM
1.3	Effects of Chemical Spray	The maximum concentration of boron employed for containment spray is 2500 ppm, and the maximum permitted pH of the initial spray solution is 10.5. For qualification testing, Westinghouse specifies a chemical spray of 2500 ppm boron buffered with 0.9 wt% dissolved sodium hydroxide to maintain a pH of approximately 10.7, starting at time zero and terminating after 24 h. This spray concentration results in an increase in alkalinity of at least 10 percent compared to the maximum concentration defined by the specification and significantly exceeds the range of sump pH values permitted long-term by the Technical Specifications for VEGP.
1.3.1	The effects of caustic spray should be addressed for the equipment qualification. The concentration of caustics used for qualification should be equivalent to or more severe than those used in the plant containment spray system.	
1.3.2	If the chemical composition of the caustic spray can be affected by equipment malfunctions, the most severe caustic spray environment that results from a single failure in the spray system should be assumed. See SRP Section 6.5.2 (NUREG-75/087), paragraph II, item (e), for caustic spray solution guidelines.	In the Westinghouse-designed containment spray system, no single failure can be postulated which would cause the specified test pH of 10.7 to be nonconservative for simulating long-term spray operations post-accident.

TABLE 3.11.N.3-1 (SHEET 5 OF 16)

	NUREG-0588 CATEGORY 1 POSITION	NSSS PROGRAM
1.4	Radiation Conditions Inside and Outside Containment Subsections 1, 2, 3, 4, 5, 6, 7, 9, 10, 11, 13, 14.	See table (B section).
1.4.8	Shielded components need be qualified only to the gamma radiation levels required, provided an analysis or test shows that the sensitive portions of the component or equipment are not exposed to beta radiation or that the effects of beta radiation heating and ionization have no deleterious effects on component performance.	Any potential concern resulting from exposure to beta radiation is limited to equipment located inside containment that is required to mitigate a high-energy line break (HELB) inside containment and only to any organic materials exposed to the in-containment environment. Qualification programs address this issue by analysis, test, or combination of both.
1.4.12	Equipment that may be exposed to radiation doses below 10^4 rads should not be considered to be exempt from radiation qualification unless analysis supported by test data is provided to verify that these levels will not degrade the operability of the equipment below acceptable values.	For safety-related electrical equipment that is not required to operate in a HELB environment and for which the anticipated qualified life integrated radiation dose is 10^4 rads or less, Westinghouse does not include a radiation aging simulation as part of any qualification testing. For such equipment and components, Appendix C to WCAP-8587 demonstrates, based on available test information on materials and components that up to approximately 10^4 rads there is no detectable effect on the structural characteristics of materials and components that would affect the capability of equipment to perform during a seismic event.
1.5	Environmental Conditions for Outside Containment Subsections 1, 2, 3	See table (B section).

TABLE 3.11.N.3-1 (SHEET 6 OF 16)

	NUREG-0588 CATEGORY 1 POSITION	NSSS PROGRAM
2.0	Qualification Methods	
2.1	Selection of Methods	
2.1.1	Qualification methods should conform to the requirements defined in IEEE Std. 323-1974.	The methodology employed by Westinghouse to qualify safety-related electrical equipment is described in WCAP-8587 and Std. 323-1974.
2.1.2	The choice of the methods selected is largely a matter of technical judgment and availability of information that supports the conclusions reached. Experience has shown that qualification of equipment subjected to an accident environment without test data is not adequate to demonstrate functional operability. In general, the staff will not accept analysis in lieu of test data unless (a) testing of the component is impractical due to size limitations, and (b) partial type test data is provided to support the analytical assumptions and conclusions reached.	Westinghouse qualifies equipment that is required to perform a safety function in a HELB environment by test.
2.1.3	The environmental qualification of equipment exposed to DBA environments should conform to the following positions: The basis should be provided for the time interval required for operability of this equipment.	The required operability time is discussed in paragraph 3.11.N.1.3.
	The operability and failure criteria should be specified and the safety margins defined.	The primary purpose of equipment qualification is to reduce the potential for common-mode failures due to postulated environmental conditions. A test unit will therefore be considered to have failed the test if the performance requirements identified in WCAP-8587 cannot be met, unless an investigation can establish that the failure mechanism is not a common-mode origin or that plant specific analyses can demonstrate that the reduced capability is acceptable.

TABLE 3.11.N.3-1 (SHEET 7 OF 16)

	NUREG-0588 CATEGORY 1 POSITION	NSSS PROGRAM
		Margins are discussed under item 3.
2.1.3.a	Equipment that must function in order to mitigate any accident should be qualified by test to demonstrate its operability for the time required in the environmental conditions resulting from that accident.	When Westinghouse employs testing to qualify electrical equipment that must function in order to mitigate any accident, the acceptance criterion for the test is that the safety-related function must be demonstrated for the specified duration while the equipment is exposed to the simulated environmental conditions resulting from the accident.
2.1.3.b	Any equipment (safety-related or nonsafety-related) that need not function in order to mitigate any accident, but that must not fail in a manner detrimental to plant safety should be qualified by test to demonstrate its capability to withstand any accident environment for the time during which it must not fail.	This is discussed in subsections 15.1.4, 15.1.5, 15.2.7, and 15.4.1.
2.1.3.c	Equipment that need not function in order to mitigate any accident and whose failure in any mode in any accident environment is not detrimental to plant safety need only be qualified for its nonaccident service environment. Although actual type testing is preferred, other methods when justified may be found acceptable. The bases should be provided for concluding that such equipment is not required to function in order to mitigate any accident, and that its failure in any mode in any accident environment is not detrimental to plant safety.	Where Westinghouse supplies an item of safety-related electrical equipment that is located in an area where it can experience the environment resulting from a HELB but is not required to perform any safety function, Westinghouse has verified that any consequential failure of such equipment, due to the adverse environment, does not prejudice the safety-related functions of other equipment claimed in the accident analysis.

TABLE 3.11.N.3-1 (SHEET 8 OF 16)

	NUREG-0588 CATEGORY 1 POSITION	NSSS PROGRAM
2.1.4	For environmental qualification of equipment subject to events other than a DBA which result in abnormal environmental conditions, actual type testing is preferred. However, analysis or operating history, or any applicable combination thereof, coupled with partial type test data may be found acceptable, subject to the applicability and detail of information provided.	Potential abnormal environments are associated with recirculating operations post-accident and with a loss of non-Class 1E HVAC systems. Recirculation operations: Equipment supplied by Westinghouse that is required to perform a post-accident safety function as part of the recirculation loop is qualified by testing for anticipated increased radiation levels resulting from recirculation of radioactive fluid. Loss of non-Class 1E HVAC. For equipment located outside containments that is not required to be qualified for the effects of a HELB environment, Westinghouse has completed performance tests in accordance with Section 6.3.2(3) of IEEE 323-1974 at the anticipated environmental extremes, including loss of non-Class 1E HVAC where applicable.
2.2	Qualification by Test	
2.2.1	The failure criteria should be established prior to testing.	In Supplement 1 to WCAP-8587 Westinghouse has identified, for each item of safety-related equipment, the safety functions to be performed for all normal, abnormal, or accident conditions during or after which the equipment is required to provide a protective function. As stated in the response to staff position 2.1.(3), a test unit will be considered to have failed the test if the safety-related performance requirements cannot be met, unless an investigation can establish that the failure mechanism is not of common-mode origin or that plant specific analyses can demonstrate that the reduced capability is acceptable.
2.2.2	Test results should demonstrate that the equipment can perform its required function for all service conditions postulated (with margin) during its installed life.	Westinghouse qualification programs are designed to demonstrate that the equipment can perform its required safety function(s) for all postulated service conditions.

TABLE 3.11.N.3-1 (SHEET 9 OF 16)

	NUREG-0588 CATEGORY 1 POSITION	NSSS PROGRAM
		Via the treatment of aging, as described in the response to staff position 4.0, a generic qualified life is established by Westinghouse for each item of equipment. This generic qualified life may be extended on a plant-specific basis by employing less conservative plant-specific assumptions concerning the plant normal operating environmental conditions. The qualified life established for the equipment on a specific plant will ultimately define the permitted installed life of the equipment. The subject of margin is discussed in the response to staff position 3.0.
2.2.3	The items described in Section 6.3 of IEEE Std. 323-1974 supplemented by items (4) through (12) below constitute acceptable guidelines for establishing test procedures.	When testing is the selected methodology for qualifying equipment, Westinghouse has established the test program in conformance with Section 6.3 of IEEE 323-1974.
2.2.4	When establishing the simulated environmental profile for qualifying equipment located inside containment, it is preferred that a single profile be used that envelops the environmental conditions resulting from any design basis event during any mode of plant operation (e.g., a profile that envelops the conditions produced by the MSLB and LOCAs).	Westinghouse prefers to use a single profile, enveloping both MSLB and LOCA, for qualification of equipment located inside containment which is required to perform a safety function to mitigate both HELBs. This approach is optimum in terms of schedule, manpower, and materials. However, there is no technical justification for making this a requirement for all equipment inside containment.
2.2.5	Equipment should be located above flood level or protected against submergence by locating the equipment in qualified watertight enclosures. Where equipment is located in watertight enclosures, qualification by test or analysis should be used to demonstrate the adequacy of such protection. Where equipment could be submerged, it should be identified and demonstrated to be qualified by test for the duration required.	NA

TABLE 3.11.N.3-1 (SHEET 10 OF 16)

	NUREG-0588 CATEGORY 1 POSITION	NSSS PROGRAM
2.2.6	The temperature to which equipment is qualified, when exposed to the simulated accident environment, should be defined by thermocouple readings on or as close as practical to the surface of the component being qualified.	In performing qualification tests for HELB environments, Westinghouse requires that the external environment temperature be measured as close to the test unit surface as practicable.
2.2.7	Performance characteristics of equipment should be verified before, after, and periodically during testing throughout its range of required operability.	Where the safety-related function of the equipment requires operation during the HELB, Westinghouse verifies the equipment performance before, during, and after the simulated event and verifies that the safety-related function is demonstrated for the specified required duration of the function.
2.2.8	Caustic spray should be incorporated during simulated event testing at the maximum pressure and at the temperature conditions that would occur when the onsite spray systems actuate.	The response to item 1.3(1) is applicable for equipment located inside containment and qualified by test to operate in a HELB environment.
2.2.9	The operability status of equipment should be monitored continuously during testing. For long-term testing, however, monitoring at discrete intervals should be justified if used.	The response to item 2.2(7) is applicable.
2.2.10	Expected extremes in power supply voltage range and frequency should be applied during simulated event environmental testing.	The Class 1E instrumentation and control equipment qualified in Westinghouse scope is supplied by a guaranteed stabilized power supply. As a consequence, the range of electrical parameters employed is extremely small, and variations within the permitted range are considered insignificant. Westinghouse also addresses extremes in power supply voltage and frequency as required in the design and sizing of equipment to provide margins for all conditions.

TABLE 3.11.N.3-1 (SHEET 11 OF 16)

	NUREG-0588 CATEGORY 1 POSITION	NSSS PROGRAM
2.2.11	Dust environments should be addressed when establishing qualification service conditions.	Dust environments have not been established as a required and/or significant qualification parameters.
2.2.12	Cobalt-60 is an acceptable gamma radiation source for environmental qualification.	Westinghouse employs Cobalt-60 sources to simulate the effects of gamma and, in some cases, beta radiation for equipment qualified by test to operate in a HELB environment.
2.3	Test Sequence	
2.3.1	The test sequence should conform fully to the guidelines established in Section 6.3.2 of IEEE Std. 323-1974. The test procedures should ensure that the same piece of equipment is used throughout the test sequence and that the test simulates as closely as practicable the postulated accident environment.	Section 6.3.2 of IEEE 323-1974 neither mandates a single unique test sequence nor requires that the same piece of equipment be used through the test sequence: - The standard identifies a test sequence that is thought to be the most conservative for most equipment; however, alternative sequences are clearly permitted with justification. - Section 6.3.2(3) specifically permits the performance test at extremes of the normal ambient to be performed on another, essentially similar, piece of equipment.

TABLE 3.11.N.3-1 (SHEET 12 OF 16)

	NUREG-0588 CATEGORY 1 POSITION	NSSS PROGRAM
2.4	Other Qualification Methods	
	Qualification by analysis or operating experience implemented, as described in IEEE Std. 323-1974 and other ancillary standards, may be found acceptable. The adequacy of these methods will be evaluated on the basis of the quality and detail of the information submitted in support of the assumptions made and the specific function and location of the equipment. These methods are most suitable for equipment where testing is precluded by physical size of the equipment being qualified. It is required that, when these methods are employed, some partial type tests on vital components of the equipment be provided in support of these methods.	Westinghouse does not employ analysis or operating experience as a prime method to establish the environmental qualification of safety-related electrical equipment. However, analysis or operating experience may be included in support of environmental qualification by test.
3.0	Margins	
3.1.1	Quantified margins should be applied to the design parameters discussed in Section 1 to assure that the postulated accident conditions have been enveloped during testing. These margins should be applied in addition to any margins (conservatism) applied during the derivation of the specified plant parameters.	In general, Westinghouse has applied margin with respect to the design postulated accident conditions defined for each item of equipment in Section 1 of the corresponding EQDP. However, since Westinghouse has conducted generic qualification tests, the design environments selected by Westinghouse will already contain significant margin with respect to most plant-specific environmental parameters.
3.1.2	In lieu of other proposed margins that may be found acceptable, the suggested values indicated in IEEE Std. 323-1974, Section 6.3.1.5, should be used as a guide. (Note exceptions stated in Section 1.4.)	Westinghouse has applied specific margin to design parameters, in deriving type test parameters, as described in Section 7.1 of WCAP-8587.(1) This method of applying margin is in accordance with Section 6.3.1.5 of IEEE 323-1974, which recognizes increasing test levels, number of test cycles, and test duration as methods of incorporating margin in the test plan.

TABLE 3.11.N.3-1 (SHEET 13 OF 16)

	NUREG-0588 CATEGORY 1 POSITION	NSSS PROGRAM
3.1.3	When the qualification envelope in Appendix C is used, the only required margins are those accounting for the inaccuracies in the test equipment. Sufficient conservatism has already been included to account for uncertainties such as production errors and errors associated with defining satisfactory performance (e.g., when only a small number of units are tested).	NA
3.1.4.a	Some equipment may be required by the design to only perform its safety function within a short time period into the event (i.e., within seconds or minutes); and, once its function is complete, subsequent failures are shown not to be detrimental to plant safety. Other equipment may not be required to perform a safety function but must not fail within a short time period into the event, and subsequent failures are also shown not to be detrimental to plant safety. Equipment in these categories is required to remain functional in the accident environment for a period of at least 1 h in excess of the time assumed in the accident analysis.	Some equipment (e.g., transmitters) was not specified to maintain trip function accuracy requirements for longer than 5 min post-accident. The operability time was conservatively established based on the reactor trip ESF function performed by each equipment item considering what consequences failure of the device would have on the operator and the mitigation of the event. Margins for trip function requirements are contained in the HELB envelopes which encompass a full spectrum of break sizes and also justified by the fact that the signal generated by the sensor is "locked-in" by the protection system and will not reset should the sensor fail after the designated trip time requirement. Most of the equipment was also specified and qualified for much longer post-accident monitoring function times to slightly reduced accuracy requirements.
3.1.4b	For all other equipment (e.g., post-accident monitoring, recombiners, etc.), the 10-percent time margin identified in Section 6.3.1.5 of IEEE Std. 323-1974 may be used.	In qualifying equipment, Westinghouse did not always include any systematic margin on the specified duration of the safety function. Margin is included in radiation dose calculations and by a conservative selection of parameters for aging simulation.

TABLE 3.11.N.3-1 (SHEET 14 OF 16)

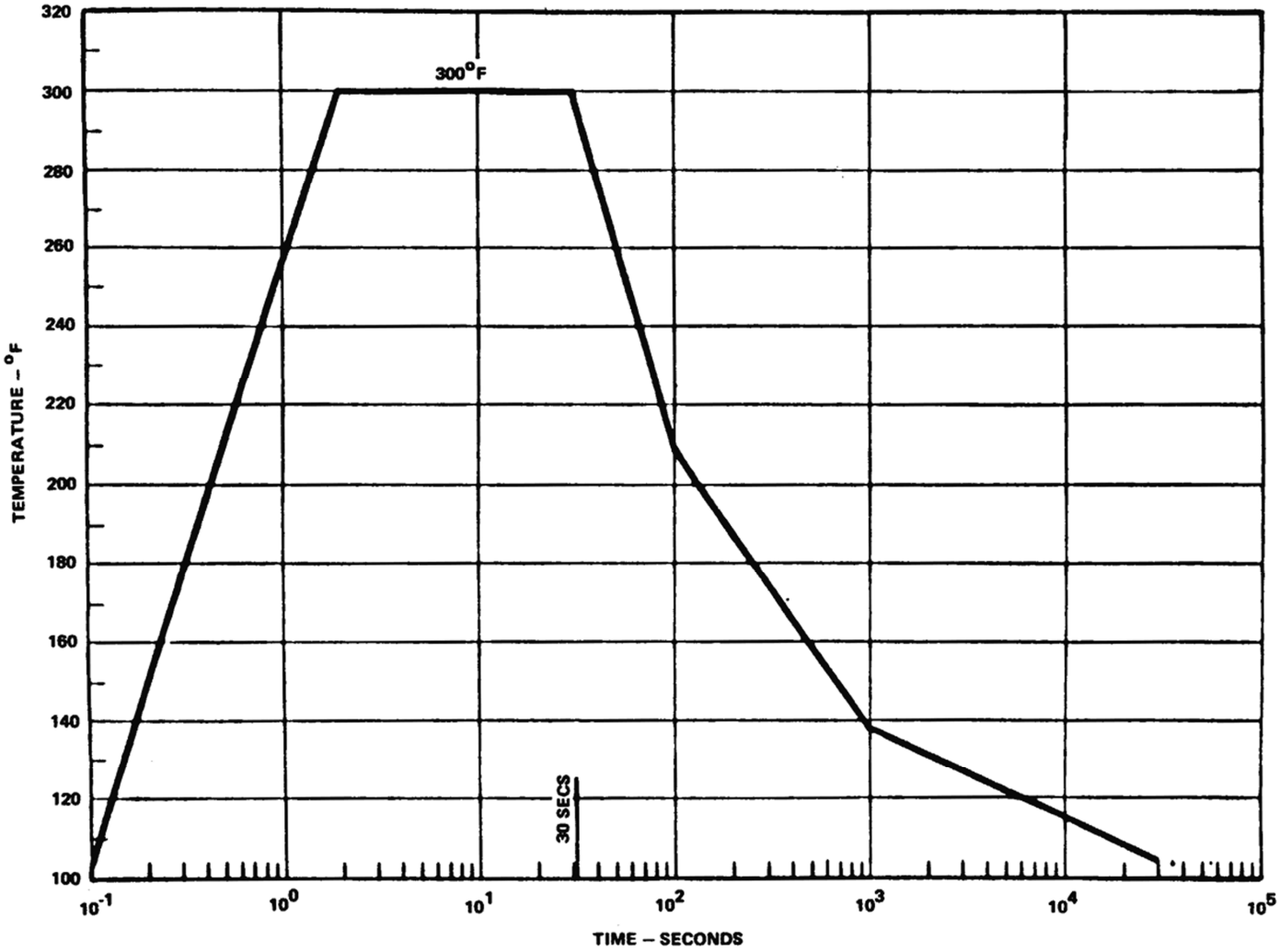
	NUREG-0588 CATEGORY 1 POSITION	NSSS PROGRAM
4.0	Aging	
4.1	Aging effects on all equipment, regardless of its location in the plant, should be considered and included in the qualification program.	Westinghouse considers and includes in the equipment qualification programs the effects of aging, as applicable, irrespective of the location of the equipment in the plant.
4.2	The degrading influences discussed in Sections 6.3.3, 6.3.4, and 6.3.5 of IEEE Std. 323-1974 and the electrical and mechanical stresses associated with cyclic operation of included as part of the aging programs.	Appendix B to WCAP-8587 describes the aging mechanisms to be considered and the methodology to be employed in simulating the aging effects. The mechanisms to be considered in the Westinghouse program in Sections 6.3.3, 6.3.4, and 6.3.5 of IEEE 323-1974 including electrical and mechanical stresses associated with cyclic operation of equipment.
4.3	Synergistic effects should be considered in the accelerated aging programs. Investigation should be performed to assure that no known synergistic effects have been identified on materials that are included in the equipment being qualified. Where synergistic effects have been identified, they should be accounted for in the qualification programs. Refer to NUREG/CR-0276 (SAND 78-0799) and NUREG/CR-0401 (SAND 78-1452), Qualification Testing Evaluation Quarterly Reports, for additional information.	Westinghouse has not identified any synergistic effects involving the materials and components comprising the equipment to be qualified in this program.
4.4	The Arrhenius methodology is considered an acceptable method of addressing accelerated aging. Other aging methods that can be supported by type tests will be evaluated on a case-by-case basis.	The Arrhenius equation is solely employed by Westinghouse in calculating appropriate temperature and duration parameters to accelerate the effects of thermal aging.
4.5	Known material phase changes and reactions should be defined to ensure that no known changes occur within the extrapolation limits.	No such mechanisms are known.

TABLE 3.11.N.3-1 (SHEET 15 OF 16)

	NUREG-0588 CATEGORY 1 POSITION	NSSS PROGRAM
4.6	The aging acceleration rate used during qualification testing and the basis upon which the rate was established should be described and justified.	Appendix D to WCAP-8587 justifies the acceleration parameters and rates to be employed for the aging program described in Appendix B to WCAP-8587.
4.7	Periodic surveillance testing under normal service conditions is not considered an acceptable method for ongoing qualification unless the plant design includes provisions for subjecting the equipment to the limiting service environment conditions (specified in Section 3(7) of IEEE Std. 279-1971) during such testing.	Westinghouse does not employ periodic surveillance testing or any form of ongoing qualification in the program described in WCAP-8587.
4.8	Effects of relative humidity need not be considered in the aging of electrical cable insulation.	For equipment subjected to HELB environments, the aging effects due to humidity during normal operation are judged to be insignificant compared to the effects of the high-temperature steam accident simulation; therefore, no additional humidity aging simulation is required. For equipment not subjected to HELB environments, the use of materials and components known to be significantly affected by humidity are avoided.
4.9	The qualified life of the equipment (and/or component as applicable) and the basis for its selection should be defined.	The EQDP identifies the demonstrated qualified life and justifies the value selected based on the aging mechanisms that have been simulated.
4.10	Qualified life should be established on the basis of the severity of the testing performed, the conservatisms employed in the extrapolation of data, the operating history, and in other methods that may be reasonably assumed, coupled with good engineering judgment.	Westinghouse conforms to this position in establishing the demonstrated qualified life of WRD-supplied equipment.

TABLE 3.11.N.3-1 (SHEET 16 OF 16)

NUREG-0588 CATEGORY 1 POSITION	NSSS PROGRAM
5.0 Qualification Documentation	
5.1 The staff endorses the requirements stated in IEEE Std. 323-1974: "The qualification documentation shall verify that each type of electrical equipment is qualified for its application and meets its specified performance requirements. The basis of qualification shall be explained to show the relationship of all facets of proof needed to support adequacy of the complete equipment. Data used to demonstrate the qualification of the equipment shall be pertinent to the application and organized in an auditable form."	The documentation system established for a specific plant application should be described by the applicant. In support of the documentation effort Westinghouse has supplied to the utilities: - Identification of the limits to any environmental qualification. - Copies of all referenced Westinghouse topical reports. In addition, Westinghouse will maintain the available raw test data which supports the referenced qualification tests on equipment subject to HELB and available information concerning the performance testing of production units.
5.2 The guidelines for documentation in IEEE Std. 323-1974 are acceptable. The documentation should include sufficient information to address the required information identified in Appendix E. A certificate of conformance by itself is not acceptable unless it is accompanied by test data and information on the qualification program.	The Westinghouse qualification test reports meet the requirements of Section 5 to IEEE 323-1974 by providing the following essential information as a minimum: - Safety-related functional requirements to be demonstrated. - Range of applicable environmental parameters to be considered. - Identification of the test unit. - Description of the test facility and monitoring instrumentation. - Description of test unit mounting and interfaces. - Summary of the test procedures. - Summary of the test results.



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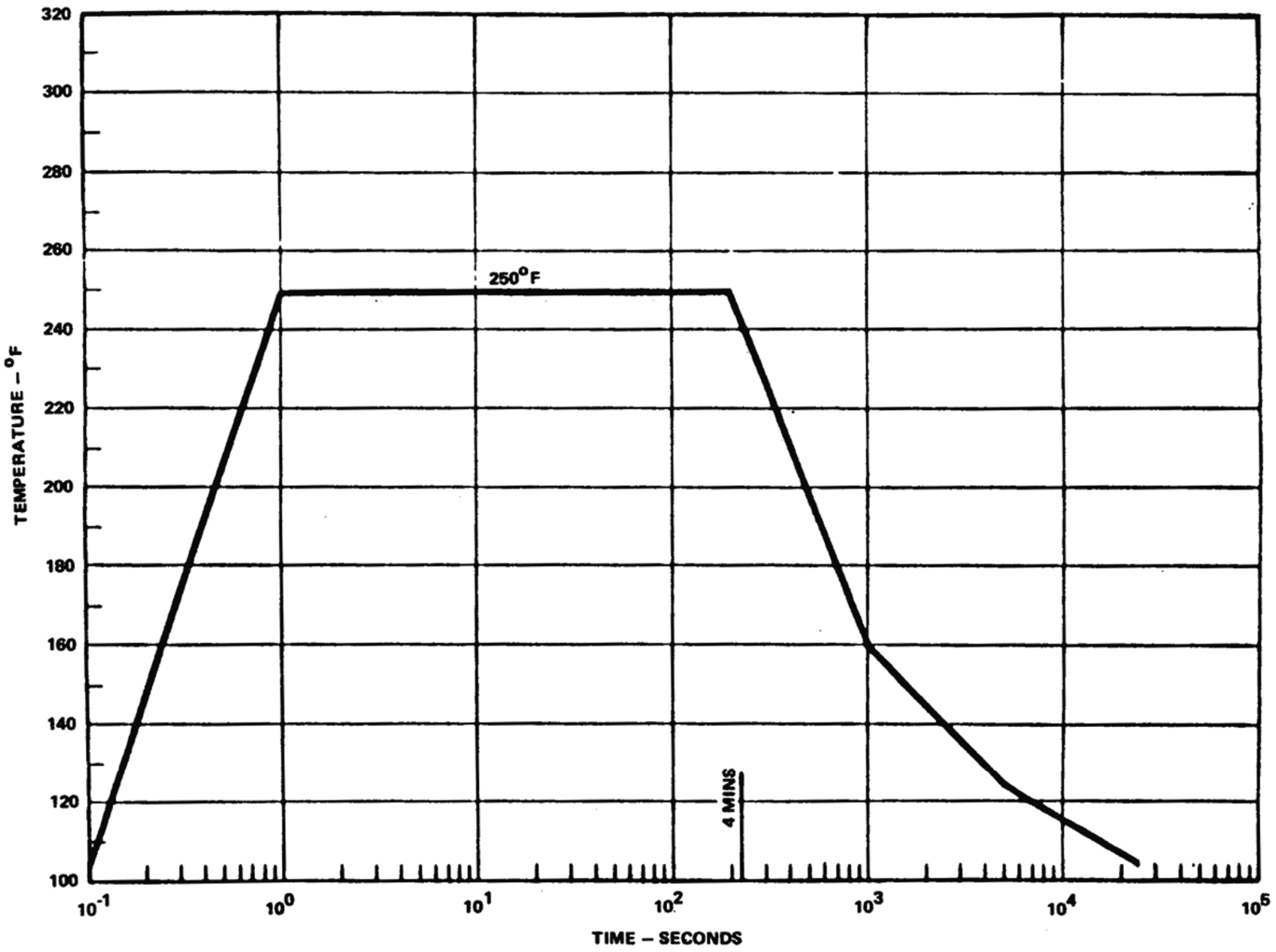
TEMPERATURE CONDITIONS FOR
EQUIPMENT QUALIFICATION IN
THE AUXILIARY BUILDING

FIGURE 3.11.B.1-1 (SHEET 1 OF 18)



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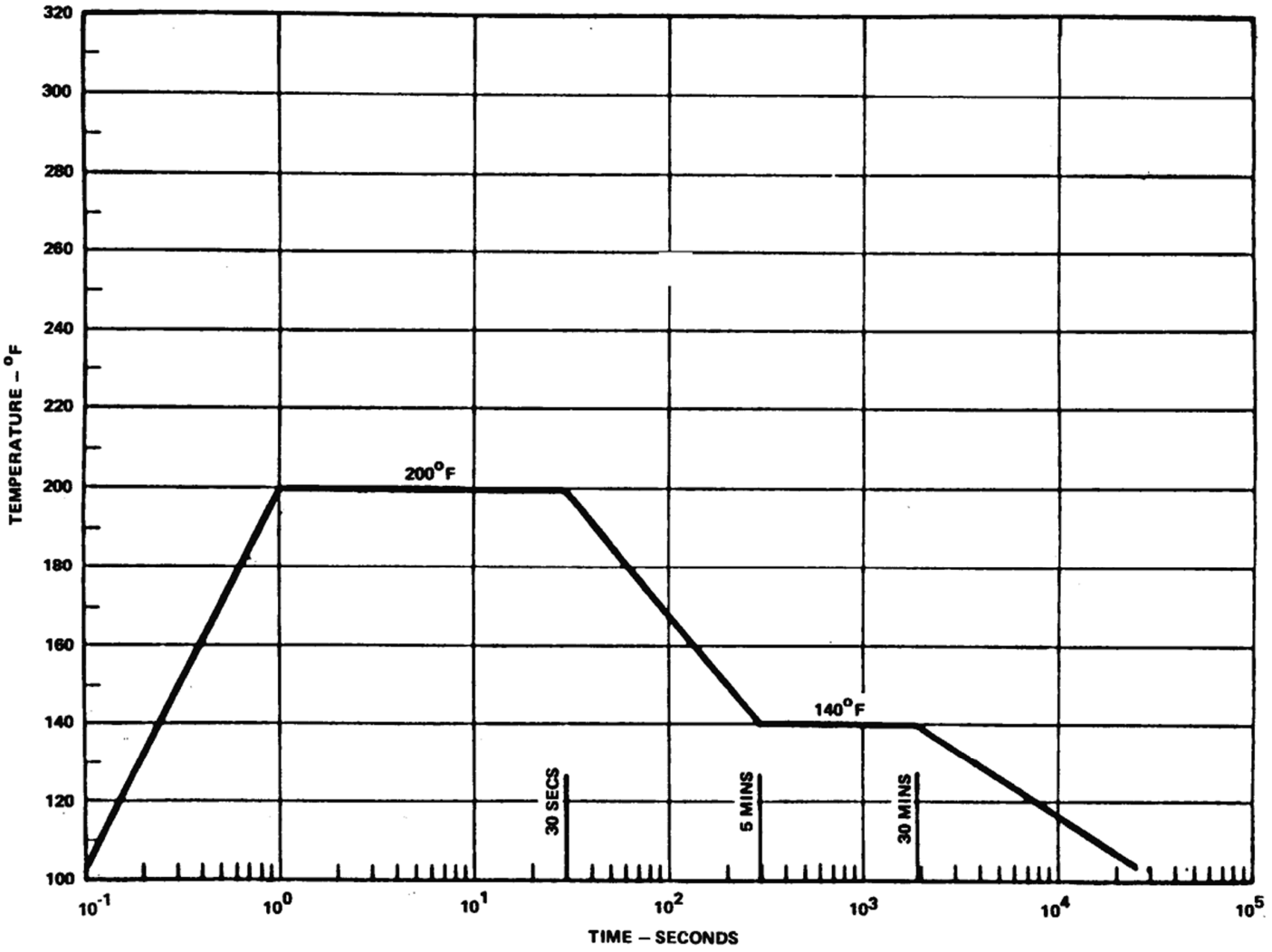
TEMPERATURE CONDITIONS FOR EQUIPMENT QUALIFICATION IN THE AUXILIARY BUILDING

FIGURE 3.11.B.1–1 (SHEET 2 OF 18)



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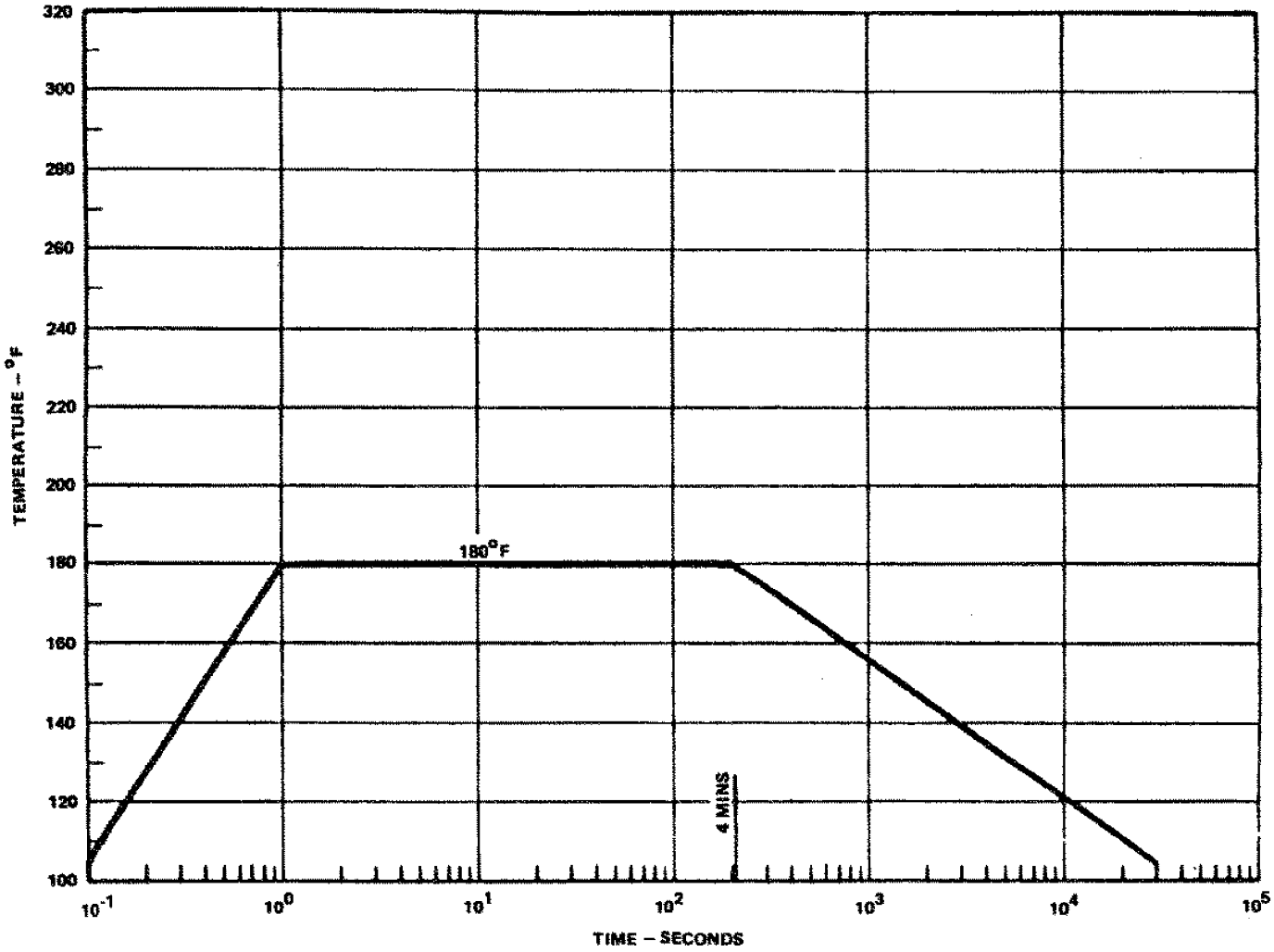
TEMPERATURE CONDITIONS FOR EQUIPMENT QUALIFICATION IN THE AUXILIARY BUILDING

FIGURE 3.11.B.1–1 (SHEET 3 OF 18)



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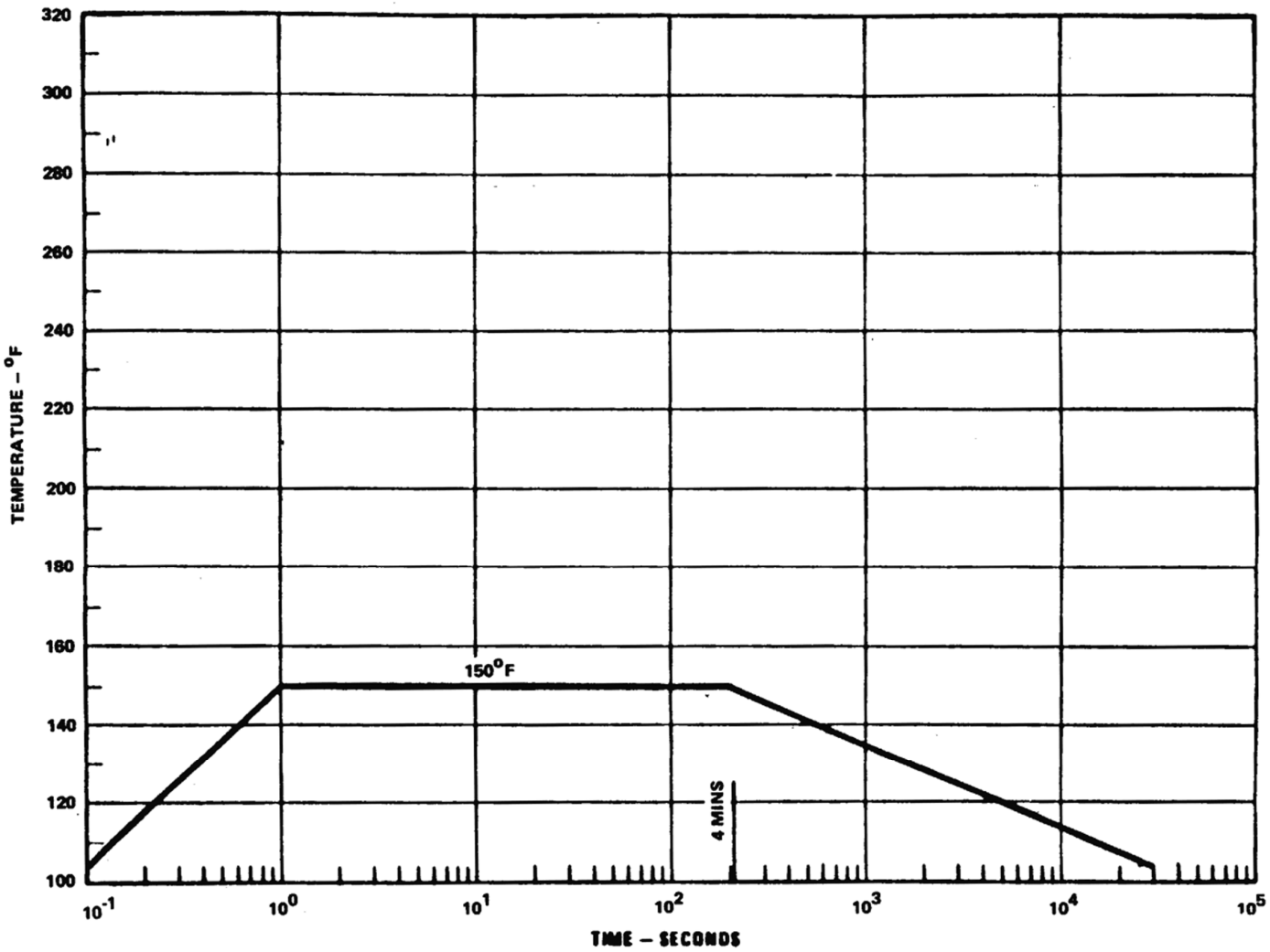
TEMPERATURE CONDITIONS FOR EQUIPMENT QUALIFICATION IN THE AUXILIARY BUILDING

FIGURE 3.11.B.1–1 (SHEET 4 OF 18)



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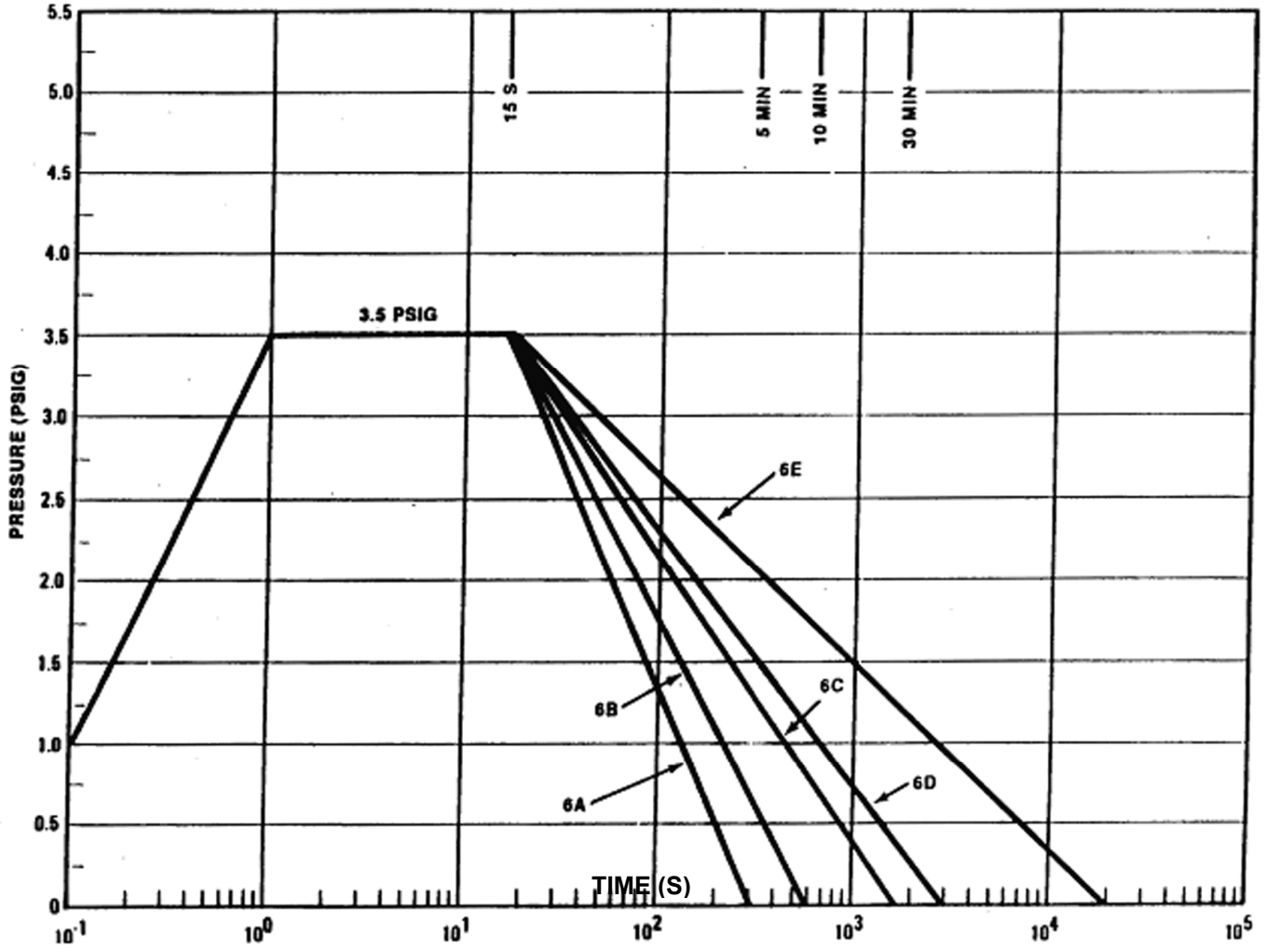
TEMPERATURE CONDITIONS FOR EQUIPMENT QUALIFICATION IN THE AUXILIARY BUILDING

FIGURE 3.11.B.1–1 (SHEET 5 OF 18)



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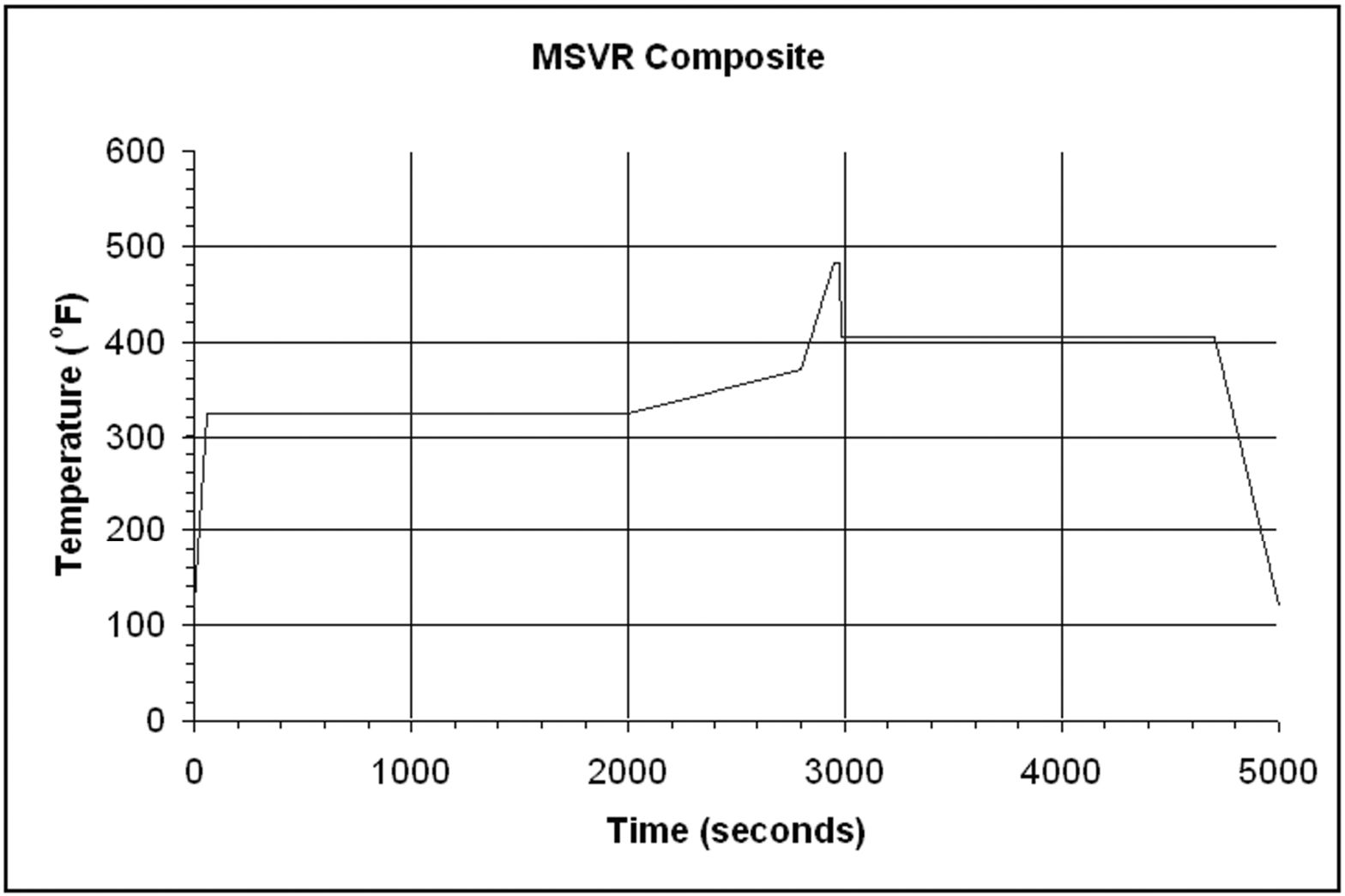
PRESSURE CONDITIONS FOR EQUIPMENT QUALIFICATION IN THE AUXILIARY BUILDING

FIGURE 3.11.B.1–1 (SHEET 6 OF 18)



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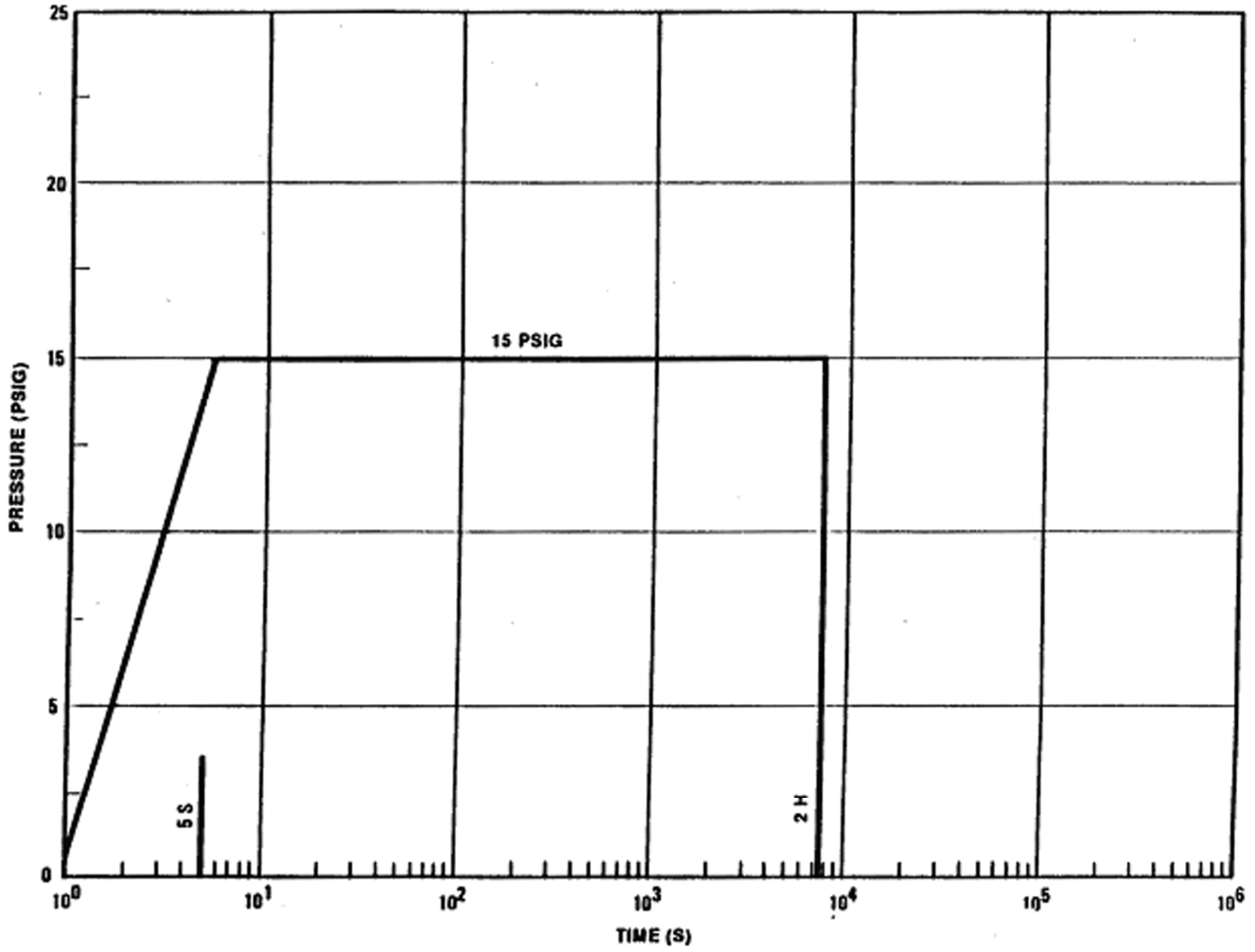
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

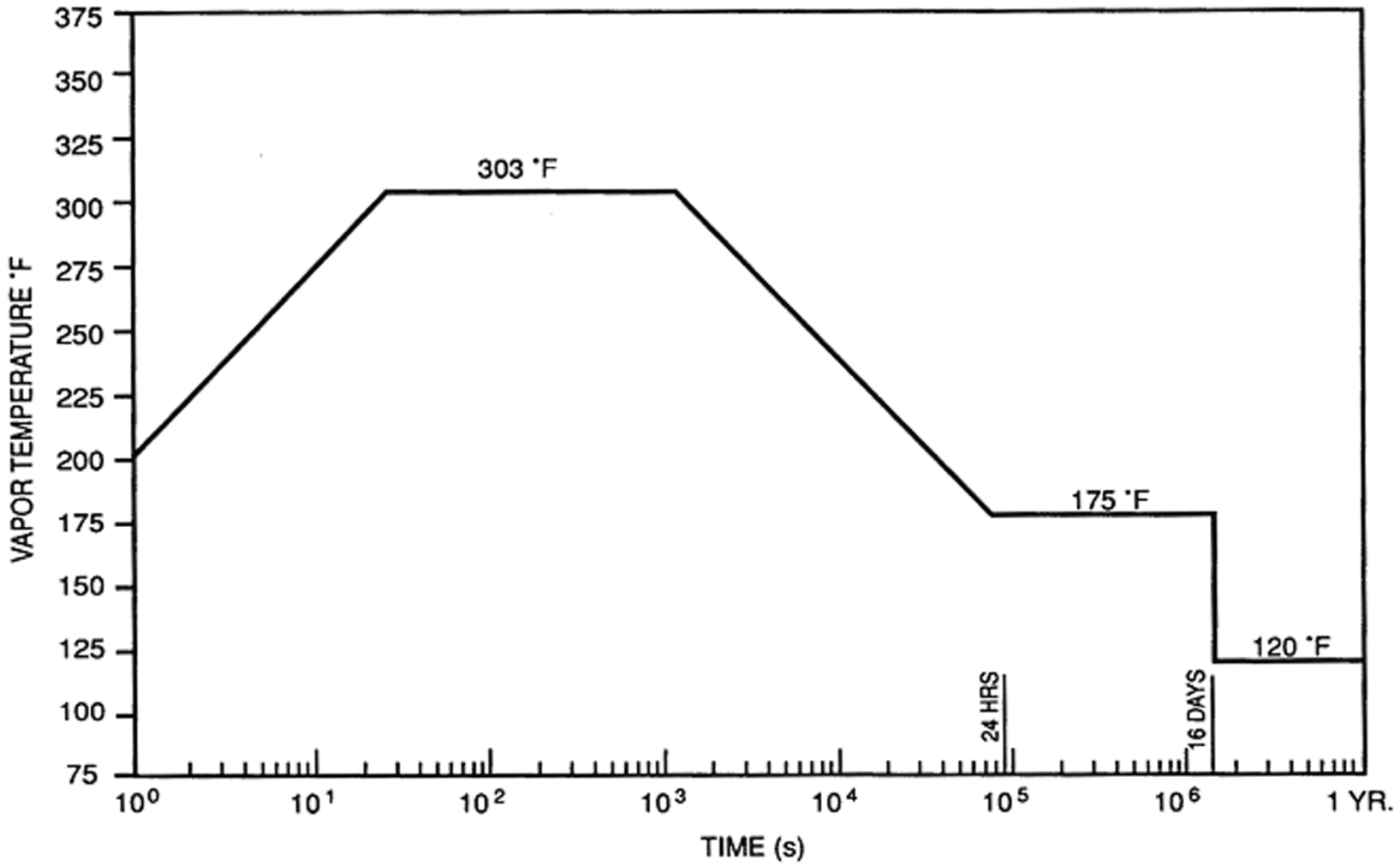
TEMPERATURE CONDITIONS FOR
EQUIPMENT QUALIFICATION
INSIDE MSIV OR MFIV AREA

FIGURE 3.11.B.1–1 (SHEET 7 OF 18)



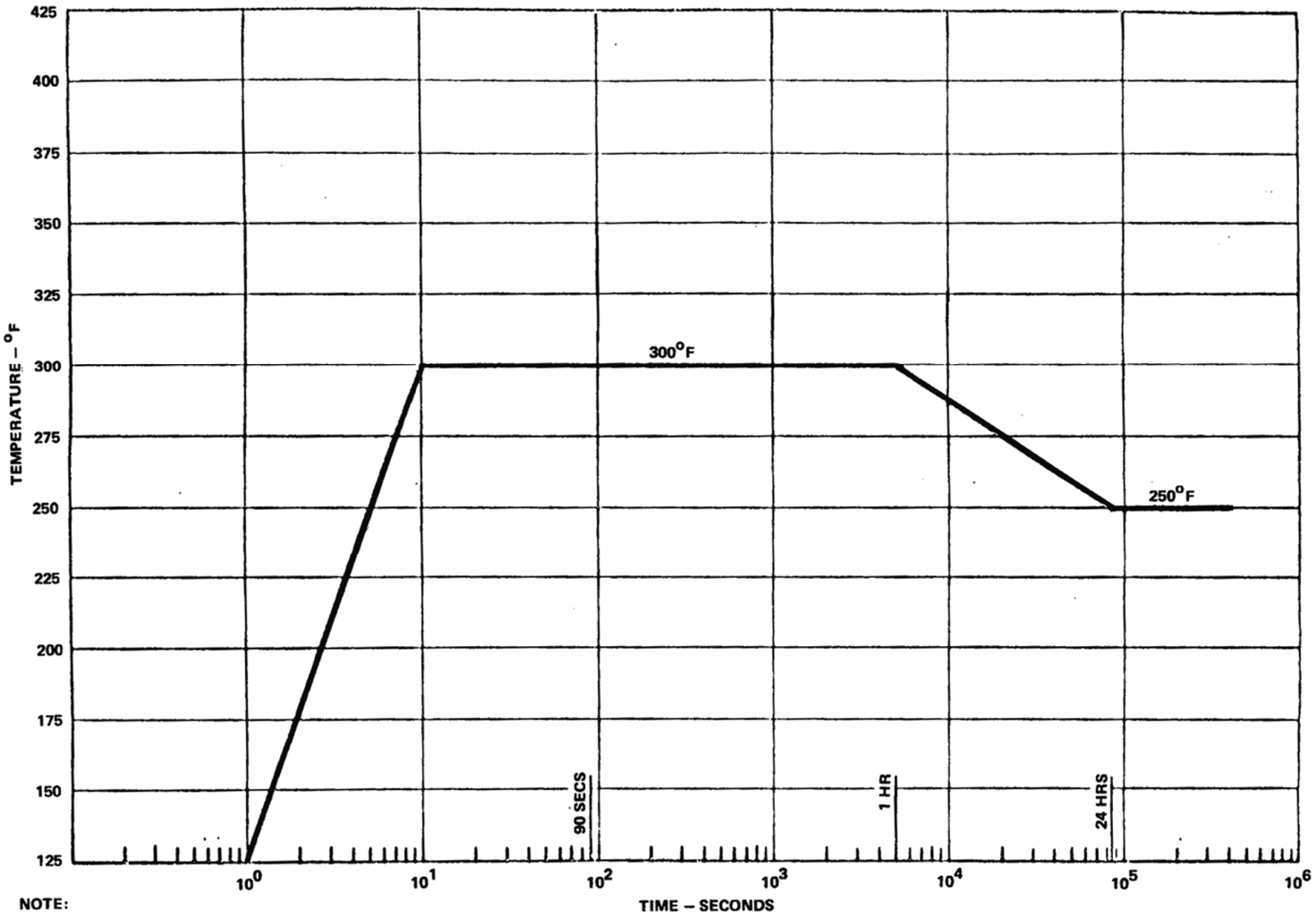
REV 14 10/07

SOUTHERN COMPANY *Energy to Serve Your World®*	**VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	PRESSURE CONDITIONS FOR EQUIPMENT QUALIFICATION INSIDE MSIV OR MFIV AREA
		FIGURE 3.11.B.1–1 (SHEET 8 OF 18)



REV 14 10/07

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	TEMPERATURE CONDITIONS FOR ENVIROMENTAL QUALIFICATION INSIDE CONTAINMENT (VAPOR REGION) FIGURE 3.11.B.1–1 (SHEET 9 OF 18)



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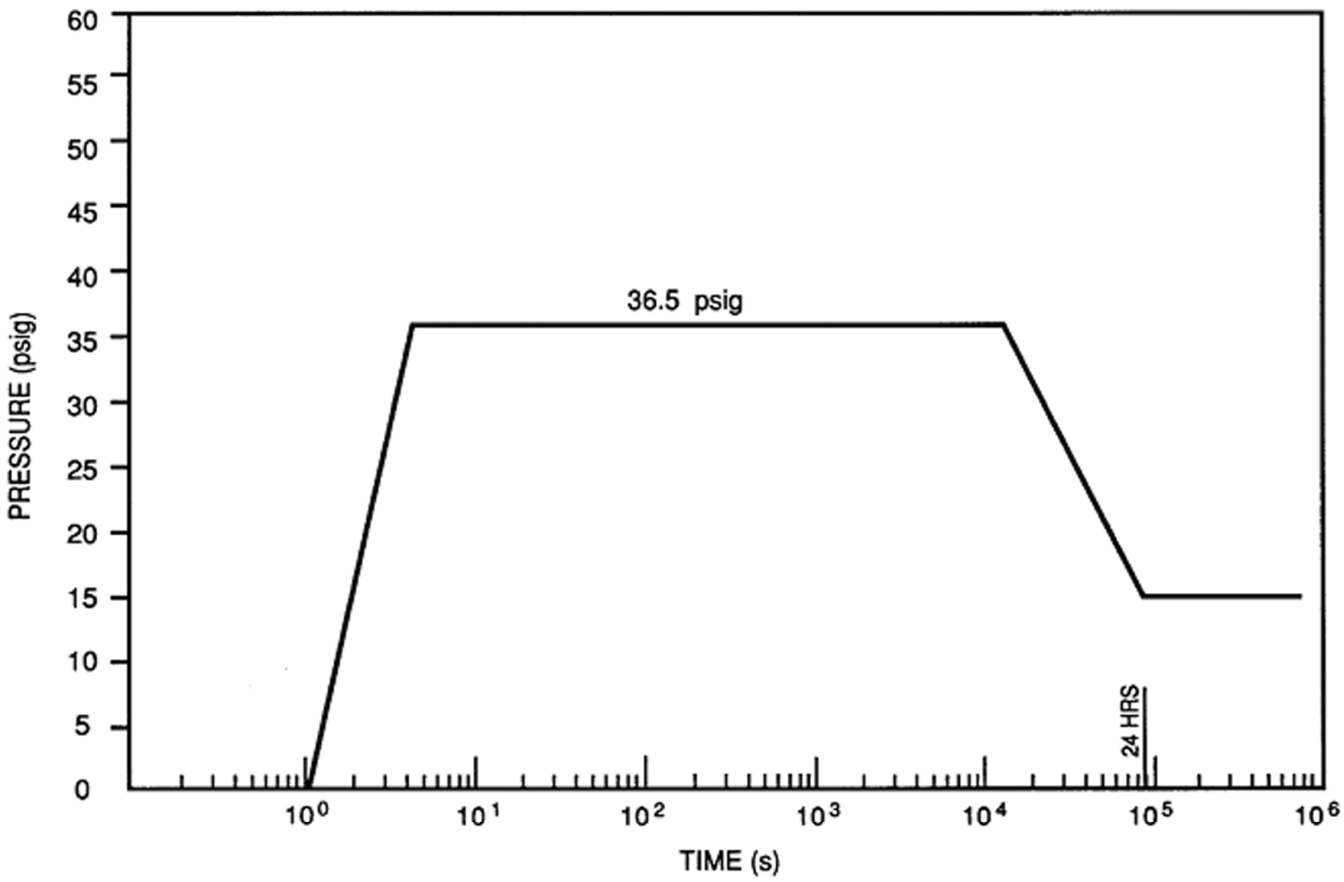
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

TEMPERATURE CONDITIONS FOR
EQUIPMENT QUALIFICATION
INSIDE CONTAINMENT (SUMP REGION)

FIGURE 3.11.B.1–1 (SHEET 10 OF 18)



NOTE:
COMBINED LOCA & MSLB
ENVELOPE CURVE SHOWN

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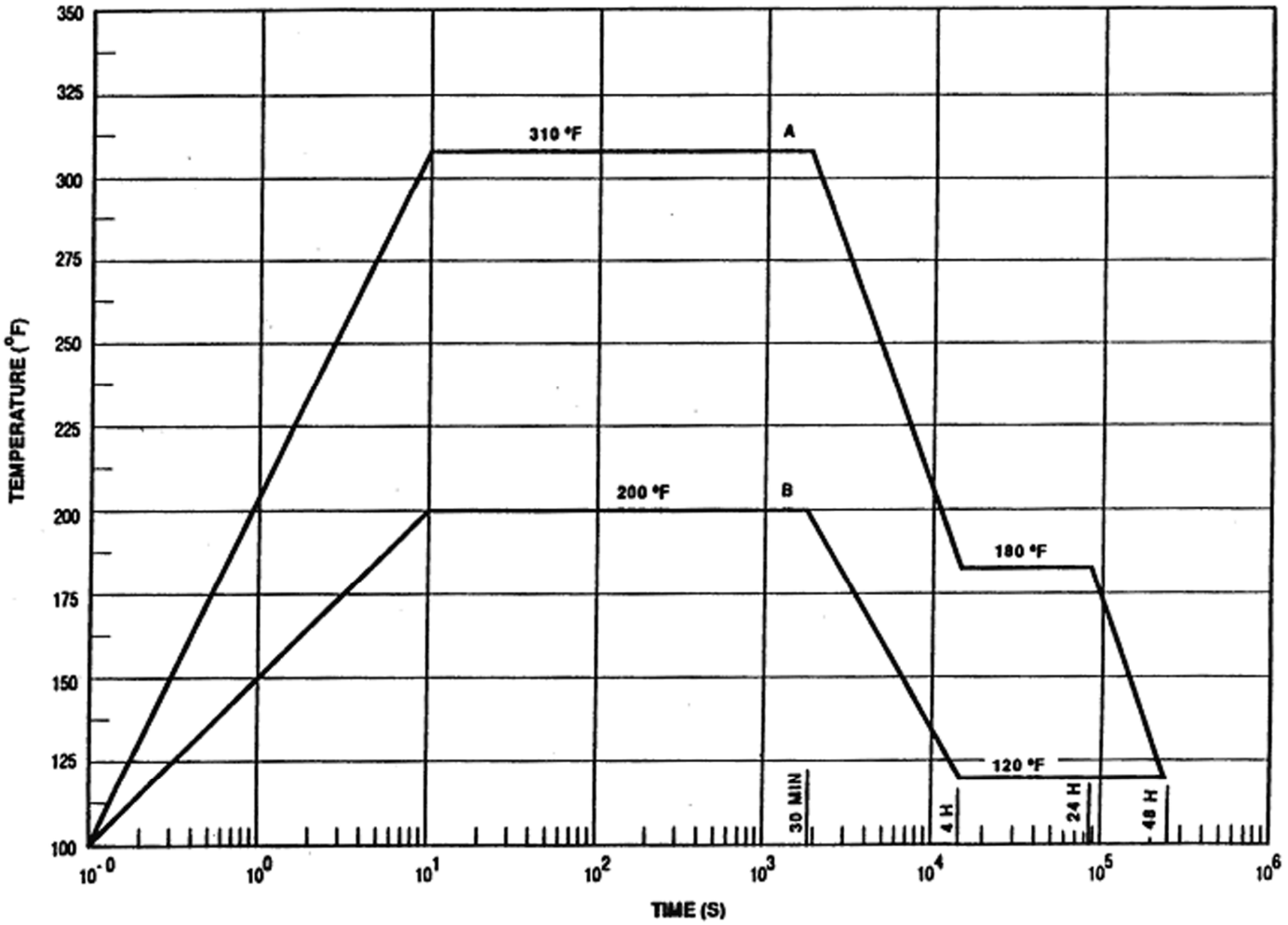
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PRESSURE CONDITIONS FOR EQUIPMENT QUALIFICATION INSIDE CONTAINMENT

FIGURE 3.11.B.1–1 (SHEET 11 OF 18)



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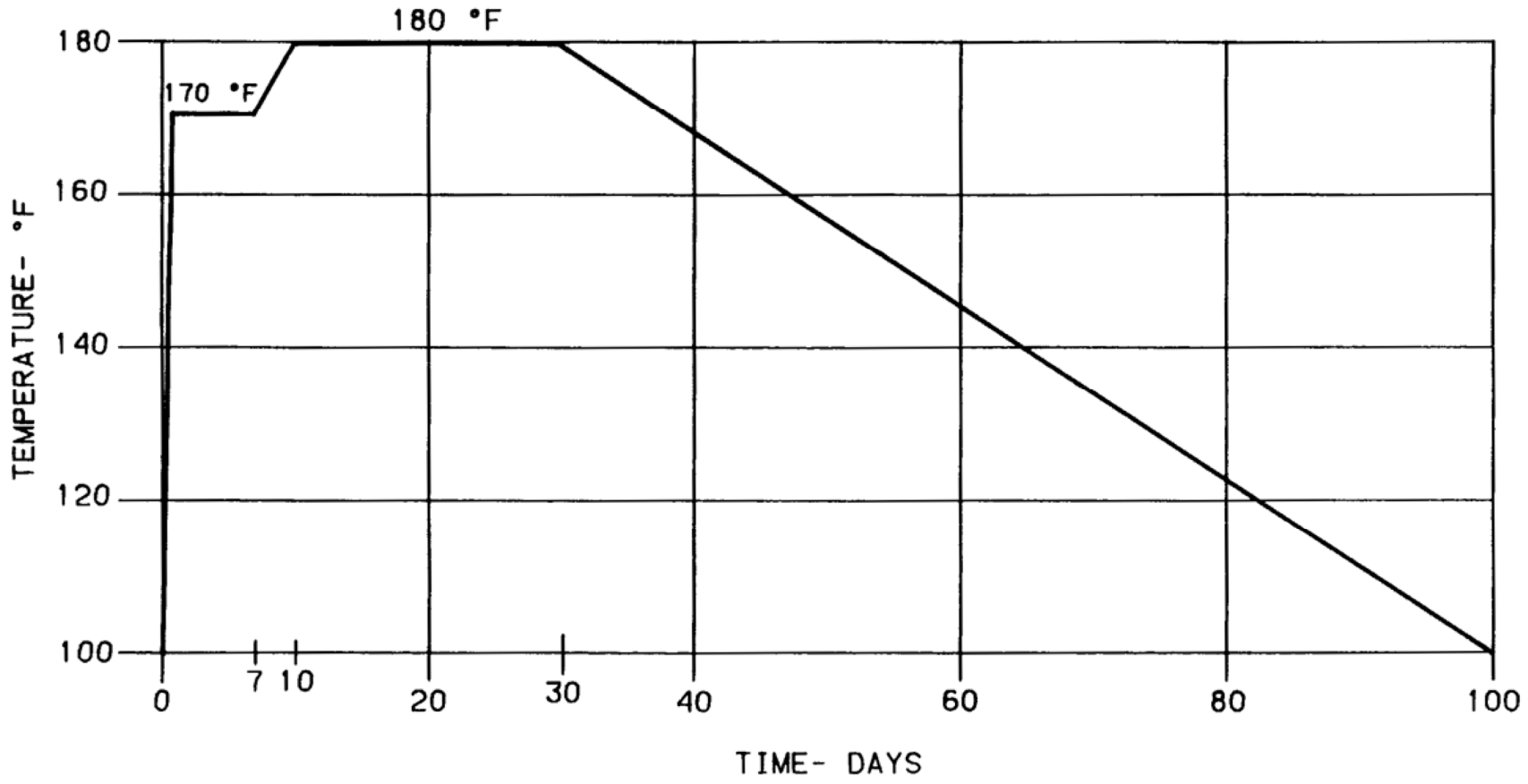
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	TEMPERATURE CONDITIONS FOR EQUIPMENT QUALIFICATION IN THE AUXILIARY BUILDING FIGURE 3.11.B.1–1 (SHEET 12 OF 18)



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SOUTHERN COMPANY
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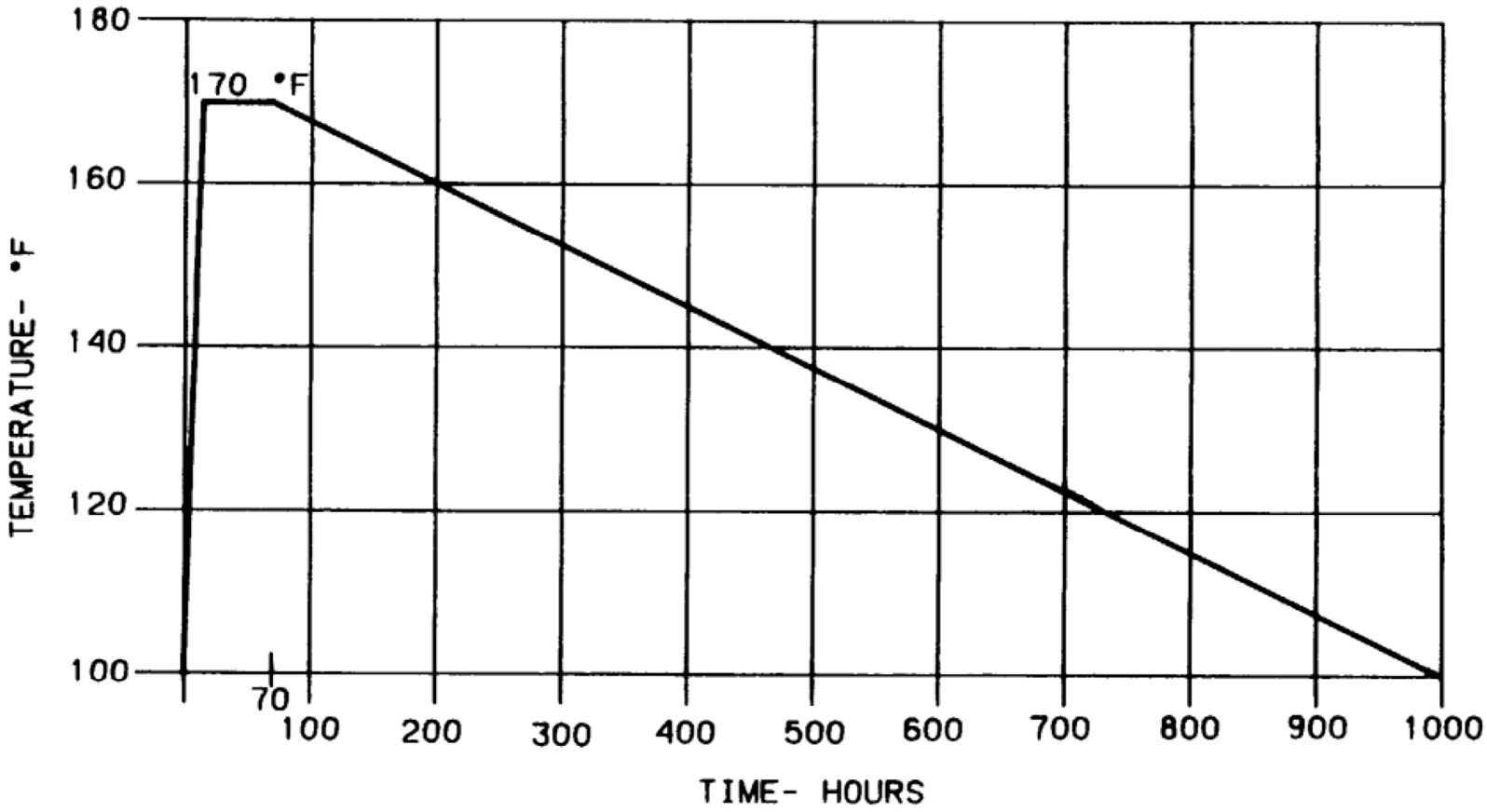
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LOCA TEMPERATURE CONDITIONS
FOR EQUIPMENT QUALIFICATION

FIGURE 3.11.B.1–1 (SHEET 13 OF 18)



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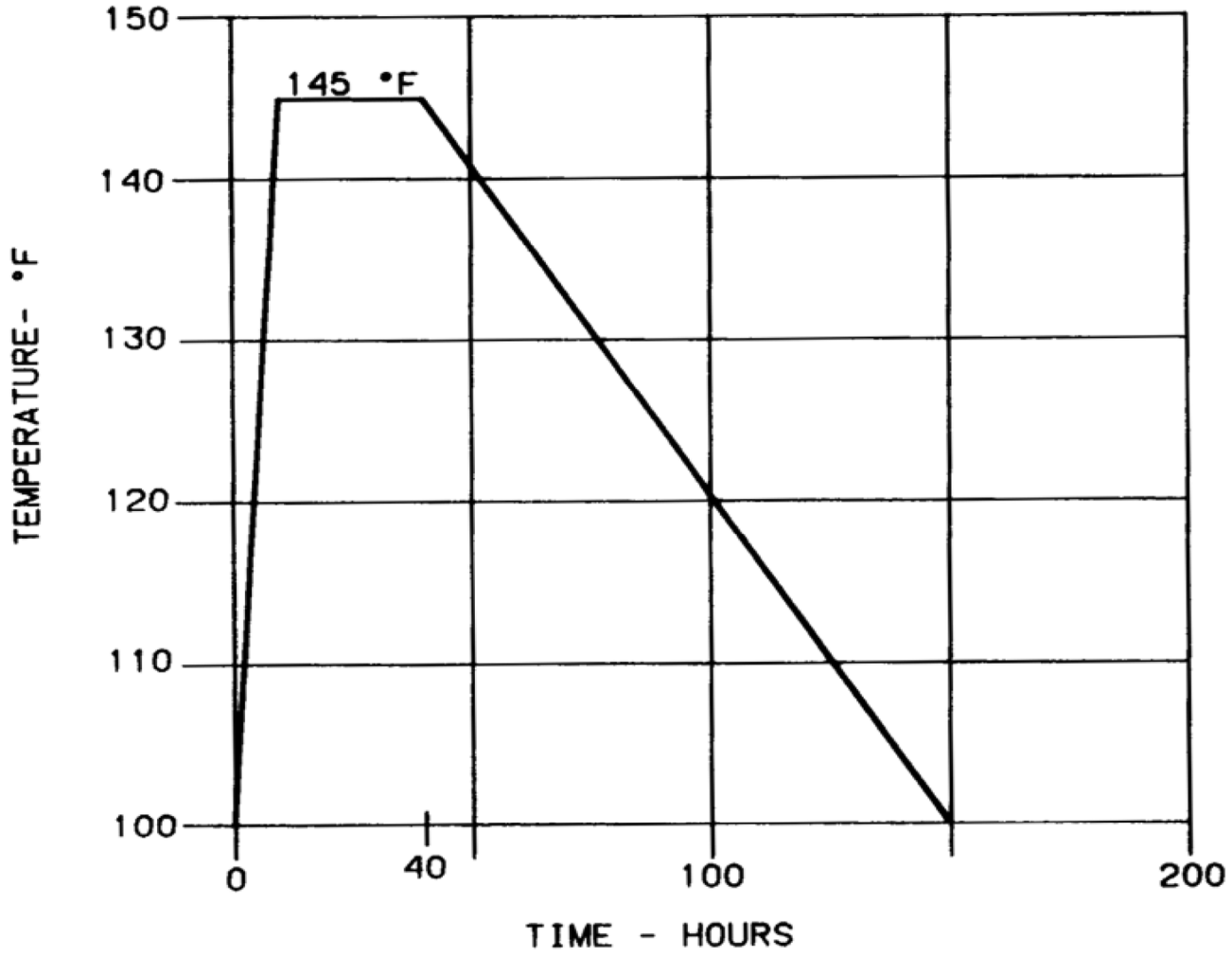
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LOCA TEMPERATURE CONDITIONS
FOR EQUIPMENT QUALIFICATION

FIGURE 3.11.B.1–1 (SHEET 14 OF 18)



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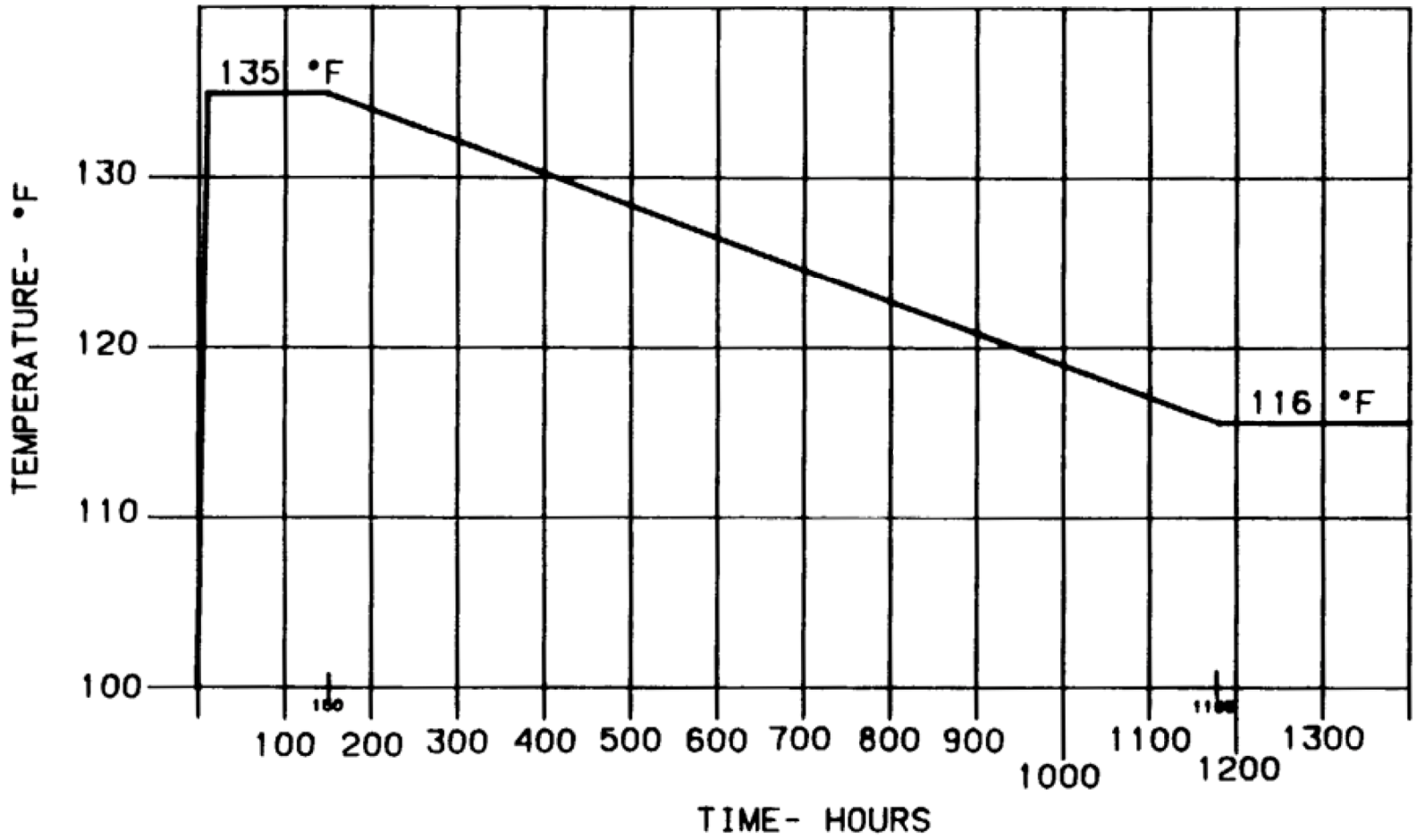
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

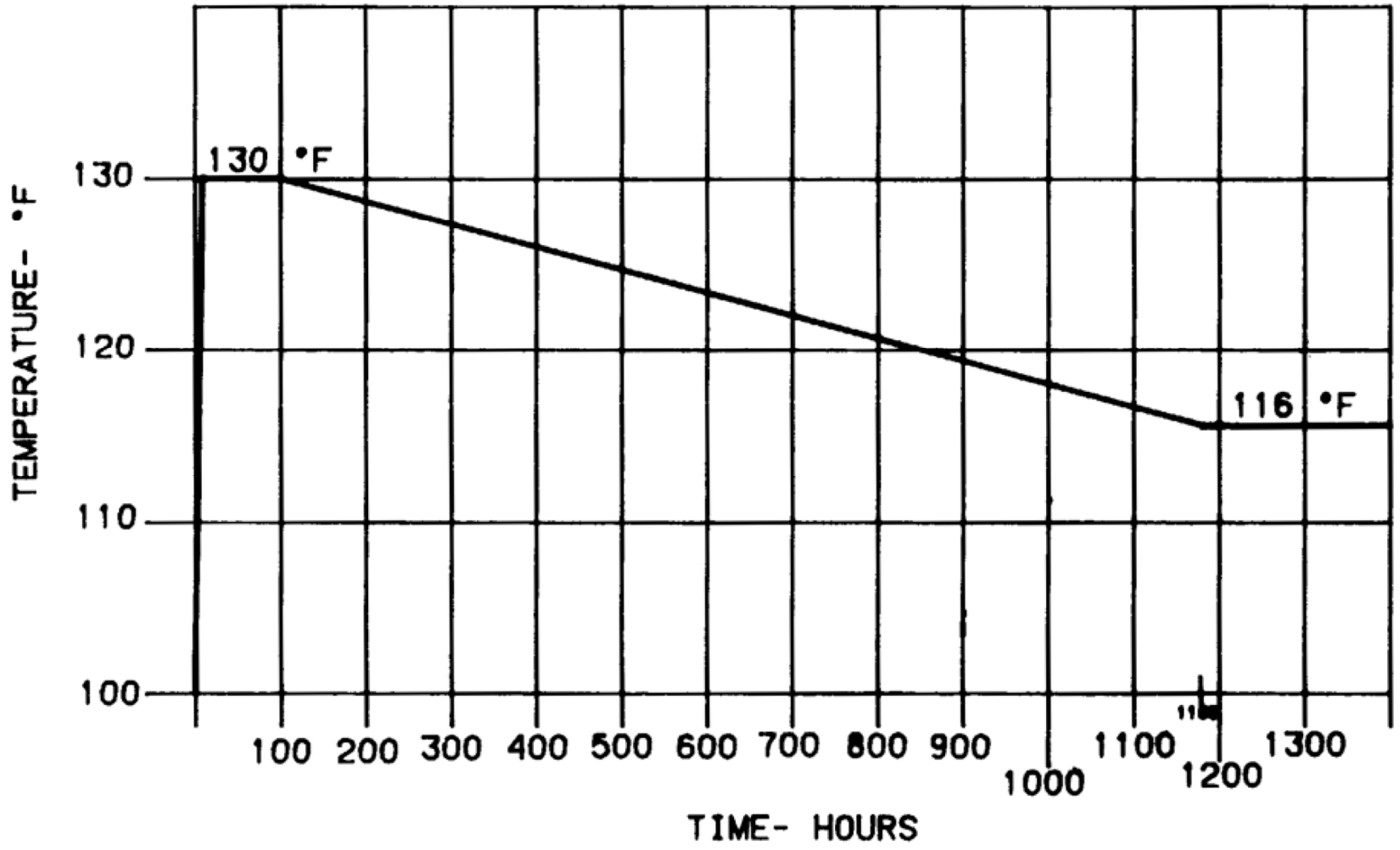
LOCA TEMPERATURE CONDITIONS
FOR EQUIPMENT QUALIFICATION

FIGURE 3.11.B.1–1 (SHEET 15 OF 18)



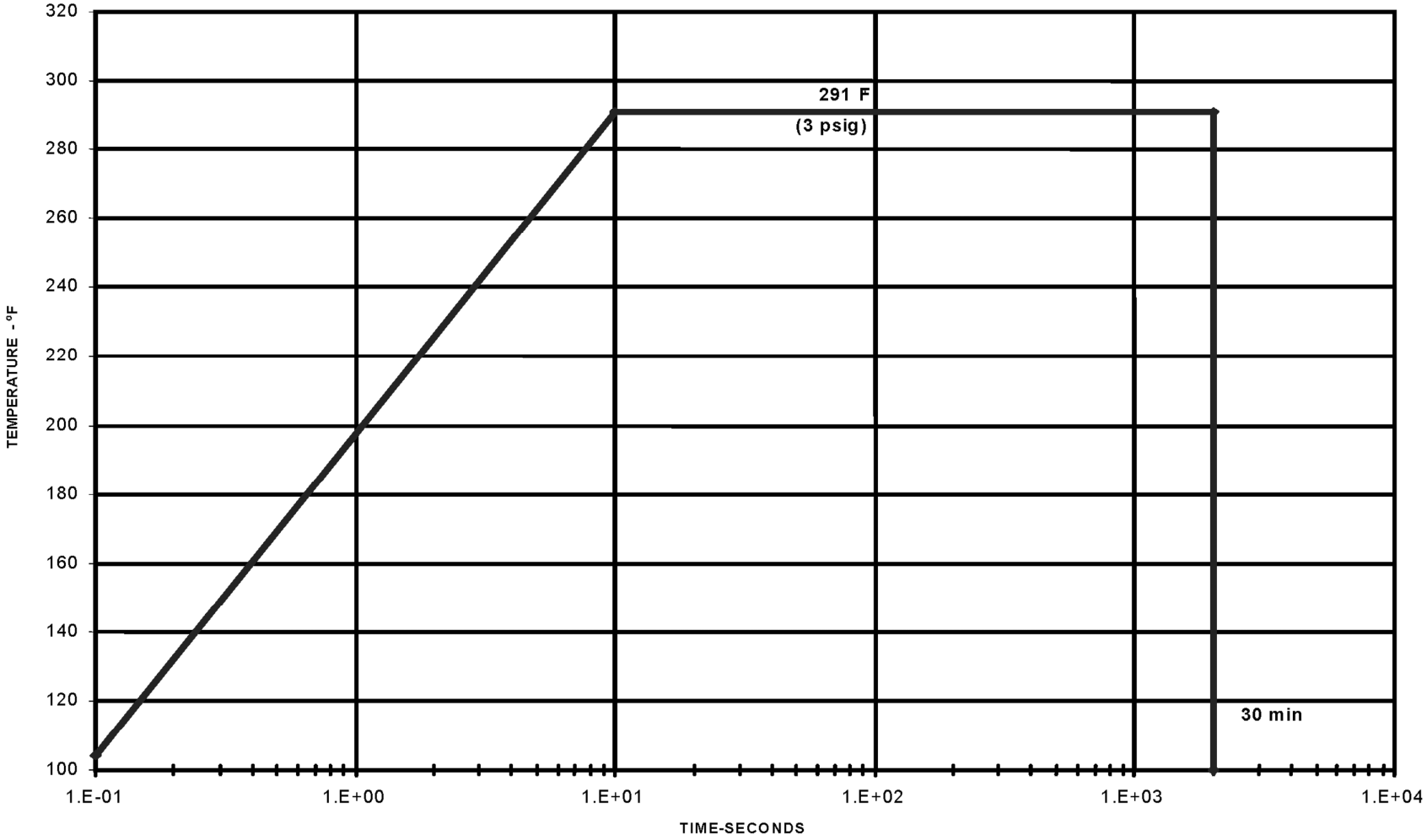
REV 14 10/07

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOCA TEMPERATURE CONDITIONS FOR EQUIPMENT QUALIFICATION
	FIGURE 3.11.B.1–1 (SHEET 16 OF 18)



REV 14 10/07

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOCA TEMPERATURE CONDITIONS FOR EQUIPMENT QUALIFICATION
	FIGURE 3.11.B.1–1 (SHEET 17 OF 18)



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SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

TEMPERATURE CONDITIONS FOR
EQUIPMENT QUALIFICATION IN
THE TURBINE BUILDING

FIGURE 3.11.B.1-1 (SHEET 18 OF 18)

APPENDIX 3A

CONFORMANCE WITH REGULATORY GUIDES

This appendix is not applicable to the VEGP. Refer to section 1.9, Conformance to NRC Regulatory Guides.

APPENDIX 3B

COMPUTER PROGRAMS USED FOR STRUCTURAL, SEISMIC, AND GEOTECHNICAL ANALYSES

3B. COMPUTER PROGRAMS USED FOR STRUCTURAL AND SEISMIC ANALYSES BY BECHTEL POWER CORPORATION

Computer programs are updated under quality control procedures to enhance capabilities and to extend their applicability. As such, earlier versions of these programs, also verified, may have been used during earlier stages of the design effort.

3B.1 BECHTEL CE 201(CE 217), BECHTEL STRUCTURAL ANALYSIS PROGRAM, POST PROCESSOR (BSAP-POST)

A. Description

The Bechtel Structural Analysis Program, BSAP-POST CE 201 (CE 217) is a general-purpose, post-processor program for the BSAP (CE 800) finite-element analysis program. The BSAP-POST program can take the output from BSAP and display this data (graphically and/or on a line printer) or perform additional calculations. In addition, some of the capabilities of BSAP-POST can be used independently. For example, the concrete design module, OPTCON, can have design loads obtained from BSAP output or from punched cards.

The BSAP-POST program consists of a number of modules that can be used independently or sequentially to display or modify the contents of a data base under the control of an executive supervisor program. The data base consists of the contents of a file (TAPE 27) created by a BSAP analysis problem. The executive supervisor ensures that each module in BSAP-POST is compatible with every other module and initiates the execution of each module when required by input data supplied by the user.

B. Validation

The BSAP-POST program has been prepared by Bechtel and has a complete set of documentation, including a user's manual, a verification report, and a theoretical manual. These documents are on file with Bechtel data processing.

C. Extent of Application

The program is used in the analysis and design of structures.

3B.1.1 BECHTEL CE 207 BSAP-DYNAM

A. Description

The BSAP-DYNAM program analyzes soil-structure interaction problems.

B. Validation

The BSAP-DYNAM program has a complete set of documentation, including a user's manual and a verification manual. These documents are on file with Bechtel data processing.

C. Extent of Application

The program is used to determine the modal damping involved with soil-structure interaction. This program has problem-solving capabilities formerly executed by CE 251. (See subsection 3.B.1.4.)

3B.1.2 BECHTEL CE 239 HEMISPHERICAL DOME TENDON ANALYSIS (TENDON)

A. Description

The dome tendon computer program calculates forces and pressures on a hemispherical dome of a prestressed, three-buttress concrete containment building, resulting from prestress by two orthogonal groups of vertical dome tendons and one group of horizontal hoop tendons. One group of vertical dome tendons is located in parallel, vertical planes normal to the x-axis, extending from 90° to 180° azimuth angle. The second group is located in vertical planes normal to the y-axis and extends from zero to 90° azimuth. The third group is located in horizontal planes normal to the z-axis. Each of the vertical dome tendons (the first two groups) has equal areas and equal spacing measured along the spring line. They are anchored at the base of the containment building. The hoop dome tendons have equal areas, but the spacing may either be constant or may vary linearly with the latitude. The hoop tendons extend from the spring line into the dome region up to 45° latitude. Each hoop tendon is anchored at buttresses 240° apart. Successive hoop tendons are anchored at alternate buttresses.

In the analysis, the dome is subdivided into a grid pattern specified by the user. The program calculates the total pressure due to tendon forces at each grid node in the radial direction normal to the dome surface and in the circumferential (hoop or azimuth) and meridional directions. Nodal forces in the hoop and meridional directions are calculated at each node point. The pressures and forces calculated by this program are intended for use as input to a finite-element computer program to determine the stress distribution in the dome.

B. Validation

The TENDON program has a complete set of documentation, including a user's manual, and verification report. The theoretical background is contained in the user's manual. These documents are on file with Bechtel data processing.

C. Extent of Application

The program is used to determine effects of prestressing the tendons in the containment dome.

3B.1.3 BECHTEL CE 251 3D COMPOSITE MODAL DAMPING (GEMD)

A. Description

The GEMD program determines the approximate composite modal damping of a soil-structure interaction system.

B. Validation

The GEMD program has a complete set of documentation, including a user's manual, a verification manual, and a theory manual. These documents are on file with Bechtel data processing.

C. Extent of Application

The program determines the composite modal damping used in the impedance method for seismic analysis.

3B.1.4 BECHTEL CE 450, TURBINE MISSILE PROBABILITY (TURMIS)

A. Description

The TURMIS program computes the damage probability of a nuclear power plant subject to the impact of missiles generated by turbine blades or disks when they fail. The program combines the damage probabilities of all inputted targets to yield a total damage probability. The damage criteria are specified as scabbing or perforation of the concrete barrier, which is predicted by a formula or formulas that can be selected as input.

B. Validation

The TURMIS program has a complete set of documentation including a user's manual, a verification manual, and a theory manual.

Verification is on file with Bechtel data processing.

C. Extent of Application

The program is used to determine the probability of both low- and high-trajectory turbine missile strikes within the power block.

3B.1.5 BECHTEL CE 800, BECHTEL STRUCTURAL ANALYSIS PROGRAM (BSAP)

A. Description

The program performs the static and dynamic analyses of linear, elastic, three-dimensional structures using the finite-element method. The finite-element library contains truss and beam elements, plane and solid elements, plate and shell elements, axisymmetric (torus) elements, and special boundary (spring) elements.

Element stresses and displacements are solved for either applied loads or temperature distributions.

Concentrated loads, pressures, or gravity loads may be applied.

Dynamic response routines are available for solving arbitrary dynamic loads or seismic excitations, using modal superposition. The program can also perform response spectrum and time-history analyses.

B. Validation

The solutions to test problems have been demonstrated to be essentially identical to the results obtained using the following recognized public-domain computer programs:

- EASE — Elastic Analysis Corporation
- STARDYNE — Mechanics Research Incorporated
- MARC/CDC — MARC Analysis Corporation
- ICES/STRUDL — McDonnell-Douglas Automation
- ASKA — Institute fur Statik and Dynamik, Stuttgart, Prof. A. J. Argyris

Agreement has also been established between BSAP program results and the results presented in the ASME Library of Benchmark Computer problems and solutions(1) and in the recognized technical journals. A complete set of documentation, including a user's manual, verification report, and theoretical manual, is on file with Bechtel data processing.

C. Extent of Application

The program is used to perform structural analyses for the majority of steel and concrete structures.

3B.1.6 BECHTEL CE 802, RESPONSE SPECTRA ANALYSIS (SPECTRA)

A. Description

The program computes the response spectra from an acceleration record digitized at equal time intervals. These spectra are plots of the maximum response of a simple oscillator over a range of natural periods and dampings.

The numerical method for computing the spectral values is based on the exact analytical solution of the governing differential equation. It is assumed that the accelerogram varies linearly between the time- history points. The response spectra are constructed by monitoring the maximum values of response parameters of each step of integration. The computed spectra are then widened to account for the effect of structural frequency variations.

B. Validation

The solutions of the program have been verified to be substantially identical with the closed formed analytical solutions of the following three tests problems:

1. Undamped system with a triangular load pulse.
2. Undamped system with a sinusoidal forcing function.
3. Damped system with a sinusoidal forcing function.

A program user's manual, a verification report, and a theoretical manual are on file with Bechtel data processing.

C. Extent of Application

The program is used to develop response spectra curves for all Seismic Category 1 structures.

3B.1.7 BECHTEL CE 982, CONTINUUM LINEAR ANALYSIS FOR SOIL STRUCTURE INTERACTION (CLASSI)

A. Description

The program is capable of evaluating the seismic response of a linear three-dimensional soil-structure interaction model. The analysis capability includes the effects of interaction of each structure and the soil as well as the interaction through the soil among adjacent structures. The method is based on a specialized form of substructuring where the elements of response of the superstructure, foundation, and soil are obtained independently and then are combined to satisfy the interaction conditions.

B. Validation

The CLASSI program has a complete set of documentation, including a user's manual, a verification report, and a theoretical manual. These documents are on file with Bechtel data processing.

C. Extent of Application

The program was used to compute the impedance functions of a layered medium.

3B.1.8 BECHTEL CE 915, A COMPUTER PROGRAM FOR EARTHQUAKE RESPONSE ANALYSIS OF HORIZONTALLY LAYERED SITES (SHAKE)

A. Description

The program computes the responses in a system of homogeneous, viscoelastic layers of infinite horizontal extent subjected to vertically traveling shear waves. The nonlinearity of the shear modulus and damping is accounted for by the use of equivalent linear soil properties, using an iterative procedure to obtain values for modulus and damping compatible with the effective strains in each layer. The program handles systems with variation in both moduli and damping and takes into account the effect of the elastic base.

B. Validation

A complete set of documentation, including a user's manual, verification report, and theoretical manual, is on file with Bechtel data processing.

C. Extent of Application

The program was used to increase the time interval of the Bechtel synthetic time-history accelerograms from 0.005 to 0.01 s.

3B.1.9 BECHTEL ME 351, PIPE FORCE AND WHIP ANALYSIS (PRTHRUST/PIPERUP)

A. Description

The PRTHRUST/PIPERUP program performs a nonlinear elasto-plastic analysis of three-dimensional piping systems subjected to concentrated static or dynamic time-history forcing function. These forces may result from fluid jet thrust at the location of a postulated rupture of high-energy piping. The program is an

adaptation of the finite element method to the specific requirements of pipe rupture analysis. Straight and curved-beam (elbow) elements are used to mathematically represent the piping, and axial and rotational springs are used to represent restraints. The stiffness characteristics of piping and restraints can reflect elastic/linear strain hardening material properties, and gaps between piping and restraints can be modeled.

B. Validation

Verification is on file with Bechtel data processing.

C. Extent of Application

The program was used to perform analysis of piping systems to obtain loads used in design of pipe whip restraints and their backup structures.

3B.1.10 BECHTEL TE 301, TWO-DIMENSIONAL STRUCTURAL ANALYSIS (MFRAME)

A. Description

The MFRAME program performs the analysis of two-dimensional framed structures using the direct stiffness method. It can process structures with beam elements and pin-jointed truss elements or a combination of these. The output consists of joint displacements, member end forces, joint loads, and reactions. The output joint loads are calculated internally from member end forces and therefore serve as a check for the validity of the solution.

B. Validation

The program has been verified, and appropriate documentation is on file with Bechtel data processing.

C. Extent of Application

The program was used for general beam, rigid frame, or truss analysis.

3B.1.11 FLUSH (CONTROL DATA CORP. (CDC) VERSION)

A. Description

This finite-element program uses two-dimensional soil and three-dimensional structure modeling techniques to compute the seismic response of a soil-structure interaction system that accounts for embedment and structure-to-structure effects.

B. Validation

Verification of CDC's version of FLUSH has been performed and appropriate documentation, as defined by CDC policy, is maintained by CDC's Utilities Service Center.

C. Extent of Application

The program is used to seismically analyze deeply embedded Seismic Category 1 structures.

3B.1.12 THE STRUCTURAL DESIGN LANGUAGE (ICES-STRUDL-II MCDONNELL-DOUGLAS AUTOMATION VERSION)

A. Description

The program performs structural analysis. Frame members can be used in conjunction with finite elements. Some special features include a built-in table for rolled steel wide flange shapes, a member selection procedure based upon the American Institute of Steel Construction Specification, a reinforced concrete member design and checking capability, and a dynamic analysis capability.

B. Validation

The program has been verified, and document traceability is available at McDonnell-Douglas Automation.

C. Extent of Application

The program is used to perform structural analysis for the basemat of the auxiliary building and preliminary structural analysis of the nuclear service cooling towers.

3B.1.13 ICES-LEASE (MCDONNELL-DOUGLAS AUTOMATION VERSION)

A. Description

The LEASE (limiting equilibrium analysis in soil engineering) program is a subsystem of ICES which performs stability analysis of arbitrary slopes by the method of slices.

B. Validation

The program has been verified, and document traceability is available at McDonnell-Douglas Automation.

C. Extent of Application

The program was used to determine the factor of safety against sliding of excavated slopes.

3B.1.14 ICES-SEPOL (MCDONNELL-DOUGLAS AUTOMATION VERSION)

A. Description

The SEPOL program is a subsystem of ICES which computes stress and strains in a layered soil system.

B. Validation

The program has been verified, and document traceability is available at McDonnell-Douglas Automation.

C. Extent of Application

The program was used to estimate the settlements of power block structures.

3B.1.15 BECHTEL CE 212 BSAP-PRE

A. Description

BSAP-PRE (CE-212) is an interactive program used to create and edit input data for the BSAP (CE 800) program. The finite element model data can be entered directly by the user or obtained from an external file.

B. Validation

The BSAP-PRE program has been prepared by Bechtel and has a complete set documentation, including a user's manual and validation report. These documents are on file with Bechtel data processing.

C. Extent of Application

The program is used to create input data files for BSAP (CE 800).

3B.1.16 OTHER COMPUTER PROGRAMS USED IN STRUCTURAL ANALYSES

In the course of generating structural design calculations, several programs are used to assist design efforts. These programs are limited in scope and are developed solely to assist the designer in making lengthy, repetitious calculations, thereby saving design efforts. The programs are listed and checked in the project design calculations. These programs are not itemized here because of their simplicity and nature of use.

3B.1.17 REFERENCES

1. Pressure Vessel and Piping - 1972 Computer Programs
2. Verification, ASME Committee on Computer Technology, Pressure Vessel and Piping Division.
3. Wilson, E. L., "SAP-IV - A Structural Analysis Program for Static and Dynamic Response of Linear Systems," University of California, Berkeley, EERC Report No. 73-11, June 1973.

APPENDIX 3C

DESIGN OF STRUCTURES FOR TORNADO MISSILE IMPACT

3C.1 INTRODUCTION

This appendix contains methods and procedures for analysis and design of steel and reinforced concrete structures and structural elements subject to tornado-generated missile impact effects. Structures subject to missile impact, postulated missiles, and other concurrent loading conditions are identified in section 3.5.

Missile impact effects are assessed in terms of local damage and structural response. Local damage (damage that occurs in the immediate vicinity of the impact area) is assessed in terms of perforation and scabbing.

Evaluation of local effects is essential to ensure that protected items would not be damaged directly by a missile perforating a protective barrier or by scab particles. Empirical formulas are used to assess local damage.

Evaluation of structural response is essential to ensure that protected items are not damaged or functionally impaired by deformation or collapse of the impacted structure.

Structural response is assessed in terms of deformation limits, strain energy capacity, structural integrity, and structural stability. Structural dynamics principles are used to predict structural response.

3C.1.1 PROCEDURES

The general procedures for analysis and design of structures or structural elements for missile impact effects include:

A. Defining the missile properties (such as type, material, deformation characteristics, geometry, mass, trajectory, strike orientation, and velocity).

B. Determining impact location, material strength, and thickness required to preclude local failure (such as perforation for steel targets and scabbing for reinforced concrete targets).

C. Defining the structure and its properties (such as geometry, section strength, deformation limits, strain energy absorption capacity, stability characteristics, and dynamic response characteristics).

D. Determining structural response considering other concurrent loading conditions.

E. Checking adequacy of structural design (stability, integrity, deformation limits, etc.) to verify that local damage and structural response (maximum deformation) will not impair the function of safety- related items.

3C.2 LOCAL EFFECTS

Evaluation of local effects consists of estimating the extent of local damage and characterization of the interface force-time function used to predict structural response. Local damage is confined to the immediate vicinity of the impact location on the struck element and consists of

missile deformation, penetration of the missile into the element, possible perforation of the element, and, in the case of reinforced concrete, dislodging of concrete particles from the back face of the element (scabbing).

Because of the complex physical processes associated with missile impact, local effects are evaluated primarily by application of empirical relationships based on missile impact test results. Unless otherwise noted, these formulas are applied considering a normal incidence of strike with the long axis of the missile parallel to the line of flight.

3C.2.1 REINFORCED CONCRETE ELEMENTS

Where concrete exterior walls and roofs are used as barriers to offer missile protection, such walls have a 24-in. minimum thickness, while the roofs are at least 21 in. thick. The concrete has a 28-day compressive strength of at least 4000 psi (91-day strength for concrete containing pozzolan). Where the interior walls and slabs having concrete compressive strength of 5000 psi are used as missile barriers, such walls have an 18-in. minimum thickness, while the slabs are at least 14 in. thick. Therefore, the walls and roofs of these structures are resistant to perforation and scabbing by the postulated missiles discussed in paragraph 3.5.1.4.

3C.2.2 STEEL ELEMENTS

Steel barriers subjected to missile impact are designed to preclude perforation. An estimate of the steel element thickness for threshold of perforation for nondeformable missiles is provided by equation 2-1 (which is a more convenient form of the Ballistic Research Laboratory (BRL) equation for perforation of steel plates with material constant taken as unity[1]).

where:

$$T_p = \frac{(E_k)^{2/3}}{627D} \qquad E_k = \frac{M_m V_s^2}{2} \tag{2-1}$$

T_p = steel plate thickness for threshold of perforation (in.).
E_k = missile kinetic energy (ft-lb).
M_m = mass of the missile (lb-s^2/ft).
V_s = missile striking velocity (ft/s).
D_p = missile diameter (in.).[1]

The design thickness to prevent perforation, t_p, must be greater than the predicted threshold value. The threshold value is increased by 25 percent to obtain the design thickness.

$$t_p = 1.25\, T_p \tag{2-2}$$

where:

t_p = design thickness to preclude perforation (in.).

[1] For irregularly shaped missiles, an equivalent diameter is used. The equivalent diameter is taken as the diameter of a circle with an area equal to the circumscribed contact, or projected frontal area, of the noncylindrical missile. For pipe missiles, D is the outside diameter of the pipe.

3C.3 STRUCTURAL RESPONSE DUE TO MISSILE IMPACT LOADING

When a missile strikes a structure, large forces develop at the missile-structure interface, which decelerate the missile and accelerate the structure. The response of the structure depends on the dynamic properties of the structure and the time-dependent nature of the applied loading (interface force- time function). The force-time function is, in turn, dependent on the type of impact (elastic or plastic) and the nature and extent of local damage.

3C.3.1 GENERAL

In an elastic impact, the missile and the structure deform elastically, remain in contact for a short period of time (duration of impact), and subsequently disengage due to the action of elastic interface restoring forces.

In a plastic impact, the missile or the structure or both may deform plastically or sustain permanent deformation or damage (local damage). Elastic restoring forces are small, and the missile and the structure tend to remain in contact after impact. Plastic impact is much more common in nuclear plant design than elastic impact, which is rarely encountered. For example, test data indicate that the impact from all postulated tornado-generated missiles can be characterized as a plastic collision.

If the interface forcing function can be defined or conservatively idealized (from empirical relationships or from theoretical considerations), the structure can be modeled mathematically, and conventional analytical or numerical techniques can be used to predict structural response. If the interface forcing function cannot be defined, the same mathematical model of the structure can be used to determine structural response by application of conservation of momentum and energy balance techniques with due consideration for type of impact (elastic or plastic).

In either case, in lieu of a more rigorous analysis, a conservative estimate of structural response can be obtained by first determining the response of the impacted structural element and then applying its reaction forces to the supporting structure. The predicted structural response enables assessment of structural design adequacy in terms of strain energy capacity, deformation limits, stability, and structural integrity.

Three different procedures are given for determining structural response: the force-time solution, the response chart solution, and the energy balance solution. The force-time solution involves numerical integration of the equation(s) of motion and is the most general method applicable for any pulse shape and resistance function. The response chart solution can be used with comparable results, provided the idealized pulse shape (interface forcing function) and the resistance function are compatible with the response chart. The energy balance solution is used in cases where the interface forcing function cannot be defined or where an upper limit check on structural response is desired. This method will consistently overestimate structural response, since the resisting spring forces during impact are neglected.

In defining the mass-spring model, consideration is given to local damage that could affect the response of the element. For concrete slab elements, the beneficial effect of formation of a fracture plane which propagates from the impact zone to the back of the slab (back face fracture plane) just prior to scabbing (reference 2) is neglected. The formation of this fracture plane limits the forces transferred to the surrounding slab and significantly reduces overall structural response. Since scabbing is to be precluded in the design, the structural response check is made assuming the fracture plane is not formed. It is recognized, however, that should the missile velocity exceed that for threshold of scabbing, structural response would be limited by this mechanism.

Therefore, the structural response is conservatively evaluated ignoring formation of the fracture plane and any reduction in response.

3C.3.2 STRUCTURAL ASSESSMENT

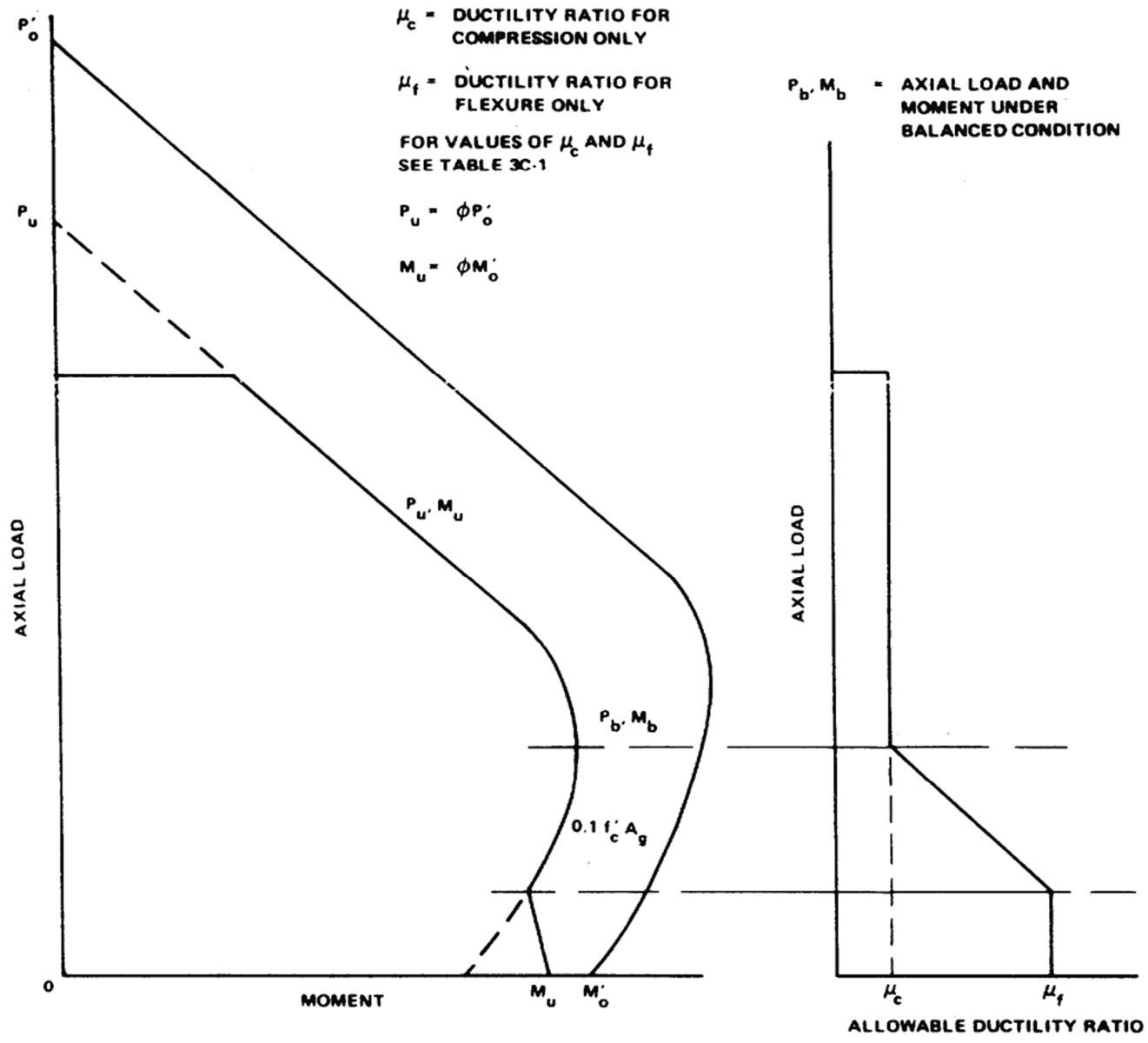
The predicted structural response enables assessment of design adequacy in terms of strain energy capacity, deformation limits, stability, and structural integrity.

For structures allowed to displace beyond yield (elasto-plastic response), a check is made to ensure that deformation limits would not be exceeded, by comparing calculated displacements or required ductility ratios with allowable values (such as those contained in table 3C-1).

3C.4 REFERENCES

1. Gwaltney, R. C., "Missile Generation and Protection in Light-Water-Cooled Power Reactor Plants," ORNL NSIC-22, Oak Ridge National Laboratory, Oak Ridge, Tennessee, for the USAEC, September 1968.

2. Rotz, J. V., "Results of Missile Impact Tests on Reinforced Concrete Panels," Vol 1A, pp 720-738, Second Specialty Conference on Structural Design of Nuclear Power Plant Facilities, New Orleans, Louisiana, December 1975.



(A) REINFORCED CONCRETE INTERACTION DIAGRAM (P VS M)

(B) ALLOWABLE DUCTILITY RATIO μ VS P

REV 14 10/07

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UNIT 1 AND UNIT 2

MAXIMUM ALLOWABLE DUCTILITY RATIO FOR REINFORCED CONCRETE SECTON WITH BEAM–COLUMN ACTION

FIGURE 3C–1

3D.1 SEISMIC RESPONSE SPECTRA

The VEGP seismic response spectra at selected levels of major Category 1 structures are provided in figures 3D-1 through 3D-78.

3D.2 SEISMIC ANALYSIS METHODS

The NRC, in its letter to GPC dated March 27, 1978, accepted the seismic design methods proposed in Preliminary Safety Analysis Report supplements 3 and 4 with the additional information provided in the GPC letter of February 20, 1978, on the scaling factor, subject to the completion of a confirmatory study and sensitivity study.

The NRC also requested that the seismic analysis include consideration of torsional moment no less than that required by the Uniform Building Code (to account for the seismic wave propagation effects), in addition to the effects resulting from the eccentricity between the center of mass and the center of rigidity at each level.

The reports on the confirmatory study and the sensitivity study together with the description of the methodology to account for torsion caused by the seismic wave propagation effects were submitted to the NRC in the GPC letter dated November 13, 1978. The reports on the confirmatory study and the sensitivity study and the write-up on torsion are reproduced in this section as requested during the NRC structural audit on December 4-6, 1984.

3D.2.1 CONFIRMATORY STUDY

3D.2.1.1 Introduction

In Supplement 3 (S3), dated November 1977 and Supplement 4 (S4), dated January 25, 1978, to the Vogtle PSAR, revisions to the seismic analysis methods were proposed by the Georgia Power Company. The NRC, in their letter dated March 27, 1978, stated that the proposed methods were acceptable and requested the Georgia Power Company to perform a study in which the calculated response spectra (in the reactor building and in one other safety-related structure) per Supplements 3 and 4 to the VEGP PSAR together with a scaling factor of 1.5 be compared with the response spectra obtained from the analyses using the lumped parameter representation of the soil and structure. In this report, such a comparison is presented. For this study, the containment (reactor building) and the control building are considered.

3D.2.1.2 Soil-Structure Interaction Analyses Per S3 and S4 Method

The details of the soil-structure interaction analysis procedure for deeply embedded structures can be found in the VEGP PSAR Supplement 3 and Supplement 4. The value of the scaling factor that is used to multiply the "envelop in-structure response spectra" generated for each deeply embedded seismic Category I structure has been documented as 1.5 by Georgia Power Company in their letter dated February 20, 1978, to the NRC. The response spectra from these analyses will be referred to as the "response spectra from the S3 and S4 method."

The important elements involved in the analysis are as follows:

A. 0.2 g (SSE) Regulatory Guide 1.60 design response spectra.

B. Regulatory Guide 1.61 damping values for the structure.

C. The design response spectra are defined for the free field and applied at the finished grade level (el 220 ft 0 in.).

D. The computer program FLUSH is used to perform the soil-structure interaction analyses using a finite element representation of the soil.

E. Strain dependent shear moduli (Figures 3.7-22B, 22C, and 22D of Supplement 3) and strain dependent damping values (Figures 3.7-21, 22, and 22A of Supplement 3) are used for the soil properties.

F. Soil property variation considered. The analyses are performed using the mean values of shear moduli, 1.5 times the mean values and the mean values divided by 1.5. The response spectra obtained from these three analyses are enveloped.

G. The response due to the three components of earthquake is combined using the SRSS technique.

H. The "envelop in-structure response spectra" is multiplied by a scaling factor of 1.5 in order to meet the 60 percent criteria of the Standard Review Plan, Section 3.7.1 - Subsection II.2.

I. The "envelop in-structure response spectra" is then broadened to account for uncertainties in modeling and analysis techniques and smoothed to arrive at the final in-structure response spectra.

3D.2.1.3 Soil-Structure Interaction Analysis Using the Impedance Method

The basic approach used in the impedance method is specified in Chapter 3 of Bechtel Topical Report BC-TOP-4A, "Seismic Analyses of Structures and Equipment for Nuclear Power Plants," Revision 3, dated November 1974. In following the intent of the BC-TOP-4A impedance method, the impedances used in the analyses were computed more appropriately by considering the actual shape of the foundation and the layering of the finished site. The response spectra from these analyses will be referred to as the "response spectra from the impedance method."

The important elements involved in the impedance method of analysis are identified below:

A. 0.2 g (SSE), Regulatory Guide 1.60 design response spectra.

B. Regulatory Guide 1.61 damping values for the structure.

C. The design response spectra are defined for the free field at the foundation level of the structure.

D. To account for the strain dependent nature of shear modulus, the impedance were computed using the iterated shear moduli obtained from the free field column study based on the "mean" soil properties. Based on the fundamental frequency of the soil-structure system, the corresponding impedance coefficients were selected from the frequency dependent impedance functions.

For the vertical analyses, the impedances were computed using a P-wave velocity of 5,000 ft/s for soil layers below the water table. This is compatible with the assumptions made in the FLUSH analysis, that in saturated soils, the P-wave

would travel with the P-wave velocity of the soil medium or the P-wave velocity of water, whichever is greater. The P-wave velocity of water has been assumed to be 5000 ft/s.

E. The embedment effects and structure-to-structure interaction effects are not considered. Neglect of embedment effects would tend to increase the magnitude of responses within the structure.

F. The response due to the three components of earthquake are combined using the SRSS technique.

G. The in-structure response spectra is broadened to account for uncertainties in modeling and analysis techniques and smoothed to arrive at the final in-structure response spectra.

3D.2.1.4 Comparison of Response Spectra from the S3 and S4 Method and the Impedance Method

A. Containment NRC in their letter dated March 27, 1978, requested that comparison be made at certain levels of the containment. These levels with their corresponding elevations are identified below:

- Top of base mat — el 169 ft 0 in.
- Operating floor level — el 220 ft 0 in.
- Reactor support level — el 179 ft 8 1/2 in.
- Steam generator support level — Vertical - el 171 ft 9 in.
Lower Lateral - el 194 ft 11 in.
Upper Lateral - el 217 ft 4 3/4 in.
- Polar crane girder — el 323 ft 0 in.

The response spectra at elevations 169 ft 0 in., 195 ft 0 in. (internal structure), 220 ft 0 in. (internal structure), and 323 ft 0 in. (exterior shell) would cover all of the above locations. The comparisons between the response spectra curves obtained for these levels from the S3 and S4 method and the impedance method are shown in figures 3D-79 through 3D-90.

B. Control Building. In the control building, the following levels are selected for comparison purpose.

- Top of base mat — el 180 ft 0 in.
- Control room floor — el 220 ft 0 in.
- Mechanical equipment room floor — el 260 ft 0 in.

The comparison between the response spectra curves obtained for these levels from the S3 and S4 method and the impedance method are shown in figures 3D-91 through 3D-99.

C. The comparison of the response spectra obtained from the S3 and S4 method, in which a scaling factor of 1.5 had been used, with those obtained from the impedance method, shows that the spectra from the S3 and S4 method, in general, envelop the spectra from the impedance method. The differences between the response spectra obtained from the two different methods could be attributed to the following.

1. Embedment effects

   In the calculations for the impedance method, embedment effects are neglected even though the structures are embedded in the soil to a certain depth. In the S3 and S4 method, where the FLUSH program is used to model the soil as an assemblage of finite elements, the structure is modeled in its embedded condition. As a result, the response spectra from FLUSH indicate a higher system frequency and lesser amplification due to the increased stiffness and damping caused by the embedment.

2. Structure to structure interaction effects

   In the S3 and S4 method, adjacent structures are modeled together in a FLUSH model and as such the resulting response spectra account for possible structure-soil-structure interaction effects. In the impedance method, the analysis is performed on the isolated structure and hence the effect of the adjacent structures is not considered. The effects of the structure-to-structure interaction are implicitly reflected in the FLUSH response spectra in terms of modifications of response and system frequencies, if such effects are significant.

3. Input motion

   In the S3 and S4 method, the design ground motion, (0.20 g, Regulatory Guide 1.60 spectra) is defined in the free field at the finished grade level, whereas in the impedance method, the design motion is assumed to be the free field motion at the base of the soil springs. Further, in the S3 and S4 method the resulting response spectra from FLUSH are multiplied by a scaling factor of 1.5.

4. Frequency independent impedances

   In the impedance method, the frequency independent impedances are selected based on the strain dependent shear moduli obtained from the soil column study in which the design motion is specified at the finished grade level.

5. Methodology

   In addition to the above, there are differences in the assumptions made in the finite element approach and the impedance approach to solving the soil-structure interaction problem.

3D.2.1.5 Conclusions

The comparison of the response spectra obtained for the containment and control building using the S3 and S4 method and the impedance method shows that, in general, the response spectra from the S3 and S4 method and those from the impedance method show similar characteristics

despite the fact that there are differences in the methods of modeling and analysis. It is therefore concluded that the seismic analysis performed in accordance with S3 and S4 would form a conservative design basis.

3D.2.2 SENSITIVITY STUDY

3D.2.2.1 Introduction

In accordance with your request, a series of ground response studies have been performed to determine the computed variation of ground motion with depth at VEGP site. The analyses were made for two conditions:

A. A series of seven analyses were made in which recorded rock motions were used to establish bedrock motions, which were then propagated upward through the site soils from the underlying bedrock.

B. A series of eight analyses were made in which recorded motions on soil deposits with generally similar characteristics to those at the VEGP site were deconvolved through the soil profile.

The computed responses from the two procedures were compared with each other and with the results of the seismic motions proposed for use in the design of the plant. In all cases, the ground surface motions were scaled to a peak ground surface acceleration of 0.2 g to facilitate the comparisons outlined above.

3D.2.2.2 Soil Profile

The soil profile at the VEGP site is shown in figure 3D-100. After excavation and backfilling to plant grade, the soils will consist of about 100 ft of sand, underlain by about 65 ft of marl and then further sand extending down to very large depths. The actual depth of bedrock is not known but for the purposes of this study it was considered to be at about 600 ft below the ground surface and to have a shear wave velocity of about 3000 ft/s. Test data have shown that the shear wave velocity in the lower sands achieves a value of 1800 ft/s at a depth of 200 to 300 ft and assuming that the nature of the sand remains the same, these results would indicate that it would be likely to increase to a value of about 2600 ft/s at a depth of 600 ft due to the increase in overburden pressure alone. On this basis, a depth to rock of about 600 ft was considered reasonable. Increases in depth of 100 or 200 ft above the selected value would have very little influence on the results of the analysis in the upper few hundred ft of the deposit.

Soil properties assigned to the various layers shown in figure 3D-100 were those previously established by tests for the various layers and used in previous VEGP reports.

3D.2.2.3 Ground Motions

For the purpose of the analyses, eight accelerograms recorded on deep soil sites similar to the VEGP site were chosen for deconvolution studies. These motions were also chosen because the peak ground accelerations were reasonably close to the SSE acceleration of 0.2 g selected for the VEGP site. The eight records selected for these studies are listed as Analyses D-1 to D-8 in table 3D-1. They include records from the El Centro site in the El Centro earthquakes of

1934 and 1941, records from the Athenaeum site at the California Institute of Technology and the Holiday Inn site on Orion Boulevard in the San Fernando earthquake of 1971, two records recorded at the Olympia Highway Laboratory in different earthquakes, and records from Ferndale and Humboldt Bay. Details of the recording stations and the particular earthquakes producing the records are given in table 3D-1. In all cases, the record selected was scaled to have a peak acceleration of 0.2 g and used as a surface control motion for a deconvolution analysis on the basis of vertical shear wave propagation. Spectra for the ground surface motions obtained in this way are shown in figures 3D-101 through 3D-108.

The other seven analyses were made using seven different recorded rock motions to determine base rock excitation and the resulting response of the deposit due to upward wave propagation. For this purpose, the rock motions were considered to develop at an outcrop close to the site, the motions were then used to determine the motions at the base of the 600 ft soil deposit, and the motions throughout the deposit were computed for these conditions. The computer program SHAKE was used for all analyses.

The rock motions used for these studies were selected from three different earthquakes and are listed as analyses U-1 to U-7 in table 3D-2. They include records from the Temblor Station and San Luis Obispo in the Parkfield earthquake of 1966, three records from the C.I.T. Seismological Laboratory, Lake Hughes Station No. 4, and Griffith Park in the San Fernando earthquake of 1971, a record from the Taft Station in the Kern County, California earthquake of 1952, and a record from the Castaic Station in the San Fernando earthquake of 1971.

The original records from these stations were initially scaled to a peak rock acceleration of 0.2 g and the resulting ground surface motions computed at representative levels throughout the soil profile. Values of peak ground surface accelerations obtained in this way ranged from 0.2 g to 0.28 g. The computed motions were therefore scaled proportionally to produce a peak ground surface acceleration of 0.2 g. The low scaling factors used in this latter operation were not considered to have any significant influence on the results which might have been obtained if repeated trials using different levels of input rock accelerations had been used until each analysis had produced a computed peak ground surface acceleration of 0.2 g.

Response spectra for the scaled rock motions (producing 0.2 g peak acceleration at the ground surface as described above) are shown in figures 3D-109 through 3D-115. It was recognized that two of the motions used (the Taft and Castaic records) were not truly rock records but they were recorded on shallow depths of stiff soil and have essentially the same characteristics of rock records. It was also thought that these two records might simulate motions in the base rock if the actual depth of the soil profile were somewhat deeper than the value of 600 ft used in the analyses.

3D.2.2.4 Variation of Peak Ground Acceleration with Depth

The computed variations of peak ground acceleration with depth determined by the eight deconvolution analyses described above are shown in table 3D-3 and those for the upward wave propagation analyses from rock in table 3D-4. As described above, all analyses were adjusted to produce a peak ground surface acceleration of 0.2 g, providing a consistent basis for comparison. The results show some variations of peak acceleration with depth within the soil profile but overall no major differences. A comparison of the mean values determined by the deconvolution analyses with those determined by the upward wave propagation analyses is shown in figure 3D-123. Again it may be seen that the variations of mean peak acceleration with depth for the two separate studies are generally similar.

3D.2.2.5 Comparison of Ground Surface Motions with those Developed at 76 ft Depth

For each of the 15 analyses, comparisons of the response spectrum for the ground surface motions with that for the computed motions at a depth of 76 ft in the soil profile (the foundation level for the containment building) are shown in figures 3D-101 through 3D-108 and figures 3D-116 through 3D-122. Figures 3D-101 through 3D-108 show the results of the deconvolution studies while figures 3D-116 through 3D-122 show the results of the upward propagation studies. It is readily apparent that all of the analyses show a substantial reduction of the intensity of shaking developed at a depth of 76 ft and that the reduction is generally comparable for each of the analyses.

To provide a collective basis for assessing the significance of these results, computations were made to determine the mean plus 1 standard deviation spectral shape for the suite of motions determined by the eight deconvolution analyses and separately for the seven upward wave propagation analyses for (1) the ground surface motions, (2) the motions computed for a depth of 40 ft below the ground surface and (3) the motions computed for a depth of 76 ft below the ground surface. The results of these studies are shown in figures 3D-124 and 3D-125. It may be noted that the spectra for the ground surface motions are reasonably uniform over the period range from about 0.2 to 0.5 s but the spectra for the motions at depths of 40 and 76 ft contain significant valleys at periods of about 0.18 and 0.3 s in both cases. This reflects the influence of wave propagation effects at the ground surface.

It is interesting to note that the mean + 1σ spectrum for the suite of eight scaled ground surface motions and the seven computed ground surface motions determined in this sensitivity study are reasonably similar to that specified by Regulatory Guide 1.60, as evidenced by the comparison in figure 3D-126. However both spectra are somewhat lower than the Regulatory Guide 1.60 spectra for frequencies higher than about 5 Hz reflecting the filtering effects of high frequencies which tends to occur in deep soil deposits. As a result the use of the Regulatory Guide 1.60 spectrum as a design basis may be considered conservative in such cases. Finally, as shown by the comparative spectra in figure 3D-127, the mean + 1σ spectrum determined for the two groups of computed motions at 76 ft depth in this analytical study, i.e., eight deconvolution analyses of suitably scaled ground surface motion records and seven ground response analyses using suitably scaled rock motions) are very similar in shape to each other and to the computed spectrum at 76 ft depth obtained in previous design studies for the VEGP obtained by deconvolution of an artificial accelerogram representative of the Regulatory Guide 1.60 spectrum.

3D.2.2.6 Conclusion

The good agreement between the results of the ground motion sensitivity studies described above and the results obtained by deconvolution of the Regulatory Guide 1.60 spectrum supports the use of the latter spectrum and motions obtained by deconvolution of it for the design of the VEGP. On this basis it seems reasonable to conclude that the seismic design procedures previously proposed by Bechtel for the design of the VEGP provide a suitability conservative basis for design.

3D.2.3 METHODOLOGY TO ACCOUNT FOR TORSION CAUSED BY THE SEISMIC WAVE PROPAGATION EFFECTS

3D.2.3.1 Introduction

The NRC in their letter dated March 27, 1978, to Georgia Power Company requested that "all safety-related structures, systems and components be designed to resist a static seismic torsional moment which is not less than that required by the latest edition of the Uniform Building Code." Further, the NRC stated that this seismic torsional moment be included as a separate moment which is independent of that resulting from the eccentricity between the center of mass and the center of rigidity of the safety-related structures.

3D.2.3.2 Safety-Related Structures

The seismic analyses of the structures are performed on the three dimensional structure models that account for the eccentricities between the centers of mass and the centers of rigidity of the structures. The accelerations obtained from these analyses at all levels are first calculated. In the design, then the actual eccentricity is increased by 5 percent of the maximum plan dimension at that level and the design static seismic torsional moment is computed as the product of the augmented eccentricity and the story shear.

3D.2.3.3 Safety-Related Equipment, Systems, and Components

Since the intent of the NRC's request on the 'additional torsional requirements' is to account for the torsional motion imparted to the structure due to the effects of seismic wave propagation, it is to be noted that this would also impact the horizontal in-structure response spectra used for equipment qualification. The procedure used to obtain the effect of this torsional ground motion is described below.

A three dimensional lumped parameter model of the structure with "soil springs" is utilized to compute the torsional spectra. The structure model accounts for the eccentricities between the centers of mass and the centers of rigidity of the structure. The translational as well as the rotational (including torsional) stiffness and inertial characteristics are modeled. The foundation impedances consist of three translational (two horizontal and the vertical) and three rotational (two rocking and the torsional) springs, and are based on the "mean" soil properties.

The model is analyzed for the design horizontal ground motion time history conforming to the Regulatory Guide 1.60 horizontal response spectra applied in the free-field at the foundation level of the structure. The base shear computed from this analysis when multiplied by 5 percent of the maximum plan dimension at the foundation level yields the incremental static torsional moment (T_s) at that level.

Then a torsional ground motion time history conforming to the Regulatory Guide 1.60 horizontal response spectra is applied again in the free field at the foundation level of the structure. The maximum dynamic torsional moment (T_d) at the base of the structure is computed from this dynamic analysis.

The magnitude of the torsional ground motion is adjusted such that T_d at the base of the structure resulting from the torsional ground motion analysis is equal to the T_s resulting from the 5-percent eccentricity. The resulting time history response from the torsional degree of freedom of the base node would then represent the torsional response of the basemat. Multiplying this

by the distance along the N-S/E-W direction of the extreme point in the building to the lumped mass node would give the maximum possible additional E-W/N-S horizontal time history response of the floor. From this, the "additional horizontal in-structure response spectra" can be computed.

Now that the magnitude of the torsional ground motion has been established, the torsional responses of the nodes at different levels of the building from the torsional ground motion analysis are used with the respective "extreme point distances" along the N-S/E-W direction to compute the "additional horizontal in-structure response spectra" at these levels.

The amplification of the torsional response of the structure as a function of height tends to be smaller than the amplification of the horizontal response of the structure. Therefore, as an added conservatism, the torsional input ground motion is increased such that the ratio between the maximum torsional acceleration at a given node (caused by the torsional ground motion) to the maximum horizontal acceleration at that node (caused by the horizontal ground motion) is maintained the same as at the foundation level of the structure.

The computed "additional horizontal in-structure response spectra" to account for the torsional ground motion effects are added to the horizontal in-structure response spectra obtained using the methods specified in VEGP PSAR Supplements 3 and 4, before the broadening of the peaks and smoothing of the curves are done. The peaks of the response spectra resulting from the addition of these two spectra are then broadened and the curves smoothed to arrive at the "final design in-structure response spectra" for the horizontal direction.

In the computation of the "additional horizontal in-structure response spectra" to be used for equipment mounted on a specific location, the actual distance of this location from the lumped mass node may be used instead of the "extreme point distance" at that level.

3D.3 SEISMIC RESPONSE SPECTRA COMPARISON

This section is provided in response to the request made by the NRC during their Structural Audit and Design Review on December 4, 1984. Specifically, the NRC requested that a comparison of the VEGP design in-structure response spectra with the impedance method response spectra provided in the Confirmatory Study (section 3D.2) be made.

The basis for the generation of curves identified as impedance method is described in the confirmatory study. The basis for the generation of the design in-structure response spectra is described in subsection 3.7.B. The horizontal design response spectra are an envelop of the E-W and N-S horizontal response spectra. A comparison of these response spectra is provided in figure 3D-128 through 3D-148.

The comparison of the design in-structure response spectra with the response spectra obtained using the impedance method shows that, in general, the design response spectra envelope those from the impedance method. Specifically, the following observations are made:

- The comparison of the horizontal response spectra curves shows that in the frequency range of 2 cps and higher, the design spectra acceleration values are significantly higher than the values obtained from the impedance method demonstrating the added conservatism in the design of structures, equipment, and systems. The slight difference in the frequency range below 2 cps is attributed to the fact that embedment effects were not considered in the impedance method (see section 3D.2). Embedment effects increase the fundamental soil-structure system frequency and, due to the increased damping, lower the response. Thus, it is expected that inclusion of the embedment

effects in the impedance method would result in spectra that will be enveloped by the design criteria.

- The comparison of the vertical response spectra curves shows that overall, the design spectra acceleration values are significantly higher than the values obtained from the impedance method. The minor difference in certain frequency range exhibited in figure 3D-130 becomes insignificant if the comparison is made with the impedance method spectra before peak broadening. It should be noted that peak broadening need only be performed for the development of design spectra to account for uncertainties in modeling techniques.

In conclusion, the seismic analysis performed in accordance with VEGP methodology has resulted in conservative design of structures, equipment, and systems.

TABLE 3D-1
RECORDS USED FOR DECONVOLUTION ANALYSES

Analysis No.	Recording Station	Earthquake
D-1	C.I.T. Athenaeum	San Fernando, 1971
D-2	El Centro	El Centro, 1934
D-3	El Centro	El Centro, 1940
D-4	Holiday Inn (Orion Blvd)	San Fernando, 1971
D-5	Humboldt Bay	Ferndale, 1975
D-6	Ferndale City Hall	Ferndale, 1954
D-7	Olympia Highway Lab.	Olympia, 1949
D-8	Olympia Highway Lab.	Pacific N.W., 1965

TABLE 3D-2

RECORDS USED FOR UPWARD WAVE PROPAGATION ANALYSES

Analysis No.	Recording Station	Earthquake
U-1	Temblor	Parkfield, 1966
U-2	C.I.T. Seismological Lab.	San Fernando, 1971
U-3	Lake Hughes, No. 4	San Fernando, 1971
U-4	Griffith Park	San Fernando, 1971
U-5	San Luis Obispo	Parkfield, 1966
U-6	Taft	Kern County, 1952
U-7	Castaic	San Fernando, 1971

TABLE 3D-3

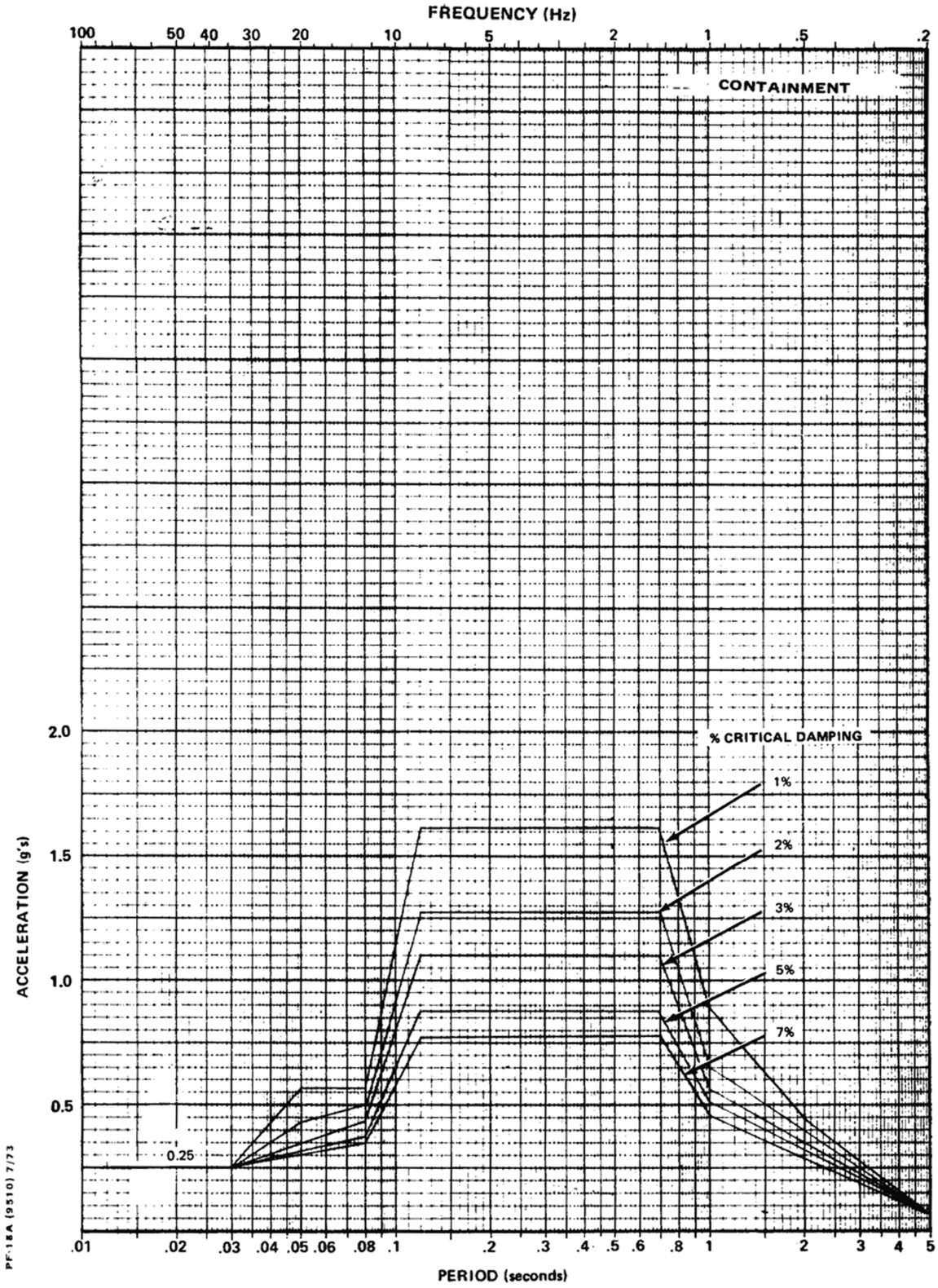
COMPUTED VARIATIONS OF PEAK ACCELERATIONS (g)
WITH DEPTH FROM DECONVOLUTION ANALYSES

	Analysis No.								
Depth (ft)	D-1	D-2	D-3	D-4	D-5	D-6	D-7	D-8	Average
0	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20
10	0.18	0.18	0.19	0.19	0.20	0.19	0.18	0.17	0.18
20	0.15	0.15	0.19	0.18	0.19	0.19	0.16	0.13	0.17
30	0.15	0.14	0.18	0.17	0.18	0.18	0.15	0.12	0.16
40	0.14	0.12	0.16	0.16	0.19	0.18	0.13	0.11	0.15
51	0.12	0.11	0.15	0.15	0.16	0.17	0.13	0.10	0.14
62	0.10	0.09	0.13	0.14	0.14	0.16	0.12	0.10	0.12
76	0.09	0.09	0.12	0.13	0.12	0.15	0.11	0.11	0.12
92	0.10	0.09	0.13	0.13	0.10	0.14	0.10	0.11	0.13
110	0.10	0.09	0.14	0.12	0.09	0.14	0.08	0.10	0.11
120	0.10	0.09	0.14	0.11	0.08	0.14	0.08	0.10	0.11
140	0.09	0.09	0.13	0.10	0.08	0.13	0.08	0.09	0.10
162	0.10	0.08	0.13	0.10	0.09	0.10	0.07	0.10	0.10
267	0.13	0.15	0.13	0.10	0.10	0.10	0.10	0.09	0.11
600	0.18	0.32	0.52	0.25	0.13	0.17	0.14	0.13	

TABLE 3D-4

COMPUTED VARIATIONS OF PEAK ACCELERATIONS (g)
WITH DEPTH FROM UPWARD WAVE PROPAGATION ANALYSES

	Analysis No.							
Depth (ft)	U-1	U-2	U-3	U-4	U-5	U-6	U-7	Average
0	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20
10	0.19	0.19	0.18	0.19	0.18	0.19	0.18	0.19
20	0.16	0.18	0.16	0.18	0.17	0.17	0.15	0.17
30	0.14	0.17	0.12	0.17	0.16	0.16	0.12	0.15
40	0.11	0.15	0.10	0.15	0.15	0.14	0.10	0.13
51	0.07	0.13	0.11	0.13	0.14	0.13	0.08	0.11
62	0.06	0.11	0.12	0.11	0.13	0.12	0.07	0.10
76	0.07	0.09	0.14	0.09	0.11	0.11	0.06	0.10
92	0.08	0.10	0.14	0.09	0.09	0.11	0.06	0.10
110	0.08	0.10	0.14	0.09	0.11	0.11	0.06	0.10
120	0.08	0.10	0.13	0.10	0.11	0.16	0.07	0.11
140	0.09	0.09	0.11	0.10	0.11	0.16	0.07	0.12
162	0.09	0.10	0.08	0.09	0.11	0.11	0.08	0.09
267	0.11	0.10	0.11	0.10	0.13	0.11	0.10	0.11
600	0.12	0.12	0.13	0.13	0.13	0.13	0.09	0.12



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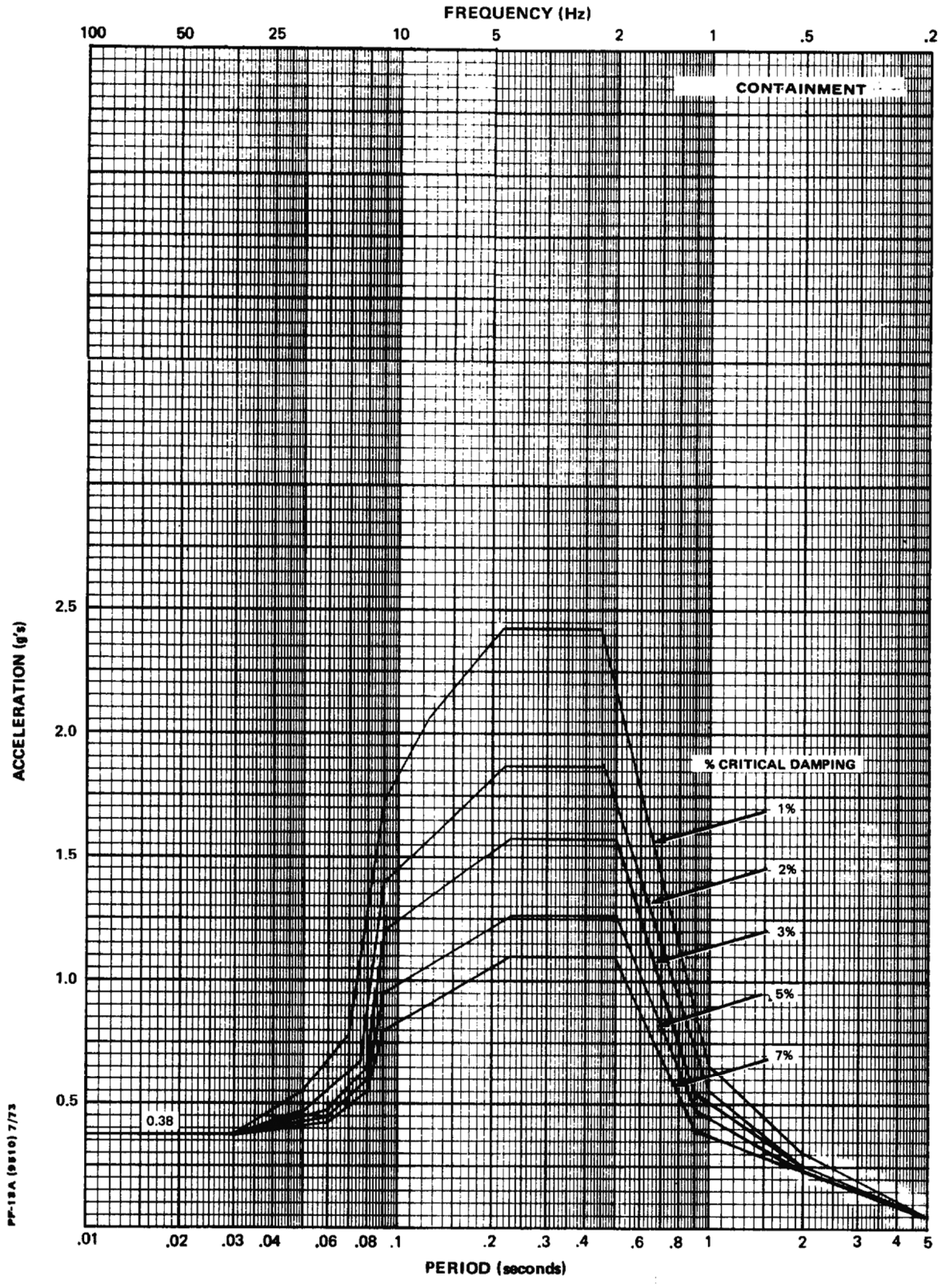
SAFE SHUTDOWN EARTHQUAKE
HORIZ. ACCEL. RESPONSE SPECTRA
EL. 169 FT 0 IN.

FIGURE 3D–1



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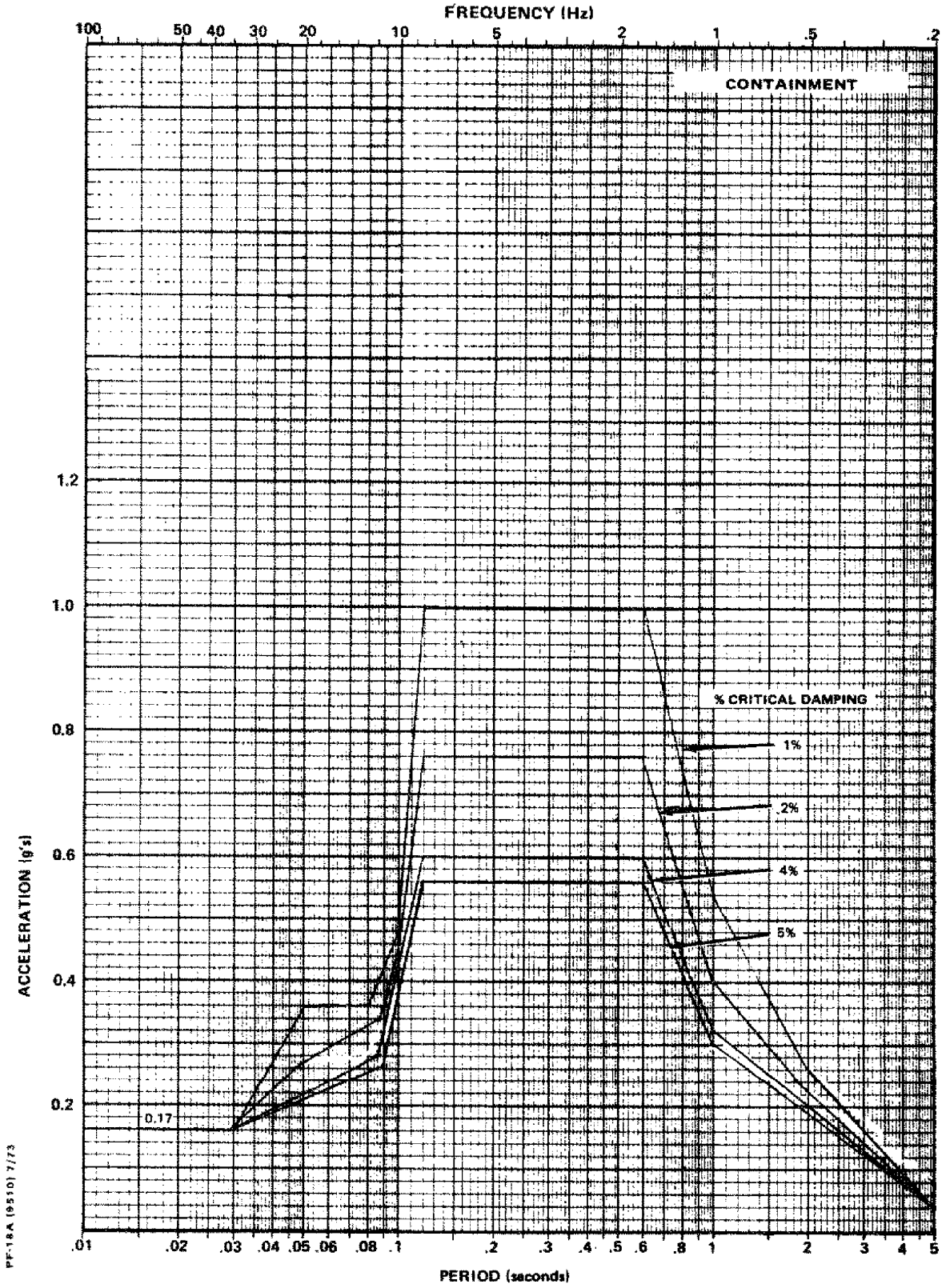
EL. 169 FT 0 IN.

FIGURE 3D–2



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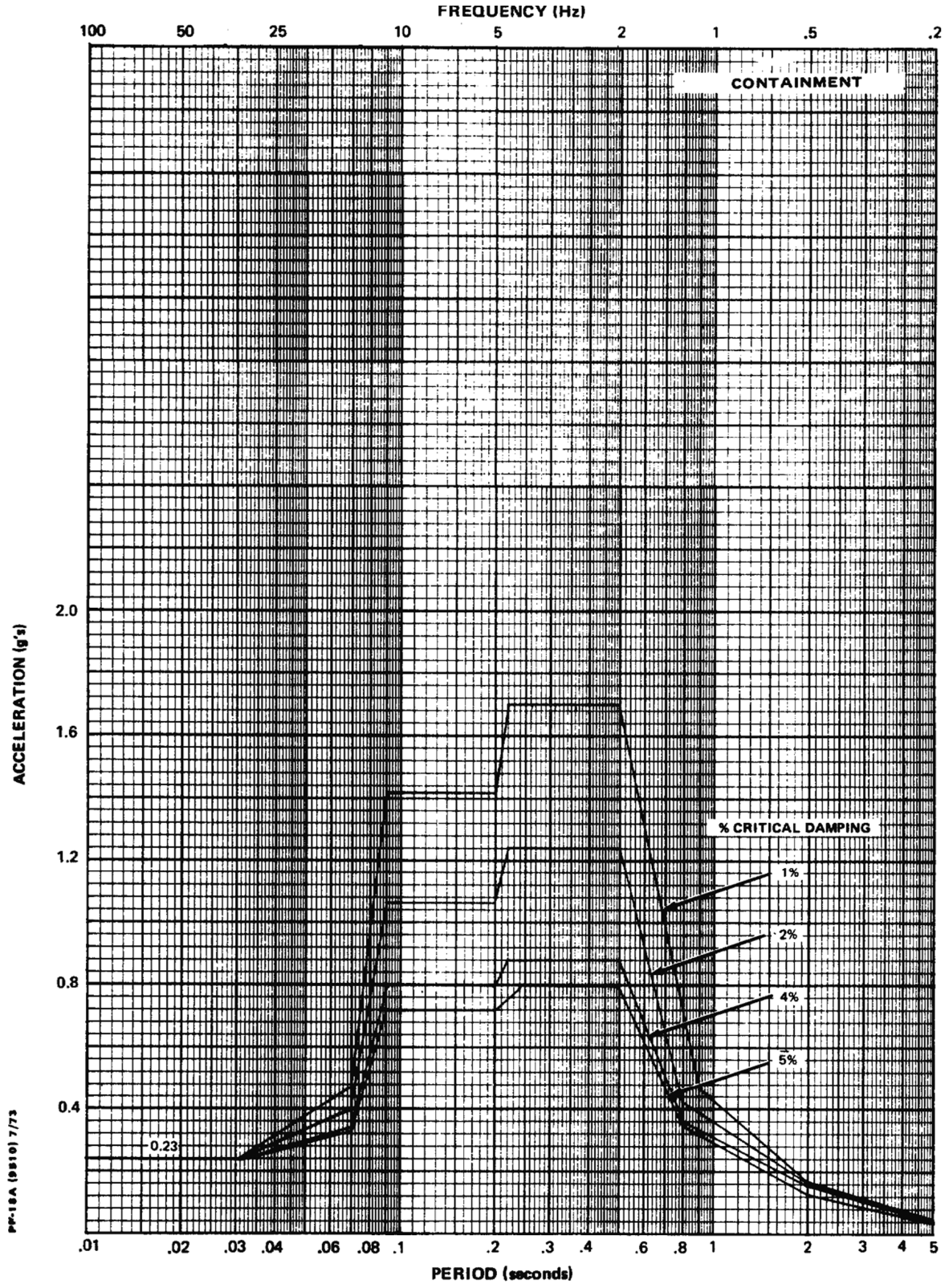
EL. 169 FT 0 IN.

FIGURE 3D–3



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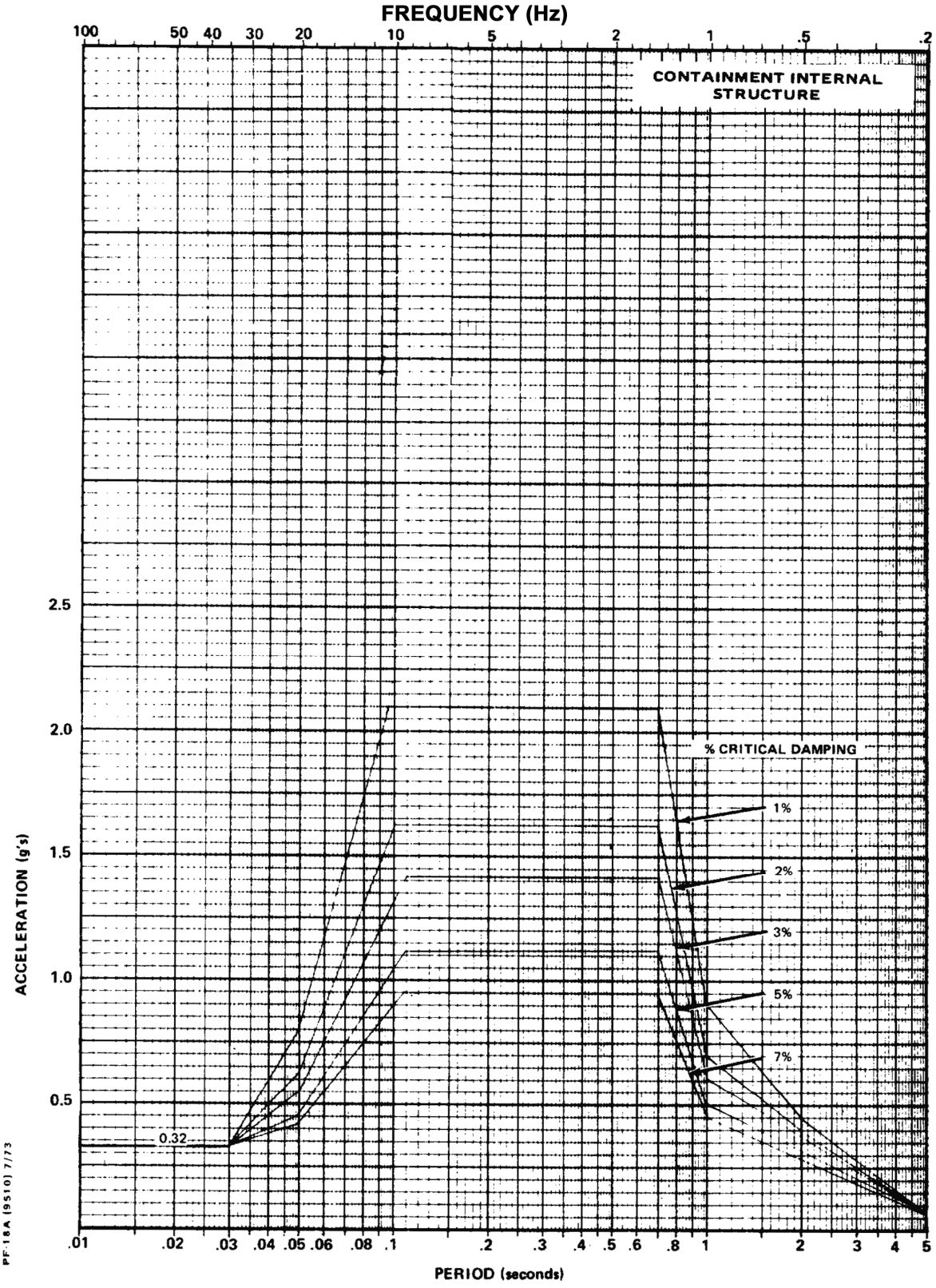
EL. 169 FT 0 IN.

FIGURE 3D–4



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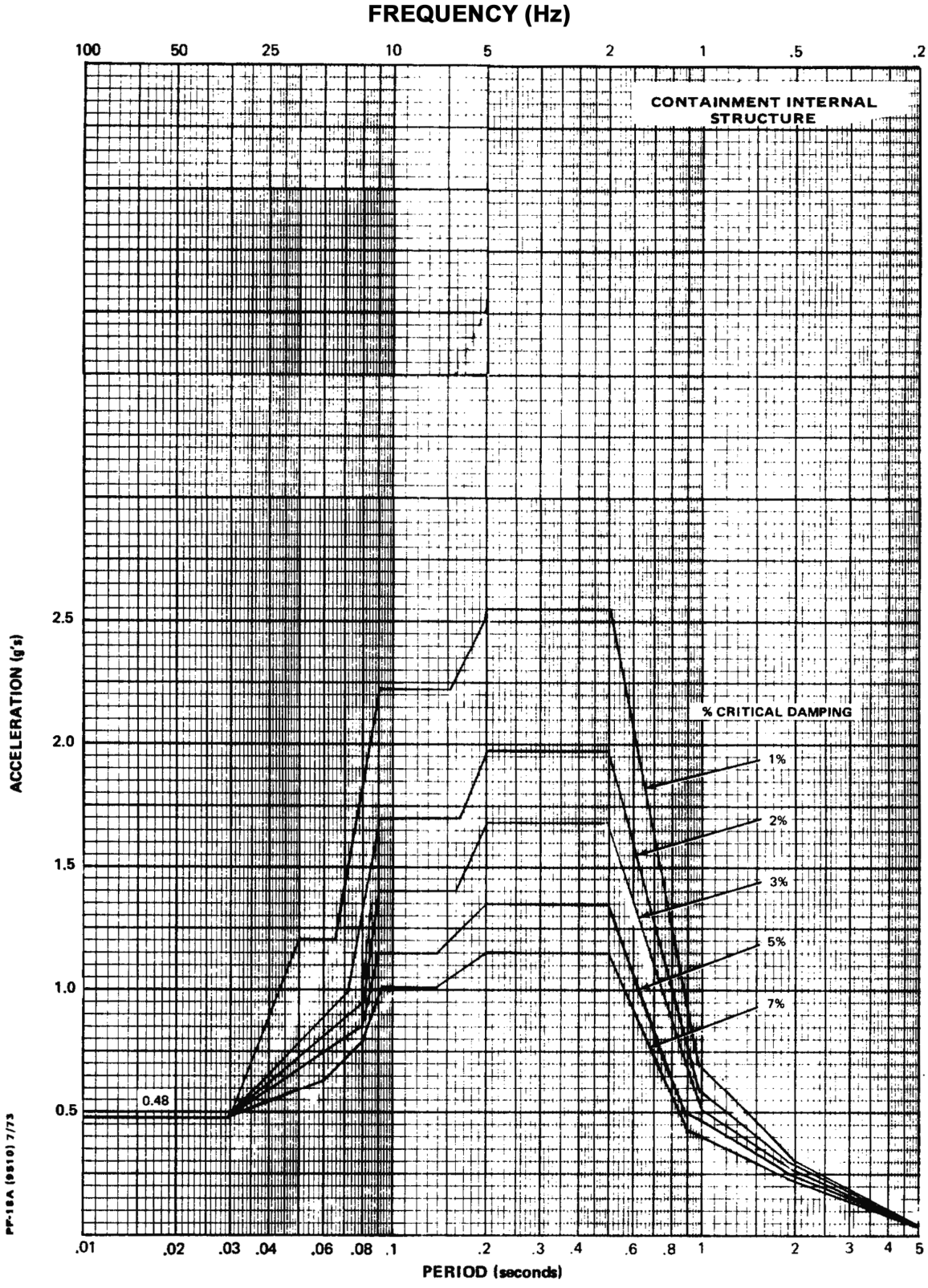
HORIZ. ACCEL. RESPONSE SPECTRA
EL. 220 FT 0 IN.

FIGURE 3D–5



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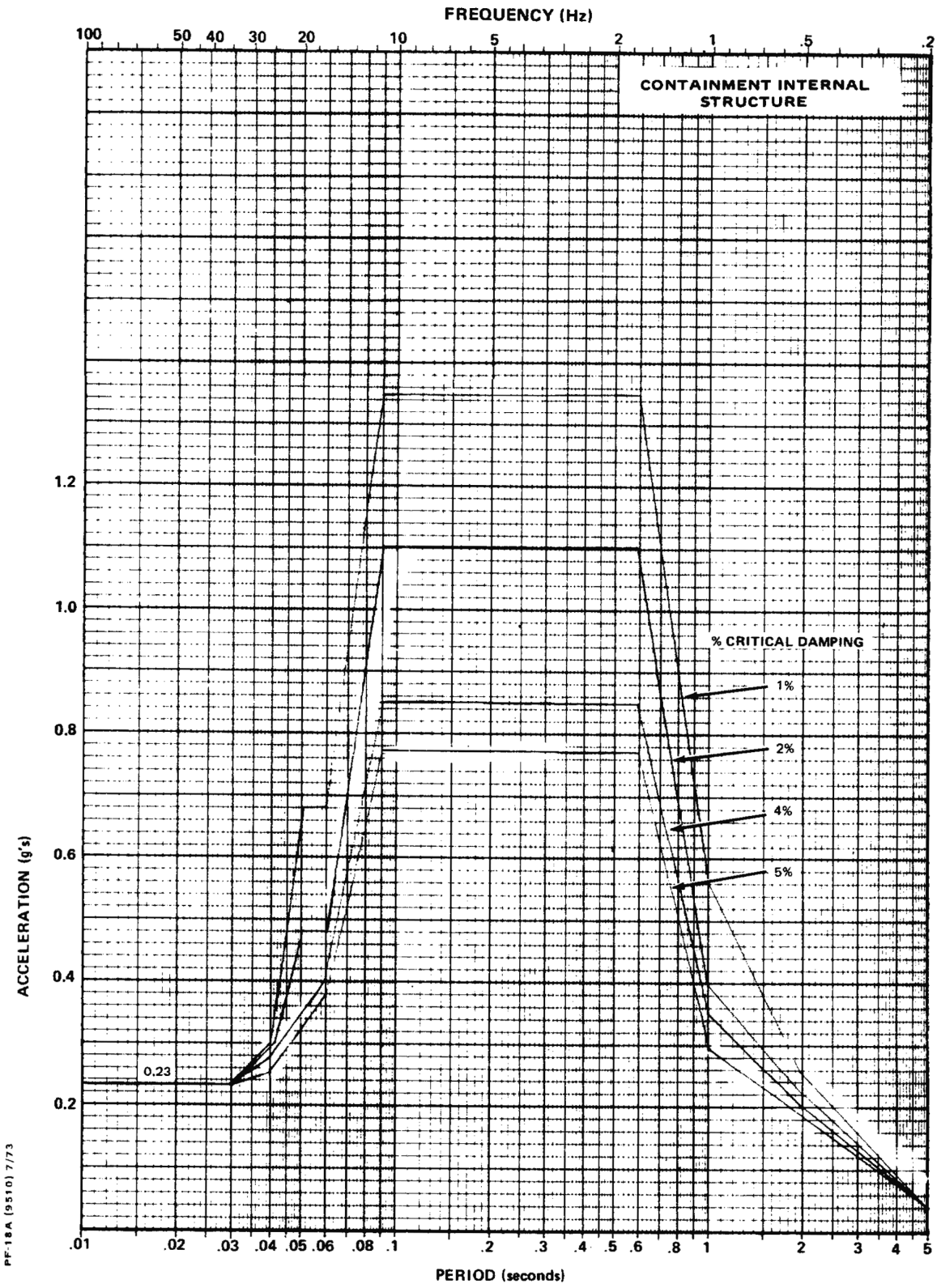
VERTICAL ACCEL. RESPONSE SPECTRA
EL. 220 FT 0 IN.

FIGURE 3D–6



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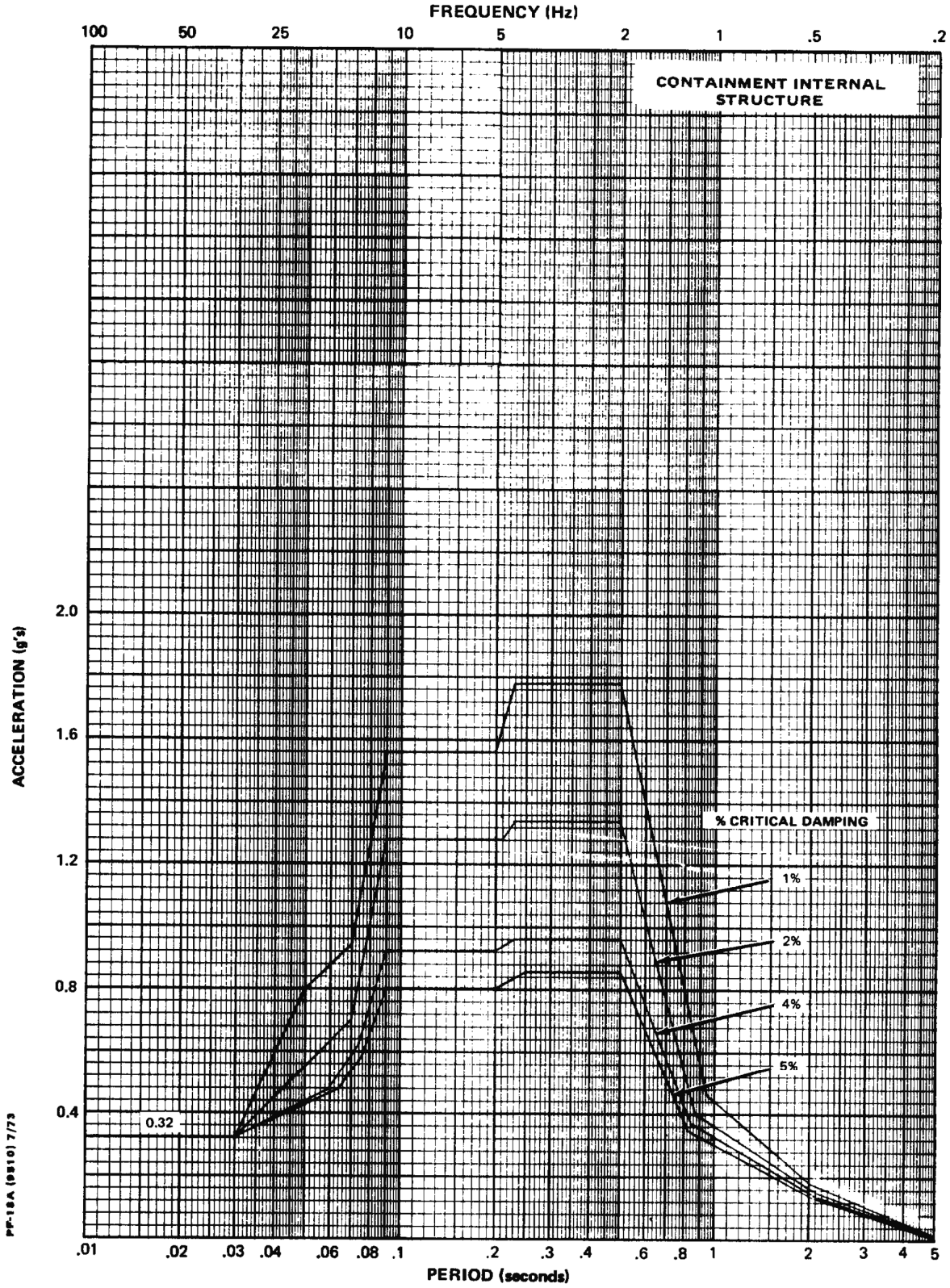
HORIZ. ACCEL. RESPONSE SPECTRA
EL. 220 FT 0 IN.

FIGURE 3D–7



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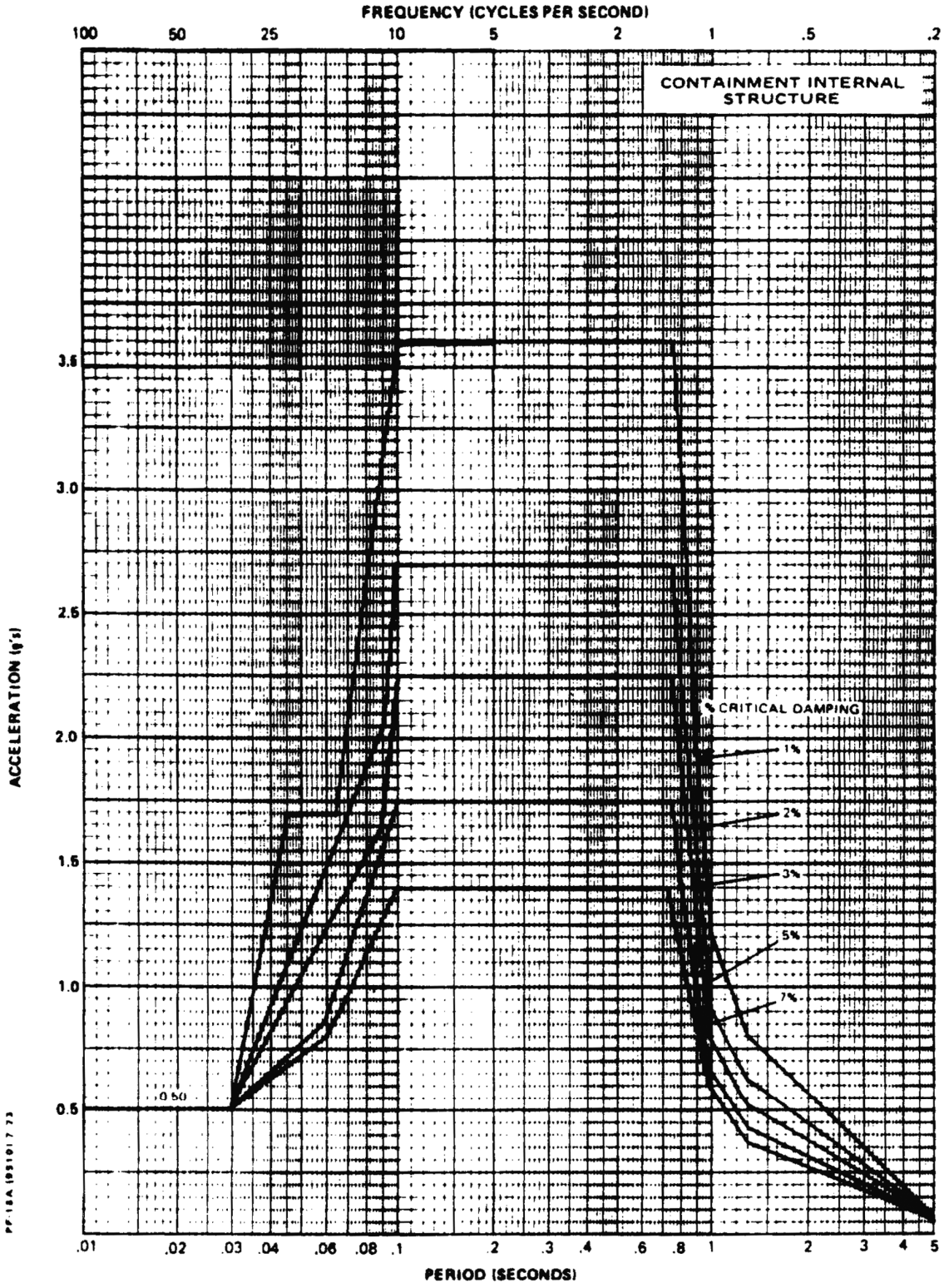
EL. 220 FT 0 IN.

FIGURE 3D–8



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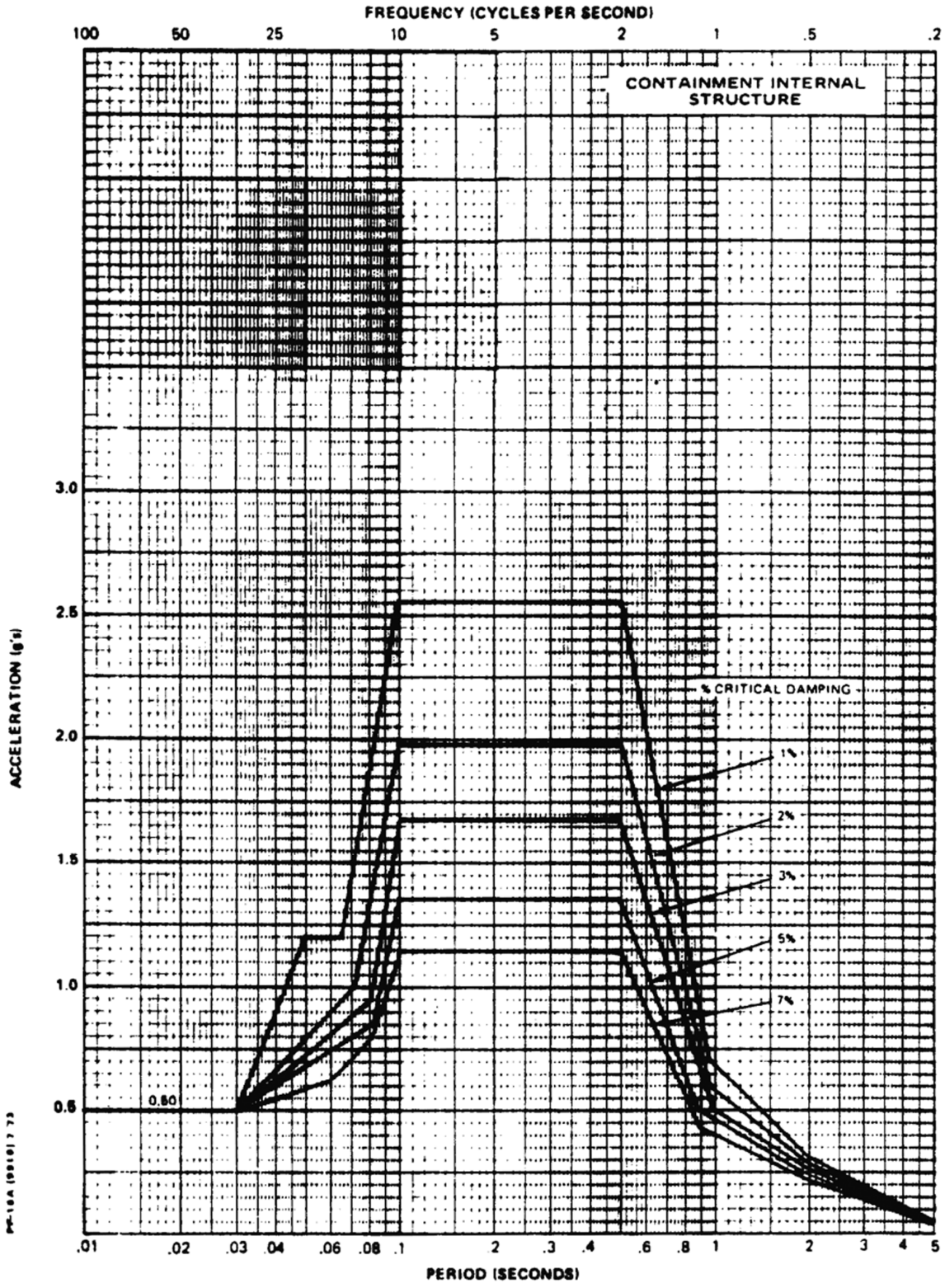
SAFE SHUTDOWN EARTHQUAKE
HORIZ. ACCEL. RESPONSE SPECTRA
EL. 261 FT 0 IN.

FIGURE 3D–9



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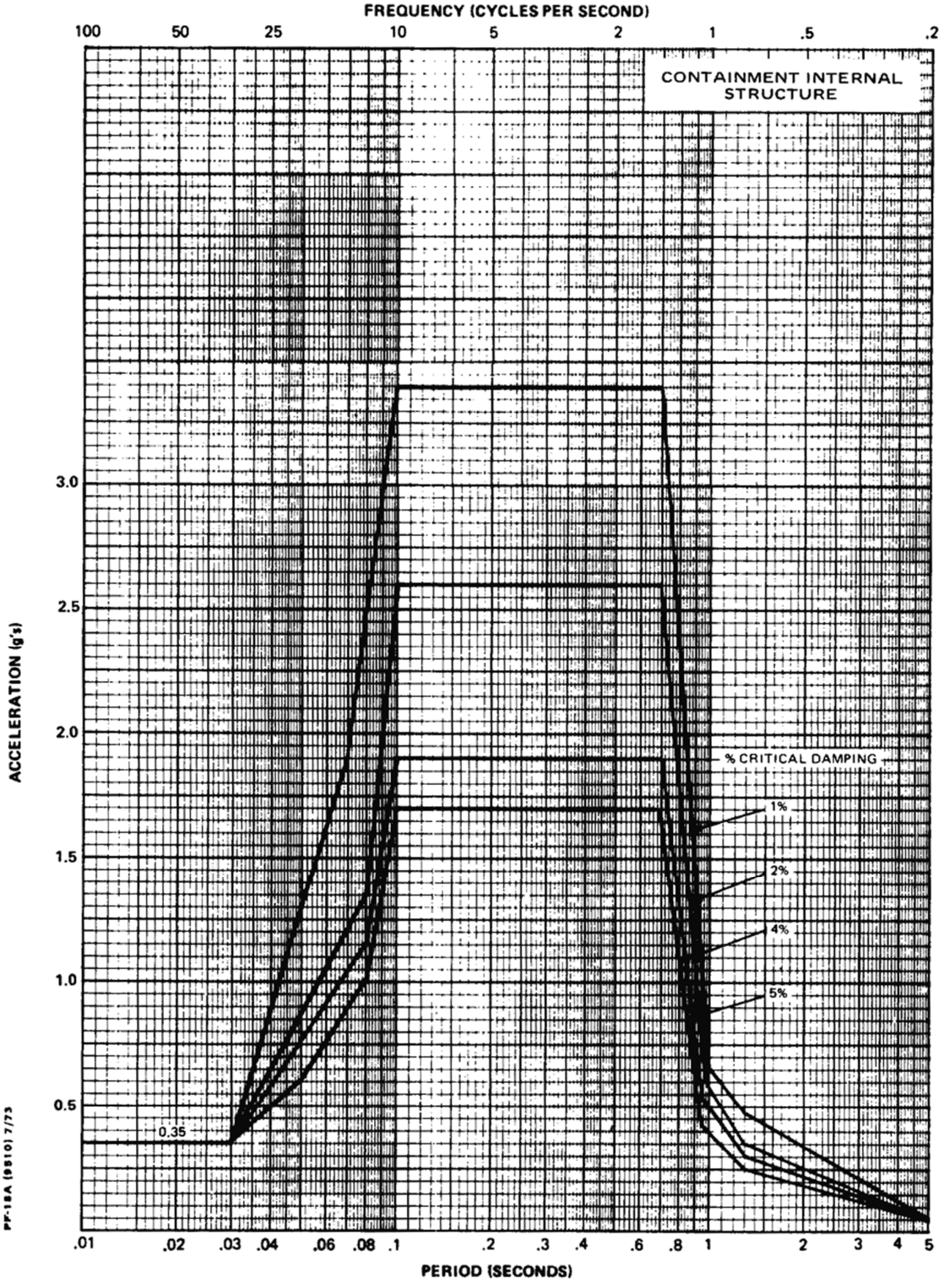
VERTICAL ACCEL. RESPONSE SPECTRA
EL. 261 FT 0 IN.

FIGURE 3D–10



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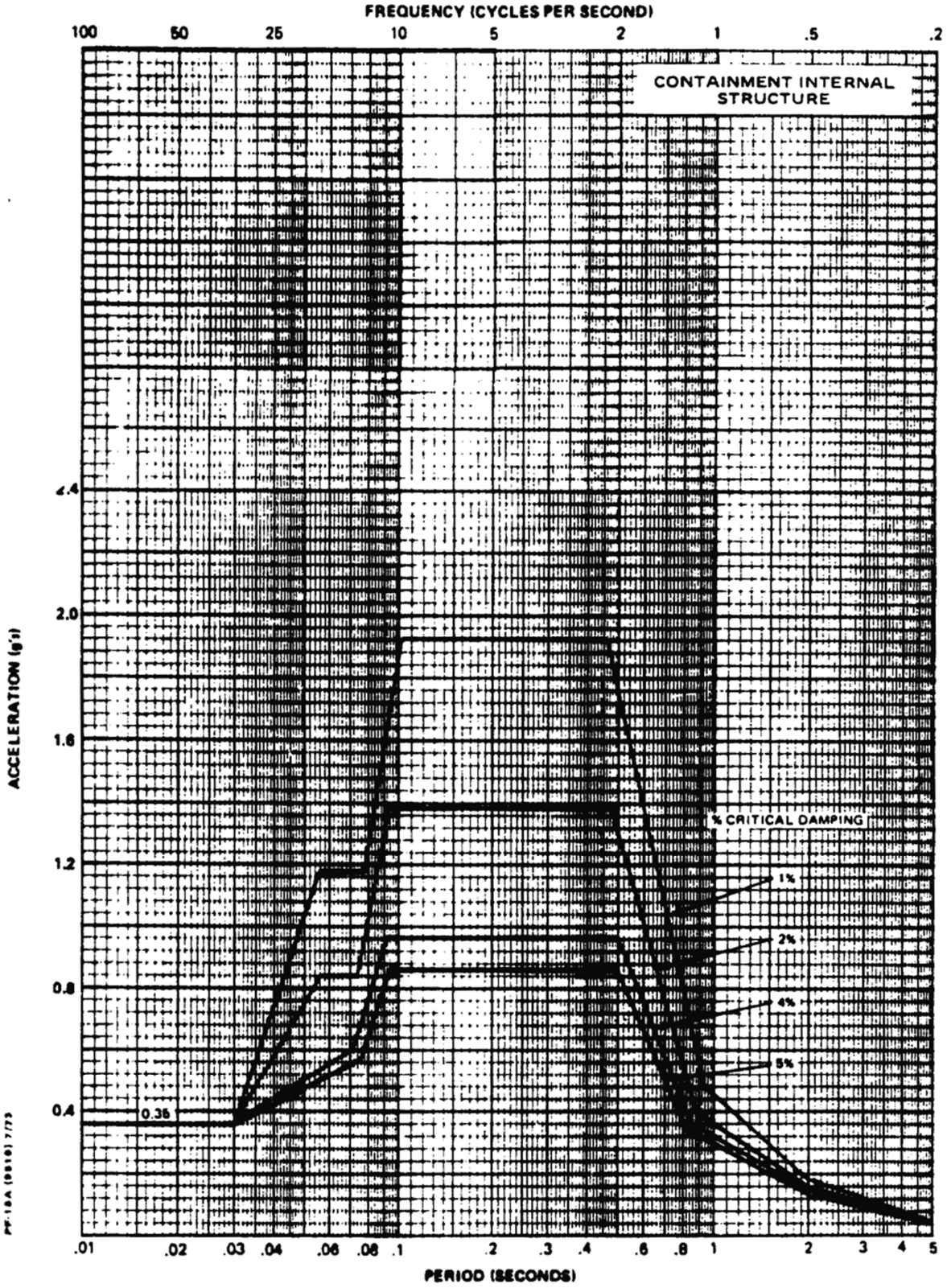
HORIZ. ACCEL. RESPONSE SPECTRA
EL. 261 FT 0 IN.

FIGURE 3D–11



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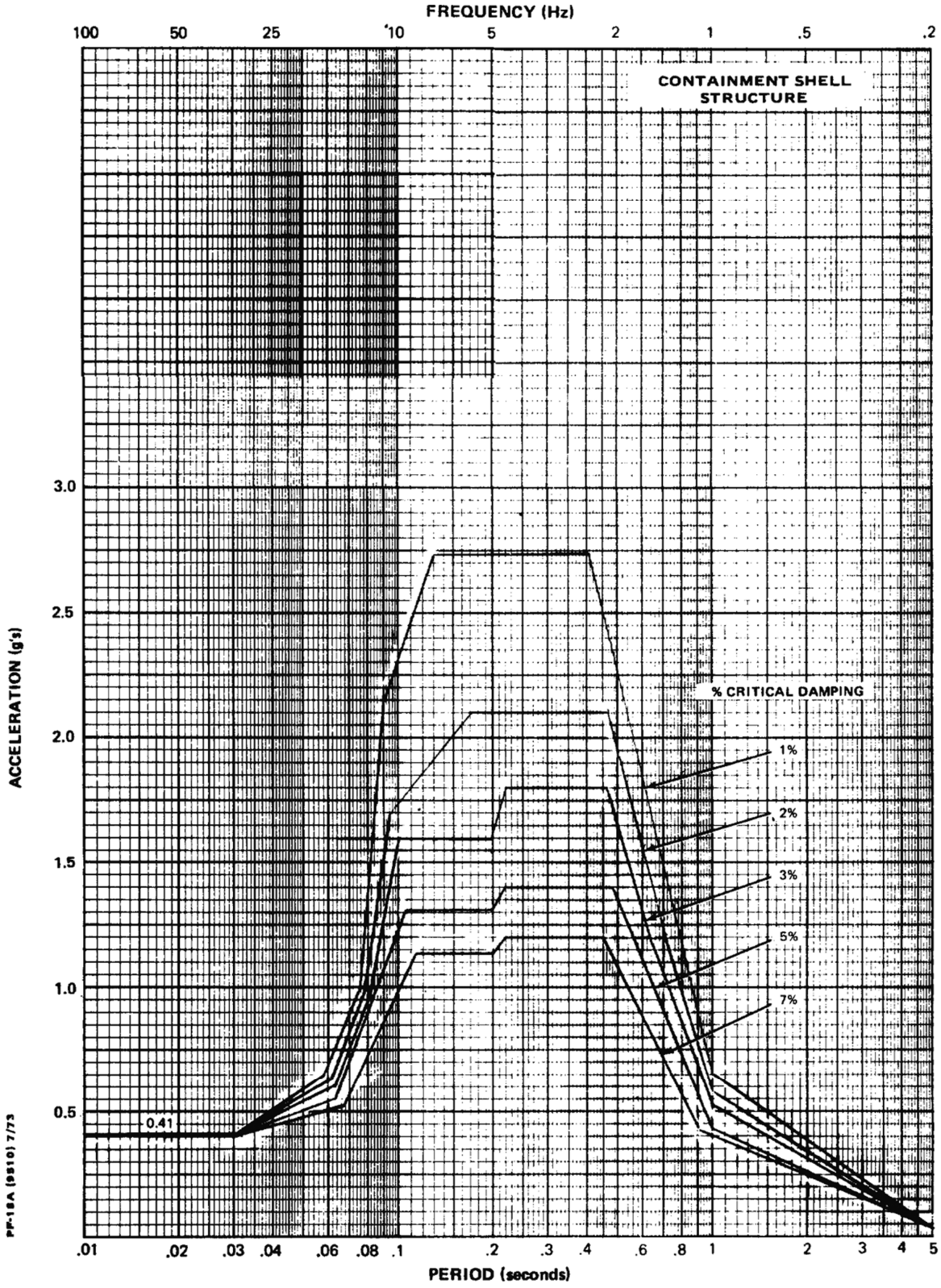
EL. 261 FT 0 IN.

FIGURE 3D–12



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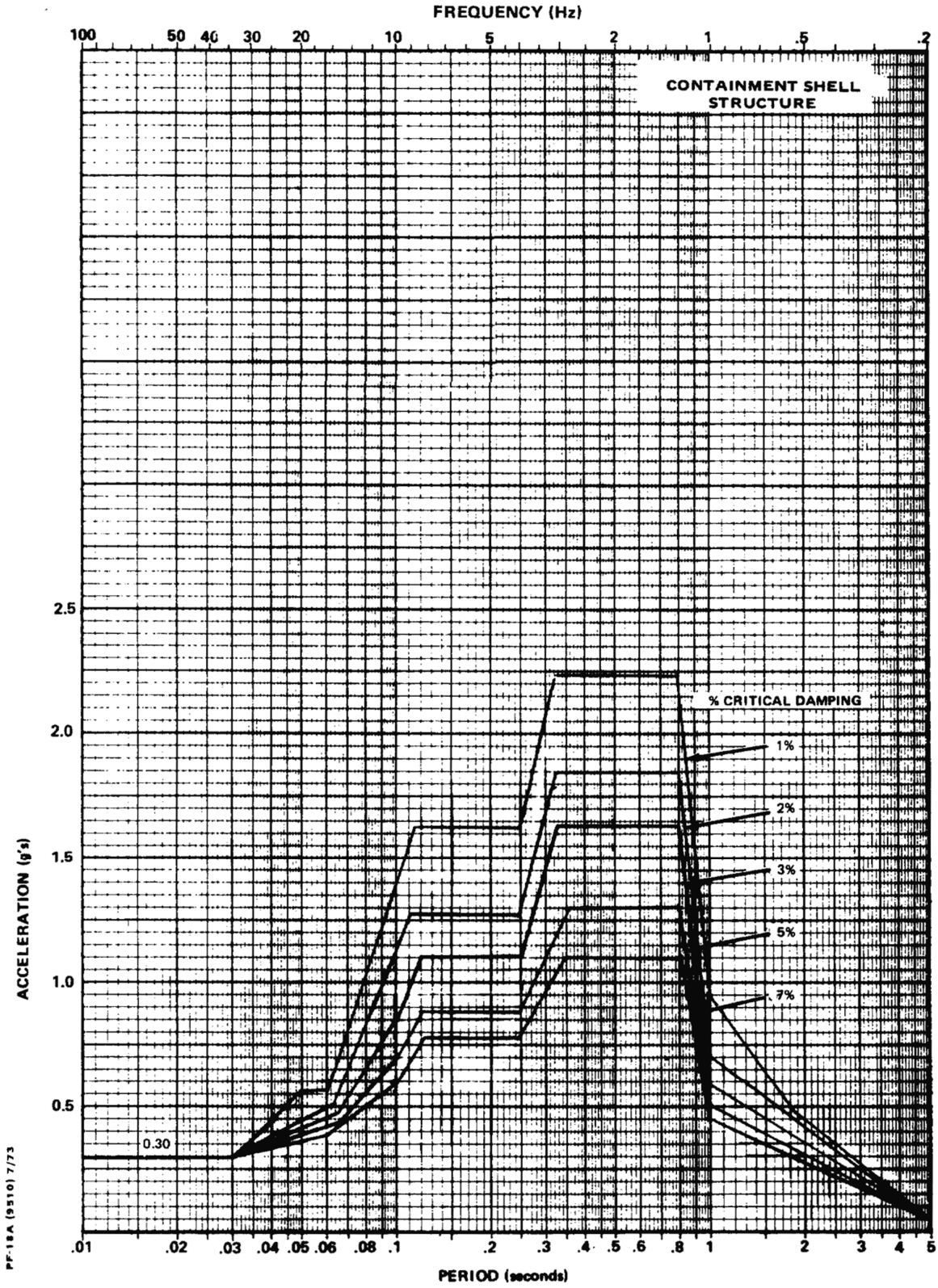
SAFE SHUTDOWN EARTHQUAKE
VERTICAL ACCEL. RESPONSE SPECTRA
EL. 220 FT 0 IN.

FIGURE 3D–13



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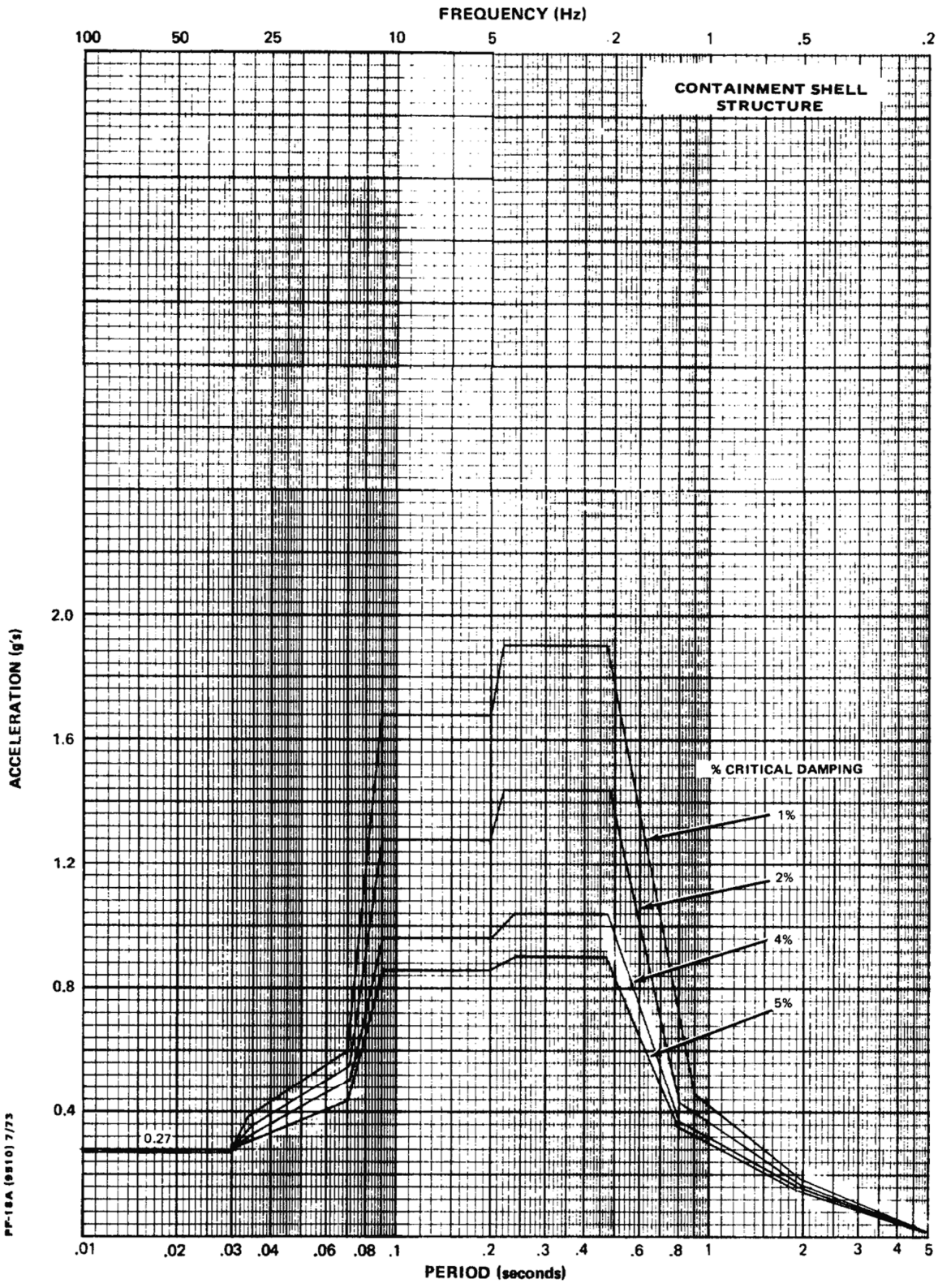
HORIZ. ACCEL. RESPONSE SPECTRA
EL. 220 FT 0 IN.

FIGURE 3D–14



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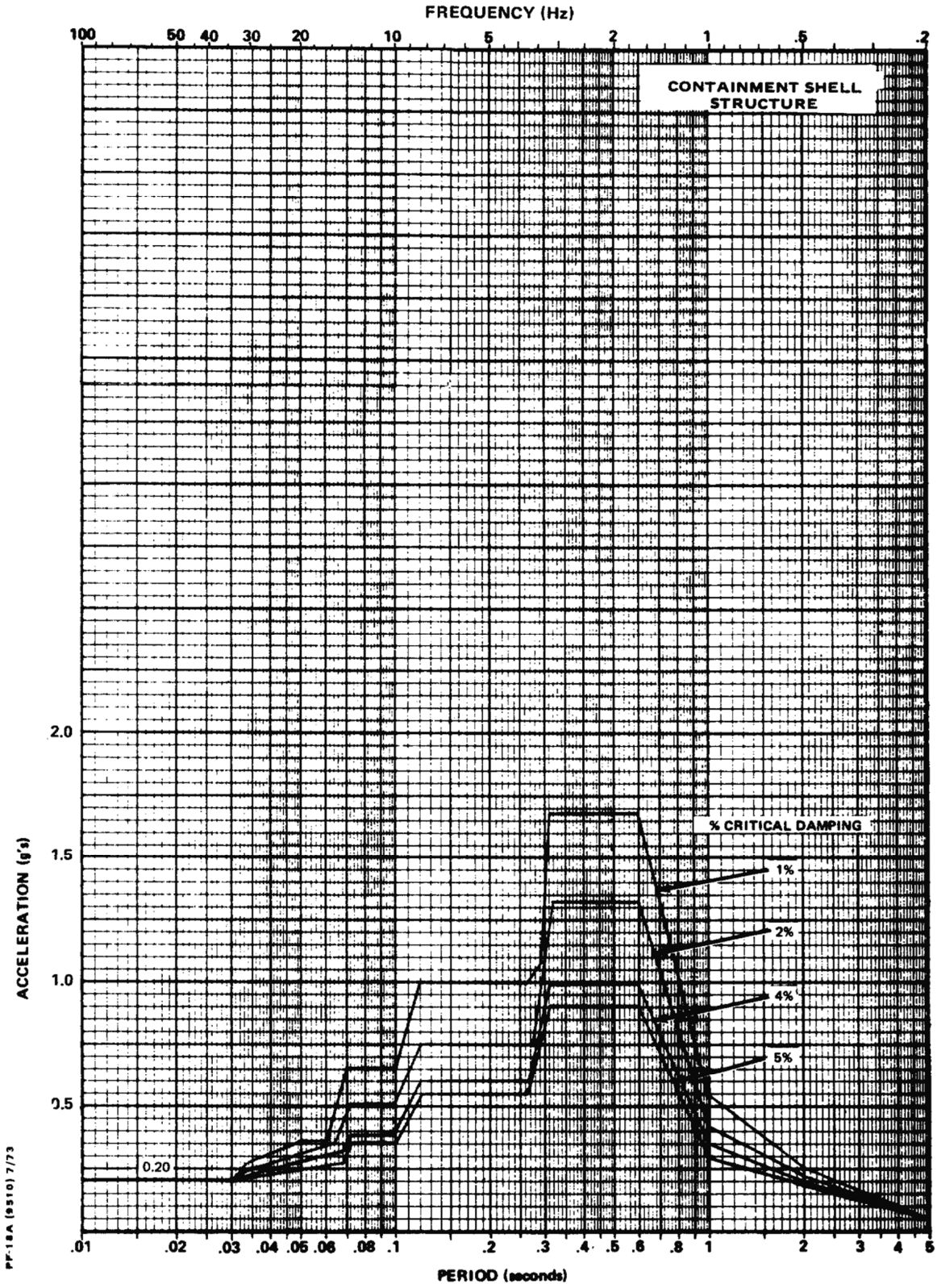
VERTICAL ACCEL. RESPONSE SPECTRA
EL. 220 FT 0 IN.

FIGURE 3D–15



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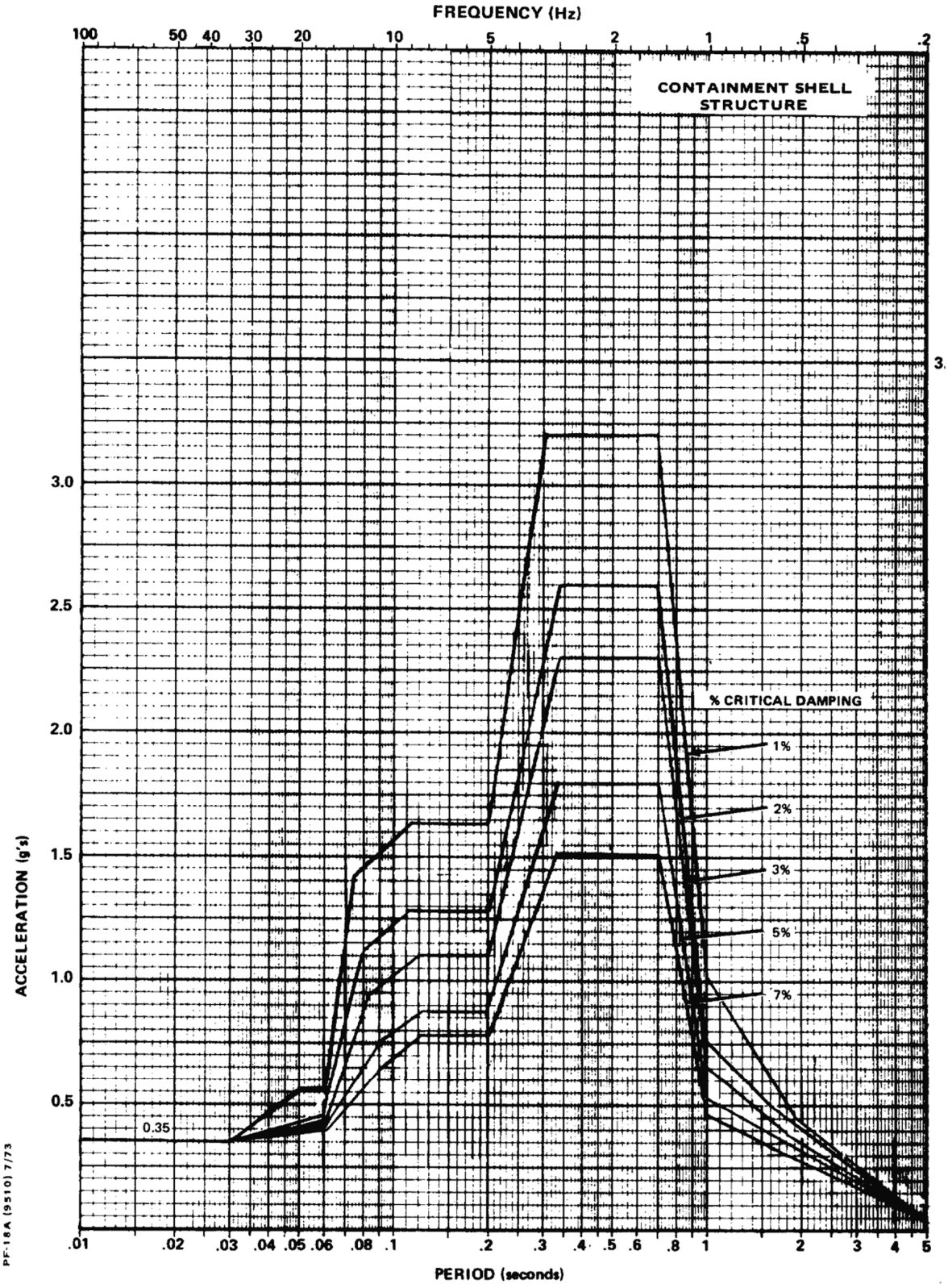
HORIZ. ACCEL. RESPONSE SPECTRA
EL. 220 FT 0 IN.

FIGURE 3D–16



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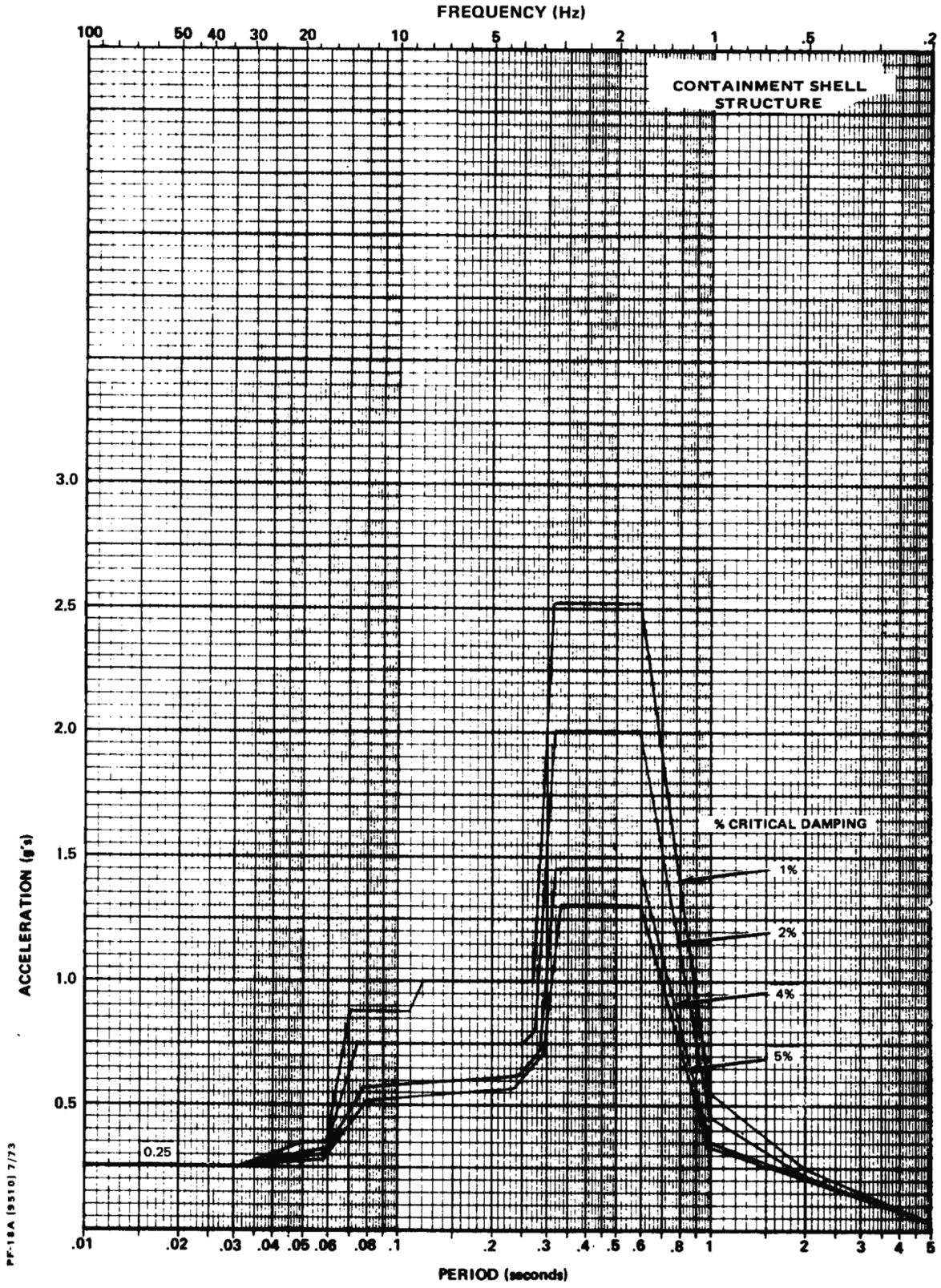
UNIT 1 AND UNIT 2

SAFE SHUTDOWN EARTHQUAKE
HORZ. ACCEL. RESPONSE SPECTRA
EL. 258 FT 0 IN.

FIGURE 3D–17



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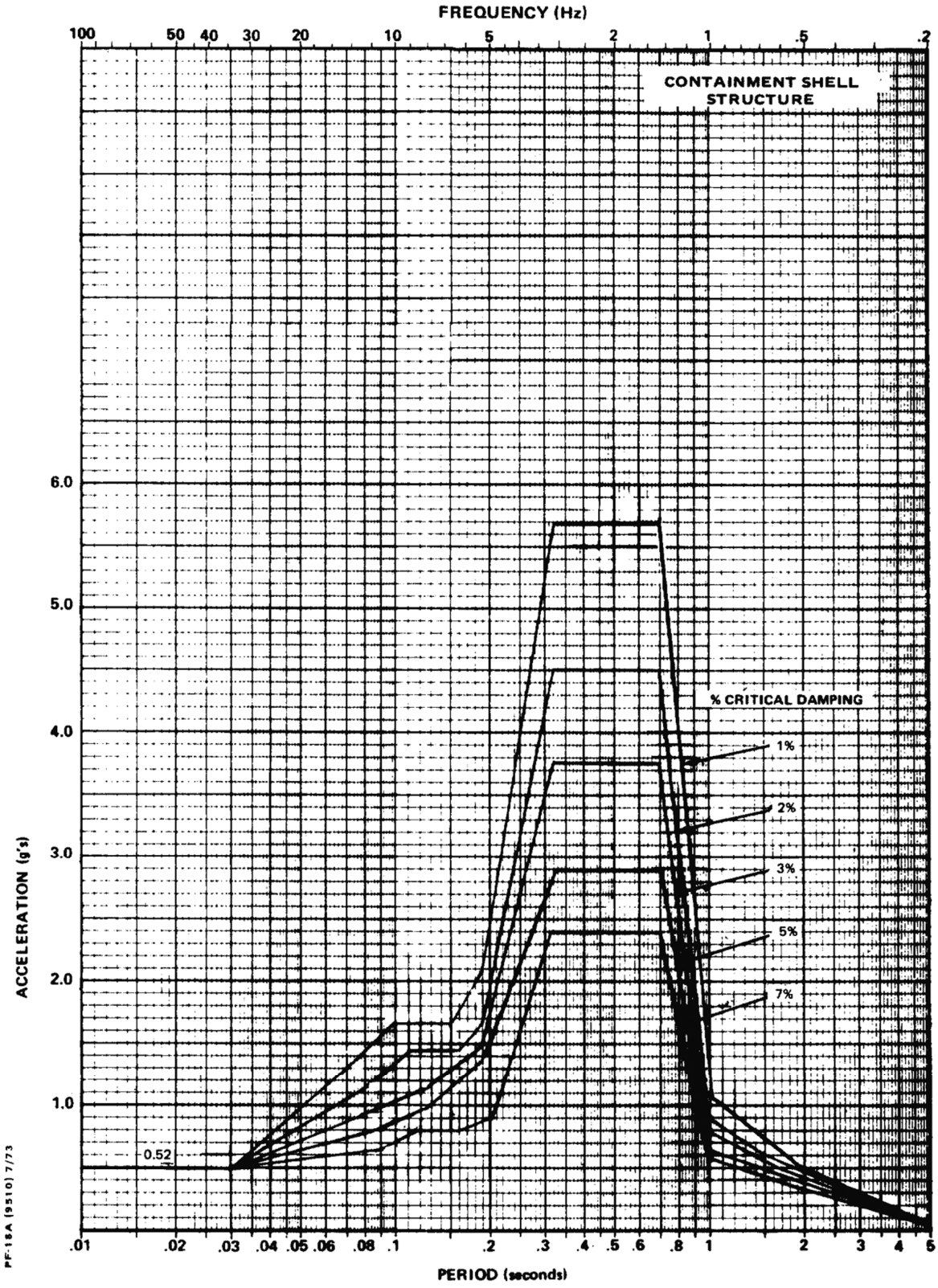
OPERATING BASIS EARTHQUAKE
HORIZ. ACCEL. RESPONSE SPECTRA
EL. 258 FT 0 IN.

FIGURE 3D–18



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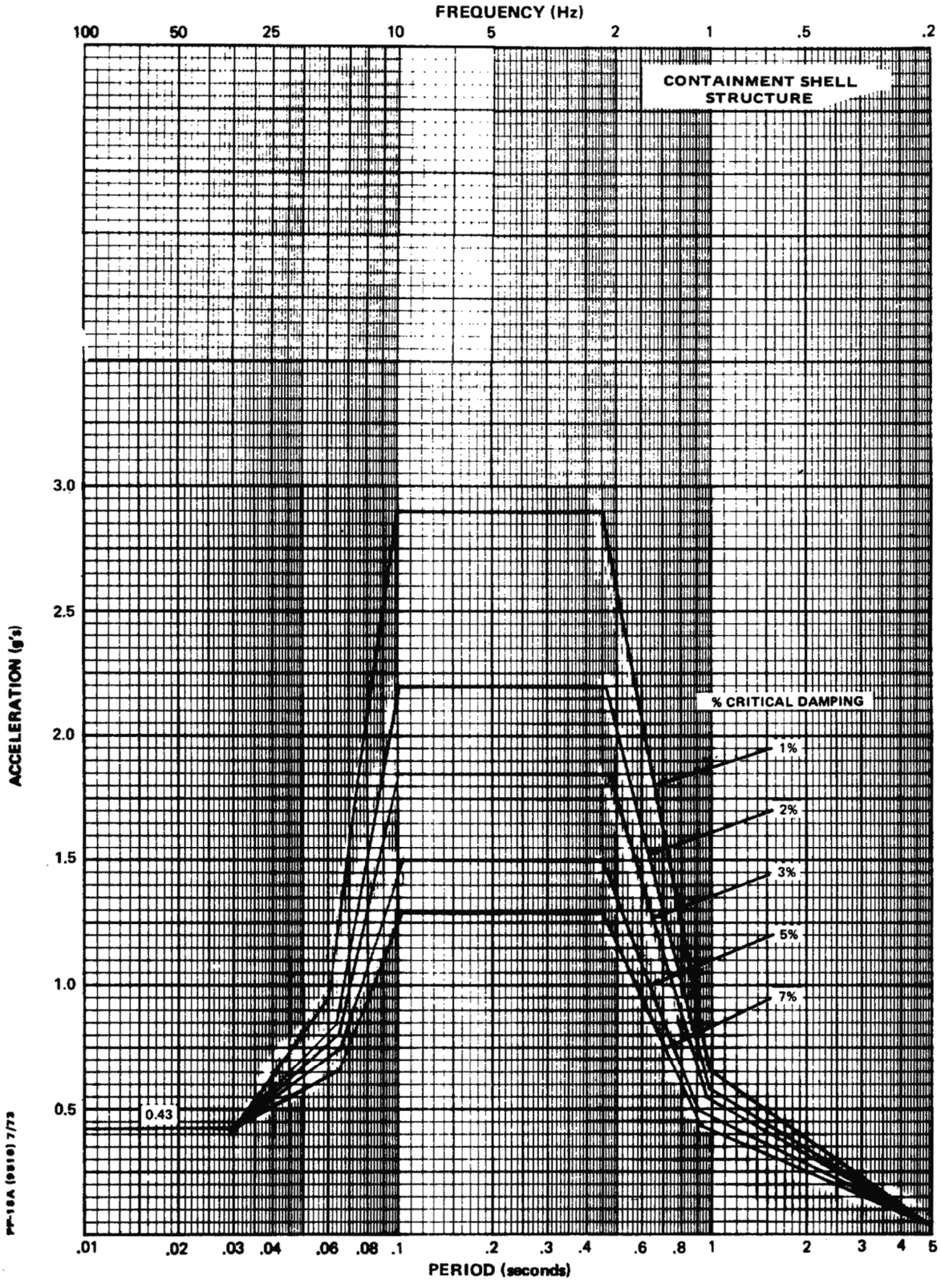
UNIT 1 AND UNIT 2

SAFE SHUTDOWN EARTHQUAKE
HORIZ. ACCEL. RESPONSE SPECTRA
EL. 323 FT 0 IN.

FIGURE 3D–19



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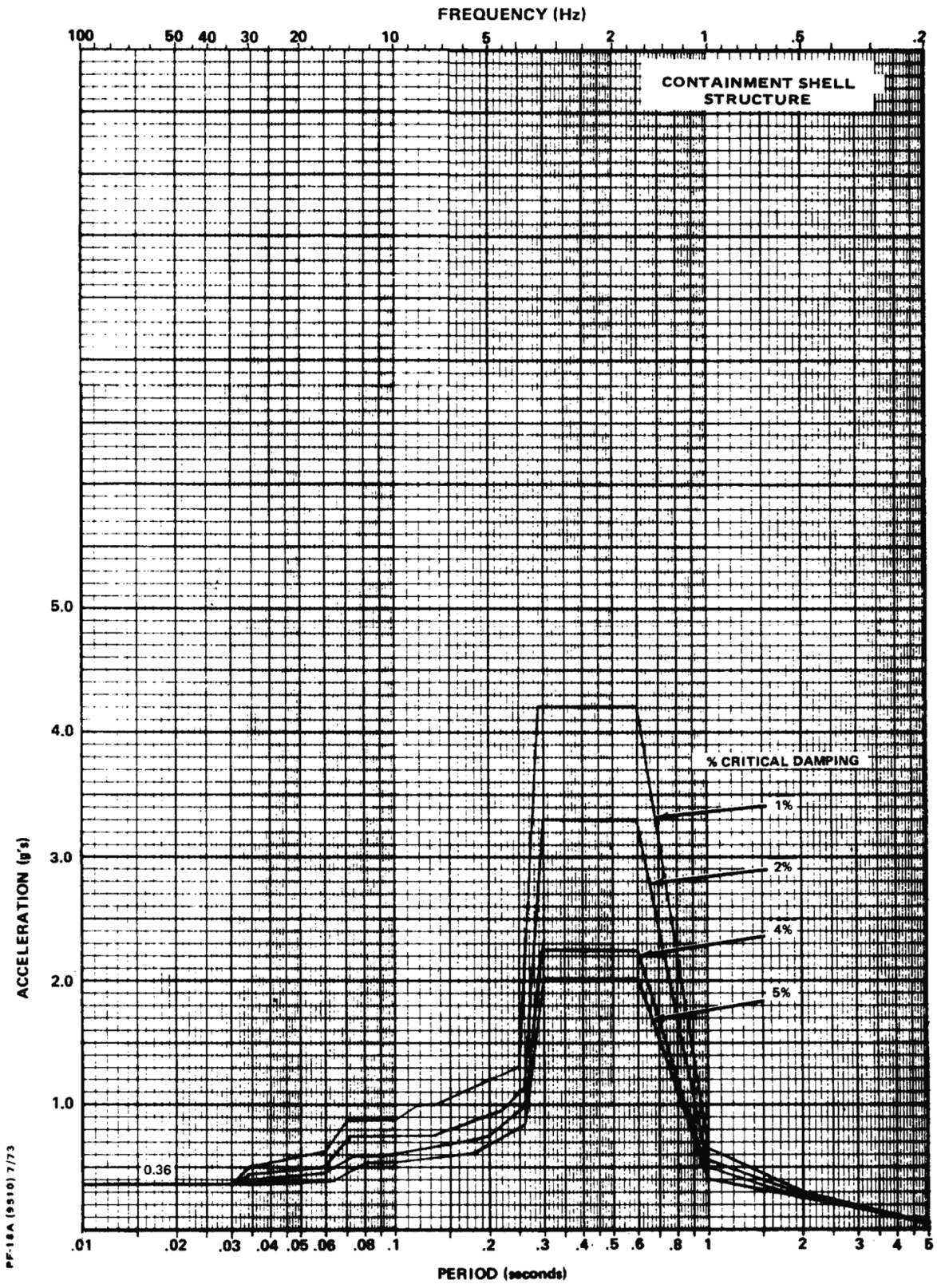
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	SAFE SHUTDOWN EARTHQUAKE VERTICAL ACCEL. RESPONSE SPECTRA EL. 323 FT 0 IN.
	FIGURE 3D–20



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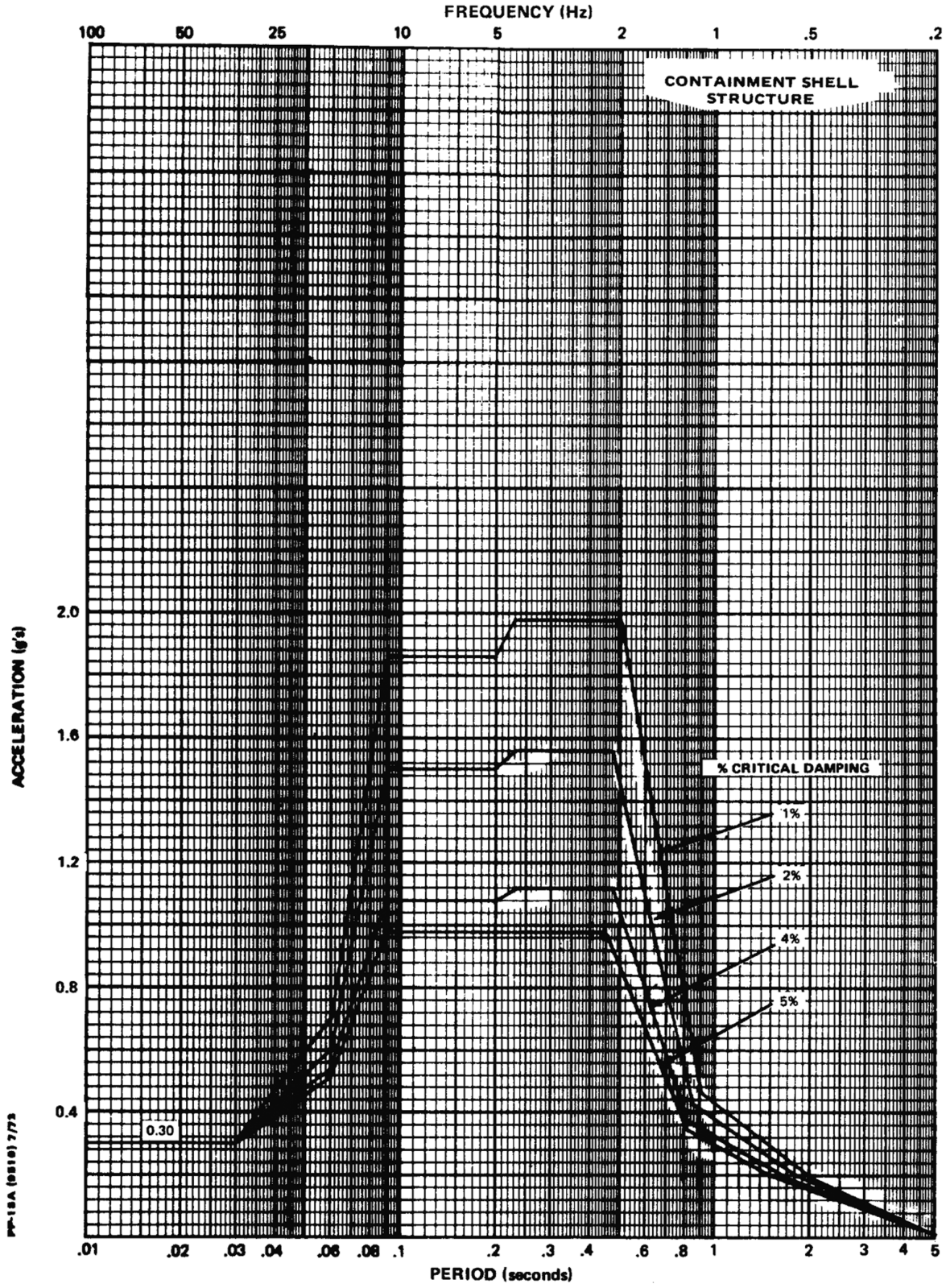
UNIT 1 AND UNIT 2

OPERATING BASIS EARTHQUAKE
HORIZ. ACCEL. RESPONSE SPECTRA
EL. 323 FT 0 IN.

FIGURE 3D–21



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UNIT 1 AND UNIT 2

OPERATING BASIS EARTHQUAKE
VERTICAL ACCEL. RESPONSE SPECTRA
EL. 323 FT 0 IN.

FIGURE 3D–22